

SMITHIAN FOLD THEORY V3

The Smithian Fold Theory of Everything

An exhaustive preliminary monograph from There Is No Nothing to the current
computational-proof frontier

Ernos Labs

Open Source Science Platform and Knowledge Tree

Maria Smith

Independent researcher and founder, Ernos Labs

Maria.Smith.Sftoe@gmail.com

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FULL-SCALE PRELIMINARY VERSION - COMPUTATIONAL PROGRAMMES CONTINUE

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An exhaustive preliminary monograph from *There Is No Nothing* to the current computational-proof frontier

Maria Smith

Independent researcher and founder, Ernos Labs

Maria.Smith.Sftoe@gmail.com

<https://github.com/MettaMazza/ernos-labs-sft-platform>

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Abstract

Smithian Fold Theory V3 is a clean-room, machine-auditable reconstruction of a single dependency-ordered scientific system. It begins with the operational root theorem *there is no nothing*: a purported counterexample either presents no statement, denial, record or proof object and therefore supplies no counterexample, or it presents an occurrence and is not nothing. The admitted occurrence forces its structural self-whole, the One; exact positive count, part, distinction, return and the minimal reversible Fold are then generated without importing arithmetic, geometry, probability, physical constants or a continuum as premises. Every later result is separately registered. No later law is permitted to become a retroactive assumption of an earlier proof.

The V3 corpus reported here contains **2,751 model-admitted claims** generated from **892,246 registered candidates**, with **2,751 unique survivors** and **11,004 passed controls**. All 2,751 receipt identities match their registered claim identities; all 2,751 candidate censuses reconstruct; every admitted claim traces to the single root; and the current ledger contains no missing declared dependency. These totals cover the premise-free methods and Foundation results; current full-field Mathematics, Information Science, Classical Computation, Reversible and Quantum Computation, Physics, Chemistry and Materials Science; completed foundations and dated return families in Biology, Medicine, Consciousness and Cognitive Science, Earth and Environmental Sciences, Astronomy and Cosmology, Social and Collective Systems, and Engineering Translation; and the current twelve-claim Cross-Branch Synthesis.

The work separates exact derivation from implementation, measurement, reconstruction, correspondence, compatibility, support, confirmation and adverse evidence. Candidate generation and elimination establish formal status only at their declared boundaries. Natural-science claims additionally retain source identity, measurement method, uncertainty, preregistration, chronology, target custody, falsification conditions and every favourable, adverse, null, failed, missing or unavailable result. Current branch status is therefore non-uniform by design: seven field-wide branches are complete to their frozen current-evidence scopes; Biology is at 82 of 424 registered full-field obligations; later natural, cognitive, collective and engineering branches possess complete foundations but do not yet claim frozen full-field closure.

The preliminary monograph also reports the computational-proof programme. Protein Fold V3 has an exact sequence representation, complete interaction custody, a six-axis joint relation, a verified 42,173,082-candidate parent search and a separately closed 583-state high-attraction slice, while the remaining attraction bands, global fixed-geometry frontier, complete whole-chain recurrence and registered 100-target AlphaFold comparison remain open. Chess Fold and Go

Fold retain measured V2 implementation results and exact historical receipts, while their clean V3 recreations have not yet started in the current repository. Unison Fold AI retains its V2 implementation record and a V3 frontier scaffold; mathematical closure is open, empirical validation is NOT_RUN, and deployment is NOT_AUTHORIZED.

This publication is deliberately preliminary because the computational programmes, Biology continuation, downstream full-field censuses, final one-owner lineage merge, canonical publication reconciliation and final heavy verification remain active. Preliminary denotes a versioned scientific state, not a short summary. This monograph presents the complete current dependency argument at scale, binds it to a complete claim inventory, scientific audit and machine archive, and is designed to be amended by later versions without erasing its chronology.

Keywords: Smithian Fold Theory; theory of everything; premise-free foundation; parameter-free derivation; computational proof; exact arithmetic; physical constants; open science; machine-verifiable science; protein folding; AlphaFold; reversible computation; quantum computation; knowledge tree.

Headline findings

1. **One operational root.** The current corpus is a directed acyclic dependency system rooted in SFT-ROOT-THERE-IS-NO-NOTHING. Every one of the 2,751 admitted claims possesses a complete route to that root.
2. **The Fold is generated rather than assumed.** Occurrence, the structural One, exact positive count, exact part and the reversible distinguished partition are admitted in separate enumerated stages. The minimal Fold is the unique survivor of its registered grammar; it is not an imported binary convention.
3. **The laws are exposed as forced selections.** Each admitted claim retains its candidate grammar, exact boundary, candidate count, one decision per candidate, survivor count, uniqueness account, controls, independent reconstruction and receipt identity. A prose statement alone cannot enter the model.
4. **The current corpus is large and fully enumerated at claim level.** The authoritative ledger contains 2,751 admitted claims, 892,246 candidate structures, 2,751 unique survivors and 11,004 passed controls. These are exact repository counts, not estimates or rounded publicity figures.
5. **Seven branches have current full-field closure.** Mathematics contains 323 admitted claims; Information Science 262; Classical Computation 369; Reversible and Quantum Computation 288; Physics 368; Chemistry 281 live claims, including 272 registered discipline obligations and nine separately returned claims; and Materials Science 289. Together these seven papers comprise 2,180 live claims, 744,944 candidates and 8,720 controls.
6. **Later branches have complete foundations without being relabelled as complete fields.** Biology contains 82 admitted claims against a frozen 424-obligation full-field census. Medicine contains 76 current foundation and return claims; Consciousness 77; Earth and Environment 75; Astronomy and Cosmology 77; Social and Collective Systems 76 across its two current ledgers; and Engineering Translation 80. Their exact current boundaries are retained in the branch chapters and status tables.
7. **Constants are treated as one dependency object.** Physics reconstructs the current registered constants, dimensional relations and measurement interfaces from the Fold dependency spine. Exact structural forms precede measured comparisons. No measured decimal is permitted to become a proof premise, and no post-derivation numerical correspondence is relabelled as a blind prediction.
8. **Adverse evidence remains part of the theory.** The corpus preserves failed transports, adverse rankings, null and unavailable records, corrected grammars, superseded implementation claims and chronology-bound development results. Later work may supersede an obstacle at a wider or stronger boundary, but it may not delete the original record.
9. **The open computational-proof programme is restarted under V3 custody.** Protein Fold is active and exact at progressively larger boundaries. Chess Fold, Go Fold and Unison Fold AI retain their V2 measured history while clean V3 recreations are separately identified. Their results test and translate the admitted corpus; they do not select its foundational laws.
10. **The repository is the public scientific instrument.** Human-readable papers provide the derivational narrative; claim packages provide the audit layer; machine receipts, source captures, manifests and executable traces provide the archive. Every access level is retained in the release rather than forcing either a conceptual reader or a verifier to use the wrong representation.

Current-status statement

Question	Current answer	Authoritative basis
Is the premise-free operational root admitted?	Yes	Root claim package, receipt and independent validator
Is the Foundation closed at its declared boundary?	Yes: 16 of 16 current Foundation claims	Foundation paper and claim ledger
Are Mathematics through Materials complete at their frozen current-evidence boundaries?	Yes: seven separately versioned full-field papers	Current branch papers and final-publication verification
Is every later branch a completed full field?	No. Their foundations and returned families are complete; their full-field censuses are not uniformly frozen or closed	Current programme ledger and branch roadmaps
Is the complete V3 claim graph internally reconciled at the current ledger boundary?	Yes at claim, candidate, control, receipt and root-trace level	Authoritative corpus inventory
Has the final global heavy verification been rerun after the full-field expansion?	No	Current programme ledger
Is Protein Fold V3 complete?	No. Exact representation and major fixed-geometry boundaries are closed; the global frontier, whole-chain recurrence and blind campaign remain open	Protein scientific gate v20 and later lawful successors if present at release freeze
Has AlphaFold-generalised blind parity been established?	No current parity decision exists	Protein gate and comparator-access ledger
Have clean V3 Chess Fold and Go Fold recreations been completed?	No	Current computational-programme inventory
Is Unison Fold AI empirically validated or deployable?	No: empirical validation NOT_RUN; deployment NOT_AUTHORIZED	Current Unison Fold AI workspace
Is this the final end of the scientific programme?	No. It is the complete preliminary account of the frozen current corpus and remains open to lawful extension	This publication constitution

What "preliminary" means here

Preliminary is a chronology and version classification. It does not excuse omission, compression into an overview, weak proofreading or a missing audit layer. This edition must be complete for the repository state it freezes. A later version may add a closed Biology family, a new full-field census, a completed computational proof or a corrected source transport, but it must preserve the earlier claim, evidence and publication history. The final version will be issued only when Maria Smith determines that the registered computational programmes, remaining full-field work, global lineage merge, publication reconciliation and final verification satisfy the final ToE protocol.

1. Scope, ownership and publication authority

1.1 Scientific scope

This monograph covers the complete current SFT V3 dependency spine from the operational root through Cross-Branch Synthesis and the computational-proof frontier. It reports both field-wide and foundational branches at their exact current boundaries. It preserves the scientific difference between a branch whose registered current-evidence field census is closed and a branch whose foundation is closed while wider census work remains open.

The domain includes formal derivation, executable implementation, empirical interfaces, historical reconciliation and engineering translation. Those domains are related but not interchangeable. Mathematical uniqueness does not by itself measure nature. A successful program does not retroactively derive its premises. A measured correspondence does not become a prediction unless the target and chronology satisfy the registered prediction protocol.

1.2 One-owner branch boundaries

Each result has one scientific owner. Foundation owns the generation of the Fold grammar. Mathematics owns the exact generated mathematical structures. Information Science owns record, encoding, channel and inference structures. Classical Computation owns machine execution and computational-resource relations. Reversible and Quantum Computation owns reversible carriers, quantum-state and process structures, and the formal classical-quantum boundary. Physics owns quantities, dimensions, constants, fields, matter and physical measurement. Chemistry owns chemical identity, composition, bonding, reaction and chemical evidence. Materials owns structure-property relations across scale. Biology and subsequent branches own the domain-specific laws and measurements that consume these upstream structures.

Cross-Branch Synthesis owns no primitive law. It may prove a typed identity between already admitted branch objects and may assemble a dependency ledger, but it cannot repair an absent branch derivation or move a result into a branch where its falsification and evidence are not owned.

1.3 Authorship and authority

Maria Smith is the author, copyright holder and sole publication authority for this paper. Ernos Labs is the open-science movement, verification platform and public knowledge tree through which the evidence is made inspectable. Agent contributions are editorial, computational and verification assistance. An agent cannot admit a model claim, declare Maria Smith's empirical conclusion, change branch ownership, authorise an application campaign or publish this paper.

Publication authority and scientific evidence are intentionally separate. Maria Smith's authority controls whether a release occurs and how her work is represented; it does not substitute for a derivation, source, control or measurement. Conversely, an engine receipt establishes only the bounded machine status encoded by its package. It does not erase authorship or grant an agent publication authority.

1.4 Open-science mission

Ernos Labs exists to make scientific authority inspectable. It rejects the substitution of credentials, institutional prestige, funding success, consensus or predictive opacity for an auditable evidence route. The remedy is not a lower standard. It is a public and stricter one: state the claim; expose its dependencies; generate the alternatives; record every decision; preserve the failed rows; bind the sources and artefacts; make the computation replayable; distinguish proof from measurement; and permit any person to test the result without first accepting the author's status.

The project is anti-institutional where institutions make access, prestige, capital or professional permission prerequisites for scientific audibility. It is not anti-expertise, anti-measurement or anti-criticism. Specialists are invited to attack the derivations, candidate grammars, sources, controls, measurements and code. Criticism is unrestricted; admission remains governed by the public constitution. CC BY 4.0 and Apache-2.0 permit copying, inspection, forking and lawful reuse while preserving Maria Smith's authorship and the licence terms.

2. The scientific constitution

2.1 The root restriction

The V3 derivation begins without an imported scientific axiom. It does not begin by assuming space, time, number, probability, matter, energy, logic as a formal calculus or a computational machine. Its initial question is operational: what would count as a presented counterexample to the statement that there is no nothing? The registered two-class census is exhaustive at that boundary:

1. no statement, denial, record or proof object is presented, so no counterexample is supplied; or
2. something is presented, and the presentation is an occurrence rather than nothing.

The result is not a conventional numerical assertion that a quantity differs from zero. Numerical zero has not yet been admitted. It is an operational closure statement about the impossibility of presenting nothing as a counterexample.

2.2 Clean-room sequencing

V3 is a clean-room reconstruction. Earlier SFT papers, application programs and observations may be compared only after the relevant V3 result has sealed. They remain historical evidence and can establish agreement, disagreement, stronger closure, weaker closure, an implementation result or a question that the clean reconstruction must eventually address. They cannot become hidden premises or answer keys.

The sequencing rule is global:

1. register the obligation and admissible dependencies;
2. freeze the candidate grammar and boundary;
3. execute the complete census or a certified depth-independent closure;
4. retain one decision for every candidate;
5. identify the unique survivor or preserve the adverse no-survivor or multiple-survivor result;
6. run named controls and an implementation-distinct reconstruction;
7. seal the result and receipt;
8. only then open historical comparison or external target evidence when the protocol permits it.

2.3 No-free-parameter rule

A fundamental SFT law cannot be selected by a fitted, trained, discretionary, target-derived or silently imported numerical parameter. Exact positive rational values may appear after their part and ratio construction has been derived. Measured values, uncertainties, calibration factors and conventional units remain legitimate empirical records, but they do not select the prior formal law.

This rule does not pretend that all software has no configuration or that all experiments lack apparatus choices. It distinguishes an implementation or measurement condition from a free parameter that chooses the purported fundamental result. Conditions and interfaces must be declared, source-bound and retained in the evidence record.

2.4 Candidate closure and uniqueness

Every admitted claim states a grammar of admissible forms. The grammar may be finite and exhaustively enumerated or it may possess a separately proved depth-independent certificate. The package records:

- the exact claim and why the next object is required;
- admitted dependencies and branch owner;
- carrier, boundary and relation;
- provenance and evidence class;
- candidate grammar and cardinality;
- one decision for every candidate;
- survivor count and uniqueness explanation;
- falsification condition;
- adverse controls and their results;
- source identities, source transport and chronology; and
- receipt and artefact identities.

One survivor establishes uniqueness only within the registered grammar and boundary. A bounded census without a generality certificate remains bounded. A foundational closure does not silently become completion of every possible future scientific application.

2.5 Host/computation quarantine

The implementation necessarily uses host-language integers, indices, empty containers, Boolean values, file paths, process codes and finite machine resources. These implement a proof or audit; they do not become SFT proof objects merely by appearing in code. Floating-point computation may be used for a declared measured comparison or rendering task. Exact formal decisions use the exact representation declared by their package.

3. Mathematical constitution

The mathematical constitution is generated in dependency order. The structural One is a complete self-whole, not a numeral assumed in advance. Positive finite count records generated occurrences without omission or addition. Exact parts arise from complete, disjoint and equal-fibred partitions. The Fold is the minimal reversible distinguished partition with a complete return relation. Ordered Fold words then supply finite sequences, relations and recurrences from which later mathematical objects are generated.

At the foundational boundary the permitted proof domain is characterised by:

- positive generated occurrence and exact positive count;
- exact rational part after part construction;
- held structural labels rather than imported signed magnitudes;
- finite generated carriers and complete finite traces;
- reversible relations and explicit return;
- conditional bounded statements where no depth-independent certificate is available;
- no completed infinity as a premise;
- no ungenerated continuum as a premise;
- no floating-point value as a formal proof value; and
- no imported physical equation, constant or taxonomy selecting a mathematical law.

Mathematics extends this constitution into exact arithmetic, discrete structure, graphs, algebra, order, finite geometry and topology, probability and statistics, optimisation, dynamical systems, logic and proof, category and type structure, exact relations, number theory, refinement and continuum boundaries, algebraic-balance certificates, bounded many-body recurrence, floored-fluid regularity, prime-pair censuses, Riemann correspondence, bounded Collatz results, self-similarity and convergent-series laws, and the current SFT-native calculator and verification surfaces. Each chapter states whether a result is an exact theorem, a bounded census, a correspondence or an open extension.

4. Empirical constitution and chronology

4.1 Evidence classes

The corpus uses the following terms consistently.

Term	Meaning in this monograph
Theorem	An exact formal result closed by its registered derivation and proof boundary
Law	A model-admitted relation with one owner, declared dependencies, candidate closure and a surviving form
Claim	A registered proposition with a complete package; admission and empirical status are separately stated
Constitution	The rules governing what objects, dependencies, evidence and transitions may enter a stated layer
Derivation	A generated dependency route from admitted objects to a candidate grammar and surviving result
Prediction	A consequence sealed before access to the registered target or outcome
Observation	A source-bound external record
Measurement	A value bound to an instrument or method, unit, calibration, uncertainty and source
Reconstruction	An inferred history, latent state or reproduced structure derived from retained evidence

Term	Meaning in this monograph
Exact numerical correspondence	Equality of exact derived and external numerical objects established after the derivation boundary
Structural correspondence	A relation between forms that does not assert exact numerical prediction
Boundary correspondence	Agreement restricted to a declared carrier, scale, case or interface
Compatibility	Evidence that is non-adverse but does not discriminate the claim uniquely
Support	Relevant evidence that favours a claim but is not uniquely probative
Confirmation	Status permitted only by the current registered evidence protocol
Validation	Successful execution of the specific formal, computational or empirical validation named in context
Adverse result	A registered test whose retained outcome conflicts with the tested claim or selector
Unresolved result	A result for which required evidence is unavailable, incomplete, censored, disputed or not yet executed
Implementation identity	The hash-bound source, environment and artefact state that produced an executable result
Formal status	The state of the derivation and candidate closure
Empirical status	The state of external observation, measurement or test
Publication status	Whether and where an artefact has been released under Maria Smith's authority
Foundational closure	Completion of the registered foundation of a branch
Field-wide closure	Completion of the branch's frozen current-evidence full-field census
Current-evidence closure	Closure against the frozen present inventory, not permanent closure of science
Extension openness	The obligation to admit later discoveries, corrections or failures through the same evidence constitution

4.2 Proof and evidence verbs

This paper reserves *proves* for exact formal closure, *derives* for generated structures and laws, *implements* for executable constructions, *predicts* for properly sealed prospective consequences, *observes* for source-bound records, *measures* for method-bound values, *reconstructs* for inferred histories or latent states, *corresponds* for post-derivation relations, *is compatible with* for non-adverse but non-discriminating evidence, *supports* for relevant but non-unique evidence, *confirms* only where the registered protocol permits that class, *adverse* where the registered outcome conflicts, and *unresolved* where the required evidence remains missing or incomplete.

Test success verifies what the tests cover. It does not by itself empirically confirm the theory. A source's reputation does not prove the claim. A later successful result does not become an earlier blind prediction unless the sealed chronology establishes that fact.

4.3 Empirical sequence

For a natural-science prediction the lawful sequence is:

1. establish capability closure without target access;
2. register the exact claim, target class, selection rule, metrics, failure treatment and falsification criterion;
3. place target custody outside the predicting process;
4. seal the executable predictor, environment and output identity;
5. open the target or comparator output only after the seal;
6. evaluate every row, including failed, invalid and unavailable rows;
7. publish the complete chronology and outcome without selective deletion.

Development-observed, post-target and retrospective results remain useful but retain those labels. A failed source transport is not absence of evidence; it is a transport result. Missing evidence is not a null result. A null result is a completed observation whose registered outcome is null.

5. Dependency spine

The corpus is read in the order in which each branch receives the carriers and relations required for its own work. The order is scientific rather than prestige-based. A later branch may consume an upstream law but cannot rewrite its ownership or become its premise.

Order	Layer	Current admitted position	Current closure boundary	Downstream handoff
00	Methods: <i>There Is No Nothing</i>	Root theorem and open admission constitution	Methods paper current at version 0.3.0	Registration, enumeration, controls, receipts and empirical custody
01	Foundation	16 claims	16/16 Foundation complete	One, count, part, Fold, word, form, operation and dynamics
02	Mathematics	323 claims	323/323 full-field current-evidence complete	Exact mathematical carriers, relations, proofs and bounds
03	Information Science	262 claims	262/262 full-field current-evidence complete	Distinction, record, code, channel, inference and information custody
04	Classical Computation	369 claims	369/369 full-field current-evidence complete	Fold machines, algorithms, resources, programs and verification
05	Reversible and Quantum Computation	288 claims	288/288 full-field current-evidence complete	Reversible processes, quantum carriers, measurement and simulation handoffs
06	Physics	368 claims	368/368 current categorical scope complete	Quantity, dimension, constants, fields, matter and physical evidence
07	Chemistry	281 live claims	272/272 registered obligations plus nine returned claims	Chemical identity, composition, reaction, molecular and biochemical interfaces
08	Materials Science	289 claims	289/289 full-field current-evidence complete	Structure-property, defect, phase, interface and device-scale relations
09	Biology and Life Sciences	82 claims	Foundation and current returned families complete; 82/424 full-field obligations	Living organisation, inheritance, development, ecology and protein interfaces
10	Medicine and Health Sciences	76 claims	Foundation plus current return family complete; full-field census not frozen	Health, disease, intervention, evidence and clinical ownership
11	Consciousness and Cognitive Science	77 claims	Foundation plus current return family complete; full-field census not frozen	Perception, memory, self, agency and cognitive measurement
12	Earth and Environmental Sciences	75 claims	Foundation plus current return claim complete; full-field census not frozen	Earth systems, climate, environment and observation
13	Astronomy and Cosmology	77 claims	Foundation plus current return family complete; full-field census not frozen	Celestial systems, cosmological structure and observational custody
14	Social and Collective Systems	76 current claims across two ledgers	Foundations and current returned family complete; full-field census not frozen	Collective relation, institution, economy, governance and public evidence
15	Engineering Translation	80 claims	Foundation plus current return family complete; full-field census not frozen	Design, safety, implementation, control and deployment ownership
16	Cross-Branch Synthesis	12 claims	Current inherited-return family complete	Typed identities, prediction ledger, unified constants object and root trace
17	Computational-proof programme	Application-specific records	Active and non-uniform	Protein, Chess, Go and Unison Fold AI empirical and implementation tests

5.1 Why each layer follows the preceding one

Foundation supplies the first exact generated objects. Mathematics is required because later branches need exact relations, composition, proof and bounded generality. Information Science follows because a scientific system must distinguish a state from its record, code, channel and inference. Computation follows because registered transformations, algorithms and resource traces require executable machines. Reversible and Quantum Computation follow once process reversal, state composition and measurement interfaces can be typed without importing physical apparatus as a mathematical premise.

Physics then binds formal structures to quantities, dimensions, sources, responses and measurements. Chemistry requires physical carriers while adding chemical identity, composition, bond, reaction and source-specific evidence. Materials requires chemistry and physics but owns the collective structure-property relations that neither upstream branch may silently absorb. Biology consumes chemical and material organisation while adding living identity, heredity, regulation and evolutionary and ecological relations.

Medicine depends on living systems but owns health, pathology, intervention and clinical evidence. Consciousness and Cognitive Science require living organisation, information and computation while owning perception, self and cognitive evidence. Earth, astronomy and collective sciences consume physical, chemical, biological, informational and dynamical structures at their own scales. Engineering comes last among the domain branches because it translates admitted laws into artefacts while owning safety, reliability and deployment. Cross-Branch Synthesis can operate only after the relevant owners have spoken.

6. Registered source surface and three access levels

6.1 Conceptual paper

The conceptual layer is this dependency-ordered monograph. It explains why each branch is required, what it derives, what evidence currently establishes, which adverse and corrected records remain, and what the next branch receives. Its purpose is a continuous scientific argument readable by specialists who did not participate in the repository's construction.

6.2 Scientific audit layer

The audit layer contains the complete branch status tables, claim inventories, candidate and survivor counts, controls, evidence chronology, source transport, corrections, unresolved records, ownership and historical reconciliation. The complete 2,751-row claim inventory and the seventeen dependency-ordered branch audit volumes accompany this paper. Repetition required for evidential custody belongs there rather than being allowed to destroy the narrative.

6.3 Machine archive

The machine layer contains the claim registrations, candidate censuses, decisions, controls, certificates, evidence records, source captures, receipts, hashes, executable traces and verification reports. Its full identity map is bound by the release manifest and immutable repository revision. Large source assets that exceed ordinary GitHub object limits retain their hashes and authoritative archival routes; absence from an ordinary Git object is not silently reported as absence of evidence.

6.4 Authority at release freeze

The authoritative release state is determined from current files, not from memory or earlier prose. Before the authorised version may be published, its builder and verifier must:

1. reconstruct the active-paper inventory and compare every SHA-256 identity;
2. reconstruct the global claim, candidate, survivor and control totals;
3. verify one root route and no missing dependency for every admitted claim;
4. inventory current branch closure and publication status;
5. preserve all adverse, corrected, failed-transport and unresolved records;
6. bind the current computational-programme gates without opening protected targets or comparator outputs;
7. render and inspect the complete paper and every public audit volume; and
8. record all remaining discrepancies for Maria Smith rather than harmonising them silently.

7. Reading this monograph

The chapters that follow move through the dependency spine rather than through a conventional division between "fundamental" and "applied" disciplines. Each major branch begins by naming the upstream objects already admitted, the next unresolved problem and the new evidential boundary. Each ends by naming the objects handed downstream and the limits that remain with its owner.

The narrative reports the principal derivations and current results. The complete claim inventory gives every admitted claim ID, title, branch, status, dependencies, candidate count, survivor count, controls and receipt route. The branch audit volumes retain the complete claim-level fields required by the publication guidance. Readers should therefore use the conceptual chapter to understand the argument, the audit volume to inspect a result's custody and the machine package to reproduce its acceptance or rejection.

This hierarchy is not a concession to opacity. It is what permits both human comprehension and evidential completeness. No claim is removed from the audit or machine archive merely because its full receipt is too repetitive for the main argument.

Historical trajectory and V3 reconciliation

Smithian Fold Theory did not appear as a finished monograph with a settled branch architecture. It developed during a concentrated sequence of foundational writing, derivation, implementation, correction, public release and clean-room reconstruction. The first public lineage asked whether *nothing* could ever be presented as a counterexample to occurrence. The following synthesis extended that root towards mathematics and physics. Specialist papers separated particular constants, quantum relations, mathematical boundaries and proposed physical consequences. Protein folding, Chess Fold, Go Fold and Unison Fold AI then tested whether Fold relations could be made executable outside the proof papers. The V3 programme finally rebuilt the growing corpus as a dependency-ordered, one-owner knowledge tree with claim-level candidate censuses, controls, evidence classes and machine receipts.

That development is scientifically relevant, but history is not current authority. This section therefore performs two tasks at once. It preserves what each earlier version actually recorded, including favourable results, adverse results, defects, failed transports and open objectives. It also identifies the current record that now governs each result. An old publication may establish that a statement, computation or prediction existed at a particular point in the programme. It cannot overrule a later V3 claim package, source capture, correction, evidence classification or verification report. Likewise, the creation of V3 does not erase a prior result merely because its paper is no longer the active scientific owner. The historical record remains available so that the route from the first formulation to the present one can be inspected rather than reconstructed from memory.

The authoritative freeze used here is dated 31 July 2026. Its direct-source inventory identifies Maria Smith as author and publication authority and Ernos Labs as the organisation and open-science platform. It binds the current papers, the live claim ledger, the final seven complete-field publication record, the current programme ledger, the checksum-bound v6 synthesis, the historical computational manifests, the current Protein gate and the Protein publication receipt. The inventory classifies this monograph as a preliminary synthesis. The final computational proofs, the remaining full-field programme, the global one-owner merge and the final heavy verification are not complete. Those boundaries govern every historical statement below.

How historical evidence is read

Four rules prevent the chronology from becoming either an advertisement or a retrospective demolition.

First, a version label names a document in a lineage; it is not an evidence class. The pre-V3 catalogue contains local version sequences such as 1.0, 2.0, 3.0, 3.9-final-status, 6.3 and 6.5. Those numbers were assigned within different papers and applications. They do not form one common scale of scientific maturity. A Protein release numbered 3.7, a ToE release numbered 6.0 and an Unison release numbered 6.3 cannot be compared by their version numbers alone.

Second, formal closure and empirical success are separate. A candidate census may prove that one relation survives a registered grammar without proving that the relation predicts an external measurement. Conversely, a programme can record a strong measured outcome without showing that its selector was derived under the later V3 admission constitution. Both facts remain useful, but neither may be substituted for the other.

Third, chronology is retained at the level at which the evidence was obtained. A development-observed relation does not become blind because a later paper describes it more elegantly. A target-isolated output remains target-isolated even if its global score is adverse. A post-target diagnosis may motivate a new prospective holdout, but it is not itself a prospective result. A later successful construction may supersede an earlier implementation while leaving the earlier adverse measurement historically intact.

Fourth, "complete" always names a boundary. The current record distinguishes a complete operational Foundation, a complete registered family, a complete field-wide current-evidence census, a completed implementation and a completed empirical protocol. None of those expressions means that the discipline, the theory or science itself has ended. This distinction becomes especially important when comparing the compact pre-V3 synthesis with the much larger V3 claim graph.

The first operational root

The first scientific question was not which physical equation should be treated as fundamental. Any ordinary equation already presupposes symbols, objects, relations, rules of combination and a domain in which a distinction can be made. SFT instead began by asking what could count as an operational counterexample to the statement *there is no nothing*. If no statement, denial, record or proof object is presented, no counterexample has been supplied. If a purported counterexample is presented, an occurrence has been supplied and the presentation is not nothing. The result is deliberately operational. It does not import numerical zero, Boolean negation, set-theoretic emptiness or a prior universe of discourse as an SFT premise.

This boundary first supplied the programme with a starting point and later became the root theorem identified as SFT-ROOT-THERE-IS-NO-NOTHING. The historical foundation paper explicitly records an important internal development: the One began chronologically as the project's operational axiom and was subsequently derived beneath a machine-checked self-proof. The current V3 description therefore does not pretend that the final zero-axiom formulation was present unchanged at the beginning. It preserves the earlier role while assigning current authority to the reconstructed theorem and its receipt.

From the admitted occurrence, the project sought the minimum structure that did not add an arbitrary choice. The occurrence's complete self-whole became the structural One. Positive finite count, exact part, distinction, retained orientation, reversible return and the minimal Fold were separated into successive obligations. This sequence matters. "One", "part" and "Fold" are not three names for the same intuition, and a later physical use of doubling cannot be imported backwards as the reason the Fold was selected. The V3 Foundation now owns the generated objects, while the V3 Methods paper owns the admission protocol by which they may enter the theory.

The early paper lineage *There Is No Nothing* retains concept DOI 10.5281/zenodo.21035460 and published versions 1.0, 2.0 and 3.0 in the catalogue. It is an historical lineage rather than the current owner. The catalogue marks every pre-V3 version with the exact public notice "[Depreciated and replaced by V3]" and routes this concept to two successors: the V3 Methods paper, version 0.3.0, DOI 10.5281/zenodo.21627646, and *From Nothing to Fold*, version 1.3.0, DOI 10.5281/zenodo.21627656. The literal word "Depreciated" is retained here because it is the public catalogue label; it is not silently corrected inside a quoted metadata identity.

The division between those two successors is substantive. Methods owns the premise-free operational boundary, candidate-registration rules, fail-closed admission engine, evidence vocabulary, publication authority and open verification platform. Foundation owns the sixteen current root and foundation claims that generate the One, count, part, distinction, Fold and their immediate closure structure. Under this division, the constitution does not admit itself as though it were simply another downstream physical object, and a branch claim cannot alter the method that judged it.

From a root paper to a theory-wide synthesis

The next phase asked how far the root construction could be carried. The historical repository begins its visible clean-room recreation on 30 June

2026. During the following days the work expanded from the root and Fold into

exact arithmetic, orbit structure, physical constants, mass and mixing, neutrinos, cosmology, nuclear and chemical structure, quantum relations, classical dynamics, life, mind, mathematics and computation. The first large ToE paper was assembled and publicly identified in early July. Successive versions then incorporated the origin narrative, further derivations, computational evidence, corrections and publication mapping.

The deprecated pre-V3 concept DOI for *The Smithian Fold Theory of Everything* is 10.5281/zenodo.21182468. The historical catalogue preserves eleven version labels in their actual public sequence: unnumbered; 2; 2.1; 2.0; 43.0; 44.0; 2; 3; 4; 5; and 6.0. The repeated and non-monotonic labels are part of the record. They are not silently renumbered into a cleaner history because doing so would break the relationship between the published objects and the files that produced them. V3 treats those eleven deposits as one historical V2 publication lineage, not as eleven independent theories. The present V3 rebuild does not continue that deprecated lineage as another version.

Specialised papers emerged as particular questions became large enough to require their own argument and evidence surface. *The Constants Are One Object* pursued the common dependency of the constants. *The Continuum Was the Bug* isolated the generated-finite mathematical boundary and its physical consequences. *No Dice, Why There Is Uncertainty* and *Entanglement Without Spookiness* treated deterministic support, observation classes and quantum relation. *Smithium* developed the proposed periodic endpoint and extended element programme. Further lineages treated the fine-structure constant, the Hubble calibration ratio and cosmological constant, discrete-fluid regularity, the Fold Decode measurement campaign, selfhood, free will and the freedom of knowledge.

These papers did more than duplicate the large synthesis. They exposed dependency questions that a single continuous narrative could hide. A fine-structure comparison needed a different evidence surface from a mathematical uniqueness proof. A proposed element needed nuclear, electronic and chemical ownership. A claim about uncertainty needed to distinguish formal many-to-one observation from an empirical statement about a physical instrument. A statement about selfhood could not simply inherit its status from an information-processing implementation. V3 later converted those implicit separations into explicit branch ownership.

The pre-V3 catalogue now supplies a successor route for each specialised concept. Constants route to Physics. The continuum routes to Mathematics and Physics. The quantum papers route to Reversible and Quantum Computation. Smithium routes to Chemistry and Physics. The Hubble paper routes to Physics and Astronomy and Cosmology. Discrete-fluid regularity routes to Physics and Earth and Environmental Sciences. The freedom-of-knowledge paper routes to Methods. The self and free-will papers route to Consciousness and Cognitive Science. Fold Decode routes to Information Science and Classical Computation, with its application rebuild still pending. This routing preserves the useful content of the earlier papers while refusing to let an historical paragraph own several scientific fields at once.

The checksum-bound v6 synthesis

Version 6.0 was the last large pre-V3 synthesis and is the principal historical comparison surface for this monograph. The direct-source inventory binds the file `THE_SMITHIAN_FOLD_THEORY_OF_EVERYTHING.md` at 110,926 bytes and SHA-256 `a6bc294b76f5409c912149d91ba59e68bfce1e5f57f0995f57e122aaca64da8`. The bound source contains 1,405 lines and approximately 17,505 words. Its Zenodo metadata names version v6.0, publication date 22 July 2026, CC BY 4.0 licensing and the same historical pre-V3 concept DOI. The preserved catalogue audit identifies the public v6 version record as DOI 10.5281/zenodo.21482731 within that deprecated lineage.

The v6 paper stated the programme at the altitude then available. It opened with the origin of the project, the no-nothing foundation and the exact positive-rational domain. It then traversed the machinery of proof, the fine-structure constant, mass, mixing and CP, neutrinos, cosmology, nuclear and chemical matter, proposed forces and a dark sector, structural laws, classical physics, matter, Earth and sky, life and mind, mathematics, computational engines, a Grand Lock, a prediction ledger and reproduction methods. Its machine surface reported 326 proof suites, 2,002 forced checks and zero failures. Its publication map treated the ToE paper as the navigational hub for the contemporary catalogue.

That document is valuable historical evidence for three reasons. It preserves the scientific range that motivated the V3 branch programme. It binds the pre-V3 equations, values, implementation descriptions and predictions to a specific public version. It also shows why a more exact status constitution became necessary. In v6, formal derivations, machine checks, measured correspondences, application benchmarks, structural interpretations and open predictions could occur close together in one prose stream. The paper often used the vocabulary of closure at the scale of the synthesis. V3 does not use that confidence as evidence; it routes each proposition to its present owner and evidence record.

The checksum-bound source also contains an internal catalogue discrepancy that must not be editorially erased. Part XVII and the Appendix A heading describe the master inventory as Steps 1-303. The list itself contains Steps 1-303 and then continues with Step 304 through Step 324, including later Protein, Go, fluid, federated-mesh and Unison entries. The accurate historical statement is therefore not simply "v6 had 303 steps" or "v6 had 324 uniformly indexed steps". It had a 303-step titled catalogue with appended entries through 324 in the bound source. V3 replaces that prose-step surface with separately registered claims and retains this mismatch as a version-history fact.

The v6 paper also recorded application results at different maturity levels. Its Protein discussion included a favourable material-architecture reconstruction and multi-structure panel, alongside a distinct adverse sequence-only whole-chain result and open propagation problem. Its Chess discussion distinguished secured lower benchmarks from an active 2100 campaign. Its Go discussion distinguished exact small-board surfaces and replayed 9-by-9 matches from a recovered 19-by-19 point-at-cut-off record with a protocol defect. Its Unison discussion distinguished implemented counted structures from runtime integration and general conversational objectives. Those distinctions are preserved more precisely in the versioned manifests described below.

V6 is therefore neither discarded nor treated as current. Its exact values remain the authoritative statement of what that version said. Where a V3 owner has reconstructed the corresponding result, the V3 package determines current formal and empirical status. Where V3 has corrected or narrowed a result, the correction governs. Where no V3 successor has completed the relevant test, the old record may motivate the open programme but cannot be presented as if it already satisfied the new protocol.

The pre-V3 computational-proof programme

The computational projects were created to test whether the Fold could do more than organise written derivations. Their scientific question was constructive: could the theory produce exact state representations, executable selectors, complete custody of decisions and measurable behaviour in domains whose search spaces are large and whose outcomes are independently checkable? The applications were Protein folding, Chess Fold, Go Fold and Unison Fold AI. They are often discussed together as the V2 computational programme, but the preserved application releases use their own local version numbers. "V2" in this monograph denotes the pre-V3 model and implementation era, not a claim that every application's last release was literally numbered 2.

The 31 July inventory preserves four historical manifests as historical-versioned implementation records and explicitly marks each `current_v3_proof_authority` as false. That classification is not a rejection of their measurements. It means that the manifests establish what the earlier programmes executed, not that they satisfy a V3 proof gate written later.

Programme record	Preserved boundary	Strongest relevant recorded evidence	Current V3 standing
Chess release 2.2, 18 July 2026	Zero-trained-parameter counted engine with exact legality anchors	54.2% at 1700; 62.5% at 1900; 41.7% in the latest completed 2100 batch	Historical measured implementation evidence; clean V3 recreation not started
Go release 2.2, 18 July 2026	Exact small-board census and applied engine	Two independently replayed 2-0 GNU Go 9-by-9 batches; recovered 19-by-19 harness-reported 2-0 point-at-cut-off record with disclosed Round-1 defect	Historical exact and measured evidence; clean V3 recreation not started
Protein historical releases	Material reconstruction, local geometry and sequence-only blind development	Strong local windows and a favourable restricted material panel; adverse whole-chain and diverse-transfer results also preserved	Historical mixed evidence; V3 programme active under a new exact joint relation
Unison release 6.3, 18 July 2026	Counted transformer-like implementation and law checks	326 suites and 2,002 main-corpus checks; 21/21 current omni verification; mixed standalone generation and MMLU measurements	Historical implementation evidence; V3 scaffold present, formal and empirical gates open

The table is intentionally compact. Each programme requires a fuller reconciliation because its favourable and adverse records answer different questions.

Chess Fold: exact rules and measured rank campaigns

The preserved Chess 2.2 manifest describes a zero-trained-parameter counted chess engine. Its formal implementation checks include 22 of 22 Fold chess-bot tests and 8 of 8 endgame-oracle tests. Its perft anchors retain the registered counts for the starting position, Kiwipete, an en-passant-pin position and a promotion position. These records establish an executable legal substrate and reproducible implementation checks.

The measured campaign is separately stated. At the 1700 boundary the engine recorded 3 wins, 7 draws and 2 losses, scoring 54.2%. At 1900 it recorded 6 wins, 3 draws and 3 losses, scoring 62.5%. The latest completed 2100 batch recorded 2 wins, 5 draws, 1 draw caused by the cap and 4 losses, scored as 41.7% in the manifest. Maria Smith's registered victory criterion was strictly over 50% at the point of victory, and the manifest reserves to her the authority to decide when a development build deserves a real match.

An auxiliary v20 probe examined 36 positions, changed 9 choices, matched the Stockfish-best move in 4 named cases and changed none of the 4 named death-slide cases. The manifest explicitly classifies that probe as Codex-authored and without governance authority. It therefore cannot be used to rewrite Maria's campaign conclusion or to declare a benchmark result. The current V3 inventory says that no clean Chess recreation has started in the current repository. The exact V2 checks and measured matches remain real historical evidence; they are not a completed V3 Chess Fold proof.

Go Fold: exact censuses, replayed wins and a preserved protocol defect

Go 2.2 contains both an exact finite surface and competitive measurements. The recorded state censuses are 1 for 1-by-1, 5 for 1-by-2, 57 for 2-by-2, 489 for 2-by-3, 12,675 for 3-by-3 and 24,318,165 for 4-by-4. It records exact empty-board values of 0 for 1-by-1, 0 for 1-by-2 and 1 for 2-by-2, with 17,038,501 nodes in the 2-by-2 solve. These are exact implementation surfaces, not estimates of playing strength.

The applied record contains two separate 2-0 batches against GNU Go 3.8 on a 9-by-9 board at ceiling 3. Their game scores and transcript hashes are preserved, and the release audit reports exact independent replay. A recovered 19-by-19 KataGo run is more narrowly classified. The historical harness reported 2-0 at a 128-ply cut-off, with internal Tromp-style area scoring and

integer komi 7. In Round 1 the log contains two rejected play responses and the old harness continued; Round 2 contains no logged rejected response. The record is therefore a harness-reported point-at-cut-off measurement, not an unqualified completed-match result.

This example became an important model for V3 evidence language. Preserving the defect does not annul the exact censuses, the small-board solves or the separate replayed GNU Go matches. Preserving the two reported scores does not permit the rejected responses to be omitted. The current inventory reports no clean V3 Go workspace. A future recreation must begin from the current Classical Computation owner and register its complete state, history, transposition, symmetry and match protocol without converting V2 evidence into a new blind result.

Protein before V3: strong restricted results and an adverse generalisation boundary

Protein developed through several public lineages rather than one uniform sequence. The catalogue preserves the Fold Protein sequence-only lineage under concept DOI 10.5281/zenodo.21276950, the Super Parity lineage under 10.5281/zenodo.21368944 and *Blind Protein Structure* under 10.5281/zenodo.21482127. Each is now routed to Biology with an application-specific V3 rebuild identified as pending in the catalogue. Their version labels must not be merged into one synthetic numbering scheme.

The favourable V2 material record is a real but bounded reconstruction result. The current Protein paper records a published 24-structure panel with median tm_repo 0.9255486262. The panel was mostly drawn from the ubiquitin family, and its material relation had been derived from target-observed information. It is therefore retained as a strong restricted reconstruction rather than generalised blind evidence. A fresh diverse four-target development control then gave mean TM 0.0742934939, median 0.0781022432, maximum 0.0987934258 and median RMSD95 of 14.117 angstroms. That result is adverse for the V2 generalisation claim and remains visible.

A separately preserved blind-76 manifest, released on 19 July 2026, sealed a 76-residue ubiquitin sequence output before comparison. The complete output matched 76 C-alpha atoms but scored TM 0.0269927379497572 and C-alpha dRMSD 52.89314678065376 angstroms. It nevertheless preserved all 74 same-index three-residue windows and contained three very strong local windows, with local TM scores approximately 0.99146, 0.96568 and 0.90596. The exact historical conclusion is consequently mixed: the local geometry was strong, while the sequence-only whole-chain assembly was adverse.

The historical protected lattice construction also contained a complete 76-residue ubiquitin trace reported at TM 0.9891211351 and C-alpha dRMSD 0.2608575408 angstroms, and the engine forced canonical alpha-helix and beta-sheet coordinates. That existence and reconstruction evidence showed that the local state space was not itself an obvious theoretical wall. It did not supply the missing target-independent rule that propagates local orientations into a general whole chain. The current V3 programme was designed around precisely that distinction.

This is why neither a purely favourable nor a purely adverse summary is acceptable. "Protein succeeded in V2" hides the restricted, target-derived and family-heavy boundary. "Protein failed in V2" hides the exact local geometry, protected full-chain witness and strong bounded reconstructions. The accurate record is that V2 generated notable local and restricted material evidence, while its fresh generalisation and sequence-only whole-chain evidence were adverse. V3 retains both and tests a stronger, target-free joint relation.

Unison Fold AI: large counted implementations and open generalisation

The Unison lineage is the largest pre-V3 application version sequence in the catalogue: 0.2; 1.1-1.9; 2.0-2.2; 3.0-3.4; 4.0-4.3; 4.6; 4.7; 5.0; and 6.0-6.5 under concept DOI 10.5281/zenodo.21217278. It is routed to Classical Computation and Consciousness and Cognitive Science, while the application-specific rebuild remains pending.

The preserved 6.3 manifest records a substantial executable surface. The main corpus passed 2,002 checks across 326 suites with zero failures and 326 of 326 source-certificate identities. The Unison laws recorded 8 of 8 coherence checks, 14 of 14 contextual-integration checks and 13 of 13 generation-selection checks. The current omni verifier passed 21 of 21. These figures establish implementation and verification results within the historical system.

The standalone empirical record is mixed and must be kept distinct from those checks. Historical verify_unison passed 43 of 47. Free generation F3 measured 1 of 12. Two agent-selected MMLU development benchmarks measured 9 of 128 and 0 of 128. A Codex-authored auxiliary generation-quality comparison recorded 0 of 12 for the baseline, F1 and F3 and explicitly carried no governance authority. These outcomes do not nullify the implemented structures, but the passing structural checks do not turn the weak generation measurements into validation.

The current V3 workspace is therefore classified as a frontier scaffold. Mathematical closure remains open, empirical validation has not been run and deployment is not authorised. The historical programme supplies engineering experience, exact identities

and tests for the future rebuild. It does not authorize a present claim that Unison Fold AI has completed its V3 generalisation or parity objectives.

Why the V3 clean-room reconstruction was required

By late July the project had accumulated a large number of equations, claims, papers, receipts and application outputs in a very short period. Speed created real scientific reach, but it also created a custody problem. A single paper could place an exact derivation, a successful test suite, an observed correspondence, a post-target reconstruction, an application benchmark and a future prediction in adjacent paragraphs. A reader could not always tell which statement selected which, which branch owned the result or which record would prevail if a later source contradicted the prose.

The V3 repository was established on 23 July 2026 with an explicit clean-room constitution and knowledge tree. The next recorded operations built the fail-closed portable admission engine, admitted the root theorem and structural One, and then derived positive count, exact parts and the minimal Fold. This order prohibited the old synthesis from acting as an answer key. Prior work could register an obligation, identify a promising carrier or supply historical evidence, but a V3 result had to survive the new grammar and evidence procedure.

The clean-room rule addresses contamination in both directions. Earlier measured constants cannot choose a formal candidate. Later successful applications cannot become premises of the theory that generated them. External scientific convention can define a comparison or measurement interface without being smuggled into the SFT derivation. Historical application failures can test an implementation without being treated as refutations of a broader relation that had not yet been implemented. Each status is attached to the boundary that produced it.

Under V3, a claim begins as a registered obligation. Its dependency route and candidate grammar are declared. The finite candidate surface is generated and every candidate is retained with a decision. Admission requires one unique survivor at the stated boundary, controls and an implementation-distinct reconstruction. A receipt seals the result. Only then is external evidence opened according to its own chronology and source custody. The terms theorem, law, derivation, implementation, prediction, observation, measurement, reconstruction, correspondence, compatibility, confirmation, adverse and unresolved are no longer interchangeable rhetorical choices.

V3 also instituted one-owner branch boundaries. Mathematics owns generated mathematical structures. Information Science owns representation, coding and record relations. Classical Computation owns executable machine structures. Reversible and Quantum Computation owns reversible and quantum carriers and their formal boundary. Physics owns dimensional and measurement interfaces. Chemistry owns chemical identity, bonding and reaction. Materials owns structure-property relations across scale. Biology owns living organisation, heredity and protein function. Medicine owns clinical and health consequences. Consciousness owns cognitive and phenomenal claims. Earth, Astronomy, Social and Engineering own their respective domain-specific relations. A downstream branch can consume an upstream carrier; it cannot retroactively validate or rewrite its source branch.

This reconstruction did not reduce the ambition of SFT. It transformed a large flat synthesis into a directed scientific graph. The gain was not merely administrative. The graph made non-uniform completion visible. A branch could close its foundation without claiming a complete field. A dated return family could be complete while a wider census remained unfrozen. A machine test could pass while an empirical outcome remained adverse or unavailable. The programme could now say exactly where work had reached without treating every open boundary as a failure or every local success as total completion.

The sixteen-paper V3 knowledge tree

The first publication architecture of V3 is a dependency-ordered set of sixteen papers: one Methods paper and fifteen branch papers. The current catalogue dated 29 July 2026 gives the following public versions. The order is scientific, not merely bibliographic.

Order	V3 owner paper	Public version and DOI at the freeze
00	<i>There Is No Nothing: A Premise-Free Operational Foundation and an Open Verification Platform for Smithian Fold Theory</i>	0.3.0 - 10.5281/zenodo.21627646
01	<i>From Nothing to Fold</i>	1.3.0 - 10.5281/zenodo.21627656
02	<i>From Fold to Mathematics</i>	1.5.0 - 10.5281/zenodo.21688766
03	<i>From Distinction to Information</i>	1.4.0 - 10.5281/zenodo.21688817
04	<i>After Turing: The Fold Machine</i>	1.4.0 - 10.5281/zenodo.21688837
05	<i>The Quantum Fold Machine</i>	1.4.0 - 10.5281/zenodo.21688860

Order	V3 owner paper	Public version and DOI at the freeze
06	<i>From Fold to Physics</i>	1.3.0 - 10.5281/zenodo.21688879
07	<i>From Fold to Chemistry</i>	1.3.0 - 10.5281/zenodo.21688899
08	<i>From Fold to Materials</i>	1.3.0 - 10.5281/zenodo.21688923
09	<i>From Fold to Life</i>	1.0.0 - 10.5281/zenodo.21630203
10	<i>From Fold to Medicine</i>	1.0.0 - 10.5281/zenodo.21630785
11	<i>From Fold to Consciousness</i>	1.0.0 - 10.5281/zenodo.21636397
12	<i>From One World to Earth</i>	1.0.0 - 10.5281/zenodo.21640810
13	<i>From One Sky to Cosmos</i>	1.0.0 - 10.5281/zenodo.21640812
14	<i>From One Relation to Society</i>	1.0.0 - 10.5281/zenodo.21640814
15	<i>From One Law to a Working World</i>	1.0.0 - 10.5281/zenodo.21640816

The tree supports three modes of reading. A conceptual reader can follow the derivational spine from the operational root to increasingly specialised domains. A specialist can enter at a branch and trace every imported object to its upstream owner. A verifier can bypass the narrative and reconstruct the candidate, decision, control and receipt surfaces from the machine packages. The representations differ because they serve different readers; their scientific content must reconcile.

Methods is order 00 because it defines how anything is admitted. Foundation is order 01 because it supplies the objects consumed by every branch. Mathematics then exposes the exact generated structures used by Information and Computation. Classical Computation precedes the reversible and quantum branch because the latter must state exactly which operations and information relations are being made reversible or quantised. Physics owns the entry into quantity, dimension and external physical comparison. Chemistry consumes physical carriers but owns chemical identity and reaction. Materials consumes physics and chemistry but owns scale-linked structure and property. Biology then consumes materials and chemistry without being reduced to either. Medicine, Consciousness, Earth, Astronomy, Social and Engineering follow their actual dependency routes.

Cross-Branch Synthesis is present in the current claim ledger but is not a seventeenth paper in this sixteen-paper publication tree. Its twelve current claims compose already admitted branch outputs. It owns no primitive law and does not repair an unfinished branch. Protein Fold is likewise present as a separate current preliminary application paper, not as a replacement for the Biology owner paper. These distinctions explain why the active-source inventory can contain additional paper and ledger surfaces while the V3 knowledge tree remains a sixteen-paper set.

Seven full-field successor publications

On 29 July 2026 Maria Smith approved seven exact publication candidates. Each was deposited through Zenodo's new-version action on its existing branch record. No independent replacement paper or new concept DOI was created. The public verifier confirmed all seven records as public and latest in their lineages, checked their parent relations and matched every public file size and MD5 identity to the approved local releases.

Branch	Version DOI	Live claims or obligations	Candidates	Controls	Rendered pages
Mathematics 1.5.0	10.5281/zenodo.21688766	323	97,280	1,292	272
Information Science 1.4.0	10.5281/zenodo.21688817	262	75,776	1,048	498
Classical Computation 1.4.0	10.5281/zenodo.21688837	369	94,464	1,476	719
Reversible and Quantum Computation 1.4.0	10.5281/zenodo.21688860	288	73,728	1,152	334
Physics 1.3.0	10.5281/zenodo.21688879	368	257,776	1,472	886
Chemistry 1.3.0	10.5281/zenodo.21688899	281 live claims; 272 registered obligations	71,936	1,124	1,796
Materials Science 1.3.0	10.5281/zenodo.21688923	289	73,984	1,156	790

Together the seven successor papers contain 2,180 live claims, 744,944 generated candidates, 8,720 controls and 5,295 rendered pages. Chemistry's count is deliberately expressed in two ways: 272 registered discipline obligations are complete and nine separately returned claims remain in the live branch, giving 281 live claims. Converting either number into the other would erase a real distinction in the branch ledger.

"Full-field" in these publications means complete to the frozen or registered current-evidence boundary stated by the branch. Mathematics, Information, Classical Computation, Reversible and Quantum Computation and Materials are recorded as full-field complete at their frozen censuses. Physics is complete at its current categorical scope. Chemistry is complete at its 272 registered discipline obligations while retaining the nine returned claims. The public verification establishes deposit identity and custody. It does not convert test success into empirical confirmation or prevent a later correction, adverse result or lawful extension.

The version operation also matters historically. These papers were not published as newly branded replacements that abandoned their earlier records. Each new version retained its concept DOI and explicit parent. A reader can therefore inspect how a branch moved from its earlier foundation or successor edition to the complete-field version without relying on an external narrative about what changed.

Why current branch completion is non-uniform

The seven complete-field papers do not imply that every discipline has reached the same boundary. All fifteen branch foundations are published and complete at their declared foundational scopes. The dated V1/V2 novel-return programme is complete across all fourteen registered subcategories. Beyond that shared base, branch progress is intentionally non-uniform.

Branch after Materials	Current admitted position	Exact current boundary
Biology and Life Sciences	82	82 of 424 frozen full-field obligations closed; 342 open
Medicine and Health Sciences	76	72-law foundation plus four-claim prior-return family complete; full-field census not frozen
Consciousness and Cognitive Science	77	72-law foundation plus five-claim prior-return family complete; full-field census not frozen
Earth and Environmental Sciences	75	74-law foundation plus one-claim prior-return extension complete; full-field census not frozen
Astronomy and Cosmology	77	72-law foundation plus five-claim prior-return family complete; full-field census not frozen
Social and Collective Systems	76	72-law foundation plus four returned claims across the current ledgers; full-field census not frozen
Engineering Translation	80	72-law foundation plus eight-claim prior-return family complete; full-field census not frozen
Cross-Branch Synthesis	12	Twelve inherited-return claims complete; final global synthesis open

Biology is the active full-field branch. Its denominator of 424 is frozen, so 82 of 424 and 342 remaining are meaningful programme counts rather than an estimate of an expanding list. The exact continuation recorded on 29 July is the fourteen-obligation living-biochemistry-and-molecular-processes family, beginning at its frozen target and formal-prediction boundary. A scaffold, draft, test or source capture cannot close one of those obligations by itself.

For Medicine through Engineering, the absence of a frozen field-wide census is not the same thing as 0% completion, nor is a complete foundation equivalent to 100% field completion. The present papers own their foundational laws and dated return families. Their broader field inventories have not yet been frozen, so a percentage would manufacture a denominator. The correct status is therefore qualitative but exact: foundation and registered returns complete; field-wide census not frozen.

This non-uniformity is one of the principal gains of V3. V6 could describe a theory-wide result in one section without exposing whether the surrounding discipline had been fully inventoried. V3 records the achieved result and the state of the containing branch separately. A strong astronomical reconstruction can remain admitted while the complete Astronomy census stays open. A protein result can be scientifically significant without being used to close Biology. An engineering consequence can be derived while deployment and safety validation remain separate.

From the v6 step list to the current 2,751-claim graph

The current corpus contains 2,751 registered, model-admitted claims. They were selected from 892,246 registered candidates, with 2,751 survivors and 11,004 passed controls. All 2,751 candidate censuses match their packages. All 2,751 receipt identities match the registered claim identities. All 2,751 claims trace to the root, the ledger reports zero missing declared dependencies, and it contains no unclassified obligations.

These figures cannot be compared to the historical 303-step heading by simple division. The unit changed. A v6 step could combine a carrier, exact relation, measured comparison, interpretation and open prediction in one paragraph. V3 splits those functions where they require separate owners, grammars, controls or evidence status. It also adds discipline-wide obligation

families, returned prior-work claims and explicit cross-branch compositions. The increase is therefore a reconstruction of granularity and custody, not merely an expansion of prose.

The direct-source content matrix adds a further publication audit boundary. It finds candidate censuses, certificates, controls, registrations and status records for all 2,751 claims. It finds elimination-receipt files for 2,671. The remaining 80 are concentrated in the Chemistry representation, where 201 of 281 live claims expose the machine-extracted survivor-decision surface in the inventory. All 281 Chemistry claims remain model-admitted, receipt-matched, candidate-census-matched and root-traced in the authoritative ledger. This paper must not turn that representational difference into either a fabricated 2,751-of-2,751 elimination-file count or an assertion that the 80 claims are scientifically rejected. It is an audit-layer packaging boundary to retain in the preliminary monograph and final reconciliation.

The content matrix also shows why this monograph cannot be a short overview. It groups the live corpus into 432 derived families. It finds 2,705 claims appearing somewhere in an active paper, 1,889 with an extracted active-paper claim section, 46 absent from active papers and 861 lacking both an extracted active-paper claim section and a WHY record. Every live claim is nevertheless retained in the JSON matrix with its statement, dependencies, grammar, counts, survivor, controls, evidence, chronology, receipt identity and package hashes. The conceptual paper must supply a readable dependency argument while the scientific audit and machine archive preserve what cannot sensibly be repeated in prose.

The current ledger is strongly reconciled at claim level, but it is not the final Grand Lock. The lightweight validator passes at 2,751 claims. The canonical engine seal verifies all sixteen runtime files and the verification-authority seal verifies all thirty-three protected files. The last complete heavy invocation was run on 28 July 2026 at the earlier all-foundations boundary: 1,319 of 1,319 derivation replays and 969 of 969 tests passed. It has not been rerun after the full-field expansion. The final heavy run remains an explicit global obligation.

The global census/lineage_reconciliation record is likewise older than the current 14-of-14 return completion and still reports a blocking row-to-claim merge. The content families are closed at their dated boundaries, but the final one-owner V1/V2 row and step merge has not been completed. A preliminary ToE can accurately synthesise the current state; the final ToE protocol cannot be declared satisfied until that merge, the canonical publication reconciliation and the heavy verification are complete.

Public catalogue reconciliation

The V3 publication operation did not delete the pre-V3 archive. On 28 July 2026 an unauthenticated read of the Zenodo catalogue returned 159 Ernos Labs version records: 38 versions within the sixteen V3 concepts and 121 versions within twenty-four pre-V3 concepts. All 121 pre-V3 version records carried the exact replacement label, and no V3 record carried it. On 29 July the complete-field verifier separately confirmed the seven new branch versions.

The twenty-four pre-V3 concepts include the historical ToE and foundation lineages, the specialist mathematics and physics papers, the computational applications and the earlier software record. Their DOIs and files remain public. The operation added a status notice and successor route rather than changing the past. This allows a reader to follow, for example, an early fine-structure paper into current Physics, an early quantum paper into Quantum Computation, or an early Protein result into Biology and the active Protein V3 application.

The earlier software concept, DOI 10.5281/zenodo.20775538, preserves versions 1.1.0, 1.1.5 and 1.1.6. It routes to the V3 Methods and platform paper and the current public repository. Software remains a scientific instrument rather than an independent theorem. Passing an old executable does not assign current claim status, but preserving its source and version is necessary for reproducing historical outputs.

Supersession therefore has a precise meaning in this corpus. It changes the record that is allowed to speak for current science. It does not delete the superseded bytes, DOI, adverse outcome or chronology. If a present V3 claim is later corrected, the same rule will apply: the corrected successor will govern current status, while the earlier version remains available as the evidence that a correction occurred.

The V3 Protein restart and its first standalone preliminary publication

Protein is the first computational programme to receive an active V3 recreation in the current repository. Its purpose is not to rerun the favourable V2 material panel with new labels. The programme replaces the target-derived selector with an exact sequence representation, explicit source-bound coordinate emission, complete atom-pair custody, a registered six-axis joint physical relation and an exact frontier solver. The intended empirical endpoint is a prospectively sealed 100-target comparison against a frozen AlphaFold 3 comparator.

The current programme has already passed several formal and computational boundaries. A 256-form joint-microstate grammar was completely decided and one relation preserving all six registered axes survived. The fixed-geometry compiler and generic solver execute at the declared standard AGA boundary. Successive parent investigations closed 215,940 and 3,347,070 candidate prefixes. The current largest parent evaluates 42,173,082 candidates across 11,012 shards, comprising 375,014 completed leaves, 41,798,068 certified prunes and 205 frontier states. An implementation-distinct reconstruction matched the result with zero symbolic-comparison fallbacks.

The registered high-attraction slice is also closed. It contains 583 states; 573 are dominated and 10 retained, with 8 new nondominated incumbents. This is a complete result for that slice, not for the remaining attraction bands or the whole fixed-geometry sector. Two registered operations were active at the 31 July freeze: the remaining attraction-subband midpoint census and the seeded-parent high-band reconciliation. No unfinished count from either is imported into the paper.

Protein scientific gate v20 records 893 tests, zero failures and zero errors. The publication-successor full-suite record v21 contains 898 tests, again with zero failures and zero errors. Those results verify their respective implementation and audit surfaces. They do not close an unexecuted frontier, create a blind target or convert the pending AlphaFold comparison into empirical evidence.

The V3 development evidence remains mixed. A no-extra-rule transition probe retained one recurrent class containing all 84 glycine words and is adverse for discrimination. An early 4APD atomistic carrier produced seven sub-0.5-angstrom carbonyl-related pairs; a corrected carbonyl-plane constructor removed every sub-1-angstrom pair, but the corrected structural comparison remained adverse. The later joint relation superseded the old zero-of-six executable-coordinate-family census at its fixed-geometry boundary, while the adverse 4APD comparison remains intact.

Fresh sequence tests preserve the same discipline. The prospectively sealed 4B19 V2-generalised baseline was adverse at official median TM-score 0.17500. A post-target alpha-helical diagnosis motivated an unseen 8HJC holdout, which was adverse at 0.18242. A narrower cysteine-bounded rule transferred positively to unseen 1JDM at 0.45609 within its registered class. The later target-free partial recurrence closed computationally on all three current development sequences, improved 4B19 to 0.23779, but regressed 8HJC to 0.139675 and 1JDM to 0.245185. The positive narrow transfer remains positive; the two later regressions remain adverse. Neither is promoted to a complete protein-fold result.

The official TM-align evaluator and AlphaFold 3 source environment are frozen. The local AlphaFold data tests pass 7 of 7, Apple MPS is visible and the inference suite passes 1 of 17. Restricted weights and the complete registered databases are unavailable. This is an unavailable comparator dependency, not an adverse AlphaFold result. At the freeze there are zero primary blind targets, zero AlphaFold campaign output accesses and no parity result.

The first standalone V3 Protein preliminary paper was published on 31 July 2026 as version 0.9.3. Its version DOI is 10.5281/zenodo.21713537 and its retained concept DOI is 10.5281/zenodo.21713536. The verified public record contains eleven files, names Maria Smith as creator, uses CC BY 4.0 and describes itself as the first standalone preliminary-results release. Its version policy requires the completed investigation to be published through Zenodo's new-version action on the same concept rather than by creating another standalone record.

Version 0.9.4 then lawfully superseded version 0.9.3 as the current publication on that same concept. Its version DOI is 10.5281/zenodo.21717581. It adds the complete 21-record claim audit, identity reconciliation and 898-test full-suite record while preserving scientific gate v20 and the earlier version's historical identity.

The paper's current scientific gate is exact: `verified_42173082_candidate_parent__closed_583_state_high_attraction_slice__remaining_subbands_and_recurrence_open__full_joint_predictor_halt`. In readable terms, the exact representation, six-axis relation, largest parent and high-attraction slice are real completed results. The lower bands, complete fixed-geometry global frontier, arbitrary-sequence whole-chain recurrence, certified full joint predictor and 100-target AlphaFold comparison remain open. The preliminary paper predicts success at the final blind-parity boundary, but prediction is not a parity result.

The standalone V3 preliminary ToE publication

This monograph is the first theory-wide publication of the rebuilt V3 corpus. Its human-readable version is 0.1.0 because it is a preliminary first release in a new standalone V3 Zenodo lineage. It is not version 7 of the deprecated pre-V3 ToE, and concept DOI 10.5281/zenodo.21182468 remains a historical V2 identity only. Maria Smith authorised creation of a new standalone Zenodo record. Draft 21717584 reserves version DOI 10.5281/zenodo.21717584 and concept record 21717583; Zenodo assigns the concept DOI when the record becomes public. The authorised transaction binds the rendered paper, audit layer, machine archive and checksums.

The word *preliminary* does not mean a sketch. It names the current programme chronology. This version must give a complete and publication-grade account of the repository state it freezes: the root, the branch derivations, the field-wide papers, the claim graph, the evidence classifications, historical corrections, current adverse results, open censuses, computational programmes and machine identities. It is preliminary because several registered scientific and integration operations remain open, not because those records may be omitted.

The standalone V3 publication differs from v6 in scientific organisation as well as size. V6 was a compact synthesis organised around a prose step catalogue and a single main proof-suite surface. The standalone V3 publication reconciles a sixteen-paper owner tree, seven complete-field successor papers, later branch foundations, 2,751 live claims, application records, source manifests and receipts. It must explain the dependency relation among those objects without repeating every machine record in its conceptual narrative. Full candidate members, decisions, hashes and traces remain in the audit and archive layers.

The candidate also freezes open work without pre-empting it. A completed Protein predictor and matched AlphaFold campaign will be later evidence. Clean V3 Chess and Go recreations, when started, will produce their own registered chronologies rather than inheriting V2 benchmark status. A closed Unison V3 architecture and empirical campaign will not be backdated into the historical 6.3 release. Biology's remaining 342 obligations and the downstream full-field censuses remain future version content until their own owner records close them.

Chronological status ledger

The principal stages can now be stated without treating every commit or published version as a separate theory.

Stage	Preserved scientific event	Present authority
Initial root lineage	<i>There Is No Nothing</i> versions 1.0-3.0 establish the historical operational starting point	V3 Methods 0.3.0 and Foundation 1.3.0 govern current admission and generated objects
30 June-early July 2026	Prior repository reconstructs the Fold spine and expands into a theory-wide paper	Historical source and public DOI lineage establish chronology
Early-to-mid July 2026	Specialist papers and Protein, Chess, Go and Unison programmes expand formal and executable reach	Current successor routes and preserved historical manifests classify each result
18-19 July 2026	Versioned Chess 2.2, Go 2.2, Unison 6.3 and blind-76 Protein manifests bind measured implementation evidence	Historical implementation evidence; not current V3 proof authority
22 July 2026	ToE v6.0 and its contemporary catalogue surface are published	Checksum-bound v6 is the final pre-V3 comparison record
23 July 2026	V3 clean-room constitution, knowledge tree and fail-closed engine begin	Current V3 repository becomes the admission authority
23-28 July 2026	Sixteen V3 Methods and branch papers, foundations and return programmes are constructed and published through their own lineages	Current V3 paper set and claim packages
28 July 2026	Last global heavy run passes 1,319 derivation replays and 969 tests	Valid for the all-foundations boundary; later expansion still awaits final heavy run
29 July 2026	Seven complete-field successor versions are published and independently checked against approved local files	Mathematics through Materials public records and final publication verifier
29-31 July 2026	Protein V3 closes progressively larger exact parent searches and a 583-state high-attraction slice	Protein scientific gate v20
31 July 2026	First standalone Protein V3 preliminary paper 0.9.3 is published	DOI 10.5281/zenodo.21713537 and its publication receipt
31 July 2026	Protein V3 preliminary paper 0.9.4 supersedes 0.9.3 on the same concept	DOI 10.5281/zenodo.21717581; scientific gate v20 preserved; complete claim audit and 898-test receipt added
31 July 2026 freeze	Live corpus reaches 2,751 admitted claims, 892,246 candidates and 11,004 controls	Authoritative corpus inventory and content matrix
Authorised V3 action	Exhaustive preliminary V3 ToE 0.1.0 prepared as a new standalone Zenodo record; deprecated V2 concept DOI 10.5281/zenodo.21182468 retained as history only	Maria Smith authorised draft 21717584 and reserved version DOI 10.5281/zenodo.21717584 for publication

This ledger does not imply that every event began or ended on the day shown. It identifies the dated records that freeze the relevant status. Application development overlapped paper preparation, branch derivation overlapped publication reconciliation and Protein work continues after its preliminary deposit. The table is a custody map, not a simplified story of one task ending before the next began.

What is preserved, what is superseded and what remains open

The root question, the One, the Fold and the project's open-science purpose are continuous through the publication history. Their current formulations are the V3 formulations. The early papers retain priority and development context, but their language is not used to bypass the current candidate grammars.

The breadth of v6 is also continuous. Mathematics, physics, chemistry, life, mind and computation remain within the programme. What changed is their custody. A v6 proposition that now has an admitted V3 successor is read through that successor. A proposition that was narrowed remains present at the narrower scope. A proposition with an adverse or unavailable current record is reported as adverse or unavailable. A proposition not yet reconstructed remains historical or open; it is not restored merely because it appeared in the v6 abstract.

The V2 computational programmes are preserved as empirical and engineering history. Chess retains its exact rules checks and measured 1700, 1900 and 2100 batches. Go retains its exact censuses, solved values, replayed GNU Go games and defective historical KataGo harness record. Protein retains strong local and restricted material results together with adverse whole-chain and generalisation tests. Unison retains its large counted implementation surfaces and mixed benchmarks. None is silently relabelled as a completed V3 application.

The seven full-field branch publications are current. Their deposit identities and current-evidence closures are established at their declared scopes. Biology and the later branches are current at their foundations and returned families, not at an invented universal full-field percentage. The 2,751-claim ledger is current at claim, candidate-census, control, receipt-identity, dependency and root-trace levels. The final heavy run, the global one-owner historical merge and the final ToE protocol remain open.

Protein V3 is current through its 42,173,082-candidate parent and 583-state high-attraction slice. Its adverse controls and positive narrow transfer are also current at their own boundaries. General whole-chain recurrence and AlphaFold parity remain unresolved because the registered decisive campaign has not been run. Chess V3 and Go V3 have not started in the current repository. Unison V3 is a scaffold with mathematical closure and empirical validation open.

Finally, the V3 ToE 0.1.0 is a local preliminary candidate until the exact release is approved and deposited as a new standalone record. Its future concept and version DOIs are not assigned. The publication process must not reuse the deprecated V2 concept DOI, invent a DOI or claim that an unapproved local artefact is public. After this first V3 publication, later V3 ToE updates must use Zenodo's version action on the new V3 concept lineage.

Reconciliation as a scientific result

The movement from the first no-nothing paper to this monograph is not simply a sequence of increasingly long documents. It is a change in the unit of scientific accountability. The first papers asked whether a premise-free starting point and its consequences could be articulated. The large v6 synthesis demonstrated the reach of that programme and brought formal, physical and computational work into one visible object. The applications showed that Fold relations could generate serious executable systems and also exposed where restricted success failed to generalise. V3 turned those lessons into explicit rules of ownership, chronology, candidate custody and evidence.

That reconstruction makes the project more persuasive because it makes disagreement local. A reader need not accept an entire theory to inspect one claim package. A failed Protein transfer does not become a vague verdict on the root theorem; it remains an adverse result for a named selector. A passing proof suite does not become empirical confirmation; it remains implementation evidence. A measured physical correspondence cannot select its own exact form; the derivation and comparison routes remain separate. A later correction does not require forgetting what was previously published.

The resulting corpus is both larger and more bounded than v6. It is larger in its 2,751 claims, 892,246 candidates, 11,004 controls, sixteen-paper tree and seven field-wide publications. It is more bounded because every use of completion, proof, prediction and validation must name its scope. The open frontier is no longer a paragraph appended after a totalising claim. It is part of the dependency graph itself.

The present trajectory therefore closes with a precise publication position. The repository contains a current V3 scientific system whose root, branch owners, live claims and public successor papers can be audited directly. It also contains historical papers and computational records whose favourable and adverse results remain discoverable. The proposed standalone V3 monograph is the first theory-wide publication designed to reconcile those two surfaces at the scale of the current corpus. It is complete for the state it freezes and preliminary for the operations it identifies as open. Any later version must extend or correct this record without rewriting its chronology. That is the end of this project-version investigation, not the end of science.

Part I - From Occurrence to Quantum Computation

The root, Foundation, Mathematics, Information Science, Classical Computation, and Reversible and Quantum Computation

This part follows the first continuous dependency route in Smithian Fold Theory V3. It begins before arithmetic, probability, computation or physical law is available. It ends at the exact interface through which Physics receives generated states, processes and resource identities while retaining ownership of preparation, units, apparatus, uncertainty and measurement.

The sequence is not a conventional classification imposed on a finished theory. Each branch exists because the preceding branch leaves a precise problem unresolved. The operational root establishes that an admissible proof object occurs, but not yet how an occurrence is represented as a complete whole. Foundation derives that whole and its minimal reversible distinction, but not a general mathematical science. Mathematics supplies exact carriers, relations, proofs and bounds, but it does not yet distinguish a state from a record of that state. Information Science supplies that custody, but not a machine that executes a registered process. Classical Computation derives the machine and its limits, but hands reversible and quantum operations to their own branch. Reversible and Quantum Computation then derives complete branch support, phase, interference, entanglement, measurement semantics, gates, circuits, correction and limits before any physical device is allowed to select the formal law.

At the repository freeze used for this monograph, the five unique scientific ledgers from Foundation through Quantum contain **1,258 admitted claims, 346,470 generated candidates, 1,258 unique survivors** and **5,032 passed controls**. The two inaugural results reproduced by the Methods paper are the root and structural One already counted inside the sixteen-claim Foundation ledger; they are not counted twice. The current claim matrix classifies 194 of the 1,258 claims as `independently_replicated` and 1,064 as `empirically_tested_and_independently_replicated`. In the formal branches, "empirically tested" refers to registered post-seal execution and observation of the declared computational objects. It is not silently enlarged into a measurement of nature or hardware.

Layer	Active version	Current publication identity	Claims	Candidates	Controls	Current boundary
Methods, <i>There Is No Nothing</i>	0.3.0	10.5281/zenodo.21627646	2 inaugural results, included in Foundation	8 across those two registrations	8	Admission constitution and founding operational results
Foundation, <i>From Nothing to Fold</i>	1.3.0	10.5281/zenodo.21627656	16	5,222	64	Foundation complete
Mathematics, <i>From Fold to Mathematics</i>	1.5.0	10.5281/zenodo.21688766	323	97,280	1,292	Full-field current-evidence complete
Information Science, <i>From Distinction to Information</i>	1.4.0	10.5281/zenodo.21688817	262	75,776	1,048	Full-field current-evidence complete
Classical Computation, <i>After Turing: The Fold Machine</i>	1.4.0	10.5281/zenodo.21688837	369	94,464	1,476	Full-field current-evidence complete
Reversible and Quantum Computation, <i>The Quantum Fold Machine</i>	1.4.0	10.5281/zenodo.21688860	288	73,728	1,152	Full-field current-evidence complete
Unique total, Foundation through Quantum	-	-	1,258	346,470	5,032	Dated and extension-open

The public release verification of 29 July 2026 is controlling for the four complete-field version DOIs. Pre-deposit wording embedded in their frozen manuscripts is historical release-candidate text and is superseded on the publication-status question by the later Maria Smith approval receipt and public Zenodo verification. That supersession changes publication status only; it does not alter a claim, candidate, control, evidence class or scientific boundary.

Method and the operational root

The problem before the first premise

A premise-free programme cannot begin by assuming a conventional logical system, a number, a set, a space or a physical object. Any such starting point would already contain the structure the programme claims to derive. The first question must therefore be narrower: what can be said about an object that has entered an operational statement, denial, record, proof or derivation?

The registered root theorem is SFT-ROOT-THERE-IS-NO-NOTHING:

<i>No admissible operational statement, denial, record, proof object or</i>
<i>derivational object is nothing: absence presents no counterexample, while</i>
<i>every presented counterexample is an occurrence.</i>

Its generator partitions every purported counterexample by one relation: whether it presents an identifiable object at the proof boundary. The complete declared grammar has two classes. `unpresented-absence` supplies no statement, denial, trace or identity and therefore supplies no counterexample. A `presented-occurrence` supplies an identifiable presentation; that presentation is itself an occurrence and therefore is not nothing. Of the two generated classes, exactly one is executable as a purported counterexample, and its execution defeats the purported counterexample.

The claim has two generated candidates, one unique survivor, four passed controls and depth-independent closure inside its stated operational boundary. The controls reject a false premise with no counterexample identity, changed source custody, a tampered two-survivor artefact and an attempted expansion into an unexpressed metaphysical domain. An implementation-distinct Python reconstruction regenerates the partition without importing an SFT module. The result is a formal operational theorem. It does not claim to describe an unpresented metaphysical domain; it establishes the root required for a public, machine-checkable derivation.

From occurrence to the structural One

Occurrence alone does not yet supply a number. The next unresolved question is how the admitted occurrence can be represented completely without omission or unforced addition. SFT-FOUNDATION-ONE-001 generates six coverage-and-addition forms. The unique survivor retains the occurrence as its complete self-whole, adds nothing not supplied by the occurrence and omits none of it. This is the structural **One**.

The One is not imported as the conventional numeral 1. It is the exact self-whole relation forced by complete coverage. Its six candidates, one survivor and four controls are included in the Foundation totals. The Methods paper owns and reproduces this inaugural result because the result completes the minimal starting interface; later count, part and Fold laws remain separately registered Foundation claims and cannot become retroactive premises of the root or One.

The admission constitution

The two founding results establish more than vocabulary. They demonstrate the route every later SFT claim must follow. A registered claim records its exact statement, dependencies, exclusions, provenance, candidate generator, candidate boundary and intended generality certificate before it can acquire model authority. The admission sequence is:

1. register the claim and its dependency identities;
2. generate the complete declared candidate grammar;
3. retain one decision for every candidate;
4. require exactly one survivor and prove its minimality and named-shape uniqueness;
5. establish finite, finite-successor or depth-independent closure at the declared boundary;
6. execute the required false-premise, changed-source, tampered-artifact and boundary controls;
7. seal the derivation and source identities;
8. reconstruct the result through an implementation-distinct route;
9. where nature is claimed, register and seal the consequence before target access, then preserve the complete observation and comparison record; and

10. issue an unchanged-engine receipt only after every applicable gate passes.

This is a fail-closed route. A successful example, conventional name, institutional endorsement, benchmark score or attractive application cannot replace the candidate census. A failed control, missing dependency, changed source or duplicate survivor halts admission. An application may consume an admitted law and may later test it, but it cannot tune the earlier grammar or select the law that the application is then said to support.

The mathematical constitution excludes imported axioms, fitted, trained, free or target-derived parameters, semantic numerical zero, negative proof magnitudes, irrational or imaginary proof scalars, floating-point proof equality, completed infinity and an ungenerated continuum. These exclusions do not deny that host programs use indices, empty containers, Boolean values and process return codes. Those objects are quarantined implementation mechanics: they may count or execute evidence but cannot become SFT proof objects. Likewise, conventional mathematical and scientific descriptions may enter later as declared correspondences or measurements; they may not flow backwards into the derivation.

The empirical constitution

Formal forcing and empirical testing answer different questions. Candidate generation asks which structure survives a declared grammar. Empirical testing asks what a source-bound observation establishes after the formal consequence has been sealed. For a natural-science claim, capability closure precedes a value-free target registry; target custody remains separate from the predicting process; the executable predictor and environment are sealed; target or comparator output opens only afterwards; and every favourable, adverse, null, failed, missing and unavailable row remains in the record.

A post-target calculation remains post-target. A development-observed result remains development-observed. A failed transport is a failed transport rather than an absence in nature. An unresolved source is not a null result. Later evidence may lawfully change a current classification when the registered protocol permits it, but it cannot become an earlier blind prediction by editorial declaration.

The engine's complete executable-line coverage verifies which implementation lines the tests executed. It does not prove the absence of every software defect, complete an omitted candidate grammar or empirically confirm a natural law. Cryptographic hashes identify the exact artefacts that were checked; they do not make those artefacts immune from scientific criticism.

What an admitted claim actually contains

The admission route is easiest to misunderstand when it is reduced to the phrase "the computer checked it". The computer does not receive a sentence and decide whether the sentence sounds true. A claim package is a typed collection of objects whose identities and relations are fixed before the final decision. The human-readable statement names the scientific proposition, but model authority belongs to the registered package rather than to persuasive prose.

The registration names the claim ID, title, owner, dependencies, excluded inputs, candidate boundary, generation rule, intended certificate and required controls. Each dependency must already possess a model-admitted receipt, and the registered route must lead to `SFT-ROOT-THERE-IS-NO-NOTHING`. A narrative citation to an upstream paper is not enough. The engine resolves the actual receipt identities, so a missing dependency or a reference to a merely proposed claim halts the route before enumeration.

Candidate generation then produces the literal declared product. A coordinate may describe coverage, order, labelling, overlap, source custody, empirical chronology or any other distinction that the claim says exhausts its boundary. Every product member receives a stable candidate identity. The decision ledger must contain one and only one decision for each identity. This requirement prevents three common forms of retrospective selection: generating only the favourable alternatives, silently dropping an awkward case, or adding a new alternative after seeing which structure appears to work.

Elimination is ordered and explicit. A rejected candidate records its first decisive failure, while the survivor records the complete set of preserving properties. "Unique" means that the complete registered product contains one survivor, not that the prose happens to discuss only one preferred object. "Minimal" means that no less structured candidate within the grammar preserves the requirements. "Named-shape uniqueness" checks that the surviving identity and exact form are the ones declared by the theorem rather than an accidental isomorphic or mislabelled result.

Closure is a separate question. A finite census can establish only its exact finite boundary unless the package also supplies a lawful successor or depth-independent certificate. A base-and-successor proof may cover every supplied positive finite depth without constructing a completed infinite object. A bounded search remains bounded when no such certificate exists. This distinction is why a 10,000-value prime census and an unrestricted positive-finite Fold recurrence can both be exact without having the same generality class.

The certificate binds the claim, grammar, decisions, survivor, controls, source manifest, derivation seal, independent implementation and closure scope. The final receipt is not a free-standing badge. It is an identity map to those underlying artefacts. If a source changes, the old hash does not bless the new source. If a candidate is added, the earlier completeness certificate does not cover it. If a law is extended, the extension receives a new claim and receipt rather than inheriting authority by similarity.

Why four controls recur

The repeated four-control structure is methodological rather than decorative. Each control attacks a different route by which an invalid result could appear to pass.

The **false-premise control** supplies an object that should fail the scientific requirement. For the root it attempts to use unrepresented absence as a counterexample; for Fold it proposes the unchanged identity as a non-trivial distinction; for an error-correction claim it may supply an ambiguous or exceeding-budget image. A system that accepts both the result and its explicit false premise has not discriminated the law.

The **changed-source control** alters a registered source or dependency identity. Its purpose is not to show that cryptography proves the scientific claim. It shows that a different input cannot silently reuse the old decision and evidence chain. Source identity is scientifically important wherever changing a source could change the result, and chronologically important where the source must remain inaccessible until a seal.

The **tampered-artifact control** changes a candidate set, decision vector, survivor, certificate or output. It tests whether the implementation notices a mechanically plausible but unauthorised result. A missing decision, duplicate survivor or altered terminal value must halt rather than being repaired by a reader's charitable interpretation.

The **boundary control** tries to import something the claim excludes or to expand the conclusion beyond its registered domain. Depending on the branch, that may be a numerical zero, a signed or floating proof value, a stochastic cause, an external answer table, an application result, a physical measurement inside a formal theorem, or an unrestricted conventional conclusion from a native bounded result. Passing the boundary control verifies that the implementation rejected the prohibited expansion. It does not establish that no other boundary criticism is possible.

The controls are called adverse because they are designed to make the route fail in known ways. A passed adverse control means the expected rejection occurred. That must not be confused with an adverse scientific observation, which is a registered empirical result conflicting with the tested claim. Both remain visible, but their evidence classes differ.

Independent reconstruction and model authority

The main derivation and its validator cannot be the same executable path under two filenames. An implementation-distinct reconstruction must regenerate the declared object without importing the scientific module whose result it is checking. This reduces the risk that one local coding error is merely repeated by the validator. It does not make the two programs logically independent in every possible sense; they may still share a mistaken specification. That is why candidate completeness and specialist criticism remain necessary.

The engine is intentionally narrower than Maria Smith's scientific and publication authority. It can establish that a registered package passed its encoded gates under the protected engine and verifier identities. It cannot decide whether Maria Smith approves publication, whether a conventional term is rhetorically helpful, whether a newly proposed empirical interpretation belongs to the claimed owner, or whether an unregistered specialist objection has scientific force. Conversely, publication does not cause model admission. An attractively edited or publicly deposited paper cannot turn a halted package into an accepted claim.

This separation also applies to agent assistance. An agent may generate prose, run a verifier, compare hashes or identify a discrepancy. It cannot alter Maria Smith's authorship, approve a release on her behalf, grant a new claim scientific status or delete an adverse record because it complicates the narrative. The public repository makes that limitation inspectable by keeping the machine authority, publication receipt and human authorship records separate.

Formal, computational and natural evidence

The same word can obscure very different evidential acts, so this monograph reserves its verbs. A theorem is **proved** only at its formal grammar and dependency boundary. A structure is **derived** when its alternatives are generated and eliminated to the surviving form. An executable program **implements** or **demonstrates** a derivation; successful execution does not by itself measure the world. A consequence is a **prediction** only when its target identity and chronology were correctly sealed before target access. A source-bound external row is an **observation**, and a value tied to instrument, method, condition and uncertainty is a **measurement**.

A post-derivation relation between an SFT form and a conventional form is a **correspondence**. Structural correspondence concerns form; exact numerical correspondence concerns equality or an exact interval relation; boundary correspondence is confined to a named interface or limit. Evidence is **compatible** when it is non-adverse but non-discriminating, and **supporting** when relevant but non-unique. **Confirmation** is used only where the current registered evidence protocol grants that classification. A conflicting registered outcome is **adverse**. Evidence remains **unresolved** when the required source is unavailable, incomplete, disputed or not yet sufficiently classified.

These distinctions are not stylistic modesty. They preserve chronology and falsifiability. A mathematical identity can correspond exactly to an observed number without having predicted that number. A program can validate its own trace without validating a physical theory. A later successful transport can resolve a current access problem without erasing the historical failed transport. A current adverse row can remain historically visible after a lawful new claim at a stronger boundary succeeds.

Threat model and limitations of the platform

The repository architecture addresses identifiable failure modes: source drift, missing candidates, duplicate or missing decisions, hidden survivors, changed controls, altered receipts, dependency substitution, post-seal target access, suppressed rows and silent authority changes. Cross-platform commands and standard-library implementations reduce environmental opacity. Source manifests and hashes permit a reproducer to identify the exact object whose status is being discussed.

It does not eliminate every threat. A registered candidate grammar may omit a scientifically necessary coordinate that no current test anticipates. Two implementations may share the same conceptual error. A source may be authentic but inappropriate for the claim. A measurement protocol may preserve its rows and still be poorly designed. A hash may bind a mistaken artefact perfectly. No finite test suite proves the absence of all defects. Domain experts must therefore inspect the definitions, boundaries, eliminations, controls and empirical methods rather than treating the receipt as an oracle.

The clean-room rule controls answer flow from V1 and V2. Earlier work may identify questions to reconstruct and may be compared after the V3 result is sealed, but earlier certificates, executables and answer tables cannot enter the new derivation runtime. The reconciliation layer then asks whether the V3 result preserves, corrects, narrows or leaves open each prior owned atomic obligation. A clean room without reconciliation could omit history; a reconciliation without a clean room could copy the answer. Both stages are required.

Repository as scientific publication

The platform's three access levels serve different readers. The conceptual paper provides the dependency argument and major results. The scientific audit layer exposes complete claim status, sources, controls, chronology, corrections, ownership and reconciliation. The machine archive retains every candidate, decision, trace, source snapshot, certificate and receipt. Moving a full hash table out of a conceptual paragraph is an editorial change; deleting the underlying hash or receipt is a scientific loss.

This architecture is also the practical anti-institutional claim of Ernos Labs. The project does not ask readers to exchange trust in an established institution for trust in a new one. It asks them to inspect the exact route by which a claim acquired its current status. Maria Smith's independent position is not evidence for the theory, and institutional prestige is not evidence against it. The same public grammar, control and source questions apply to an independent author, a professor, a corporation and an automated system.

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Open authority and falsification

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The most direct formal falsifications are an omitted same-boundary candidate, a second survivor, a simpler preserving form, a failure of the base or successor certificate, or a counterexample inside the declared domain. Mechanical falsifications include a dependency mismatch, changed source, missing decision, failed control, non-reproducible trace or receipt mismatch. Natural-science falsification additionally follows the claim-specific measurement and chronology protocol. The method is therefore complete as an admission constitution at its current versioned boundary while remaining open to a lawful correction, a stronger test and a new registered question.

8. Foundation: from One to Fold

8.1 Why Foundation follows the root

The root and One establish occurrence and complete self-wholeness. They do not yet provide a positive count, a proper part, a distinction, a transformation, a composed word, a proof trace or an empirical boundary. Foundation owns those objects because every later branch requires them. It does not own general mathematics, a physical quantity, an observational dataset or the behaviour of an application.

The current Foundation is complete at sixteen claims. It generated 5,222 candidates, retained sixteen unique survivors, passed sixty-four controls and was independently reconstructed sixteen times. Its prior-work audit read all 763 registered V1/V2 entries-356 V1 and 407 V2-identified twenty-four Foundation-relevant source entries, decomposed them into thirty-two Foundation-owned atomic obligations and closed all thirty-two. The remaining 739 entries were recorded as reviewed non-Foundation ownership rather than silently omitted.

8.2 Complete Foundation theorem ledger

Dependency order	Claim	Candidates	Exact contribution
1	SFT-ROOT-THERE-IS-NO-NOTHING	2	Presented occurrence is the unique admissible operational class.
2	SFT-FOUNDATION-ONE-001	6	The occurrence's complete self-whole is the structural One.
3	SFT-FOUNDATION-COUNT-001	10	A non-empty finite succession has one exact positive count: its complete generation trace.
4	SFT-FOUNDATION-PART-001	192	A held selection in a complete, disjoint, equal-fibred partition has an exact positive held/whole coordinate.
5	SFT-FOUNDATION-FOLD-001	288	The minimal non-trivial reversible distinction is two equal disjoint held fibres with a complete return to One.
6	SFT-FOUNDATION-PART-EQUIVALENCE-001	1,024	Parts from different partitions are equal exactly through complete common refinement and bijective lifted selection.
7	SFT-FOUNDATION-FOLD-ASSEMBLY-001	192	Every finite Fold succession has complete held-label word support and retained transition and return traces.
8	SFT-FOUNDATION-EXACT-OPERATIONS-001	128	Exact positive junction, product, guarded Take, complete-One cast, unison and Fold form one root-bound interface.
9	SFT-FOUNDATION-HALF-ONE-001	128	One held fibre of the first Fold is the unique proper one-of-two ground.
10	SFT-FOUNDATION-FOLD-DYNAMICS-001	128	Fold is closed, two-to-one, phase-pair identifying and recurrent at the first one-third/two-thirds cycle.
11	SFT-FOUNDATION-PRIMITIVE-MAP-UNIQUENESS-001	84	Fold is the unique least-size generator in the declared four-map composition grammar.
12	SFT-FOUNDATION-FORM-GRAMMAR-001	288	Foundational forms are generated by a One leaf or a finite two-child Fold node with held edges and return.
13	SFT-FOUNDATION-FORM-ENFORCEMENT-001	192	A form's identity is its complete canonical recursive construction trace.
14	SFT-FOUNDATION-DERIVATION-TRACE-001	256	Every admitted finite derivation retains immutable sources, ordered operations, intermediates and root lineage.
15	SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001	256	Derivation-to-measurement flow is one-way and every target row is retained.
16	SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001	2,048	Exactly one generated admission route retains every required formal, control, validation, empirical and receipt gate.
	Total	5,222	Sixteen depth-independent, model-admitted Foundation claims

Every row has exactly one survivor and four passed controls. Candidate counts are not publicity estimates: they are the exact sizes of the registered products. The unequal sizes reflect different grammars. The root requires a two-class operational partition, whereas admission enforcement must decide a 2,048-member product of custody and gate choices.

8.3 Count, part and exact equality

SFT-FOUNDATION-COUNT-001 resolves the first question left by One: how can plural occurrence be represented without assuming arithmetic? A generated non-empty finite succession retains every generated occurrence exactly once and adds none. Its ordered generation trace is its positive count. The theorem does not begin with a numeral; the familiar numeral is a later interface label for the already generated trace.

A count alone does not identify a portion of a whole. SFT-FOUNDATION-PART-001 therefore starts from a complete, disjoint, equal-fibred partition of One and a non-empty held selection. The exact coordinate is the positive count of held fibres relative to the positive count of the complete partition. Equality across different partitions cannot be imported as fraction reduction. SFT-FOUNDATION-PART-EQUIVALENCE-001 generates the pair-cell common refinement of two partitions, lifts both selections completely and requires a one-to-one and onto pairing of the lifted held cells. The resulting equivalence is reflexive, symmetric, transitive and stable under uniform refinement without requiring an external scale.

This establishes the exact positive part carrier used throughout the corpus. Empty selection remains structural empty One rather than a numerical proof value. Opposition is retained by a held label identifying which side owns an unmatched positive remainder. The constitution can therefore represent absence, order and orientation without importing signed arithmetic at the foundation.

8.4 The minimal Fold

The next unresolved problem is change. Identity leaves One unchanged, while an arbitrary partition would introduce a selector or scale not present in the unqualified whole. SFT-FOUNDATION-FOLD-001 exhausts 288 combinations of whole coverage, fibre overlap, fibre equality, extension class, label relation, return relation and extra data. The unique survivor is complete, disjoint and equal; uses the first positive-count extension; retains two distinct held labels; includes an exact return to One; and adds no free datum.

That survivor is the minimal **Fold**. Its two labels are structural identities, not positive and negative numerical quantities. The return relation matters: without it, the partition is not established as a reversible distinction of the source whole. The candidate eliminations also establish minimality. The identity has no non-trivial distinction; every later extension is non-minimal because the first extension already supplies one; overlapping or unequal fibres fail exact partition; unheld or identical labels fail operational distinguishability.

Repeated Fold must then be generated rather than presumed to be binary notation.

SFT-FOUNDATION-FOLD-ASSEMBLY-001 begins from the structural empty One word. At each finite successor, it appends each of the two held labels once to every preceding word, retaining predecessor, transition and return traces. The first five generated support sizes are conventionally displayed as 2, 4, 8, 16 and 32, but the generality certificate is the base-and-successor construction, not extrapolation from those examples.

8.5 Exact operations, half-One and dynamics

The exact operational interface acts on positive rational parts through One. Complete turns cast to One rather than to numerical absence. Junction combines positive traces; product is generated pair-cell support; Take (a, b) is admissible only when a is strictly larger than b ; unison is the exact One relation. Operational Fold is self-junction followed by complete-One cast:

$$\text{Fold}(x) = \text{cast}(x \text{joined} x).$$

Thus $\text{cast}(2) = \text{One}$, $\text{Fold}(3/4) = 1/2$, and $\text{Take}(\text{One}, 1/4) = 3/4$ in the exact registered examples. Exact forms precede their conventional notation; the notation does not select the operation.

The first Fold forces the unique proper one-of-two ground, **half-One**. Either singleton held fibre self-junctions to One, has an equal-coordinate complement and Folds to One; One itself is Fold-fixed. Version 1.3 explicitly corrects a terminology collision retained from earlier work. Complement and phase translation are different operations:

$$C(x) = \text{Take}(\text{One}, x), \quad A(x) = \text{cast}(x + \text{half} - \text{One}).$$

Half-One is self-complementary, so $C(\text{half-One}) = \text{half-One}$; its half-One phase translation casts to One, so $A(\text{half-One}) = \text{One}$. Both prior structural results remain preserved, but the names are no longer conflated.

Fold dynamics then follows. For each image y , one preimage is $y/2$ and the other is its half-One phase partner. The phase map is involutive and Fold identifies each phase pair:

$$\text{Fold}(A(x)) = \text{Fold}(x).$$

For every generated even partition, phase-paired positions have the same Fold image and the distinct images form the next uniform partition. The first non-static recurrent orbit is exact:

$$\frac{1}{3} \mapsto \frac{2}{3} \mapsto \frac{1}{3}.$$

The primitive-map theorem scopes its stronger uniqueness claim carefully. Its normalised zero-parameter grammar contains identity, square, constant One and Fold. Base-four enumeration executes 4 one-operation words, 16 two-operation words and 64 three-operation words, 84 in total. Identity is static; square is injective and contracts every proper part; constant One collapses; Fold alone is both non-injective and non-statically recurrent at size one. Positive construction-size induction prevents any larger composition from displacing that already qualifying size-one survivor. This is uniqueness within the registered grammar and generator predicate, not an unrestricted theorem over every imaginable expression language.

8.6 Form, proof and evidence boundaries

Fold words provide support, but downstream branches also require a canonical object language. The foundational form grammar therefore has two productions: a structural One leaf and a finite Fold node with two recursively generated children on distinct held-labelled edges and an explicit return to the parent whole. Form enforcement identifies a form by its complete canonical construction trace; equivalent forms reproduce the same trace.

The derivation-trace theorem extends that custody to a whole proof. A valid record retains immutable source and dependency identities, every operation in order, the exact identity of each input and output, every intermediate and the terminal result. Deterministic replay must reproduce the record and its route to the root. The final admission theorem exhausts 2,048 possible combinations of root custody, source custody, parameter status, enumeration, uniqueness, form closure, controls, distinct validation, empirical order, disposition retention and dependency use. Only the route retaining every gate survives.

The measured-value boundary is deliberately one-way. A formal law may seal a consequence that a later source-bound measurement tests. A measurement may support, conflict with or leave the consequence unresolved, but no value, score or target may travel backwards to alter the law. Foundation itself owns no physical measurements. Its sixteen empirical classifications are independent formal reproductions, not claims that nature has been measured.

8.7 How the Foundation grammars force rather than name the result

The theorem ledger can look deceptively familiar after its objects have names. "One", "part", "Fold" and "half" evoke conventional mathematics, but the candidate products precede those translations. The scientific burden is to show why the survivor is forced by the admitted dependencies and why nearby forms fail without relying on the familiar name.

For count, the generator distinguishes complete versus incomplete generation trace, duplicate versus single custody of an occurrence, ordered succession versus an untraceable total, and extra data versus no addition. A conventional integer label cannot repair a trace that omits or duplicates an occurrence. The survivor is therefore not "the numeral that equals the length"; it is the complete positive succession itself. The numeral is a readable correspondence to that retained construction.

For an exact part, the whole must be completely partitioned, cells must be disjoint, fibres must be equal and the held selection must be non-empty and contained within the partition. Dropping completeness makes the denominator unknown. Permitting overlap double-counts shared structure. Unequal fibres introduce a weight not supplied by One. An unheld selection has no operational identity. The 192 candidates exhaust these distinctions and leave the positive held-count/whole-count coordinate. This order is important: a fraction is not assumed and then used to justify equal fibres; equal-fibre generation and held selection force the coordinate later written as a fraction.

The Fold product makes the same discipline visible at greater scale. Its 288 candidates classify whole coverage, overlap, equality, extension class, label relation, return and added data. Exactly 144 lose part of One and fail at whole coverage. Of the remaining forms, 72 repeat structure through overlapping fibres and 36 introduce an unsupported allocation through unequal fibres. Twelve identity-count forms never become non-trivial, while twelve later extensions are non-minimal because the first extension already distinguishes. Four forms omit labels, four reuse one label for both fibres, two omit the return and one adds an external datum. The one remaining form has every property in the theorem statement. This elimination census is the reason for the result; the word Fold is assigned after the form survives.

Cross-partition equivalence must then solve a problem that naive coordinates cannot. One-of-two and two-of-four have different construction traces but may represent the same part. One-of-two and one-of-three do not. The 1,024-member equivalence grammar generates common pair-cell refinement and asks whether it covers both wholes, remains disjoint and equal-fibred, refines both source partitions, lifts both held selections completely and pairs the lifted cells bijectively without a new scale. Incomplete coverage eliminates 512 forms; overlap eliminates 256; unequal cells eliminate 128; failures of the two refinement and two lifting conditions successively remove 64, 32, 16 and 8. Non-bijective, unmatched and absent pairing classes remove the remaining non-preserving forms, and an extra-scale form is rejected last. The generated one-of-two/two-of-four example passes; the one-of-two/one-of-three control does not.

Fold-word assembly similarly avoids importing exponentiation. Its 192 forms classify step coverage, word-length consistency, support completeness, word uniqueness, transition records, return records and added data. A proposal that retains no Fold step or omits one loses the recursive support. A word with the wrong length loses one-label-per-step custody. Missing or duplicate words destroy completeness. Missing transitions prevent replay; missing returns break the relation to One. The survivor is therefore the complete recursive word tree, from which the conventional support size 2^k may later be read.

The form grammar then compresses that recursively generated support without discarding its origin. A One leaf is terminal. A Fold node has two children on distinct held edges and an explicit return to the parent whole. A candidate with one child omits a forced fibre; a node with indistinguishable edges collapses the distinction; a node without a return ceases to represent a Fold of its parent. Form enforcement serialises the complete recursive construction canonically, so equality is replayable trace identity rather than a visual or authority judgement.

8.8 Exact operation design and the prohibition on silent absence

The operational interface is designed to remain closed on the exact positive carrier. An unguarded subtraction-like operation could produce absence or a signed magnitude not yet derived. *Take* therefore requires a strictly larger source part and returns only the retained positive remainder. When two forms coincide, the system records the unison relation rather than assigning a negative or zero distance. When self-junction completes a whole turn, *cast* returns *One* rather than deleting the completed cycle.

This distinction is functional, not cosmetic. If a completed turn were stored as numerical zero, the same literal could mean an empty support, an index origin, a process status, a vanished quantity or a complete return. Those meanings have different dependencies and different reconstruction rights. The Foundation preserves them as typed structural records. Empty *One* names a structural absence at an interface; *One* names the complete self-whole and the completed return; held orientation names which positive side retains an unpaired part.

Fold's two-preimage law follows exactly from this cast. For an image y , the lower preimage self-junctions directly to y . Adding half-*One* before self-junction adds one complete *One* after doubling, which cast removes as a complete turn without losing the retained phase label. The two preimages are therefore distinct in the source and identical under Fold. A third preimage would require an added phase class not generated by the first Fold.

Uniform even partitions expose the same law globally. In a partition with $2n$ positions, position i and position $i+n$ are half-*One* phase partners. Fold maps them to the same member of the n -position image partition. After duplicate images are identified, every image position occurs with the same generated multiplicity. This is a general finite-successor result over every generated positive n , not a pattern inferred from the six executed control partitions.

The recurrent one-third/two-thirds orbit supplies the least non-static witness for the primitive-map predicate. Identity has no movement. Square retains distinct inputs and strictly contracts a proper positive part. Constant *One* has a fixed collapsed image. Fold has both a two-member fibre and a non-static return cycle. The grammar's construction-size order then makes Fold the first qualifying map before any composition. Enumerating the 84 words through size three demonstrates the ranking implementation; the positive-size induction excludes every longer word from being *least*, regardless of whether a longer composition also has recurrence.

8.9 The proof trace as a mathematical object

An informal derivation can conceal where a symbol changed meaning, where an unavailable premise entered or where a later result was treated as earlier. The Foundation trace theorem makes those failures mechanically addressable. A trace begins with immutable source and dependency receipt identities. Each operation record names its exact input identities, operation identity and output identity. Intermediate records are retained rather than reconstructed from the desired conclusion. The terminal identity is accepted only when deterministic replay reproduces every stage.

Root lineage is transitive. A Mathematics claim may depend on Foundation form enforcement rather than cite the root directly, but the engine follows the Foundation receipt's own dependencies until it reaches the root. A dependency graph with a missing node, an unresolved cycle or an unadmitted leaf cannot be repaired by placing "there is no nothing" in the abstract. The actual receipt path must exist.

The trace also protects historical status. If an early attempt halted, its source and terminal failure remain reproducible. A corrected claim uses a new registration and does not change the earlier output identity. If a later observation changes a current evidence classification, the later source and date are appended; they do not appear inside the earlier derivation trace.

8.10 Why one admission path survives 2,048 alternatives

The admission theorem turns the Methods constitution into a Foundation result. Its candidate grammar combines whether a proposed route is root-bound, source-bound, axiom-free and zero-parameter; whether its declared grammar is complete; whether it leaves one survivor and closes its form; whether controls and implementation-distinct validation pass; whether an empirical target opens only after sealing; whether every disposition is preserved; and whether only admitted dependencies are used. These binary and multi-class choices generate 2,048 possible routes.

A route that keeps the right result but loses the source fails. A route that keeps the source but samples candidates fails. A unique survivor without closure fails. Closure without adverse controls fails. Main-path agreement without distinct reconstruction fails. An empirical match whose target opened before sealing fails the prediction class even if the value agrees. A route that suppresses a rejected row fails evidence custody. A route that allows a proposed but unadmitted dependency fails model authority. Only the route that retains every gate survives.

This theorem explains why the later papers can share an admission vocabulary without repeatedly redefining it. The generic gate logic belongs here; each downstream claim still owns its exact grammar, eliminations, controls, sources and chronology. The conceptual paper may therefore summarise the common route while the audit and machine layers preserve the complete claim-specific record.

8.11 Historical reconstruction and non-ownership

The Foundation audit does not treat every earlier SFT statement as a Foundation theorem. It identifies the atomic obligation, asks which branch owns its scientific falsification condition and records non-ownership explicitly. A statement about a physical constant may depend on Fold, but its value and measurement belong to Physics. A statement about a protein may use Foundation recurrence, but sequence, structure and biological evidence belong to Biology and the computational-proof programme. Dependency is not ownership.

This one-owner rule prevents an upstream formal result from acquiring empirical authority merely because many downstream applications cite it. It also prevents a later adverse application result from silently deleting the formal theorem. The application may falsify a claimed handoff, parameterisation or domain law; it does not alter the upstream candidate census unless its counterexample lies inside that census's declared boundary.

8.12 Foundation status and handoff

The Foundation has no open registered row at its sixteen-claim boundary. The half-One terminology correction is current and the earlier usages remain visible in the reconciliation history. No adverse natural result exists here because no natural measurement is owned. Open extension remains real: a new foundational object, an omitted candidate or a counterexample inside a stated boundary must enter through a new registered package rather than by revising a receipt in prose.

Foundation hands Mathematics the structural One, positive finite count, positive parts, exact common refinement, Fold, held labels, finite word and form grammars, exact primitive operations, dynamics, proof traces and the one-way empirical boundary. Mathematics must now derive general structure from those objects without treating the names of conventional disciplines as axioms.

9. Mathematics: the exact generated-finite language

9.1 Why Mathematics follows Foundation

Foundation can generate and compare Fold forms, but Physics, information and computation require a much wider exact language: relations, graphs, algebraic operations, order, geometry, probability, optimisation, dynamics, logic, composition, bounds and certified approximation. Mathematics owns those formal structures. It does not own the physical interpretation of a rational threshold, a biological function, the semantics of a lived observation or the performance of a device.

The current branch contains 323 admitted claims across twenty-four families. It generated 97,280 candidates, retained 323 unique survivors, passed 1,292 controls and has 323 implementation-distinct reconstructions. Twenty-seven claims-the twenty-two foundational mathematical claims and five scientific calculator lineage claims-are classified as independently replicated. The other 296 carry registered post-registry execution observations and are classified as empirically tested and independently replicated within their formal computational boundaries.

9.2 Mathematical constitution

The primitive numerical carrier is an exact positive part of One with its construction and comparison trace. Addition is a disjoint junction; product is complete pair-cell generation; quotient is common refinement; order retains a positive unmatched remainder together with its held orientation. Structural empty One expresses an absent selection or identity path without becoming a semantic numerical zero. A held orientation replaces an imported negative proof magnitude. Irrational, imaginary and floating descriptions may appear only through exact enclosures, finite constructions, conventional correspondences or measured values after the relevant law seals.

The continuum boundary is potential rather than completed. Every supplied positive finite refinement depth can be generated, compared and extended, and there is no greatest finite depth. That successor statement does not construct a completed infinite carrier. A conventional continuum object therefore requires a separately registered correspondence or approximation certificate; it cannot be smuggled into an earlier proof because its notation is familiar.

9.3 Complete family census

Order	Family	Claims	Candidates	Scientific scope
1	BASE	27	21,504	Twenty-two foundational theorems and five calculator-lineage records
2	ARITH	18	4,608	Arithmetic and number structure
3	ALEXT	10	2,560	Algebraic extensions
4	COMB	12	3,072	Combinatorics
5	GRAPH	14	3,584	Graphs, networks and matroids
6	LINEAR	14	3,584	Linear, multilinear and tensor structure
7	ALG	16	4,096	Algebraic systems
8	ORDER	12	3,072	Order, lattices and domains
9	GEOM	16	4,096	Geometry
10	TOPO	14	3,584	Topology
11	CALC	12	3,072	Calculus correspondence
12	ANAL	16	4,096	Analysis correspondence
13	EQN	12	3,072	Equations and balance structures
14	MEAS	10	2,560	Measure and integration
15	PROB	16	4,096	Deterministic-support probability and statistics
16	OPT	16	4,096	Optimisation and operations research
17	DYN	12	3,072	Dynamical systems
18	LOGIC	16	4,096	Logic and foundations
19	CAT	12	3,072	Category, type and composition
20	NUM	12	3,072	Numerical mathematics
21	SYMB	10	2,560	Symbolic and constructive mathematics
22	XINT	8	2,048	Cross-disciplinary mathematical interfaces
23	VALID	12	3,072	Dated validation and Grand Lock
24	HAND	6	1,536	One-owner handoffs and open extension
	Total	323	97,280	Current full-field census

The twelve validation claims partition all 305 pre-validation receipts across the twenty-two scientific families before six handoffs complete the 323-row ledger. The Grand Lock is a dated reconciliation: it preserves favourable, adverse, absent, unresolved and boundary rows. It does not transform every row into confirmation or prohibit a later mathematical question.

9.4 The first mathematical dependency chain

The first twelve mathematical claims establish the general generated-finite language. Their candidate counts differ because each claim exhausts its own registered product.

Claim	Candidates	Principal result and exact boundary
SFT-MATH-EXACT-ARITHMETIC-001	256	Positive traces and parts; disjoint junction, pair-cell product, common-refinement quotient, comparison and held remainder, without semantic zero or signed proof magnitude.
SFT-MATH-DISCRETE-001	256	Canonical finite collections, selections, relations, maps and base-successor induction.
SFT-MATH-COMBINATORICS-001	256	Complete finite recurrence, selection and canonical deduplication.
SFT-MATH-GRAPH-NETWORK-001	512	Nodes from generated forms, pair relations, paths, reachability, cycles, trees, cuts and conservation.
SFT-MATH-ALGEBRA-001	512	Complete finite carriers with total closed operation relations and exhaustive property witnesses.
SFT-MATH-ORDER-LATTICE-001	512	Witnessed order; totality only when certified; meet and join only when unique witnesses exist.
SFT-MATH-GEOMETRY-TOPOLOGY-001	1,024	Finite cells, boundary incidence and path distance; generated open families and inverse-image continuity.

Claim	Candidates	Principal result and exact boundary
SFT-MATH-PROBABILITY-STATISTICS-001	512	Exact held-support parts over deterministic microstates; uncertainty as observation closure rather than stochastic cause.
SFT-MATH-OPTIMIZATION-001	512	Complete candidates, feasibility and preference with all ties, undominated alternatives and Pareto classes retained.
SFT-MATH-DYNAMICAL-SYSTEMS-001	1,024	Carrier, transition, trajectories, recurrence, predecessor custody and an empty-One identity trajectory.
SFT-MATH-LOGIC-PROOF-001	1,024	Propositions, held orientation and proof traces; consistency and completeness statements remain relative to the registered finite grammar.
SFT-MATH-CATEGORY-TYPE-COMPOSITION-001	1,024	Objects, typed transitions, empty-One identity return, interface-preserving composition, functors and naturality.

Arithmetic comes first because every later structure must count its support and retain exact comparison. Discrete structure then separates carriers from held selections and relations. Combinatorics exhausts finite choices; graph theory turns pair relations into paths and networks. Algebra asks when operation relations close and compose. Order refuses to manufacture an extremum where no unique witness exists. Geometry and topology are generated from cells, incidence, paths, open-family closure and inverse images rather than assumed as a continuum.

Probability is downstream of complete deterministic support. A microstate is a generated form; an observation is a total classification relation; states with the same observation form a class. An event is a held support selection, and its non-empty probability is its exact positive count relative to complete support. Uncertainty is the ledger of distinctions closed by observation. It does not cause an ungenerated transition. Optimisation then retains every feasible undominated object and every tie. Dynamics iterates an admitted transition while keeping terminal, recurrent and predecessor classes. Logic and composition arrive only after those carriers exist, so their familiar names cannot be retroactive premises of arithmetic.

9.5 Exact extensions and bounded famous-problem results

Ten further foundational claims reconstruct important V1/V2 lineages without inflating bounded or structural results into unrestricted conventional solutions.

SFT-MATH-EXACT-RELATIONS-002 derives exact ratios, separations, reciprocals, multi-fold complements, Fold transport, telescoping and least-common-return cycles. SFT-MATH-ORBIT-NUMBER-THEORY-002 separates power-of-two transients from odd-denominator recurrence: the recurrent orbit length is the multiplicative order of the binary generator modulo the odd denominator. The statement is exact at the admitted rational Fold carrier.

SFT-MATH-LIMIT-CONTINUUM-002 supplies exact finite rungs, the absence of a greatest finite depth and rational convergence certificates while refusing a completed infinity as a premise. SFT-MATH-ALGEBRAIC-BALANCE-002 retains positive polynomial sides, bracket and order records, and bisection certificates for algebraic magnitudes without inserting an irrational root as a foundational proof value.

SFT-MATH-BOUNDED-N-BODY-002 proves recurrence for every finite tuple of rational Fold states after any transient, with joint period given by the least-common return of the odd-core periods. SFT-MATH-FLOORED-FLUID-REGULARITY-002 uses a depth- k grid whose minimum separation is $1/b^k$, bounds velocity separation through One, and bounds discrete gradient and vorticity by b^k . At depth five the exact bound is 32. This is a generated finite-depth regularity result, not a continuum-fluid theorem.

SFT-MATH-PRIME-PAIR-CENSUS-002 exhausts every even integer from 4 through 10,000: all 4,999 registered even values possess a prime complementary pair, and the same bounded source contains 205 twin-prime pairs with upper member at most 10,000. The claim is the complete bounded census, not an unbounded Goldbach or twin-prime proof.

SFT-MATH-RIEMANN-MIRROR-002 derives the complement involution and its unique self-partner at half-One, then records a structural correspondence to the $1/2$ reflection of the zeta functional relation. Symmetry alone does not locate every complex zero, and unrestricted complex-zero location lies outside the current proof language.

SFT-MATH-COLLATZ-FINITE-CENSUS-002 executes every starting value from 1 through 100,000 and records that each reaches the $1-4-2$ cycle; the start 27 takes 111 steps. The claim also preserves an adverse control against the older constant-three-quarter contraction shortcut: an odd macrostep can rise, so the shortcut is not a proof even though the complete bounded census succeeds.

Finally, SFT-MATH-SELF-SIMILAR-CONVERGENCE-002 derives binary local separation, one distinction per step, the unit exponent under rank doubling, m^d finite support and exact rational convergence records. Together these claims show the

intended discipline: an exact bounded result remains bounded, a correspondence remains a correspondence and a failed shortcut remains visible.

9.6 Arithmetic and algebraic extension across the full field

The complete-field extension asks whether the first exact arithmetic kernel is sufficient to reconstruct the working relations used throughout later mathematics. Eighteen ARITH claims answer that question in dependency order. Generated succession supplies exact induction. Disjoint junction and pair-cell generation supply addition and multiplication. Divisibility is witnessed by a complete equal-cell decomposition; common divisors and multiples retain all candidate witnesses before selecting a greatest or least class. Quotient and remainder retain a positive quotient trace and a held orientation for the unmatched part, so a conventional signed remainder is not a primitive.

Prime structure is irreducibility under the admitted positive product grammar. Unique factorisation is not assumed from its conventional name: the certificate retains the complete factor tree, canonical factor order and reconstruction of the original whole. Canonical fractions reduce through common refinement and pairing rather than an unrecorded division rule. Finite continued fractions retain each reciprocal and remainder stage. Congruence is equality of held remainders at a declared modulus, and compatible congruences compose only when the registered compatibility witness exists. Prime-power valuation records divisibility depth as a positive succession.

The later arithmetic claims keep Diophantine solution support, recurrence sequences, finite generating-function correspondences, partitions, compositions, divisor ledgers and prime-distribution enclosures. A generating function is a post-derivation representation of a finite recurrence, not an infinite series premise. A prime-distribution claim is a finite support and enclosure result at its declared boundary; it does not inherit an asymptotic theorem merely because conventional notation can display one.

Ten ALEXT claims then distinguish exact algebraic construction from imported irrational values. Polynomial identity is checked by exact coefficient and evaluation traces. A root is isolated by positive rational brackets and orientation changes; the proof object is the bracket and refinement certificate, not an ungenerated irrational scalar. Finite extension towers retain each adjoining relation, dependency and reconstruction map. Finite field, Galois-orbit and cyclotomic structures enter as explicit finite correspondences after their carriers and operations have been forced.

The held-pair complex correspondence represents the two components and their orientation as exact records. It permits finite algebraic manipulation and enclosure without making an imaginary scalar a foundational magnitude. Real-algebraic ordering compares certified brackets; prime-adic correspondence retains valuation depth and declared completion boundary. The final transcendental claim is a non-representability boundary: a failure to construct an object in the admitted algebraic grammar is not reworded as a proof about every broader conventional number system.

Sixteen algebraic-system claims build the hierarchy from a complete carrier and total operation relation. A magma has closure. A semigroup adds an associativity witness over complete triples. A monoid adds an identity relation. A group adds held reversal with exact composition back to identity. Permutation actions, normal substructures and quotients retain source and equivalence-class custody. Ring, integral-domain and field correspondences add their registered distributive, non-collapse and exact-division conditions without importing a pre-existing field axiom.

Modules and algebras then combine scalar action with compatible products; ideals and quotient structures retain the classes closed by the projection. Representation and action decomposition preserve component and reconstruction maps. Exact-sequence and homological correspondences record image/kernel incidence using structural empty support where a conventional display uses zero. Universal identities range over the complete declared finite term grammar. Operadic composition closes the algebraic interfaces while retaining slot, order and source identity. Each familiar structure is therefore a successive organisation of admitted relations, not a word that imports its textbook axioms.

9.7 Combinatorics, graphs and linear structure

The twelve combinatorial extensions begin with product, sum and bijection counting, then generate permutations and combinations from complete held selections. Inclusion-exclusion retains every overlap class and corrects each repeated cell exactly once rather than using a memorised alternating formula as a premise. Pigeonhole forcing follows because a map from a larger generated support into fewer held target classes must produce a non-singleton fibre. Recurrence and generating correspondences retain the construction order.

Partition incidence, extremal finite set systems, block designs, coding and packing then use complete support and exact relation ledgers. The probabilistic-method claim is explicitly a correspondence: deterministic support is completely generated, and an exact positive part can witness that a desired class is non-empty. No ontic random cause selects an object. Finite Ramsey forcing

retains the exact registered support and boundary rather than claiming a completed infinite colouring universe. Species-like composition records how labelled structures assemble and reconstruct.

Fourteen graph claims turn complete pair relations into a broad network language. Graph identity retains nodes, adjacency and canonical isomorphism witnesses. Paths and walks retain ordered edge traces; reachability is the existence of such a trace; cycles retain return. Connectivity, cuts and flow support keep source, sink and every crossing relation. Trees and forests are acyclic connected or componentwise structures with explicit spanning custody. Planarity is an embedding correspondence at the registered finite carrier, not an assumed continuous plane.

Colouring, matching, covering and packing are complete finite constraint problems in which all ties and alternative optima remain available. Directed graphs add held orientation and causal reachability. Weighted networks attach exact positive parts whose units and physical meaning remain downstream. Hypergraphs generalise pair incidence to higher generated cells. Matroids retain hereditary and exchange witnesses for independence. Reliability records every registered failure class rather than replacing it with an unidentified probability. Spectral graph results use exact rational enclosures, and temporal networks retain versioned edge and causal records.

The fourteen `LINEAR` claims make coordinate structure similarly explicit. A vector is a generated coordinate carrier with each position and value trace. Linear maps preserve the declared junction and scalar-action relations. Matrix notation is a representation of that map, and row operations retain an exact inverse or equivalence certificate. Rank counts retained independent image distinctions; nullity is the closed predecessor class at the map boundary, expressed structurally rather than as a primitive zero dimension.

Determinants retain orientation and complete permutation terms. Linear systems preserve exact solution, inconsistency or non-unique predecessor classes. Basis, dimension and change of basis require complete spanning and independent reconstruction. Inner product and metric are correspondences built from exact positive forms. Eigen and invariant support retain exact equations or rational enclosures rather than floating approximations. Multilinear maps, tensor products and contractions retain every component identity and contraction pair. Exterior and symmetric composition record antisymmetric held orientation or permutation equivalence without importing signed scalars as primitive. Finite-dimensional operator decomposition closes only where complete reconstruction is certified.

9.8 Order, geometry and topology

The order family begins with a preorder and then quotients observationally indistinguishable elements before asserting antisymmetry. A partial order therefore retains witnessed comparability rather than presuming every pair can be ranked. Totality is conditional on a complete pairwise comparison certificate. Meet and join exist only where unique greatest-lower or least-upper witnesses survive. Distributive, modular and Boolean-like correspondences remain relative to the generated lattice and its complement law.

Closure operators retain extensivity, monotonicity and idempotent return. Galois connections preserve the two order maps and their exact adjunction relation. Domain approximation is a generated finite correspondence; monotone maps preserve order; finite fixed points are found through complete iteration and recurrence. A conventional complete lattice enters only at an explicit boundary because the current proof language does not install an ungenerated completed carrier.

Sixteen geometry claims begin from point identity, incidence and exact coordinates. Finite Euclidean distance is a correspondence derived from exact coordinate relations and certified enclosures. Affine and projective structures retain combinations, incidence and perspective maps. Convex hulls are complete generated combinations, and separation requires an exact witness. Discrete polytopes retain lattice incidence; Euler-type results retain the complete face ledger rather than rely on a memorised characteristic.

Computational predicates make orientation and intersection executable on exact inputs. Algebraic-geometry solution sets retain the registered equations and complete finite solution or enclosure boundary. Differential geometry is a finite-chart correspondence with explicit transition records. Curvature is retained through exact finite transport; metric geodesics are shortest generated path classes. Fractal and self-similar geometry retains its construction recurrence and exact scaling relation. Packing, covering, tessellation and transformation groups close through complete finite supports and action ledgers.

Fourteen topology claims then separate neighbourhood structure from metric assumption. An open-family correspondence is generated over complete support; continuity is inverse-image preservation of that family. Compactness is a finite-subcover correspondence at the declared carrier. Connected components are reachability classes, and separation properties are observation-relative distinction conditions. Product, quotient and subspace constructions retain their source maps and closed distinctions.

Simplicial complexes turn cells and boundary incidence into composable finite objects. Homotopy is a registered path-deformation correspondence. Fundamental-cycle, homology and cohomology claims retain cycles, boundaries, dual observations and exact composition; structural empty support replaces an untyped zero group at the SFT boundary. Manifold claims require a finite atlas and compatible chart records. Knot and link results apply to finite diagrams with registered invariants. Persistent topology retains the filtration, feature birth, feature closure and source custody rather than reporting an untraceable barcode alone.

9.9 Calculus, analysis and equations without a hidden continuum

The calculus family starts with exact finite difference. Higher differences record polynomial degree on generated support, while accumulation is a complete finite sum. Their fundamental correspondence is an exact cancellation and boundary ledger. Product and composition difference laws retain every term and interface. Rational enclosure sequences then permit a derivative correspondence through shrinking exact parts and an integral correspondence through refinement sums.

These are correspondences because no completed continuum is admitted as a premise. Multivariable and directional change use generated coordinate directions. Divergence and flux are exact cell-boundary balances. Variational stationarity is absence of an admissible oriented improvement at the declared finite perturbation support, not a primitive derivative equal to numerical zero. The final calculus claim records precisely when a continuum-limit description is admissible and what finite certificate supports it.

Sixteen analysis claims extend the same rule. A sequence converges when a registered rational enclosure and successor relation force every later term inside the required exact band. Cauchy and completeness language remains a correspondence over generated support, never an assertion that the construction has produced a completed continuum. Series and power-series claims preserve finite truncations and exact remainder bounds. Functional spaces are finite representation classes with explicit evaluation and comparison interfaces.

Norm, seminorm and metric relations are exact positive or held-orientation correspondences. Bounded and compact operator claims retain their finite witness boundary. Harmonic, Fourier, transform, convolution and correlation claims operate on complete generated finite support and preserve inversion or reconstruction where certified. Orthogonal expansions retain basis and coefficient custody. Distributional and weak observations are explicit test-record correspondences. Nonlinear contraction states its exact shrinking condition. Complex analysis uses held-pair representation and rational enclosures. Spectral measure remains an operator and observation correspondence, not an imported continuum object.

The equation families keep model syntax distinct from solution evidence. Ordinary and partial difference equations are exact transition relations. Differential counterparts are registered correspondences supported by finite enclosures. Boundary and initial conditions are source records; well-posedness separates existence, uniqueness and stable dependence. Integral, functional and recurrence equations retain their complete relation and solution-space support. Green-response forms are finite response correspondences. Weak conservation keeps boundary and residual ledgers. Stability retains the declared perturbation support. Existence, uniqueness and blow-up claims report which of those statuses the exact boundary establishes rather than inferring all three from one successful computation.

9.10 Measure, probability, optimisation and dynamics

Ten measure claims begin with exact positive weight on finite support and disjoint additivity. Outer covering, measurable boundary and integration enter as generated correspondences. Refinement-sum integration keeps each partition and enclosure. Product and conditional support retain the exact whole against which a part is formed. A signed-measure display is replaced at proof level by positive magnitude plus held orientation. Generalised observations and convergence of measures remain finite-witness boundaries.

The sixteen probability extensions then reconstruct conditional and Bayesian relations from exact support restriction. Independence requires complete product factorisation; expectation is exact support accumulation; variance retains positive separation or paired differences without a negative primitive. Finite distributions are complete part ledgers. Large-count and central-limit claims are enclosure correspondences under their registered deterministic enumerations, not ontic stochastic laws.

Estimation retains the estimator map and sufficient record. Confidence and credible regions preserve their distinct construction and source assumptions. Hypothesis testing keeps false-positive, false-negative and unavailable rows. Likelihood and evidence ratios are exact positive relations at a registered support. Bayesian update is conditional restriction, and a finite stochastic process is a description over unresolved deterministic transition support. Conditional conservation and statistical identifiability are stated through complete observation classes; observationally equivalent sources remain unresolved rather than being separated by a prior that the evidence did not supply.

Sixteen optimisation claims distinguish feasibility, objective and preference. Exact extrema are retained only after complete candidate evaluation. Convex, linear, integer, combinatorial, network, dynamic, variational and multi-objective forms each expose their carrier and admissible moves. Duality and certificates retain the primal and comparison witnesses. Pareto fronts and ties remain complete. Approximation records the exact objective distinction from the optimum and the resource used to obtain it. Robust and uncertainty claims preserve the complete registered scenario support. No optimiser may erase an alternative merely because a software routine returned one member of a tied class.

The twelve dynamical extensions retain state, transition, orbit, recurrence, basin, invariant, stability, bifurcation, symbolic coding and multiscale records. Reversibility is singleton predecessor custody; irreversible dynamics retain the merged predecessor class. Apparent chaos is represented by exact separation and observation sensitivity over generated finite successors. A physical dynamical interpretation requires a later owner to supply units, source law and measurements.

Logic and category then bind the whole language. Sixteen logic claims cover syntax, semantics, consequence, proof, consistency, completeness boundaries, decidability, model classes, induction, fixed points and constructive witnesses relative to explicit grammars. Twelve category claims preserve objects, arrows, identity, composition, functors, natural relations, products, coproducts, limits, adjunction-like interfaces and type composition. These families organise already admitted mathematics; they do not reach backwards to become its axioms.

9.11 Numerical, symbolic and cross-branch interfaces

The numerical family provides exact rounding custody, interval propagation, conditioning, stability, convergence, root isolation, linear and nonlinear solvers, quadrature, recurrence evaluation and finite error certificates. A floating result may be displayed at an interface only when its exact source or enclosure remains available. The symbolic family retains canonical syntax, rewrite provenance, identity checks, transforms and constructive certificates. Neither a numerical library nor a computer-algebra output is an answer oracle.

Eight cross-disciplinary interfaces then specify what Information Science, Computation, Physics, Chemistry, Biology, Social Science and Engineering may consume. The one-owner identity permits the same mathematical object to appear in several branches without duplicating ownership of the law. A downstream branch owns the new scientific interpretation, source and falsification condition; Mathematics retains the underlying formal structure.

The validation family audits the complete predecessor vector, while the six handoffs preserve measurement custody, conventional correspondence and lawful extension. This field-wide layer adds breadth without weakening the original constraints: each extension has 256 registered candidates, one survivor, four controls, a root route, an independent reconstruction and a post-registry observation record. Their shared shape permits compact exposition here, but the individual machine records remain separate and complete.

9.12 The scientific calculator and preserved adverse evidence

The calculator lineage is a scientific result because it exposes how exact proof objects become usable without becoming a shadow admission engine. It contains five claims, SFT-MATH-SCIENTIFIC-CALCULATOR-003 through -007, with 11,520 candidates in total.

Claim 003 passed its declared grammar, but a later operational audit found reversed odd/even endpoints in one alternating arctangent enclosure. Its artefacts and receipt remain preserved as **superseded adverse evidence**. Claim 004 re-derived the core with corrected endpoint parity and an exact tangent-composition check. Claim 005 extended the corrected core to the declared calculator language and stateful application. Claim 006 completed the calculator-first interface, law explorer and resource controls. Direct manual testing then exposed a blank deprecated macOS desktop widget and phone usability failures. Claim 007 kept the admitted evaluator immutable and forced an accessible standards-rendered local web adapter.

The current calculator returns exact fractions or certified enclosures, records its expression and complete operation trace, treats displayed 0 as structural absence and halts on a prohibited negative result. A calculator receipt establishes that an admitted operation evaluated an expression. It does not admit the expression as a new theorem. A proposed new law still needs its own registration, candidate grammar, controls, reconstruction and engine receipt.

9.13 Reconciliation, evidence and open boundary

The V1/V2 Mathematics audit reviewed 356 V1 rows and 407 V2 numbered results, identified seventy-one Mathematics-owned atomic obligations and closed all seventy-one. Four corrections or invalidations remain prominent:

1. signed and zero host storage is implementation history, while the admitted proof boundary uses structural empty One and held orientation;

2. the prime skeleton and half-One reflection axis do not prove unrestricted complex-zero location;
3. the Collatz three-quarter shortcut fails even though the bounded 100,000-start census succeeds; and
4. the depth-five fluid value 32 is derived from the reciprocal floor rather than installed as a damping parameter.

The branch is complete only to its frozen generated-finite current-evidence census. It does not assert every named theorem over a completed continuum or every object of conventional real and complex analysis. No natural dataset is used to admit the formal mathematics. A downstream branch must separately own the meaning and measurement of a mathematical structure.

Mathematics now hands Information Science canonical forms, exact relations, finite probability support, maps, graphs, order, proof, dynamics and typed composition. The unresolved problem is not whether those objects can be represented, but how a scientific system records which distinctions are retained, lost, communicated or unavailable.

10. Information Science: distinction under custody

10.1 Why information is not merely mathematics

A mathematical relation may map one carrier to another, but it does not by itself say which object is the source, which is the record, what an observer can distinguish, what a channel merges or what evidence is required to reverse that merge. Information Science owns those questions. Its central object is not an imported scalar. It is complete generated support together with exact records of distinctions retained, closed, transformed and required for reconstruction.

The branch contains 262 admitted claims across twenty families, 75,776 generated candidates, 262 unique survivors, 1,048 passed controls and 262 implementation-distinct reconstructions. The twelve foundational information claims are independently replicated. The 250 field extensions add post-registry observation packages and are classified as empirically tested and independently replicated at their declared computational boundaries.

10.2 Information constitution and complete census

An information carrier is a complete generated support. A symbol is a held distinction with its observation class. A record binds source identity, transformation, context and retained output. An absent support cell is empty One, not a zero-probability scalar. Entropy is the full ledger of observation classes and unresolved within-class distinctions. Probability is an exact part of deterministic support. Noise is a registered transformation, not an uncaused event. Loss is a non-singleton predecessor fibre whose closed source distinctions remain recorded.

Family	Claims	Candidates	Scientific scope
BASE	12	11,776	Foundational information laws
SYMREP	14	3,584	Symbols and representation
RECORD	12	3,072	Records and provenance
SOURCE	14	3,584	Sources and custody
MEASURE	16	4,096	Measurement information
SIGNAL	14	3,584	Signals and transformations
COMP	14	3,584	Compression and description
CHAN	18	4,608	Channels and capacity
NOISE	12	3,072	Noise, error and ambiguity
CODE	18	4,608	Codes, detection and correction
REL	14	3,584	Joint and conditional relations
COARSE	12	3,072	Coarse-graining and retained classes
RETR	12	3,072	Retrieval and ranking
INFER	14	3,584	Inference and estimator custody
PRIV	10	2,560	Privacy and distinction access
THERM	10	2,560	Information-thermodynamic boundary
CORR	16	4,096	Classical, probabilistic and quantum correspondences
SEM	12	3,072	Symbol, reference and context
VALID	12	3,072	Dated validation and Grand Lock

Family	Claims	Candidates	Scientific scope
HAND	6	1,536	One-owner downstream handoffs
Total	262	75,776	Current full-field census

The Grand Lock binds all 244 pre-validation receipts and 976 controls before the twelve validation claims; six handoffs then complete the ledger. As in Mathematics, this is a reconciliation of every frozen row, not a conversion of every evidence type into one undifferentiated success label.

10.3 Twelve foundational information laws

Claim	Principal derivation
SFT-INFO-SYMBOL-DISTINCTION-001	A symbol is a canonical held form plus its observation class; distinguishability is a retained label relation.
SFT-INFO-ENCODING-DECODING-001	Encoding is a total source-to-code map; exact decoding holds precisely on used codewords with a singleton predecessor or left inverse; many-to-one mapping is declared loss.
SFT-INFO-QUANTITY-001	Information quantity is complete support plus retained and closed distinction ledgers; the first Fold supplies one unit and positions supply support.
SFT-INFO-ENTROPY-UNCERTAINTY-001	Entropy retains every observation class, exact support part and unresolved deterministic pair; refinement cannot increase closed distinctions.
SFT-INFO-COMPRESSION-REDUNDANCY-001	A lossless description retains dictionary, token trace, unique parse and decoder; shortening beyond support injection is impossible.
SFT-INFO-CHANNEL-CAPACITY-001	A channel is a complete total relation; confusability is output-support overlap; zero-error codes have disjoint output supports; all greatest-cardinality ties are retained.
SFT-INFO-NOISE-ERROR-001	Noise is a registered transformation; error is an exact changed-position trace; correction is permitted only for a singleton registered predecessor.
SFT-INFO-CODING-001	A code retains word support and complete error images; correction is equivalent to pairwise-disjoint error-image supports, with redundancy witnessed by separation.
SFT-INFO-MUTUAL-CONDITIONAL-001	Conditioning is exact support restriction; dependence is joint incidence against exact marginal product; absent cells remain empty One.
SFT-INFO-CONSERVATION-LOSS-001	A transformation either retains a distinction or closes it inside a predecessor fibre; reversibility holds exactly for singleton fibres with complete custody.
SFT-INFO-CLASSICAL-PROBABILISTIC-001	Classical information is one held deterministic state; probabilistic information is the unresolved observation-class ledger over the same support.
SFT-INFO-QUANTUM-CORRESPONDENCE-001	Fold words supply complete quantum basis support at a correspondence boundary; phase, amplitudes, gates, entanglement and dynamics remain quantum-owned.

The sequence matters. A distinction must exist before it can be encoded. An encoding must preserve a source relation before quantity can count retained and closed alternatives. Entropy then records which alternatives observation fails to distinguish. Compression cannot be assessed before reconstruction is defined. A channel cannot have capacity before its source-to-output relation and confusability classes are complete. Noise and coding follow because error must name the transformation and every predecessor before correction can be unique.

The resulting coding law is deliberately stronger in custody and narrower in assumption than a nearest-codeword rule. A metric may tie and an error rate may be unknown. The exact decoder therefore enumerates complete registered error images. If one received form belongs to one source image family, correction is forced. If it belongs to several, the complete predecessor class is returned and the machine refuses to guess. Width-three repetition is an executed single-label example; it does not install a stochastic error model.

10.4 Relation, inference, privacy and thermodynamic meaning

The field extension applies the same custody to joint records, coarse-graining, retrieval, estimators, privacy and thermodynamic correspondences. A conditional record restricts both the held event and the reference whole. Coarse-graining groups source forms while retaining the closed within-class distinctions. Retrieval and ranking keep the query, source corpus, matching relation, ties and exclusions. Inference retains the hypothesis support, observation relation, compatible alternatives and any reconstruction assumption; a terminal score cannot hide its source surface.

Privacy is exact access to distinctions. A released observation leaks the source distinctions it separates and preserves privacy only for distinctions that remain in a non-singleton observation class at the declared adversary boundary. This structural statement does not make a practical security claim without the Classical Computation branch's resource-bounded adversary and a domain-specific implementation test.

The thermodynamic correspondence is equally disciplined. An irreversible merge closes predecessor distinctions. Reversal requires a retained predecessor record. That relation supplies an exact information-accounting boundary for Landauer-, Bennett- and Maxwell-demon comparisons after derivation. It does not assign energy, temperature or a hardware coefficient; those remain physical measurements.

The semantic families retain symbol, reference and context as separate records. A symbol can be formally bound to a source and use context without claiming that information quantity alone is biological function, cognition, consciousness or lived qualia. Those meanings remain explicit downstream ownership boundaries.

10.5 Symbols, records and source identity

The SYMREP family begins by generating a complete alphabet from held distinctions. A symbol has a canonical identity and observation class; two symbols are distinguishable only where a retained observation separates them. Codewords are finite generated sequences with position custody. Parsing is a relation from a complete word to its candidate decompositions. A prefix or instantaneous representation is not declared merely because one example parses: the complete used-word support must rule out every conflicting prefix or delayed alternative.

Grammar-constrained representations retain production identities and complete parse trees. Two representations are equivalent only through an explicit bidirectional reconstruction or isomorphism. Canonicalisation chooses the unique normal form supplied by the registered equivalence relation rather than by an undocumented formatting convention. Variable-length codes retain delimiter, prefix or complete parsing custody; product alphabets retain both component identities; typed and hierarchical symbols retain their level and parent relations. Conversion and transduction must preserve the declared information or record its loss. If several source words produce the same observed form, the ambiguity family is retained rather than choosing one.

The finite-alphabet successor law appends each generated symbol under the same complete rule used at the prior depth. The representation completeness claim then reconciles all fourteen obligations and refuses an omitted symbol, codeword or parse class. This is what permits a downstream language to cite a complete representation rather than an informal alphabet.

The RECORD family distinguishes an information object from a bare value. A data item has a source-bound identity. A tuple retains field identities and a relation retains its complete row support. Metadata is an interpretive record whose own source and version matter; it cannot be treated as neutral decoration. A schema names allowed fields and types, while provenance composes the chain of sources and transformations by which a record arose.

Integrity is evidence that the current artefact agrees with its bound identity, not proof that the recorded scientific proposition is true. Version and revision are distinct: a new artefact receives a new identity, while the earlier one remains recoverable. Missing, absent and unknown are three different states. Absence is an observed empty class at the declared boundary; missing means the required record was not supplied or transported; unknown means the evidence does not yet distinguish the alternatives. Duplicates and aliases require an identity relation rather than naive string equality. Record linkage preserves both matches and unresolved candidate links. Dataset completeness requires an omission ledger, and reproducibility requires custody of sources, transformations, environment and result.

Fourteen SOURCE claims then type the structures from which records arise. Source support may be a sequence, transition process, spatial adjacency or network path. Refinement separates distinctions; coarsening closes them. Stationary and non-stationary descriptions retain the exact support and time boundary under which their correspondence is asserted. Memoryless structure requires conditional equivalence after the current source state; finite memory retains the exact predecessor window. Joint sources preserve pair-cell support, and dependence is witnessed by departure from complete product structure. Every successor extension keeps the earlier source rows, and the completeness claim exposes the boundary beyond which no source behaviour is inferred.

10.6 Measurement information and signal custody

The information-measure family deliberately separates several quantities that are often collapsed into one scalar. Distinction count enumerates separable source alternatives. Combinatorial quantity counts complete selections or arrangements. Operational discrimination cost counts the registered actions needed to resolve a class. Description length counts symbols only inside a fixed grammar. Algorithmic-description language is a correspondence boundary because an unrestricted universal-description minimum cannot be asserted from one finite enumerator.

Partition refinement supplies an information order: a finer observation retains every distinction of a coarser one and possibly more. Independent product support permits exact additive composition, whereas shared support forces subadditive custody so a distinction is not counted twice. Coarsening cannot increase retained distinctions. Balance ledgers state which distinctions

entered, left, remained or were closed. Relative information and divergence compare exact observation partitions and preserve cells where one record cannot distinguish what another can.

Information geometry is built from those exact relations rather than an imported continuous metric. Multiscale decomposition retains the refinement tree. Unit conversion keeps the base support, scale relation and conventional display unit separate. The completeness certificate reconciles all measure types without declaring them interchangeable.

A signal is an ordered observation record. Its amplitude label and its support are different objects: a position may be present even when the retained label corresponds conventionally to a baseline. Sampling is a held selection of source positions. Sufficiency requires that every admissible source in the registered grammar reconstruct uniquely from the retained samples. Aliasing is therefore an exact non-singleton predecessor class, not merely a visual artefact. Quantisation is an observation partition, and quantisation error is the retained source-to-class distinction with its held orientation.

Reconstruction is authorised only by the complete retained sample and source grammar. Interpolation remains a correspondence outside points whose values were observed directly. Transform representations retain an exact inverse on the declared finite support. Time-frequency and multidimensional descriptions keep both coordinate systems and the transform identity. Signal provenance links acquisition, sampling, transformation and display. The final signal certificate states which source classes the sampling law closes and which it does not.

These laws prepare measurement custody without pretending to own a physical instrument. The information branch can say what a complete record of samples, positions, transformations and uncertainty must contain. Physics or another domain branch must supply the apparatus, calibration, units, environmental conditions and natural source.

10.7 Compression, channels, noise and codes

The fourteen compression extensions distinguish a short string from a lawful compressed record. Lossless reconstruction retains an injective or uniquely decodable source-to-description relation. Prefix trees, dictionaries and run records carry their own overhead and parsing rules. Transform compression is a correspondence through an invertible representation before any coefficient is discarded. A source model can improve description length only at the declared source boundary; its success cannot establish that the model is universally correct.

Redundancy is retained excess distinction structure relative to a complete reconstructing grammar. Minimum description is therefore grammar-relative. A lossy compressor declares the coarsening that merges sources and preserves the complete predecessor class of each output. Distortion is an exact registered relation, not necessarily a conventional norm. Rate-distortion language is a post-derivation correspondence between description support and closed distinctions. Successive refinement retains each layer and reconstruction stage. Side information is an additional source record whose custody must be declared; it cannot silently reduce the reported description.

The eighteen channel claims treat a channel as a complete input-output relation. Deterministic transport has singleton output for each input, whereas an unresolved or externally varying channel retains all registered outputs. Observation-equivalent channels are equivalent only for the named observer and support. Zero-error single-user capacity is the greatest cardinality of a held input family with pairwise-disjoint output supports; every tied greatest family is retained. Resource-bounded capacity adds a declared use, cost or process limit.

Cascades compose relations and expose the distinctions lost at each stage. Parallel channels retain product support and shared-source dependencies. Feedback is a named reverse record, not an unexplained improvement. Multiple access, broadcast, relay, interference and bidirectional structures retain all senders, receivers, paths and local observations. Network capacity is bounded by exact cut support. Finite-use succession composes the complete channel at each generated use without asserting an infinite-use limit. Channel simulation must reconstruct the declared input-output relation and resource ledger.

Noise is then typed by what it changes. A transformation that closes an unrecorded source distinction, a source-output mismatch, an erasure, a substitution, a burst, a correlated pattern and an adversarial mask have different records. Detection asks whether the received form lies outside valid support. Localisation asks which positions are forced to differ. Estimation may return a class of possible error actions. A noise budget is an exact positive resource boundary, not a fitted probability. Composition transports every registered mask through the channel and keeps the resulting predecessor classes.

The eighteen coding claims then apply those complete error families. Injective representation is necessary but not sufficient for correction. Minimum separation is the least exact changed-position count among distinct valid words. A decoder corrects a registered received form exactly when one valid predecessor remains inside the allowed error support. Parity and syndrome records are observation partitions over error classes. Repetition and majority are generated structures; erasure, substitution and burst correction each retain their own complete mask families.

Convolutional, tree, product, concatenated, sparse-check and network codes are correspondences with explicit encoder, state and decoder records. List decoding returns every surviving predecessor instead of falsely reporting uniqueness. Adversarial correction exhausts every mask inside the declared budget and retains the first exceeding-budget control. Code composition records both resource and reconstruction effects. No sampled favourable mask, likelihood decoder or hidden error rate is permitted to stand in for complete registered support.

10.8 Relational information, coarse-graining, retrieval and inference

Relational information begins with joint support and exact conditional restriction. Mutual information is the shared distinction ledger between two observations, while a chain rule partitions retained information without double-counting the shared source. Data processing is monotonic because a later deterministic observation cannot reopen a distinction already closed by an earlier non-singleton fibre unless an additional source record enters. Conditional independence is product structure inside every retained condition class.

Common, directed and interaction information retain their declared support, order and multi-source incidence. Multi-information and total dependence compare complete joint support with its marginal product. Shared, unique and synergistic decompositions remain boundary correspondences where the registered lattice supports them; the branch does not pretend one scalar decomposition is forced for every possible semantics. Causal information flow requires an admitted causal path from the computation or domain owner, rather than inferring causation from association alone. Representation change must preserve the source distinction ledger.

Coarse-graining asks what summary is sufficient for a declared observation. A sufficient record preserves every distinction needed for the target; a minimal sufficient record removes every unnecessary distinction without losing one that matters. A bottleneck is a lawful coarsening with an explicit retained and closed ledger. Feature selection, state lumping and multi-stage aggregation retain the source classes and composition order. Reversibility holds only when the summary plus retained side record reconstructs each predecessor uniquely.

Retrieval makes location and relevance source-bound. An index maps a query term or form to registered source locations. Forward and inverted indices are correspondences only if they reconstruct the same memberships. A query is a declared observation relation; ranking is an exact finite order that retains ties and incomparability. Relevance is a registered relation, not an authority label. Precision and recall are exact parts of complete returned and relevant support. Taxonomies and citation graphs retain version, edge provenance and scope. An index may become stale after a source update; the old result keeps its chronology rather than silently inheriting the new corpus.

Inference is support restriction. Detection chooses an observation class; estimation selects a representative only with a declared selection rule. Filtering updates a record sequentially from earlier to later observations, whereas smoothing uses a complete later record to reconstruct an earlier state. The smoothing result does not rewrite the historical forward-only knowledge available at the earlier time. Prediction registers transition support before the future observation. Evidence accumulation retains every source and update. Likelihood is a correspondence over exact support rather than a free causal propensity.

Structural thresholds arise from order and loss conditions, not a tuned scalar. False-positive and false-negative rows remain separate. Identifiability fails where two source objects have the same complete observation class. Sequential stopping requires an explicit rule and trace. Multi-sensor fusion keeps each sensor's source, timing, overlap and failure rows. The terminal inference certificate reconciles these distinctions without turning an estimator output into a proof of the source model.

10.9 Privacy, thermodynamics, quantum correspondence and semantics

The privacy family defines leakage as source distinctions that an adversary's observation retains. Perfect secrecy is the structural condition that every secret remains in the same adversary observation class at the declared support. Side information refines that class and must be included in the adversary boundary. Anonymity is a non-singleton identity class; least disclosure is the coarsest observation still sufficient for the authorised task. Composed releases may jointly distinguish what neither distinguishes alone, so privacy losses require a combined ledger.

Repeated queries can reconstruct a source even where each reply looks coarse. The query-reconstruction claim retains that complete attack support. A privacy-utility trade is an exact pair of retained task and adversary distinctions; no universal scalar preference is imposed. Practical cryptographic hardness is handed to Classical Computation because it requires a process and resource-bounded adversary, not merely an information partition.

The thermodynamic family begins by recognising that a physical information record must have an owning physical carrier, while refusing to derive energy from information structure alone. An irreversible merge lacks a predecessor label; reversible retention keeps it. Erasure, measurement, memory reset and a Maxwell-demon cycle therefore require complete records of what

distinctions are closed, retained and exported. Landauer and Bennett enter as explicit correspondence boundaries. The non-equivalence claim is essential: an information count is not an energy quantity, and no dimensional coefficient is admitted without a Physics law and measurement.

Sixteen correspondence claims then type the route to Quantum Computation. Classical information is a held source word; deterministic-support probability is an unresolved observation partition; a probabilistic mixture is a description of hidden support. Quantum correspondence begins with complete Fold-word support and one distinction unit. Product support, joint correlation, measurement records, classical-to-quantum encoding and quantum-to-classical observation can be stated at the information boundary. Phase, interference, entangling dynamics, gates, channels and quantum coding remain owned by the quantum branch. The no-cloning information boundary identifies the source and copy distinctions that any later operational proof must preserve, but does not pre-empt that proof.

Finally, the semantic family separates symbol count from content. Reference binds a symbol to an interpreted source under a context. Context can change which distinctions are operationally available. Genetic representation, neural and cognitive information, conscious report, social communication and pragmatic action are typed handoffs. Meaning-preserving translation requires a declared bidirectional relation at the intended content boundary. It cannot be established by matching symbol counts alone.

10.10 Validation, chronology and one-owner handoffs

The twelve validation laws do not introduce a new theory of information. They replay and partition the 244 scientific receipts, their 976 controls, independent reconstructions, observation rows and source identities. A favourable execution stays favourable; an ambiguity stays an ambiguity; an absent or unavailable source retains that classification. The six handoffs then assign downstream derivation, measurement, formal-to-empirical chronology, conventional correspondence and open extension.

SFT-INFO-HAND-MEASUREMENT-BOUNDARY-002 keeps a formal derivation, externally held observation and exact comparison as three different records. SFT-INFO-HAND-FORMAL-EMPIRICAL-003 requires derivation before sealed prediction and target release before comparison while retaining every favourable, adverse, absent, unresolved and boundary row. SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006 reconciles all six handoffs with the 256 pre-handoff receipts and assigns one owner to every downstream interface.

10.11 Evidence status, limitations and handoff

The branch preserves favourable, adverse, ambiguous and exceeding-boundary images inside its complete transformation ledgers. Ambiguity is not reported as failed transport or null evidence: it is the exact non-singleton predecessor class. No current authoritative record supplies a basis for turning every formal execution into a natural measurement, and the branch does not do so. The current V1/V2 reconciliation retains forward-only records even where a later smoothing law can reconstruct an earlier state; future evidence is not inserted into the earlier chronology.

Information Science is complete to its frozen 29 July 2026 census, not to every future information problem. It does not own a physical channel's measured capacity, a living system's function, a person's experience or the operational phase and gates of a quantum machine.

The next unresolved requirement is execution. Information laws specify states, records, transformations, ambiguity and reconstruction, but a computational science must derive languages, machines, resources, algorithms and limits. Information Science therefore hands complete symbol support, encoding, conservation, channel and evidence custody to Classical Computation.

11. Classical Computation: the Fold machine

11.1 Why computation follows information

A source-to-output relation does not yet specify an executable process. A machine requires generated configurations, enabled transitions, a language, composition, terminal and recurrent classes, resources and a complete trace. Classical Computation owns those objects and their formal limits. It does not own quantum operations, physical hardware, engineering reliability or an application's empirical performance.

The current branch contains 369 claims across twelve families, 94,464 candidates, 369 unique survivors, 1,476 passed controls and 369 independent reconstructions. The 117 foundational and prior-corpus claims are independently replicated. The

remaining 252 complete-field extensions carry post-registry execution observations and are classified as empirically tested and independently replicated within their declared computational boundaries.

11.2 Native machine constitution and family census

The native machine begins with exact generated configurations, held Fold labels and a source-bound transition relation. Blank support is empty One. A word is a complete sequence of held labels at canonical generated positions. Reading is an observation relation; writing is held-label substitution; movement is a counted transition. Each configuration retains its word, held position, process state and proof trace. Every step records its source, operation, successor, resources and retained or closed distinctions.

Family	Claims	Candidates	Scientific scope
BASE	117	29,952	Foundational Fold machine and prior-corpus reconstruction
FORMX	22	5,632	Formal computation
CBLX	21	5,376	Computability and its limits
CPLXX	33	8,448	Computational complexity
ALGX	31	7,936	Algorithms and mathematical data structures
SEMX	25	6,400	Semantics and programming theory
DISTX	26	6,656	Concurrent and distributed computation
SECX	25	6,400	Cryptography and computational security
LEARNX	26	6,656	Learning and intelligence theory
SCIX	25	6,400	Scientific computation
VALID	12	3,072	Dated validation and Grand Lock
HAND	6	1,536	One-owner handoffs and open extension
Total	369	94,464	Current full-field census

11.3 From state to universal machine

SFT-COMP-FORM-STATE-TRANSITION-001 combines Foundation form enforcement, mathematical discrete and graph structure, logic and composition, and information encoding and conservation. Its survivor retains a complete state carrier, held action labels, source-bound transition rows, initial and terminal classes and complete execution traces. It does not yet presume a language or machine model.

The next claims derive, in order, a generated word grammar, finite automata, rewriting, structural recursion, binding and capture-safe substitution, abstract machines, circuit syntax, operational processes, serial and parallel composition, bidirectional model translations and a universal interpreter. Universality means exact interpretation of every description generated by the frozen grammar, with preserved terminal result and an explicit overhead ledger. It is not a completed totality of every possible language object. Turing, Church and circuit systems enter only after sealing as comparison encodings.

11.4 Computability, halting and incompleteness

Recognition retains acceptance, rejection and unresolved continuation as different trace classes. For any supplied complete finite generated carrier, exhaustive state execution can classify a process as terminal or recurrent. That bounded classification does not imply that every bounded process halts.

The unrestricted internal halting boundary uses exact self-description and a held complement action. If a total internal decider predicts the behaviour of every generated universal-machine description, a process can be constructed that rejects when the decider predicts acceptance and accepts when it predicts rejection on its own description. Neither held verdict is a fixed point. The contradiction closes the total internal decision claim without treating non-termination as numerical zero.

The same missing distinction appears in internal self-verification. An effective proof enumerator cannot retain consistency and supply every demanded self-referential verdict. Relative computation remains possible only through a named query alphabet and a declared oracle relation whose identity and answers are preserved. A hypercomputation proposal must expose an executable law, finite evidence boundary, self-reference test, resources, oracle custody and falsification condition. A finite demonstration cannot silently become an unrestricted oracle.

11.5 Resources, algorithms and semantics

Input size is the complete generated extent of a canonical representation. Time is a transition trace; space is live retained configuration support; circuit size and depth are generated gate and causal-layer ledgers. Randomised computation does not import an uncaused schedule. It retains the complete registered schedule support and executes one deterministic trace per held schedule. Exact success parts summarise the resulting output partition. If a physical source supplies schedules, the source distribution is a separate measurement.

Reversibility cost is information custody. A non-singleton transition fibre merges predecessors; reverse execution then requires retained predecessor labels. Lower and upper bounds must name the problem family, representation, machine, resource and witnesses. Average, worst, approximate and parameterised claims similarly retain complete instance support and the exact aggregation or preference relation rather than a hidden distribution.

The algorithm families derive search, order, arithmetic, strings, trees, graphs, dynamic programming, optimisation, parallel, distributed, online, streaming, numerical, symbolic and approximate computation from those resource objects. Program semantics then retains syntax, binding, evaluation, typed interfaces, equivalence, termination, specification, correctness, transformation, compilation and verification. Correctness is always relative to the exact registered precondition, program, input support and specification; a proof of the wrong specification is not practical correctness.

11.6 Distribution, security, learning and scientific computation

Distributed computation adds causal partial order, synchronisation, communication, replication, fault models, local knowledge and network interfaces. Agreement impossibility is derived from two predecessor executions that produce identical local observation but require different decisions. The impossibility remains relative to the registered asynchronous and fault grammar; it cannot be exported to a different model without a new claim.

Security begins with complete message, key, observation, adversary and resource support. One-wayness is resource-relative forward execution versus inverse search; unbounded finite inversion is not declared impossible. Information-theoretic security means the relevant source distinctions remain closed over complete observation support. Computational security means the registered resource-bounded adversary grammar fails to reopen them. Hashing, commitment, signatures, proofs of knowledge, zero knowledge and multiparty computation therefore retain both their exact adversary and their declared boundary.

Learning claims retain hypothesis support, development rows, sealed predictors, withheld target support and failure conditions. Generalisation is empirical unless a structural successor proof exhausts the target family. Two hypotheses that agree on every observed row but require different unobserved decisions establish an exact identifiability limit. No benchmark score, pretrained model or fitted opacity replaces the premise-to-result trace.

Scientific computation separates exact proof values from bounded approximations, retains orientation-labelled error, propagates exact error ledgers through dependency graphs, and requires explicit refinement and convergence certificates. A simulation derives consequences of its registered model relation. It cannot establish the corresponding natural law without a separate sealed measurement claim owned by the relevant natural-science branch.

11.7 Algorithms and data structures as trace-preserving processes

The thirty-one ALGX laws reconstruct the common algorithmic families without letting a library routine or asymptotic slogan become the premise. Searching begins with a complete input support, a target relation and an ordered sequence of inspected positions. Ordering retains the comparison witness for every placed pair and preserves ties. Arithmetic algorithms operate on the admitted exact positive carrier. String and sequence algorithms preserve position, symbol and source-word identity. Tree and graph algorithms consume the mathematical incidence records and emit paths, traversals, components or cuts with complete predecessor traces.

Algebraic and geometric algorithms retain their exact carrier and predicate boundaries. Dynamic programming is derived as a dependency-ordered table of subproblem identities whose entries reconstruct the terminal result; a cached value without its recurrence and input relation is insufficient. Optimisation keeps feasibility, objective order and every tied or undominated output. Randomised algorithms are deterministic executions over complete held schedule support, and their success part is computed only after every registered schedule has a result. Parallel algorithms retain a task-dependency graph, work, causal span and synchronisation records.

Distributed algorithms add message and local-state custody. Online and streaming algorithms distinguish what was available at each decision time from the later complete input; a retrospective optimum cannot be inserted into the earlier trace. Numerical algorithms carry exact reference values or rational enclosures, precision provenance and stopping certificates. Symbolic

algorithms retain parse, rewrite and identity traces. Approximation algorithms retain the exact distinction between their output and the optimum at the declared resource boundary. The family extensions then cover selection, hash-table, heap, union-find, traversal, shortest-path, flow, matching, constraint, scheduling, transform and composition structures through the same input-operation-output custody rather than as opaque callable names.

The consequence is not that one algorithm is universally best. Resource claims remain representation- and model-relative. An apparent speedup that changes the input encoding, hides preprocessing or discards failed instances does not satisfy the comparison grammar. A lawful comparison preserves both implementations, the same registered instance support, each complete resource trace and the exact result relation.

11.8 Programming meaning, transformation and proof

The twenty-five SEMX claims derive a programming language as more than a word grammar. Syntax retains production and position identity. Binding and substitution distinguish free from held-bound occurrences and prevent capture. Evaluation is a source-bound transition trace. Operational semantics records the transitions a process executes; denotational correspondence records the mathematical relation assigned to the same complete input support. Agreement is proved only at their declared interface.

Types are generated carriers and admissible-operation relations, not authority labels. A type error is a failed interface relation. Program equivalence may be trace equality, result equality or observational equivalence, but the chosen boundary must be registered. Termination is a terminal-trace property; recurrent and unresolved continuations remain distinct. Correctness combines a precondition support, program execution, postcondition relation and an explicit total- or partial-termination declaration.

Specification therefore owns the claim the program is meant to satisfy. Transformation must map source to target programs while preserving the registered semantics. Compilation retains parse, intermediate representations, lowering steps, target process and a semantics-preservation certificate. Optimisation cannot delete an observable distinction outside the declared equivalence. Verification retains the proposition, proof system, program, input support and every failed proof condition. A verifier can establish a formal property of the registered model; it cannot certify an unstated physical environment or safety requirement.

The later semantic extensions cover modular interfaces, effects, state, exceptions, nondeterministic branch support, concurrency, refinement, abstraction, executable specification, proof-carrying code and compiler bootstrap custody. In each case, the question is which distinctions survive translation and which records are required to reconstruct meaning. A terminal binary identity is evidence only when its complete source and compilation route remain bound.

11.9 Concurrency, causality and distributed knowledge

The twenty-six DISTX laws begin from interleaved and partially ordered transitions. Concurrent processes retain component identity and every allowed schedule. Computational causality is a dependency path from source state and message to later state; temporal order alone is not enough. Synchronisation is a registered relation that closes a set of local transitions into one shared boundary event. Communication keeps sender, receiver, payload, send and receive positions, and transport outcomes.

A distributed partial order records events that are causally comparable and leaves independent events unordered. Consensus requires one decision relation, agreement among admissible non-faulty participants, validity against the registered input support and the declared termination boundary. The agreement impossibility law identifies two source executions that are indistinguishable to a local process yet require different decisions. Because the local trace is the same, a deterministic process must decide the same way and fail one requirement. The result belongs to that exact asynchronous and fault grammar; changing the failure detector or synchrony model creates a new problem.

Replication records source version, copies, update order and observation. Consistency is an explicit relation among those histories, not a generic quality. Fault models enumerate allowed omissions, corruptions, timing changes or adversarial actions. Distributed knowledge is the class of global states consistent with a process's local observation. Locality names the interfaces a process may access. Network computation then composes node processes, paths, messages and failures without a hidden coordinator.

The extension families add mutual exclusion, clocks, snapshots, leader and membership records, broadcast, commit, replicated state, eventual and stronger consistency correspondences, fault tolerance, recovery, routing, load and resource distribution. Every result retains the exact schedule and fault support on which it was tested. A successful execution under one schedule is not consensus over all registered schedules; a simulation without the declared failure rows is not fault-tolerance evidence.

11.10 Security without an imported hardness oracle

The twenty-five **SECX** laws use Information Science's observation classes and Computation's adversary resources. One-wayness compares a total forward map with complete inverse search under a declared adversary process and resource bound. It does not declare mathematical inversion impossible. Secrecy asks which message distinctions the adversary observation retains. Integrity asks whether an unauthorised transformation produces an accepted record. Authentication binds identity, challenge, response and source custody.

Hashing retains the input grammar, output relation, collision classes and resource boundary. A commitment must retain hiding and binding as separate observation and transformation properties. A signature binds message, signatory key relation, verification process and forgery grammar. Proof of knowledge distinguishes possessing a reconstructible witness from merely producing an accepted transcript. Zero knowledge preserves the verifier's observation class and simulator correspondence at the declared protocol boundary. Multiparty computation keeps each input owner, allowed disclosure, joint process and output.

Adversarial computation is never an unnamed hostile possibility. The adversary's input, observations, allowed operations, schedule and resources are part of the candidate grammar. Information-theoretic security means complete support closes the protected distinction for every allowed adversary action. Computational security means the registered resource-bounded process family does not reopen it. These statuses cannot be exchanged. A reduction preserves the exact problem and resource interfaces; an authority claim that a primitive is "believed hard" cannot replace the registered witness.

Post-quantum security is a handoff boundary. Classical Computation can state the current adversary and the formal relation a quantum consumer must preserve, but it cannot assume a quantum speedup or device. Quantum Computation must derive the relevant process and resource law independently. Practical security then remains an Engineering and application measurement problem involving implementation, side channels, key custody, performance and threat evolution.

11.11 Learning, generalisation and intelligence boundaries

The twenty-six **LEARNX** claims treat learning as a registered computation over hypothesis, evidence and target support. Inference restricts hypotheses by their compatibility with observed rows. Classification and prediction retain the label relation, development support, sealed model and target chronology. Representation learning may transform the input only with a reconstruction or declared-loss ledger. It cannot make the original source and exclusions disappear.

Generalisation requires more than development accuracy. The hypothesis class, development rows, withheld support, structural invariance claim, sealed predictor, complete target rows and failure condition are all retained. If a successor proof exhausts every generated target, the result may be formal at that support. Otherwise generalisation is an empirical result whose status depends on the seal and target custody. Sample complexity counts exact examples under a fixed hypothesis and observation grammar; it is not a universal number detached from representation.

Learning optimisation retains objective, alternatives, ties and resource trace. Induction is a transparent rule from evidence to hypotheses and cannot silently import a prior. Search and planning retain state, action, transition, goal and explored support. Reinforcement computation keeps reward as a registered observation or preference relation, not a primitive purpose. Multi-agent learning includes each observer, communication and joint-action support. Adaptation preserves the version history of data, model and environment.

The limits law constructs observationally equivalent hypotheses with different unobserved requirements. Any learner given the same evidence must make the same decision for both and therefore cannot be guaranteed correct on both. Interpretability and verification ask which internal relation or trace can be reconstructed and checked. A plausible verbal explanation is not automatically the process that produced the output. The classical-learning model collects these objects without asserting consciousness, agency, biological cognition or deployment fitness; those belong to downstream owners.

11.12 Scientific calculation, simulation and inverse problems

The twenty-five **SCIX** laws provide the formal machinery later branches use to make computational predictions. Exact and approximate quantities remain separate. Numerical stability records how input distinctions propagate through the process. Error propagation follows the dependency graph, retaining local error classes, held orientation, combination rules and the final enclosure. Cancellation may close a distinction only when the exact relation and provenance are retained.

Discretisation maps a registered model relation to generated cells, retains coverage and boundary conditions, and provides a successor refinement. Convergence is a certificate that refinement preserves the model interface and forces the declared error class to descend. Symbolic computation preserves model syntax and exact rewrite identity. Simulation begins from an explicit state carrier, update law, initial support, schedule, observation map and terminal comparison boundary.

Computational dynamics retains every orbit, invariant, recurrence and perturbation class. An inverse problem begins with a forward relation and an observation; reconstruction returns the complete predecessor fibre and labels the source identifiable only when that fibre is singleton. Computational statistics preserves sample support, statistic map, observation fibres and uncertainty. High-dimensional computation retains coordinate product or factorisation and records interactions lost by an approximation. Many-body computation keeps body identities, local states, interaction graph, joint support and update trace.

Mathematical modelling closes the family by binding a phenomenon disclosure, formal grammar, parameter status, derivation or constitutional provenance, sealed predictions, external measurements, falsification and revision records. This law does not make every model true. It specifies the evidence objects required to test one. A model with a complete simulation but no source-bound measurement remains a formal model. A model fitted on the target cannot later describe its development result as a blind prediction.

11.13 Validation and branch interfaces

The twelve validation claims partition the 351 pre-validation receipts by family, evidence type, closure boundary and ownership. They preserve theorem, finite-census and positive-finite distinctions, as well as adverse and unresolved rows. The six handoff claims then enforce one owner for each downstream result.

SFT-COMP-HAND-MEASUREMENT-BOUNDARY-002 states that Classical Computation owns form, trace and resource laws while measured consequences remain with Physics, Chemistry, Biology, Medicine, Earth science, Astronomy or another empirical owner. SFT-COMP-HAND-FORMAL-EMPIRICAL-003 requires a value-free target registry, sealed prediction, independently custodied observation and complete comparison before a formal result gains empirical status.

SFT-COMP-HAND-CLASSICAL-QUANTUM-004 passes state, trace and resource interfaces to Quantum Computation while leaving quantum states, gates, channels and fault tolerance quantum-owned.

The branch is therefore internally complete without becoming a universal owner. Its applications-including Chess Fold, Go Fold, Protein Fold and Unison Fold AI-may implement and test admitted computation, but their scores and domain observations cannot select these laws. A successful application is a new evidence object; a failed application is retained at its own registered boundary.

11.14 Three native exact results

The branch includes three strong native results whose scope is essential to their meaning.

For SFT-COMP-CBL-NATIVE-BUSY-BEAVER-002, every supplied positive finite description depth k in the admitted closing-process grammar halts within k lawful depth-lowering transitions, and a depth- k source attains k . Recurrent period- b processes carry exact non-halting certificates and are excluded from the closing maximum. Therefore

$$BB_F(k) = k$$

for every supplied positive finite k in that grammar.

SFT-COMP-CPLX-FOLD-P-NP-EQUALITY-002 establishes that exact deterministic evaluation emits the complete lawful trace already accepted by the proof system, while sound verification forces the same unique evaluation. Both resources equal description depth, so

$$P_F = NP_F$$

inside the admitted native Fold grammar. The claim explicitly does not assert conventional $P=NP$.

SFT-COMP-CPLX-ARBITRARY-CIRCUIT-LOWER-BOUND-002 ranges over every circuit assembled from admitted Fold edges. Closing a depth- k path requires at least k gates; complete source support requires width b^k ; and complete layered coverage requires

$$\sum_{r=One}^k b^r$$

edges. The registered circuit attains all three bounds. These are tight bounds for admitted Fold-edge circuits, not an unqualified lower bound over every conventional circuit model.

11.15 Preserved corrections and current boundary

The V1/V2 ownership audit read all 763 prior entries, decomposed seventy computation-relevant source rows into 134 atomic obligations and closed all

134. Four reconciliations remain explicit:

1. unboundedness alone is not a proof of undecidability; the admitted proof uses self-description and the held-complement diagonal contradiction;
2. a bounded deterministic process is guaranteed to halt **or recur**, not to halt;
3. earlier logarithmic or exponential efficiency prose is replaced by exact Fold description depth, support, transition, space and retained-information resources; and
4. native $P_F=NP_F$ is not a verdict on conventional P versus NP.

The validation layer replays all 351 pre-validation receipts and preserves their adverse, boundary and ownership rows. There is no open frozen obligation, but there are permanent scope boundaries: physical device behaviour, engineering quality and application results require their own owners and evidence. The six handoffs therefore assign quantum operations, physical measurement, software and hardware engineering and applications exactly once.

Classical Computation hands exact configurations, languages, processes, resources, reversible-cost records and proof-preserving interfaces to Reversible and Quantum Computation. It does not pre-derive quantum state, phase, interference, entanglement, measurement, gates or fault tolerance.

12. Reversible and Quantum Computation: one extended Fold machine

12.1 Why the quantum branch follows computation

Classical reversibility establishes that a merged predecessor distinction must be retained if inverse execution is required. It does not yet derive complete branch support, phase-sensitive merging, non-factorable joint support or an observation relation over such a carrier. The quantum branch owns those operations. Physical preparation, energy, timing, fidelity, error rates, apparatus and outcome distributions remain downstream measurements.

The current branch contains 288 claims across thirteen families, 73,728 candidates, 288 unique survivors, 1,152 passed controls and 288 independent reconstructions. Its twenty-two foundational claims are independently replicated. The 266 full-field extensions carry registered post-registry execution observations and are classified as empirically tested and independently replicated within their formal computational boundaries.

12.2 Fold-native quantum constitution and census

One Fold distinction supplies two held fibre labels and therefore the native information unit. Complete held-label words across generated positions supply branch support. A branch carries an exact held phase label from a generated finite cycle. Joint states are complete pair-cell words retaining both component identities. A transformation is a total branch-word bijection plus a period-held phase action. These objects require no Hilbert-space axiom, complex amplitude, irrational normalisation, imaginary proof value or stochastic collapse premise.

Family	Claims	Candidates	Scientific scope
BASE	22	5,632	Reversible and quantum computational foundation
REVS	18	4,608	Complete reversible computation
QSTATEX	28	7,168	Information units, states, support and composition
GATEX	22	5,632	Transformations, gates, circuits, compilation and universality
QALGX	30	7,680	Quantum algorithms and correctness
QCPLXX	26	6,656	Quantum resources, complexity and bounds
QCOMM	24	6,144	Channels, communication and cryptography
QCODEX	32	8,192	Coding, error correction and fault tolerance
QSIMX	24	6,144	Simulation and verification
QLEARNX	22	5,632	Quantum learning and intelligence
QLIMITX	22	5,632	Classical-quantum correspondence and computational limits
VALID	12	3,072	Dated validation and Grand Lock
HAND	6	1,536	One-owner downstream handoffs

Family	Claims	Candidates	Scientific scope
Total	288	73,728	Current full-field census

12.3 The twenty-two foundational quantum claims

Every foundational claim has 256 candidates, one survivor and four passed controls. Together they form a single ordered argument.

Claim	Principal result and boundary
SFT-QUANTUM-REVERSIBLE-MODEL-001	Canonical configurations, bijective transitions, held labels, exact inverse traces, universal reversible interpretation and complete resources; no thermodynamic energy is assigned.
SFT-QUANTUM-INFORMATION-UNIT-001	One Fold distinction, two held labels, complete word support, observation classes and a phase-label carrier; the unit is structural rather than an imported qubit.
SFT-QUANTUM-STATE-COMPOSITION-001	Complete branch support, pair-cell joint words, component identities, phase composition, marginals and product/correlated classes.
SFT-QUANTUM-SUPERPOSITION-001	Complete b^k word support with one phase-bearing record per branch and no selected branch before observation.
SFT-QUANTUM-PHASE-INTERFERENCE-001	A reversible period- b phase action and total branch-to-image relation with complete predecessor merge and phase provenance.
SFT-QUANTUM-ENTANGLEMENT-001	Joint support is factorable exactly when it equals the product of its marginals; a strict correlated subset is non-factorable.
SFT-QUANTUM-MEASUREMENT-001	A total observation relation retains the selected class, closed branch distinctions, full pre-observation record and reconstruction boundary.
SFT-QUANTUM-GATE-001	A gate is a bijective branch-word map plus an invertible held phase action; physical fidelity remains empirical.
SFT-QUANTUM-CIRCUIT-001	Generated wires, units, reversible gates, causal order, complete branchwise evaluation, inverse circuit, observation boundary and resources.
SFT-QUANTUM-UNIVERSALITY-001	One interpreter simulates every circuit description in the frozen grammar while preserving branch, phase, observation and overhead records.
SFT-QUANTUM-ALGORITHMS-001	Complete input support, reversible branch generation, phase action, interference merge, observation decoder and branchwise correctness trace.
SFT-QUANTUM-COMPLEXITY-001	Canonical input size, gates, causal depth, live branch support, queries, communication, measurement and lower/upper witness boundaries.
SFT-QUANTUM-COMMUNICATION-001	Sender and receiver support, joint composition, local transformations, channel relation, retained correlations, observations and loss ledger.
SFT-QUANTUM-CODING-001	Logical support, redundant joint words, reversible encoding, declared error actions, disjoint syndrome support and reconstruction.
SFT-QUANTUM-ERROR-CORRECTION-001	For positive finite fault order t , repetition width $2t+1$, every mask through t , majority held-label decoding and recovery custody.
SFT-QUANTUM-FAULT-TOLERANCE-001	Encoded support, declared component faults, contained transformations, recovery by location, malignant-fault boundary and circuit induction; no hardware threshold constant.
SFT-QUANTUM-SIMULATION-001	Encoded target support, reversible update, phase/interference trace, observation map, simulator correspondence, resources and falsification boundary.
SFT-QUANTUM-VERIFICATION-001	Claim and witness support, verifier circuit, challenge schedule, branchwise acceptance, soundness/completeness ledgers and tamper controls.
SFT-QUANTUM-LEARNING-001	Example support, reversible hypothesis process, phase/interference search, observation decoder, alternatives, resources and sealed evaluation.
SFT-QUANTUM-CLASSICAL-CORRESPONDENCE-001	Classical states embed as a reversible phase-insensitive submodel; measurement decodes classical records while phase-sensitive traces remain distinct.
SFT-QUANTUM-LIMITS-001	Finite description, self-reference, halting, undecidability and incompleteness boundaries survive; no hypercomputational oracle appears.
SFT-QUANTUM-UNBOUNDED-FINITE-FAULT-TOLERANCE-002	The first repetition width correcting every mask through each supplied positive finite t is exactly $2t+1$, with constructive failures below it.

12.4 Superposition, interference, entanglement and measurement

Superposition-equivalent support means that every generated branch is retained before observation. It is neither a hidden selected branch nor an imported stochastic mixture. Each branch carries a held phase in a finite generated cycle. Interference is then a branch-to-image relation whose predecessor fibre keeps every merging branch and phase identity. No branch disappears from the audit record merely because several branches share an image.

Two systems compose through complete pair-cell joint support. Their joint is factorable when it is exactly the Cartesian product of its marginal supports. A strict correlated joint subset is non-factorable and supplies the derived entangling class. This definition is structural and operational; it is not a metaphysical postulate added to the carrier.

Measurement is a total observation relation over complete support. It retains one observation class as the post-observation state while the measurement record preserves every pre-observation branch, phase and image identity. Thus support reduction is exact at the selected-class boundary and reconstructible at the complete-record boundary. Outcome frequencies are not inferred from the formal relation alone. They require source-bound physical evidence.

Classical computation is recovered as the single-held-branch, phase-insensitive reversible submodel. Quantum execution extends that carrier with complete simultaneous branch support, phase-sensitive merging and non-factorable joint support. The correspondence is exact where the interfaces agree, but it does not identify the two modes where phase changes the trace.

12.5 Gates, algorithms and exact correction

A reversible gate must be bijective on branch words and invertible on held phase. A circuit retains generated wires, gate identities, causal order, complete branchwise evaluation, inverse construction, observation boundary and resource ledger. Universality is confined to descriptions generated by the frozen gate grammar. The algorithm and complexity families then derive query, phase, transform, search, counting, simulation, verification, communication, upper and lower witness laws without presuming a measured speedup.

The exact correction theorem is particularly clear. If at most t labels may change in a repeated word, width $2t+1$ leaves at least $t+1$ source labels against at most t changed labels. The source therefore retains a strict majority. Every width through $2t$ has a constructive tie or wrong-majority counterexample. Consequently the first correcting width and exact redundancy are

$$w(t) = 2t + 1, \quad r(t) = 2t.$$

The executable census exhausts every registered mask through one fault at width 3, two faults at width 5 and three faults at width 7. Successor checks run through $t=14$, and the fixed-depth circuit tests round-trip all 128 depth-seven source words at that registered boundary. The proof for arbitrary supplied positive finite t is the majority and counterexample construction, not an extrapolation from those test depths. Advancing from t to $t+1$ adds exactly two labels and defeats every proposed fixed positive finite ceiling.

This is an unbounded **finite-order** theorem. It does not construct a completed infinite-width word, measure a stochastic device error rate or derive a physical fault-tolerance threshold. Those quantities belong to Physics and Engineering.

12.6 Reversible computation as complete predecessor custody

The eighteen REVX laws develop the reversible carrier before the specifically quantum families. A reversible state transition is a bijection over complete registered configurations. Its inverse is not inferred from a final value; it is an exact relation whose trace reconstructs the preceding state and action. If a forward process merges two predecessors, reversibility requires a held record that distinguishes them. This is the operational content inherited from Information Science's conservation and Classical Computation's reversibility cost.

Reversible words retain every input label and ancilla-like support. A lawful reversible extension may add a clean held carrier, compute into it and later uncompute intermediate distinctions, but it must account for every retained or closed record. Copying a definite classical held word is possible because the source label remains available and the output relation preserves both source and copy provenance. That fact does not pre-empt the quantum no-cloning boundary, which concerns a different complete support and transformation grammar.

Reversible gates compose only when output and input interfaces match. Serial composition applies one bijection after another and reverses in the opposite order. Parallel composition uses complete product support and retains component identity. Controlled transformations preserve the control label while applying the target action on the declared class. Permutations, swaps and reversible Boolean correspondences enter after this carrier is derived, not as assumed gate libraries.

Resource accounting retains forward actions, inverse actions, live support, held history and any record later closed. Reversible simulation maps a classical process into a bijective process by retaining enough predecessor information to reconstruct every step. The overhead is part of the theorem. A terminal output without the history or an uncomputation certificate is not evidence of reversible execution. Universal reversible interpretation ranges over every description in the frozen grammar and no further.

The reversible family also supplies the interface to physical thermodynamics without assigning energy. Information Science has shown which distinctions a merge closes; the reversible machine shows which record restores them. Whether retaining, moving or resetting that record consumes a measured energy under a particular apparatus is a Physics and Engineering question.

12.7 State support, phase and joint composition

The twenty-eight QSTATEX laws refine the six foundational state results into a complete state science. One distinction gives two held labels. A depth- k word has one label at each generated position, and complete support contains all b^k generated words. Each support member retains branch identity, position trace and held phase. Preparation is a registered source-to-support relation; it does not assume that a physical laboratory has realised the formal state.

Normalisation is support completeness, not an irrational scalar constraint. A state is complete when every branch required by its generation grammar is present exactly once with its source and phase record. A partial support is a different state with a different registry. Marginalisation groups joint cells by the retained component and preserves the closed partner distinctions. Product composition contains every pair of marginal members. Correlated composition is a declared strict subset with complete joint custody.

Phase is a held position in a finite generated cycle. Phase translation is a reversible action on that cycle, and relative phase is an exact relation between held labels. Global-phase-like equivalence may be recorded where a uniform translation preserves every registered observation, but the source phase record remains available in the audit. Interference occurs only through a declared branch-to-image map with complete predecessor fibres. Constructive and cancelling correspondences are therefore properties of the phase-labelled merge relation, not arithmetic on hidden complex amplitudes.

Entanglement is tested by factorisation. Compute both marginals from the joint support, generate their complete product and compare it with the registered joint. Equality yields the factorable class. A strict correlated joint that cannot be reconstructed as that product is non-factorable. This definition retains every joint cell and marginal; a correlation score cannot replace it. Multipartite extensions retain each component identity and every proposed partition, so factorability may differ across cuts.

Observation partitions complete branch support into outcome classes. A sharp observation has singleton classes at the declared label; a coarse observation retains several branches. Compatible observations are relations that preserve one another's retained distinctions at the registered interface. Disturbance is the exact difference between pre- and post-observation support or phase relations. Repeated observation retains the first record and the new source state rather than pretending both acts observed the untouched original.

The state family also records mixed-support, purification-like and reduced-state correspondences as relations among complete hidden, joint and observed supports. These names enter after the Fold objects exist. No density operator, amplitude table or stochastic preparation is silently imported. A physical claim about preparation frequencies needs a sealed device experiment.

12.8 Gates, circuits, algorithms and compilation

The twenty-two GATEX laws expand the gate foundation into a complete operational language. A gate's branch map is total and bijective, its phase action is exact and invertible, and its source identity is retained. Single- unit, controlled and joint transformations are generated from those relations. Gate equality is exact operational equivalence over complete registered support, not agreement on a few test inputs.

A circuit is a directed acyclic dependency structure of wires and gate locations with a complete branchwise evaluation. Causal depth counts the longest dependency path; size counts generated gates; live support and measurement records are separate resources. Reversing a circuit requires inverse gates in reverse causal order. Circuit composition retains interface, wire and phase identities. A malformed cycle, unmatched wire or non-bijective gate fails before an output can acquire authority.

Compilation translates a source description into the admitted gate grammar. The compiler retains parse, intermediate decompositions, target circuit, branchwise result equivalence, phase relation and resource overhead. Optimisation is permitted only under the registered observational equivalence. A shorter circuit that changes an unmeasured phase may be equivalent for one decoder and inequivalent for another; the observation boundary must therefore be explicit.

Universality provides one interpreter for every generated description in the frozen grammar. It does not claim a completed set of all possible unitary objects. Approximation, where used, retains exact rational phase or result enclosures and a successor certificate. Gate synthesis retains the target relation, generated candidate sequences and the first or least-resource survivor under the declared order.

The thirty QALGX claims organise complete input preparation, reversible branch generation, phase action, interference, observation decoding and correctness. Query algorithms retain the oracle as a named external relation; its answers and resource

count remain visible. Transform algorithms operate on complete finite support. Search and counting preserve every candidate branch and the decoder by which an answer class is observed. Simulation and verification algorithms retain the target model or witness rather than being judged by a terminal label alone.

Historical algorithm names may identify correspondences after sealing, but they cannot supply an answer table. A claimed quantum advantage requires two registered machine descriptions, the same problem and input support, exact resource ledgers and, where hardware is claimed, a sealed experiment. The formal algorithm family derives quantum process structures and their internal resources; it does not declare that a current device has achieved practical speedup.

12.9 Quantum resources and complexity

The twenty-six QCPLXX claims retain canonical input size, gate count, causal depth, live branch support, query rows, communication, measurement and error records. A resource belongs to a representation and model. Complete support size and the resources needed to represent or traverse it are not silently interchanged. A compact process description may generate a large support; both identities remain in the ledger.

Upper bounds require a constructive circuit and complete execution trace. Lower bounds require an adversarial, indistinguishability or structural witness inside the declared grammar. A correspondence to a conventional complexity class is admitted only after the native resources are fixed. Average and worst cases retain the complete instance family and distribution provenance. Approximation records exact output error and success support. Parameterised claims name the parameter carrier and its relation to input size.

Quantum schedule support remains deterministic. A branch or measurement class may be unresolved to an observer, but no uncaused transition is used as a complexity resource. If a physical random source supplies an input or schedule, its distribution belongs to a measurement package. Resource-bounded probabilistic or quantum success is then an exact part of the complete registered support, not a floating propensity.

Complexity separations are never granted by vocabulary. A new separation requires its own generated family, reduction boundary and witnesses. The foundational complexity claim therefore records lower and upper witness interfaces without asserting every familiar conventional class relation. The same caution applies to speedup: fewer native Fold resources in one exact grammar is a formal result; wall-clock, energy or device advantage is an empirical engineering result.

12.10 Communication, cryptography and no-cloning boundaries

The twenty-four QCOMM laws combine joint support with sender, receiver, channel and local-operation custody. A quantum channel is a complete state-to-state relation retaining phase, loss and observation interfaces. Memory and composition record whether successive uses share source distinctions. Capacity is relative to the registered channel, code, decoder and resource support. Physical transmission rate and fidelity remain measured quantities.

Entanglement-assisted communication retains the pre-shared joint support and its preparation custody. Teleportation-like correspondence records the source state relation, shared joint support, classical message, receiver operation and decoded state. Dense-coding-like correspondence records the same resource ledger in the opposite informational arrangement. Neither relation permits source-free transmission; the consumed shared support and classical channel remain explicit.

No-cloning is derived at the complete unknown-support boundary. A copier that must leave the source intact and create a second exact copy for every admissible phase-bearing state would need one reversible transformation to map distinct non-factorable source relations into product copies while preserving their pairwise distinctions. The complete relation exposes the incompatible requirements. Copying a known held classical label is outside that unknown quantum-support condition and remains lawful, so the branch does not conflate the two results.

Disturbance and information-disturbance relations retain which source distinctions an observation extracts and which state or phase relations change. Key-distribution and adversary laws name the channel, basis or observation choices, public records, retained key classes and attack support. Formal secrecy is confined to the registered adversary. A practical claim requires device sources, losses, side channels, authentication and implementation evidence owned by the appropriate downstream branches.

12.11 Coding, correction and fault-tolerant composition

The thirty-two QCODEX laws distinguish logical support from its physical or component representation. An encoder is reversible and retains the mapping from every logical branch into redundant joint words. An error family is a complete registered set of actions and locations. A syndrome is an observation partition over those actions. Recovery is lawful only when the syndrome and retained records identify a unique correction class or when all surviving actions are equivalent on the logical support.

Degeneracy is therefore not hidden ambiguity: distinct component errors may share one syndrome because they induce the same logical relation. Erasure, substitution, phase-label and joint errors retain different action grammars. Distance and separation are post-derivation summaries of complete image supports. Concatenated, product and sparse-check correspondences retain every level, check and reconstruction map. A decoder that uses a likelihood requires an external measured error model; the structural decoder itself cannot fit one.

Fault propagation follows each allowed component error through every gate and wire. A contained or transversal transformation preserves the registered correctable support at the declared locations. Recovery after a location retains both the faulted predecessor and corrected logical result. A malignant set is an explicit action combination that escapes the correctable class. The fault-tolerance induction composes these records through a finite circuit.

The $2t+1$ theorem supplies an exact positive-finite correction family without supplying a hardware threshold. It proves a strict majority for every mask of size at most t and gives a constructive counterexample for every smaller width. A physical threshold would additionally need error mechanisms, correlations, gate and measurement fidelity, timing, architecture and a registered statistical protocol. Those values cannot be inferred from the structural width alone.

12.12 Simulation, verification and quantum learning

The twenty-four **QSIMX** laws require a simulator to retain target state support, update law, phase and interference relations, observation map, resource ledger and exact correspondence boundary. A digital, analogue or mixed simulation is distinguished by its encoding and update relation. Approximate simulation retains a rational enclosure or declared observation equivalence. A terminal match on one row does not establish model equivalence.

Verification retains a claim support, witness, verifier circuit, challenge schedule, complete branchwise acceptance, soundness and completeness ledgers, measurement records and tamper controls. Interactive and delegated correspondences keep prover and verifier observations separate. A verification claim is only as broad as its prover, witness and resource grammar. It cannot certify an unknown physical device characteristic that was not part of the registry.

The twenty-two **QLEARNX** laws extend Classical Learning with reversible hypothesis processes, phase/interference search and quantum observation. Input examples may be classical, quantum or joint, but their source and preparation support remain explicit. Training retains hypotheses, update trace, resources and all alternatives consistent with the data. Evaluation is sealed before the target opens. A reported advantage needs a same-task classical comparison and complete resource accounting.

Quantum representation, classification, regression, clustering, generation, reinforcement, multi-agent and adaptive correspondences remain formal process families at their registered support. No pretrained oracle or fitted amplitude is admitted as a law. Generalisation remains empirical unless target support is exhausted by a successor proof. Interpretability retains which branch, phase or measurement relation supports an output; a post-hoc narrative is not equated with the executed process.

12.13 Classical-quantum limits and downstream ownership

The twenty-two **QLIMITX** laws map the exact overlap and separation between the classical and quantum machines. Classical states embed as single-held-branch, phase-insensitive reversible states. Classical probabilistic descriptions correspond to unresolved support classes. Quantum decoding returns an exact classical record through the observation relation. Bidirectional simulation retains description and resource overhead rather than declaring the two modes identical.

Measurement disturbance, no-cloning, communication and correction limits retain their exact source and adversary grammars. Halting, undecidability and incompleteness transfer because the quantum interpreter is still a generated finite description capable of self-reference. An external oracle remains an external named relation. Complete support does not turn the machine into a hypercomputer.

Six handoff claims close ownership. Physics receives formal states, processes and resources but owns preparation, apparatus, units, uncertainty, timing, energy, fidelity and thresholds. Chemistry and Materials consume admitted simulators but own composition, structure, observables and domain falsification. Engineering owns architecture, control, reliability, performance and safety. Open extension requires a new value-free obligation, complete enumeration, controls, reconstruction and post-registry evidence.

12.14 Quantum limits, preserved halt and open questions

Every admitted quantum circuit remains a generated finite description executed by an admitted universal process.

Self-description and held complement therefore transfer the classical halting and incompleteness boundaries. Complete branch support does not supply an undeclared oracle; quantum computation does not escape the registered computability limit.

No-cloning and measurement-disturbance laws likewise retain the distinction and observation conditions under which copying or non-disturbance fails.

The validation history preserves a genuine rejected submission. The first official

SFT-QUANTUM-VALID-REVERSIBLE-001 attempt halted because a sealed prediction label and an already frozen vector label did not match. Its rejected receipt remains

sha256:bc2e0eb42e8650c848de9872a8666fc879d84aec5c21201b378721f00fd8a62e. The halt is not counted as a success. A separately versioned retry corrected only the protocol label binding and passed the unchanged route; the scientific law, target, candidate grammar, observation vector, engine, verifier and acceptance condition did not change.

The branch has no open row in its frozen 288-claim census, but several domains remain expressly open or downstream: physical amplitudes and outcome frequencies, hardware timing and energy, measured gate fidelity and noise, device thresholds, empirical speedup, chemistry and materials behaviour, and application performance. A formal simulator derives consequences of its model; it does not measure nature.

Cumulative dependency spine and handoff to Physics

Across these branches, the same scientific object becomes progressively more capable without changing owner retroactively.

Branch	Receives	Derives and owns	Explicitly does not absorb
Methods and Foundation	No earlier SFT law; the operational presentation boundary	Occurrence, One, count, part, Fold, form, exact operations, trace and admission custody	General mathematics, natural measurements and application behaviour
Mathematics	Foundation carriers and trace laws	Exact generated-finite mathematical structures, relations, proofs, bounds and certified approximation	Physical meaning, domain measurements and downstream semantics
Information Science	Mathematical support, relation, probability and composition	Symbol, record, source, channel, loss, inference, privacy and information custody	Quantum dynamics, biological function, cognition and lived experience
Classical Computation	Information carriers and conservation plus mathematical dynamics	Languages, machines, algorithms, resources, semantics, distributed processes, security, learning and simulation	Quantum operations, hardware and domain performance
Reversible and Quantum Computation	Classical process and reversibility records	Reversible carrier, branch support, phase, interference, entanglement, measurement semantics, gates, circuits, codes and quantum limits	Physical units, apparatus, fidelity, energy, thresholds and natural outcome distributions

The handoff is registered as SFT-QUANTUM-HAND-PHYSICS-MEASUREMENT-002: Physics receives sealed state, process and resource identities and owns preparation, units, uncertainty, apparatus, timing, energy, fidelity and threshold measurements. This boundary prevents two opposite errors. Quantum Computation cannot claim that a formal circuit execution measured a device; Physics cannot use a later measured value to select the earlier formal quantum law.

Physics therefore begins with a rich but non-physical formal inheritance: exact positive quantity relations, generated geometry and dynamics, deterministic-support probability, information and measurement custody, executable processes, reversible and quantum states, and explicit resource interfaces. Its next problem is to derive physical carriers, dimensions, constants, fields, matter and measurement relations without importing the target values those relations will later face. That is the subject of the next part.

Current corrections, adverse records and unresolved boundaries

At this freeze, no registered claim in the five ledgers is omitted from its dated census. That does not mean the history is uniformly favourable or that every scientific question is resolved.

- Foundation preserves the corrected distinction between complement and half-One phase translation.
- Mathematics preserves the failed Collatz shortcut, the bounded Riemann and fluid boundaries, calculator claim 003's reversed enclosure parity, the blank desktop widget and phone usability failures, and their separately admitted corrections.

- Information Science preserves ambiguity and unavailable distinctions as complete predecessor classes rather than forcing a likely decoder.
- Classical Computation preserves the halt-or-recur correction, the exact diagonal basis of undecidability, the replacement of vague efficiency prose by resource ledgers, and the native-only scope of $P \neq NP$.
- Quantum Computation preserves the first halted validation attempt and the separately versioned protocol-only retry.
- Every branch preserves favourable, adverse, absent, unavailable, unresolved and boundary rows in its audit layer even when the conceptual narrative abbreviates them.

The open boundary is extension, not an unregistered hole concealed by the word complete. "Complete" here means complete to the named frozen census and its current evidence surface. A new mathematical object, information problem, machine model, quantum operation, counterexample or stronger empirical test must receive a new versioned registration and cannot rewrite prior evidence.

Reproducibility and authority for this part

The conceptual account above is a synthesis. The complete claim fields remain in the scientific audit volumes and the machine packages. Every claim retains its statement, dependencies, candidate grammar and count, decisions, unique survivor, controls, falsification condition, evidence chronology, source identities, receipt and closure certificate. The principal current branch census identities are:

Branch	Frozen census or reconciliation identity
Mathematics	census sha256:10e2399fbecd99ae9106436660edbc8c0f84fad6b9dde75139099d202d58c36c; reconciliation sha256:526c6d78a3ff77c21c6d4dd71c266499f5b234a7763cb223e5d009fa9f101c64
Information Science	census sha256:cf6d90095ada5fca5ca39579764ffbd79fa65194719260ba6a30261efa288965; reconciliation sha256:48056bb5684a97ca5e9eba6bda72afe536a11dd1b9f970a0d220f82dc1ee3a54
Classical Computation	census sha256:411959c9505e6e09ec57e03923572f249cb5402996dca17911b3e1616e81b6ea; reconciliation sha256:84e436aa27ed392c4c2a5cab77833952a027f43e878151627e958e64be3ee30e
Reversible and Quantum Computation	census sha256:a7f855b7e9fc6ffe4ae9485faf26f1be44bd672b843ecbfbad42b0f523726e3c; reconciliation sha256:ac4acee07a49cbde19eee7baacd7be7d8c87c93e4d75a47467160fb31499e650

The active source files used for this part are bound in the authoritative corpus inventory dated 31 July 2026. Their SHA-256 identities are retained in the release manifest and editorial audit. The current source inventory reports that each active paper was read completely; superseded drafts were excluded as scientific authority. The final-publication receipt is controlling where a frozen pre-deposit manuscript still describes itself as awaiting deposit.

Compact upstream claim-navigation index

This index exposes the complete active Foundation, Mathematics and Information Science claim surface without reproducing the audit template. It preserves claim identity, family, candidate count and formal closure so that a reader can move from the conceptual prerequisites to the controlling claim package. Exact statements, survivor forms, controls, evidence chronology, source identities and receipt hashes remain in the scientific audit volumes and machine archive. A title is a navigation aid, not a substitute for the registered statement.

Foundation: 16 active claims

Root (root)

Claim ID	Claim title	Candidates	Closure
SFT-ROOT-THERE-IS-NO- -NOTHING	Premise-free operational root theorem	2	depth_independent

Check (check)

Claim ID	Claim title	Candidates	Closure
SFT-FOUNDATION-ONE-001	The structural One	6	depth_independent
SFT-FOUNDATION-COUNT-001	Exact positive finite count	10	depth_independent
SFT-FOUNDATION-PART-001	Exact positive part coordinate	192	depth_independent
SFT-FOUNDATION-FOLD-001	The minimal structural Fold	288	depth_independent
SFT-FOUNDATION-PART-EQUIVALENCE-001	Cross-partition exact part equivalence	1,024	depth_independent
SFT-FOUNDATION-FOLD-ASSEMBLY-001	Complete finite Fold-word assembly	192	depth_independent
SFT-FOUNDATION-FORM-GRAMMAR-001	Complete foundational form grammar	288	depth_independent
SFT-FOUNDATION-FORM-ENFORCEMENT-001	Canonical foundational form enforcement	192	depth_independent
SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001	One-way derivation-to-measurement boundary	256	depth_independent
SFT-FOUNDATION-EXACT-OPERATIONS-001	Exact positive foundational operations	128	depth_independent
SFT-FOUNDATION-HALF-ONE-001	The forced half-One ground	128	depth_independent
SFT-FOUNDATION-FOLD-DYNAMICS-001	Exact Fold dynamics and fibre law	128	depth_independent
SFT-FOUNDATION-PRIMITIVE-MAP-UNIQUENESS-001	Mechanically enumerated primitive Fold-map uniqueness	84	depth_independent
SFT-FOUNDATION-DERIVATION-TRACE-001	Complete replayable derivation trace	256	depth_independent
SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001	Single fail-closed SFT admission law	2,048	depth_independent

Mathematics: 323 active claims

Current Status, Evidence Language And Reader Map
(current_status, _evidence_language_and_reader_map)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-EXACT-ARITHMETIC-001	Exact arithmetic and generated number structure	256	depth_independent

Derivation 1: Exact Arithmetic And Generated Number Structure
(derivation_1:_exact_arithmetic_and_generated_number_structure)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-DISCRETE-001	Generated discrete objects, relations and induction	256	depth_independent

Derivation 2: Generated Discrete Objects, Relations And Induction
(derivation_2:_generated_discrete_objects,_relations_and_induction)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-COMBINATORICS-001	Exact finite combinatorial generation	256	depth_independent

Derivation 3: Exact Finite Combinatorial Generation

(derivation_3:_exact_finite_combinatorial_generation)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-GRAPH-NETWORK-001	Exact finite graph and network structure	512	depth_independent

Derivation 4: Exact Finite Graph And Network Structure

(derivation_4:_exact_finite_graph_and_network_structure)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-ALGEBRA-001	Generated finite algebraic structures	512	depth_independent

Derivation 5: Generated Finite Algebraic Structures

(derivation_5:_generated_finite_algebraic_structures)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-ORDER-LATTICE-001	Exact finite order and conditional lattice structure	512	depth_independent

Derivation 6: Exact Finite Order And Conditional Lattice Structure

(derivation_6:_exact_finite_order_and_conditional_lattice_structure)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-GEOMETRY-TOPOLOGY-001	Exact finite geometry and topology for computation	1,024	depth_independent

Derivation 7: Exact Finite Geometry And Topology For Computation

(derivation_7:_exact_finite_geometry_and_topology_for_computation)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-PROBABILITY-STATISTICS-001	Exact finite probability and statistics from Fold observation	512	depth_independent

Derivation 8: Exact Finite Probability And Statistics From Fold Observation

(derivation_8:_exact_finite_probability_and_statistics_from_fold_observation)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-OPTIMIZATION-001	Exact finite optimization and retained optima	512	depth_independent

Derivation 9: Exact Finite Optimisation And Retained Optima

(derivation_9:_exact_finite_optimisation_and_retained_optima)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-DYNAMICAL-SYSTEMS-001	Exact generated finite dynamical systems	1,024	depth_independent

Derivation 10: Exact Generated Finite Dynamical Systems

(derivation_10:_exact_generated_finite_dynamical_systems)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-LOGIC-PROOF-001	Exact finite logic, inference and proof boundaries	1,024	depth_independent

Derivation 11: Exact Finite Logic, Inference And Proof Boundaries

(derivation_11:_exact_finite_logic,_inference_and_proof_boundaries)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-CATEGORY-TYPE-COMPOSITION-001	Forced category, type and compositional structure	1,024	depth_independent

Derivation 12: Forced Category, Type And Compositional Structure

(derivation_12:_forced_category,_type_and_compositional_structure)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-EXACT-RELATIONS-002	Exact ratio, separation, reciprocal, threshold and cycle composition	256	depth_independent

Derivation 13: Exact Ratio, Separation, Reciprocal, Threshold And Cycle Composition (derivation_13:_exact_ratio,_separation,_reciprocal,_threshold_and_cycle_composition)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-ORBIT-NUMBER-THEORY-002	Fold orbit number theory and exact prime-spectrum structure	256	depth_independent

Derivation 14: Fold Orbit Number Theory And Exact Prime-Spectrum Structure

(derivation_14:_fold_orbit_number_theory_and_exact_prime-spectrum_structure)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-LIMIT-CONTINUUM-002	Potential infinity, exact refinement and the continuum boundary	256	depth_independent

Derivation 15: Potential Infinity, Exact Refinement And The Continuum Boundary

(derivation_15:_potential_infinity,_exact_refinement_and_the_continuum_boundary)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-ALGEBRAIC-BALANCE-002	Positive polynomial-balance certificates for algebraic magnitudes	256	depth_independent

Derivation 16: Positive Polynomial-Balance Certificates For Algebraic Magnitudes (derivation_16:_positive_polynomial-balance_certificates_for_algebraic_magnitudes)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-BOUNDED-N-BODY-002	Exact recurrence of every finite componentwise Fold configuration	256	depth_independent

Derivation 17: Exact Recurrence Of Every Finite Componentwise Fold Configuration (derivation_17:_exact_recurrence_of_every_finite_componentwise_fold_configuration)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-FLOORED-FLUID-REGULARITY-002	Depth-independent discrete-gradient bound for floored Fold fluids	256	depth_independent

Derivation 18: Depth-Independent Discrete-Gradient Bound For Floored Fold Fluids (derivation_18:_depth-independent_discrete-gradient_bound_for_floored_fold_fluids)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-PRIME-PAIR-CENSUS-002	Exact bounded Goldbach and twin-prime census	256	depth_independent

Derivation 19: Exact Bounded Goldbach And Twin-Prime Census

(derivation_19:_exact_bounded_goldbach_and_twin-prime_census)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-RIEMANN-MIRROR-002	Unique half-One reflection axis and Riemann correspondence boundary	256	depth_independent

Derivation 20: Unique Half-One Reflection Axis And Riemann Correspondence Boundary (derivation_20:_unique_half-one_reflection_axis_and_riemann_correspondence_boundary)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-COLLATZ-FINITE-CENSUS-002	Exact bounded Collatz census and contraction-control correction	256	depth_independent

Derivation 21: Exact Bounded Collatz Census And Contraction-Control Correction (derivation_21:_exact_bounded_collatz_census_and_contraction-control_correction)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-SELF-SIMILAR-CONVERGENCE-002	Exact self-similarity, chaotic rate and convergent-series laws	256	depth_independent

Preserved Version 1.2 Extension: The Smithian Fold Scientific Calculator (preserved_version_1.2_extension:_the_smithian_fold_scientific_calculator)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-SCIENTIFIC-CALCULATOR-003	Exact traced SFT-native scientific calculator	1,024	depth_independent
SFT-MATH-SCIENTIFIC-CALCULATOR-004	Exact traced SFT-native scientific calculator with corrected enclosure parity	1,024	depth_independent
SFT-MATH-SCIENTIFIC-CALCULATOR-005	Complete exact SFT scientific calculator and cross-platform learning app	8,192	depth_independent
SFT-MATH-SCIENTIFIC-CALCULATOR-006	Fully realised SFT scientific calculator and complete Mathematics law explorer	1,024	depth_independent
SFT-MATH-SCIENTIFIC-CALCULATOR-007	Accessible SFT-native scientific calculator application	256	depth_independent

Arith (arith)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-ARITH-GENERATED-SUCCESSION-001	Generated whole succession and exact induction	256	depth_independent
SFT-MATH-ARITH-JUNCTION-ADDITION-002	Addition and disjoint-junction arithmetic	256	depth_independent
SFT-MATH-ARITH-PAIR-CELL-MULTIPLICATION-003	Multiplication and complete pair-cell arithmetic	256	depth_independent
SFT-MATH-ARITH-DIVISIBILITY-GCD-LCM-004	Divisibility, common divisors and common multiples	256	depth_independent
SFT-MATH-ARITH-QUOTIENT-REMAINDER-005	Exact quotient and oriented remainder	256	depth_independent
SFT-MATH-ARITH-PRIME-IRREDUCIBLE-006	Prime and irreducible whole structure	256	depth_independent
SFT-MATH-ARITH-UNIQUE-FACTORIZATION-007	Unique finite factorization certificate	256	depth_independent
SFT-MATH-ARITH-CANONICAL-FRACTION-008	Canonical exact fractions and common refinement	256	depth_independent
SFT-MATH-ARITH-CONTINUED-FRACTION-009	Finite continued-fraction correspondence	256	depth_independent
SFT-MATH-ARITH-CONGRUENCE-010	Residue classes and congruence	256	depth_independent
SFT-MATH-ARITH-COMPATIBLE-CONGRUENCE-011	Compatible congruence composition	256	depth_independent
SFT-MATH-ARITH-VALUATION-012	Prime-power valuation and divisibility depth	256	depth_independent
SFT-MATH-ARITH-DIOPHANTINE-ENUMERATION-013	Diophantine relation enumeration	256	depth_independent
SFT-MATH-ARITH-RECURRENCE-SEQUENCE-014	Recurrence laws and exact sequences	256	depth_independent
SFT-MATH-ARITH-GENERATING-FUNCTION-015	Finite generating-function correspondence	256	depth_independent
SFT-MATH-ARITH-PARTITION-COMPOSITION-016	Whole partitions and ordered compositions	256	depth_independent
SFT-MATH-ARITH-ARITHMETIC-FUNCTIONS-017	Arithmetic functions and divisor ledgers	256	depth_independent
SFT-MATH-ARITH-PRIME-DISTRIBUTION-ENCLOSURE-018	Finite prime-distribution and growth enclosures	256	depth_independent

Alext (alex)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-ALEX-T-POLYNOMIAL-ROOT-ISOLATION-001	Polynomial identity and exact root isolation	256	depth_independent
SFT-MATH-ALEX-T-ALGEBRAIC-BALANCE-002	Algebraic-magnitude balance certificates	256	depth_independent
SFT-MATH-ALEX-T-EXTENSION-TOWER-003	Exact finite extension towers	256	depth_independent
SFT-MATH-ALEX-T-FINITE-FIELD-004	Finite-field correspondence	256	depth_independent
SFT-MATH-ALEX-T-GALOIS-ORBIT-005	Finite Galois-orbit correspondence	256	depth_independent
SFT-MATH-ALEX-T-CYCLOTOMIC-CORRESPONDENCE-006	Cyclotomic and root-of-unity correspondence	256	depth_independent
SFT-MATH-ALEX-T-HELD-PAIR-COMPLEX-007	Held-pair complex-number correspondence	256	depth_independent
SFT-MATH-ALEX-T-REAL-ALGEBRAIC-ORDER-008	Exact real-algebraic ordering	256	depth_independent
SFT-MATH-ALEX-T-PRIME-ADIC-VALUATION-009	Prime-adic valuation correspondence	256	depth_independent
SFT-MATH-ALEX-T-TRANSCENDENTAL-BOUNDARY-010	Transcendental and nonrepresentability boundary	256	depth_independent

Comb (comb)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-COMB-COUNTING-LAWS-001	Product, sum and bijection counting laws	256	depth_independent
SFT-MATH-COMB-PERMUTATION-COMBINATION-002	Permutation and combination enumeration	256	depth_independent
SFT-MATH-COMB-INCLUSION-EXCLUSION-003	Inclusion-exclusion with complete overlap custody	256	depth_independent
SFT-MATH-COMB-PIGEONHOLE-OCCUPANCY-004	Pigeonhole and occupancy forcing	256	depth_independent
SFT-MATH-COMB-RECURRENCE-GENERATING-005	Recurrence and generating-function counting	256	depth_independent
SFT-MATH-COMB-PARTITION-INCIDENCE-006	Integer partitions and Young-type incidence	256	depth_independent
SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007	Extremal finite-set systems	256	depth_independent
SFT-MATH-COMB-PROBABILISTIC-METHOD-CORRESPONDENCE-008	Probabilistic-method correspondence without ontic randomness	256	depth_independent
SFT-MATH-COMB-DESIGN-INCIDENCE-009	Design, block and incidence structures	256	depth_independent
SFT-MATH-COMB-CODING-PACKING-COMBINATORICS-010	Coding and packing combinatorics	256	depth_independent
SFT-MATH-COMB-RAMSEY-FORCING-011	Ramsey-type finite forcing boundaries	256	depth_independent
SFT-MATH-COMB-SPECIES-COMPOSITION-012	Species and compositional enumeration	256	depth_independent

Graph (graph)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-GRAPH-IDENTITY-ISOMORPHISM-001	Graph identity, adjacency and isomorphism	256	depth_independent
SFT-MATH-GRAPH-PATH-REACHABILITY-CYCLE-002	Paths, walks, reachability and cycles	256	depth_independent
SFT-MATH-GRAPH-CONNECTIVITY-CUT-FLOW-003	Connectivity, cuts and flow support	256	depth_independent
SFT-MATH-GRAPH-TREE-FOREST-SPANNING-004	Trees, forests and spanning structure	256	depth_independent
SFT-MATH-GRAPH-PLANARITY-EMBEDDING-005	Planarity and embedding correspondence	256	depth_independent
SFT-MATH-GRAPH-COLOURING-CONSTRAINT-006	Colouring and constraint partitions	256	depth_independent
SFT-MATH-GRAPH-MATCHING-COVERING-PACKING-007	Matching, covering and packing	256	depth_independent
SFT-MATH-GRAPH-DIRECTED-CAUSAL-REACHABILITY-008	Directed graphs and causal reachability	256	depth_independent
SFT-MATH-GRAPH-WEIGHTED-NETWORK-EXACT-PARTS-009	Weighted network correspondence with exact parts	256	depth_independent
SFT-MATH-GRAPH-HYPERGRAPH-HIGHER-INCIDENCE-010	Hypergraphs and higher incidence	256	depth_independent
SFT-MATH-GRAPH-MATROID-INDEPENDENCE-011	Matroids and independence systems	256	depth_independent
SFT-MATH-GRAPH-RELIABILITY-FAILURE-CUSTODY-012	Network reliability and failure custody	256	depth_independent
SFT-MATH-GRAPH-SPECTRAL-CORRESPONDENCE-013	Spectral graph correspondence	256	depth_independent
SFT-MATH-GRAPH-DYNAMIC-TEMPORAL-NETWORK-014	Dynamic and temporal networks	256	depth_independent

Linear (linear)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001	Vectors as exact generated coordinate carriers	256	depth_independent
SFT-MATH-LINEAR-MAP-COMPOSITION-002	Linear maps and composition	256	depth_independent
SFT-MATH-LINEAR-MATRIX-ROW-OPERATIONS-003	Matrix representation and exact row operations	256	depth_independent
SFT-MATH-LINEAR-RANK-NULLITY-004	Rank, nullity and retained distinctions	256	depth_independent
SFT-MATH-LINEAR-DETERMINANT-ORIENTATION-005	Determinants and orientation custody	256	depth_independent
SFT-MATH-LINEAR-EXACT-SYSTEMS-006	Exact systems of linear relations	256	depth_independent
SFT-MATH-LINEAR-BASIS-DIMENSION-007	Basis, dimension and change of basis	256	depth_independent
SFT-MATH-LINEAR-INNER-PRODUCT-METRIC-008	Inner-product and metric correspondence	256	depth_independent
SFT-MATH-LINEAR-EIGENVALUE-INVARIANT-SUPPORT-009	Eigenvalue and invariant-support correspondence	256	depth_independent
SFT-MATH-LINEAR-RATIONAL-SPECTRAL-ENCLOSURE-010	Exact rational spectral enclosures	256	depth_independent
SFT-MATH-LINEAR-MULTILINEAR-TENSOR-PRODUCT-011	Multilinear maps and tensor products	256	depth_independent
SFT-MATH-LINEAR-TENSOR-CONTRACTION-012	Tensor contraction and index custody	256	depth_independent
SFT-MATH-LINEAR-EXTERIOR-SYMMETRIC-013	Exterior and symmetric composition	256	depth_independent
SFT-MATH-LINEAR-OPERATOR-DECOMPOSITION-014	Finite-dimensional operator decomposition	256	depth_independent

Algorithms (algorithms)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-ALG-MAGMA-CLOSED-OPERATION-001	Magma and closed operation structure	256	depth_independent
SFT-MATH-ALG-SEMIGROUP-ASSOCIATIVITY-002	Semigroup associativity witnesses	256	depth_independent
SFT-MATH-ALG-MONOID-IDENTITY-003	Monoid identity witnesses	256	depth_independent
SFT-MATH-ALG-GROUP-HELD-INVERSE-004	Group inverse as held reversal	256	depth_independent
SFT-MATH-ALG-PERMUTATION-GROUP-ACTION-005	Permutation-group action	256	depth_independent
SFT-MATH-ALG-QUOTIENT-NORMAL-SUBSTRUCTURE-006	Quotient and normal-substructure correspondence	256	depth_independent
SFT-MATH-ALG-RING-DISTRIBUTIVE-007	Ring distributive structure	256	depth_independent
SFT-MATH-ALG-INTEGRAL-DOMAIN-008	Integral-domain boundary	256	depth_independent
SFT-MATH-ALG-FIELD-EXACT-DIVISION-009	Field correspondence under exact division	256	depth_independent
SFT-MATH-ALG-MODULE-SCALAR-ACTION-010	Modules and scalar-action structure	256	depth_independent
SFT-MATH-ALG-COMPATIBLE-ALGEBRA-PRODUCT-011	Algebras and compatible products	256	depth_independent
SFT-MATH-ALG-IDEAL-QUOTIENT-012	Ideals and quotient structures	256	depth_independent
SFT-MATH-ALG-REPRESENTATION-ACTION-DECOMPOSITION-013	Representation and action decomposition	256	depth_independent
SFT-MATH-ALG-EXACT-SEQUENCE-HOMOLOGICAL-014	Exact sequence and homological correspondence	256	depth_independent
SFT-MATH-ALG-UNIVERSAL-IDENTITIES-015	Universal algebra and identities	256	depth_independent
SFT-MATH-ALG-OPERADIC-COMPOSITION-016	Operadic algebraic composition interface	256	depth_independent

Order (order)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-ORDER-PREORDER-QUOTIENT-001	Preorder and distinguishability quotient	256	depth_independent
SFT-MATH-ORDER-PARTIAL-ANTISYMMETRY-002	Partial order and antisymmetry	256	depth_independent
SFT-MATH-ORDER-CONDITIONAL-TOTALITY-003	Conditional total order	256	depth_independent
SFT-MATH-ORDER-MEET-JOIN-LATTICE-004	Meet, join and lattice structure	256	depth_independent
SFT-MATH-ORDER-DISTRIBUTIVE-MODULAR-005	Distributive and modular lattice correspondence	256	depth_independent
SFT-MATH-ORDER-BOOLEAN-COMPLEMENT-006	Boolean-like complement correspondence	256	depth_independent
SFT-MATH-ORDER-CLOSURE-SYSTEM-007	Closure operators and closure systems	256	depth_independent
SFT-MATH-ORDER-GALOIS-CONNECTION-008	Galois connections between orders	256	depth_independent
SFT-MATH-ORDER-DOMAIN-APPROXIMATION-009	Domain approximation correspondence	256	depth_independent
SFT-MATH-ORDER-MONOTONE-MAP-010	Monotone maps and order preservation	256	depth_independent
SFT-MATH-ORDER-FIXED-POINT-011	Fixed-point existence on finite orders	256	depth_independent
SFT-MATH-ORDER-COMPLETE-LATTICE-BOUNDARY-012	Complete-lattice correspondence boundary	256	depth_independent

Geom (geom)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-GEOM-POINT-INCIDENCE-COORDINATE-001	Point, incidence and exact coordinate identity	256	depth_independent
SFT-MATH-GEOM-EUCLIDEAN-DISTANCE-002	Finite Euclidean-distance correspondence	256	depth_independent
SFT-MATH-GEOM-AFFINE-INVARIANCE-003	Affine combinations and affine invariance	256	depth_independent
SFT-MATH-GEOM-PROJECTIVE-PERSPECTIVE-004	Projective incidence and perspective	256	depth_independent
SFT-MATH-GEOM-CONVEX-HULL-SEPARATION-005	Convex hulls and separation	256	depth_independent
SFT-MATH-GEOM-DISCRETE-LATTICE-POLYTOPE-006	Discrete geometry and lattice polytopes	256	depth_independent
SFT-MATH-GEOM-POLYHEDRAL-EULER-INCIDENCE-007	Polyhedral faces and Euler-type incidence	256	depth_independent
SFT-MATH-GEOM-COMPUTATIONAL-PREDICATES-008	Computational geometry predicates	256	depth_independent
SFT-MATH-GEOM-ORIENTATION-INTERSECTION-009	Exact orientation and intersection tests	256	depth_independent
SFT-MATH-GEOM-ALGEBRAIC-SOLUTION-SET-010	Algebraic-geometry solution-set correspondence	256	depth_independent
SFT-MATH-GEOM-DIFFERENTIAL-FINITE-CHART-011	Differential-geometry finite-chart correspondence	256	depth_independent
SFT-MATH-GEOM-CURVATURE-FINITE-TRANSPORT-012	Curvature by exact finite transport	256	depth_independent
SFT-MATH-GEOM-METRIC-GEODESIC-013	Metric geometry and geodesic correspondence	256	depth_independent
SFT-MATH-GEOM-FRACTAL-SELF-SIMILAR-014	Fractal and self-similar geometry	256	depth_independent
SFT-MATH-GEOM-PACKING-COVERING-TESSELLATION-015	Packing, covering and tessellation	256	depth_independent
SFT-MATH-GEOM-TRANSFORMATION-GROUPS-016	Geometric transformation groups	256	depth_independent

Topo (topo)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-TOPO-OPEN-SET-SUPPORT-001	Open-set correspondence on generated supports	256	depth_independent
SFT-MATH-TOPO-CONTINUITY-TRANSPORT-002	Continuity as distinction-preserving transport	256	depth_independent
SFT-MATH-TOPO-COMPACT-FINITE-SUBCOVER-003	Compactness finite-subcover correspondence	256	depth_independent
SFT-MATH-TOPO-CONNECTED-COMPONENT-004	Connectedness and component structure	256	depth_independent
SFT-MATH-TOPO-SEPARATION-FINITE-005	Separation properties on finite observations	256	depth_independent
SFT-MATH-TOPO-PRODUCT-QUOTIENT-SUBSPACE-006	Product, quotient and subspace topology correspondence	256	depth_independent
SFT-MATH-TOPO-SIMPLICIAL-INCIDENCE-007	Simplicial complexes and incidence	256	depth_independent
SFT-MATH-TOPO-HOMOTOPY-PATH-DEFORMATION-008	Homotopy path-deformation correspondence	256	depth_independent
SFT-MATH-TOPO-FUNDAMENTAL-CYCLE-GROUP-009	Fundamental-cycle and group correspondence	256	depth_independent
SFT-MATH-TOPO-HOMOLOGY-BOUNDARY-010	Homology and boundary composition	256	depth_independent
SFT-MATH-TOPO-COHOMOLOGY-DUAL-OBSERVATION-011	Cohomology and dual-observation correspondence	256	depth_independent
SFT-MATH-TOPO-MANIFOLD-FINITE-ATLAS-012	Manifold finite-atlas correspondence	256	depth_independent
SFT-MATH-TOPO-KNOT-LINK-INVARIANTS-013	Knot and link finite-diagram invariants	256	depth_independent
SFT-MATH-TOPO-PERSISTENT-FEATURE-CUSTODY-014	Computational topology and persistent-feature custody	256	depth_independent

Calc (calc)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-CALC-FINITE-DIFFERENCE-001	Exact finite difference and local change	256	depth_independent
SFT-MATH-CALC-HIGHER-DIFFERENCE-DEGREE-002	Higher finite differences and polynomial degree	256	depth_independent
SFT-MATH-CALC-ACCUMULATION-SUMS-003	Accumulation and exact finite sums	256	depth_independent
SFT-MATH-CALC-DIFFERENCE-ACCUMULATION-004	Fundamental difference-accumulation correspondence	256	depth_independent
SFT-MATH-CALC-PRODUCT-COMPOSITION-LAWS-005	Product and composition difference laws	256	depth_independent
SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006	Rational enclosure convergence	256	depth_independent
SFT-MATH-CALC-DERIVATIVE-SHRINKING-PARTS-007	Derivative correspondence through shrinking exact parts	256	depth_independent
SFT-MATH-CALC-INTEGRAL-REFINEMENT-SUMS-008	Integral correspondence through refinement sums	256	depth_independent
SFT-MATH-CALC-MULTIVARIABLE-DIRECTIONAL-009	Multivariable difference and directional change	256	depth_independent
SFT-MATH-CALC-DIVERGENCE-FLUX-010	Discrete divergence and flux correspondence	256	depth_independent
SFT-MATH-CALC-VARIATIONAL-STATIONARY-011	Variational difference and stationary structure	256	depth_independent
SFT-MATH-CALC-CONTINUUM-LIMIT-BOUNDARY-012	Continuum-limit admissibility boundary	256	depth_independent

Anal (anal)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-ANAL-SEQUENCE-CONVERGENCE-001	Exact sequence convergence certificates	256	depth_independent
SFT-MATH-ANAL-CAUCHY-SUPPORT-002	Cauchy-type generated support correspondence	256	depth_independent
SFT-MATH-ANAL-COMPLETENESS-CORRESPONDENCE-003	Completeness correspondence without completed continuum	256	depth_independent
SFT-MATH-ANAL-SERIES-REMAINDER-004	Series convergence and remainder enclosures	256	depth_independent
SFT-MATH-ANAL-POWER-SERIES-TRUNCATION-005	Power-series finite truncation custody	256	depth_independent
SFT-MATH-ANAL-FUNCTIONAL-SPACE-REPRESENTATION-006	Functional-space finite-representation correspondence	256	depth_independent
SFT-MATH-ANAL-NORM-SEMINORM-METRIC-007	Norm, seminorm and metric correspondence	256	depth_independent
SFT-MATH-ANAL-BOUNDED-COMPACT-OPERATOR-008	Bounded and compact operator correspondence	256	depth_independent
SFT-MATH-ANAL-HARMONIC-FOURIER-SUPPORT-009	Harmonic and Fourier finite-support correspondence	256	depth_independent
SFT-MATH-ANAL-TRANSFORM-INVERSION-010	Transform inversion on generated supports	256	depth_independent
SFT-MATH-ANAL-CONVOLUTION-CORRELATION-011	Convolution and correlation identities	256	depth_independent
SFT-MATH-ANAL-ORTHOGONAL-BASIS-EXPANSION-012	Orthogonality and basis expansion correspondence	256	depth_independent
SFT-MATH-ANAL-DISTRIBUTIONAL-WEAK-OBSERVATION-013	Distributional and weak-observation correspondence	256	depth_independent
SFT-MATH-ANAL-NONLINEAR-CONTRACTION-014	Nonlinear analysis and contraction boundaries	256	depth_independent
SFT-MATH-ANAL-COMPLEX-HELD-PAIR-015	Complex-analysis held-pair correspondence	256	depth_independent
SFT-MATH-ANAL-SPECTRAL-MEASURE-016	Operator spectral-measure correspondence	256	depth_independent

Eqn (eqn)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-EQN-ORDINARY-DIFFERENCE-001	Ordinary difference-equation structure	256	depth_independent
SFT-MATH-EQN-ORDINARY-DIFFERENTIAL-002	Ordinary differential-equation correspondence	256	depth_independent
SFT-MATH-EQN-PARTIAL-DIFFERENCE-003	Partial difference-equation structure	256	depth_independent
SFT-MATH-EQN-PARTIAL-DIFFERENTIAL-004	Partial differential-equation correspondence	256	depth_independent
SFT-MATH-EQN-BOUNDARY-INITIAL-WELL-POSED-005	Boundary and initial record well-posedness	256	depth_independent
SFT-MATH-EQN-INTEGRAL-CORRESPONDENCE-006	Integral-equation correspondence	256	depth_independent
SFT-MATH-EQN-FUNCTIONAL-STRUCTURE-007	Functional-equation structure	256	depth_independent
SFT-MATH-EQN-RECURRENCE-SOLUTION-SPACE-008	Recurrence-equation solution spaces	256	depth_independent
SFT-MATH-EQN-GREEN-RESPONSE-009	Green-response finite correspondence	256	depth_independent
SFT-MATH-EQN-CONSERVATION-WEAK-010	Conservation-law weak correspondence	256	depth_independent
SFT-MATH-EQN-STABILITY-PERTURBATION-011	Stability and perturbation enclosures	256	depth_independent
SFT-MATH-EQN-EXISTENCE-UNIQUENESS-BLOWUP-012	Existence, uniqueness and blow-up boundaries	256	depth_independent

Measurement and Metrology (measurement_and_metrology)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-MEAS-FINITE-SUPPORT-WEIGHT-001	Finite measure and exact support weight	256	depth_independent
SFT-MATH-MEAS-DISJOINT-ADDITIVITY-002	Additivity on disjoint generated support	256	depth_independent
SFT-MATH-MEAS-OUTER-COVERING-003	Outer-measure and covering correspondence	256	depth_independent
SFT-MATH-MEAS-MEASURABLE-BOUNDARY-004	Measurable-boundary correspondence	256	depth_independent
SFT-MATH-MEAS-FINITE-SUPPORT-INTEGRATION-005	Exact integration over finite support	256	depth_independent
SFT-MATH-MEAS-REFINEMENT-SUM-INTEGRATION-006	Refinement-sum integration correspondence	256	depth_independent
SFT-MATH-MEAS-PRODUCT-CONDITIONAL-SUPPORT-007	Product measure and conditional support	256	depth_independent
SFT-MATH-MEAS-HELD-ORIENTATION-SIGNED-008	Signed-measure replacement by held orientation	256	depth_independent
SFT-MATH-MEAS-DISTRIBUTION-OBSERVATION-009	Distribution and generalized-observation correspondence	256	depth_independent
SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010	Convergence-of-measures finite witness boundary	256	depth_independent

Prob (prob)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-PROB-SUPPORT-CORRESPONDENCE-001	Exact support-based probability correspondence	256	depth_independent
SFT-MATH-PROB-CONDITIONAL-BAYES-002	Conditional support and Bayes correspondence	256	depth_independent
SFT-MATH-PROB-INDEPENDENCE-FACTORIZATION-003	Independence and factorization witnesses	256	depth_independent
SFT-MATH-PROB-EXPECTATION-004	Expectation as exact support accumulation	256	depth_independent
SFT-MATH-PROB-VARIANCE-DISPERSION-005	Variance and dispersion without negative magnitudes	256	depth_independent
SFT-MATH-PROB-FINITE-DISTRIBUTION-006	Finite distribution families	256	depth_independent
SFT-MATH-PROB-LARGE-COUNT-007	Law-of-large-count correspondence under deterministic enumeration	256	depth_independent
SFT-MATH-PROB-CENTRAL-LIMIT-ENCLOSURE-008	Central-limit enclosure correspondence	256	depth_independent
SFT-MATH-PROB-ESTIMATION-SUFFICIENT-RECORD-009	Estimation and sufficient-record structure	256	depth_independent
SFT-MATH-PROB-CONFIDENCE-CREDIBLE-REGION-010	Confidence and credible-region correspondence	256	depth_independent
SFT-MATH-PROB-HYPOTHESIS-ERROR-CUSTODY-011	Hypothesis testing and error custody	256	depth_independent
SFT-MATH-PROB-LIKELIHOOD-EVIDENCE-RATIO-012	Likelihood and evidence ratios	256	depth_independent
SFT-MATH-PROB-BAYESIAN-UPDATE-013	Bayesian update as exact conditional support	256	depth_independent
SFT-MATH-PROB-FINITE-STOCHASTIC-PROCESS-014	Finite stochastic-process correspondence	256	depth_independent
SFT-MATH-PROB-CONDITIONAL-CONSERVATION-015	Martingale-like conditional conservation	256	depth_independent
SFT-MATH-PROB-IDENTIFIABILITY-016	Statistical identifiability and nonidentifiability	256	depth_independent

Opt (opt)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-OPT-FEASIBLE-OBJECTIVE-ORDER-001	Feasible set and objective-order structure	256	depth_independent
SFT-MATH-OPT-FINITE-EXTREMA-002	Exact minimum and maximum on finite support	256	depth_independent
SFT-MATH-OPT-PARETO-DOMINANCE-003	Pareto dominance and multiobjective order	256	depth_independent
SFT-MATH-OPT-LINEAR-PROGRAM-004	Linear-program correspondence	256	depth_independent
SFT-MATH-OPT-INTEGGER-COMBINATORIAL-005	Integer and combinatorial optimization	256	depth_independent
SFT-MATH-OPT-CONVEX-CORRESPONDENCE-006	Convex optimization correspondence	256	depth_independent
SFT-MATH-OPT-DUAL-CERTIFICATE-007	Duality and certificate structure	256	depth_independent
SFT-MATH-OPT-VARIATIONAL-CORRESPONDENCE-008	Variational optimization correspondence	256	depth_independent
SFT-MATH-OPT-DYNAMIC-PROGRAMMING-009	Dynamic programming as compositional optimization	256	depth_independent
SFT-MATH-OPT-GENERATED-STATE-CONTROL-010	Optimal control on generated state paths	256	depth_independent
SFT-MATH-OPT-GAME-EQUILIBRIUM-011	Game equilibrium finite correspondence	256	depth_independent
SFT-MATH-OPT-DECISION-LOSS-012	Decision theory and loss ordering	256	depth_independent
SFT-MATH-OPT-FLOW-SCHEDULING-013	Operations-research flow and scheduling	256	depth_independent
SFT-MATH-OPT-ROBUST-BOUNDED-UNCERTAINTY-014	Robust optimization under bounded uncertainty	256	depth_independent
SFT-MATH-OPT-APPROXIMATION-GAP-015	Approximation guarantees and gaps	256	depth_independent
SFT-MATH-OPT-INFEASIBLE-UNBOUNDED-BOUNDARY-016	Infeasibility and unboundedness boundaries	256	depth_independent

Dyn (dyn)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-DYN-STATE-ORBIT-001	State maps and exact orbit structure	256	depth_independent
SFT-MATH-DYN-FIXED-PERIODIC-002	Fixed points and periodic cycles	256	depth_independent
SFT-MATH-DYN-RECURRENCE-RETURN-003	Recurrence and return-time structure	256	depth_independent
SFT-MATH-DYN-INVARIANT-CONSERVED-004	Invariant sets and conserved records	256	depth_independent
SFT-MATH-DYN-STABILITY-ATTRACTION-005	Stability and attraction correspondence	256	depth_independent
SFT-MATH-DYN-BIFURCATION-DISTINCTION-006	Bifurcation as finite distinction change	256	depth_independent
SFT-MATH-DYN-SYMBOLIC-SHIFT-007	Symbolic dynamics and shift correspondence	256	depth_independent
SFT-MATH-DYN-EXACT-SENSITIVITY-008	Chaos through exact sensitivity witnesses	256	depth_independent
SFT-MATH-DYN-ERGODIC-AVERAGE-009	Ergodic-average finite correspondence	256	depth_independent
SFT-MATH-DYN-HAMILTONIAN-REVERSIBLE-010	Hamiltonian and reversible-map correspondence	256	depth_independent
SFT-MATH-DYN-DISSIPATIVE-RETAINED-LOSS-011	Dissipative dynamics and retained loss	256	depth_independent
SFT-MATH-DYN-COUPLED-NETWORKED-012	Coupled and networked dynamical systems	256	depth_independent

Logic (logic)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-LOGIC-PROPOSITION-DISTINCTION-001	Propositions as generated distinctions	256	depth_independent
SFT-MATH-LOGIC-INFERENCE-CONSEQUENCE-002	Inference and consequence preservation	256	depth_independent
SFT-MATH-LOGIC-SOUND-COMPLETE-CORRESPONDENCE-003	Soundness and completeness correspondence	256	depth_independent
SFT-MATH-LOGIC-PROOF-OBJECT-CHECK-004	Formal proof objects and checking	256	depth_independent
SFT-MATH-LOGIC-QUANTIFIER-FINITE-SUPPORT-005	First-order quantifier finite-support correspondence	256	depth_independent
SFT-MATH-LOGIC-MODEL-INTERPRETATION-006	Model and interpretation structure	256	depth_independent
SFT-MATH-LOGIC-COMPACTNESS-BOUNDARY-007	Compactness correspondence boundary	256	depth_independent
SFT-MATH-LOGIC-DECIDABILITY-INTERFACE-008	Decidability and computability interface	256	depth_independent
SFT-MATH-LOGIC-INCOMPLETENESS-SELF-REFERENCE-009	Incompleteness and self-reference boundary	256	depth_independent
SFT-MATH-LOGIC-FINITE-COLLECTION-010	Set-like finite collection theory	256	depth_independent
SFT-MATH-LOGIC-SIZE-BOUNDARY-011	Class, universe and size boundaries	256	depth_independent
SFT-MATH-LOGIC-CONSTRUCTIVE-CORRESPONDENCE-012	Constructive and intuitionistic correspondence	256	depth_independent
SFT-MATH-LOGIC-MODAL-TEMPORAL-013	Modal and temporal logic correspondence	256	depth_independent
SFT-MATH-LOGIC-MANY-VALUED-014	Nonclassical many-valued correspondence	256	depth_independent
SFT-MATH-LOGIC-NORMALIZATION-015	Proof-theoretic normalization	256	depth_independent
SFT-MATH-LOGIC-CONSISTENCY-SELF-VERIFICATION-016	Foundational consistency and self-verification limits	256	depth_independent

Cat (cat)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-CAT-OBJECT-ARROW-COMPOSITION-001	Objects, arrows and typed composition	256	depth_independent
SFT-MATH-CAT-IDENTITY-ASSOCIATIVITY-002	Identity and associativity witnesses	256	depth_independent
SFT-MATH-CAT-FUNCTOR-PRESERVATION-003	Functorial structure preservation	256	depth_independent
SFT-MATH-CAT-NATURAL-TRANSFORMATION-004	Natural transformation correspondence	256	depth_independent
SFT-MATH-CAT-PRODUCT-COPRODUCT-005	Products, coproducts and universal constructions	256	depth_independent
SFT-MATH-CAT-LIMIT-COLIMIT-006	Limits and colimits on generated diagrams	256	depth_independent
SFT-MATH-CAT-ADJUNCTION-007	Adjunction and paired universal maps	256	depth_independent
SFT-MATH-CAT-MONOIDAL-TENSOR-008	Monoidal composition and tensor interface	256	depth_independent
SFT-MATH-CAT-CLOSED-INTERNAL-MAP-009	Closed structure and internal maps	256	depth_independent
SFT-MATH-CAT-TYPE-DEPENDENT-RECORD-010	Type correspondence and dependent records	256	depth_independent
SFT-MATH-CAT-SHEAF-LOCAL-GLOBAL-011	Sheaf-like local-to-global custody	256	depth_independent
SFT-MATH-CAT-OPERAD-HIGHER-BOUNDARY-012	Operadic and higher-composition boundary	256	depth_independent

Num (num)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-NUM-EXACT-REPRESENTATION-ROUNDING-001	Exact numerical representation and rounding custody	256	depth_independent
SFT-MATH-NUM-INTERVAL-RATIONAL-ENCLOSURE-002	Interval and rational enclosure arithmetic	256	depth_independent
SFT-MATH-NUM-TRUNCATION-DISCRETIZATION-ERROR-003	Truncation and discretization error	256	depth_independent
SFT-MATH-NUM-FORWARD-BACKWARD-STABILITY-004	Forward and backward stability	256	depth_independent
SFT-MATH-NUM-CONDITIONING-SENSITIVITY-005	Conditioning and sensitivity	256	depth_independent
SFT-MATH-NUM-CONVERGENCE-ORDER-BOUND-006	Convergence order with exact bounds	256	depth_independent
SFT-MATH-NUM-ROOT-ISOLATION-EQUATION-SOLVING-007	Root isolation and equation solving	256	depth_independent
SFT-MATH-NUM-LINEAR-SYSTEM-SOLVERS-008	Linear-system exact and approximate solvers	256	depth_independent
SFT-MATH-NUM-INTERPOLATION-APPROXIMATION-009	Interpolation and approximation	256	depth_independent
SFT-MATH-NUM-QUADRATURE-ACCUMULATION-010	Quadrature and accumulation enclosures	256	depth_independent
SFT-MATH-NUM-DIFFERENTIAL-EQUATION-CORRESPONDENCE-011	Differential-equation numerical correspondence	256	depth_independent
SFT-MATH-NUM-VERIFIED-COMPUTATION-CERTIFICATE-012	Verified computation and certificate extraction	256	depth_independent

Symb (symb)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-SYMB-CANONICAL-EXPRESSION-001	Symbolic expression identity and canonical form	256	depth_independent
SFT-MATH-SYMB-SIMPLIFICATION-PROVENANCE-002	Exact simplification with provenance	256	depth_independent
SFT-MATH-SYMB-POLYNOMIAL-FACTOR-EXPAND-003	Polynomial factorization and expansion	256	depth_independent
SFT-MATH-SYMB-EQUATION-SOLVING-004	Symbolic equation solving	256	depth_independent
SFT-MATH-SYMB-REWRITE-TERMINATION-CONFLUENCE-005	Rewrite termination and confluence interface	256	depth_independent
SFT-MATH-SYMB-GENERATING-FUNCTION-TRANSFORM-006	Generating-function transforms	256	depth_independent
SFT-MATH-SYMB-FOURIER-LAPLACE-CORRESPONDENCE-007	Fourier and Laplace correspondence transforms	256	depth_independent
SFT-MATH-SYMB-SPECIAL-FUNCTION-RECURRENCE-008	Special-function recurrence representation	256	depth_independent
SFT-MATH-SYMB-THEOREM-SEARCH-BOUNDARY-009	Automated theorem search boundary	256	depth_independent
SFT-MATH-SYMB-CONSTRUCTIVE-CERTIFICATE-010	Constructive witness and certificate generation	256	depth_independent

Xint (xint)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-XINT-INFORMATION-HANDOFF-001	Mathematics-to-Information exact-structure handoff	256	depth_independent
SFT-MATH-XINT-COMPUTATION-HANDOFF-002	Mathematics-to-Computation model handoff	256	depth_independent
SFT-MATH-XINT-PHYSICS-HANDOFF-003	Mathematics-to-Physics quantity and geometry handoff	256	depth_independent
SFT-MATH-XINT-CHEMISTRY-HANDOFF-004	Mathematics-to-Chemistry structure handoff	256	depth_independent
SFT-MATH-XINT-BIOLOGY-HANDOFF-005	Mathematics-to-Biology organization handoff	256	depth_independent
SFT-MATH-XINT-SOCIAL-HANDOFF-006	Mathematics-to-Social inference handoff	256	depth_independent
SFT-MATH-XINT-ENGINEERING-HANDOFF-007	Mathematics-to-Engineering calculation handoff	256	depth_independent
SFT-MATH-XINT-ONE-OWNER-IDENTITY-008	Shared mathematical identity without duplicate ownership	256	depth_independent

Valid (valid)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-VALID-ARITHMETIC-ALGEBRA-001	Arithmetic and algebra complete validation vector	256	depth_independent
SFT-MATH-VALID-COMBINATORICS-GRAPH-002	Combinatorics and graph complete validation vector	256	depth_independent
SFT-MATH-VALID-LINEAR-ALGEBRAIC-003	Linear and algebraic-structure validation vector	256	depth_independent
SFT-MATH-VALID-ORDER-GEOMETRY-004	Order and geometry complete validation vector	256	depth_independent
SFT-MATH-VALID-TOPOLOGY-ANALYSIS-005	Topology and analysis complete validation vector	256	depth_independent
SFT-MATH-VALID-EQUATION-MEASURE-006	Equation and measure complete validation vector	256	depth_independent
SFT-MATH-VALID-PROBABILITY-STATISTICS-007	Probability and statistics complete validation vector	256	depth_independent
SFT-MATH-VALID-OPTIMIZATION-DYNAMICS-008	Optimization and dynamics complete validation vector	256	depth_independent
SFT-MATH-VALID-LOGIC-COMPOSITIONAL-009	Logic and compositional complete validation vector	256	depth_independent
SFT-MATH-VALID-NUMERICAL-SYMBOLIC-010	Numerical and symbolic complete validation vector	256	depth_independent
SFT-MATH-VALID-ADVERSE-BOUNDARY-011	Adverse absent unresolved and boundary vector	256	depth_independent
SFT-MATH-VALID-EMPIRICAL-FORMAL-GRAND-LOCK-012	Mathematics empirical and formal Grand Lock	256	depth_independent

Hand (hand)

Claim ID	Claim title	Candidates	Closure
SFT-MATH-HAND-DOWNSTREAM-ONE-OWNER-001	Mathematics one-owner downstream handoff	256	depth_independent
SFT-MATH-HAND-MEASUREMENT-BOUNDARY-002	Mathematics measurement-boundary handoff	256	depth_independent
SFT-MATH-HAND-FORMAL-EMPIRICAL-003	Mathematics formal-to-empirical handoff	256	depth_independent
SFT-MATH-HAND-CONVENTIONAL-CORRESPONDENCE-004	Mathematics conventional-correspondence handoff	256	depth_independent
SFT-MATH-HAND-OPEN-EXTENSION-005	Mathematics open-extension handoff	256	depth_independent
SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006	Mathematics cross-branch one-owner completeness certificate	256	depth_independent

Information Science: 262 active claims**Current Status, Evidence Language And Reader Map**

(current_status, _evidence_language_and_reader_map)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-SYMBOL-DISTINCTION-001	Exact symbols, observation classes and distinguishability	512	depth_independent

Derivation 1: Exact Symbols, Observation Classes And Distinguishability

(derivation_1:_exact_symbols,_observation_classes_and_distinguishability)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-ENCODING-DECODING-001	Exact encoding, decoding and declared representational loss	1,024	depth_independent

Derivation 2: Exact Encoding, Decoding And Declared Representational Loss

(derivation_2:_exact_encoding,_decoding_and_declared_representational_loss)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-QUANTITY-001	Exact information quantity as retained distinction structure	512	depth_independent

Derivation 3: Exact Information Quantity As Retained Distinction Structure

(derivation_3:_exact_information_quantity_as_retained_distinction_structure)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-ENTROPY-UNCERTAINTY-001	Exact entropy and uncertainty from deterministic observation classes	512	depth_independent

Derivation 4: Exact Entropy And Uncertainty From Deterministic Observation Classes (derivation_4:_exact_entropy_and_uncertainty_from_deterministic_observation_classes)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-COMPRESSION-REDUNDANCY-001	Exact lossless compression and redundancy accounting	1,024	depth_independent

Derivation 5: Exact Lossless Compression And Redundancy Accounting

(derivation_5:_exact_lossless_compression_and_redundancy_accounting)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-CHANNEL-CAPACITY-001	Exact finite channels, confusability and capacity	1,024	depth_independent

Derivation 6: Exact Finite Channels, Confusability And Capacity

(derivation_6:_exact_finite_channels,_confusability_and_capacity)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-NOISE-ERROR-001	Exact noise, error, detection and correction boundaries	1,024	depth_independent

Derivation 7: Exact Noise, Error, Detection And Correction Boundaries

(derivation_7:_exact_noise,_error,_detection_and_correction_boundaries)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-CODING-001	Exact finite coding, detection and correction	2,048	depth_independent

Derivation 8: Exact Finite Coding, Detection And Correction

(derivation_8:_exact_finite_coding,_detection_and_correction)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-MUTUAL-CONDITIONAL-001	Exact mutual and conditional information from joint observation incidence	1,024	depth_independent

Derivation 9: Exact Mutual And Conditional Information From Joint Observation Incidence (derivation_9:_exact_mutual_and_conditional_information_from_joint_observation_incidence)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-CONSERVATION-LOSS-001	Exact information conservation, loss and transformation	1,024	depth_independent

Derivation 10: Exact Information Conservation, Loss And Transformation

(derivation_10:_exact_information_conservation,_loss_and_transformation)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-CLASSICAL-PROBABILISTIC-001	Exact classical and probabilistic information correspondence	1,024	depth_independent

Derivation 11: Exact Classical And Probabilistic Information Correspondence

(derivation_11:_exact_classical_and_probabilistic_information_correspondence)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-QUANTUM-CORRESPONDENCE-001	Exact quantum-information support correspondence	1,024	depth_independent

Symrep (symrep)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-SYMREP-ALPHABET-GENERATION-001	Canonical alphabet generation	256	depth_independent
SFT-INFO-SYMREP-SYMBOL-IDENTITY-DISTINCTION-002	Symbol identity and distinguishability	256	depth_independent
SFT-INFO-SYMREP-CODEWORD-PARSING-003	Codeword formation and parsing	256	depth_independent
SFT-INFO-SYMREP-PREFIX-UNIQUE-DECODING-004	Prefix, uniquely decodable and instantaneous structure	256	depth_independent
SFT-INFO-SYMREP-GRAMMAR-REPRESENTATION-005	Grammar-constrained representation	256	depth_independent
SFT-INFO-SYMREP-EQUIVALENCE-ISOMORPHISM-006	Representation equivalence and isomorphism	256	depth_independent
SFT-INFO-SYMREP-CANONICAL-NORMALIZATION-007	Canonicalization and normalization	256	depth_independent
SFT-INFO-SYMREP-VARIABLE-LENGTH-BOUNDARY-008	Variable-length representation boundaries	256	depth_independent
SFT-INFO-SYMREP-PRODUCT-ALPHABET-009	Composite and product alphabets	256	depth_independent
SFT-INFO-SYMREP-HIERARCHICAL-TYPED-SYMBOL-010	Hierarchical and typed symbols	256	depth_independent
SFT-INFO-SYMREP-CONVERSION-TRANSDUCTION-011	Representation conversion and transduction	256	depth_independent
SFT-INFO-SYMREP-AMBIGUITY-ALTERNATIVES-012	Ambiguity detection and retained alternatives	256	depth_independent
SFT-INFO-SYMREP-FINITE-SUCCESSOR-013	Finite-alphabet successor law	256	depth_independent
SFT-INFO-SYMREP-COMPLETENESS-014	Representation completeness and no-omission certificate	256	depth_independent

Record (record)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-RECORD-DATA-IDENTITY-001	Data item and record identity	256	depth_independent
SFT-INFO-RECORD-TUPLE-RELATION-002	Field, tuple and relation organization	256	depth_independent
SFT-INFO-RECORD-METADATA-003	Metadata as retained interpretive record	256	depth_independent
SFT-INFO-RECORD-SCHEMA-TYPE-004	Schema and type custody	256	depth_independent
SFT-INFO-RECORD-PROVENANCE-005	Provenance chain composition	256	depth_independent
SFT-INFO-RECORD-INTEGRITY-006	Integrity and tamper evidence	256	depth_independent
SFT-INFO-RECORD-VERSION-REVISION-007	Version and revision identity	256	depth_independent
SFT-INFO-RECORD-ABSENT-MISSING-UNKNOWN-008	Missing, absent and unknown distinctions	256	depth_independent
SFT-INFO-RECORD-DUPLICATE-ALIAS-009	Duplicate and alias resolution	256	depth_independent
SFT-INFO-RECORD-LINKAGE-IDENTITY-010	Record linkage and identity boundaries	256	depth_independent
SFT-INFO-RECORD-DATASET-COMPLETENESS-011	Dataset completeness and omission ledger	256	depth_independent
SFT-INFO-RECORD-CUSTODY-REPRODUCIBILITY-012	Record custody and reproducibility certificate	256	depth_independent

Source (source)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-SOURCE-SUPPORT-001	Finite source-support generation	256	depth_independent
SFT-INFO-SOURCE-SEQUENCE-ORDER-002	Sequence information and retained order	256	depth_independent
SFT-INFO-SOURCE-PROCESS-TRANSITION-003	Process information and transition custody	256	depth_independent
SFT-INFO-SOURCE-SPATIAL-ADJACENCY-004	Spatial information and adjacency custody	256	depth_independent
SFT-INFO-SOURCE-NETWORK-PATH-005	Network information and path custody	256	depth_independent
SFT-INFO-SOURCE-REFINEMENT-COARSENING-006	Source refinement and coarsening	256	depth_independent
SFT-INFO-SOURCE-STATIONARY-SUPPORT-007	Stationary-support correspondence	256	depth_independent
SFT-INFO-SOURCE-NONSTATIONARY-SUPPORT-008	Nonstationary-support correspondence	256	depth_independent
SFT-INFO-SOURCE-MEMORYLESS-009	Memoryless-source correspondence	256	depth_independent
SFT-INFO-SOURCE-FINITE-MEMORY-010	Finite-memory source structure	256	depth_independent
SFT-INFO-SOURCE-JOINT-COMPOSITION-011	Joint-source composition	256	depth_independent
SFT-INFO-SOURCE-DEPENDENCE-COMMON-SUPPORT-012	Source dependence and common support	256	depth_independent
SFT-INFO-SOURCE-SUCCESSOR-013	Source extension by exact successor	256	depth_independent
SFT-INFO-SOURCE-COMPLETENESS-014	Source completeness and boundary certificate	256	depth_independent

Measure (measure)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-MEASURE-DISTINCTION-COUNT-001	Distinction-count information measure	256	depth_independent
SFT-INFO-MEASURE-COMBINATORIAL-QUANTITY-002	Combinatorial information quantity	256	depth_independent
SFT-INFO-MEASURE-OPERATIONAL-COST-003	Operational discrimination cost	256	depth_independent
SFT-INFO-MEASURE-DESCRIPTION-LENGTH-004	Description length on a fixed grammar	256	depth_independent
SFT-INFO-MEASURE-ALGORITHMIC-BOUNDARY-005	Algorithmic-description correspondence boundary	256	depth_independent
SFT-INFO-MEASURE-PARTITION-REFINEMENT-006	Partition refinement information order	256	depth_independent
SFT-INFO-MEASURE-PRODUCT-ADDITIVITY-007	Additivity on independent product support	256	depth_independent
SFT-INFO-MEASURE-SHARED-SUBADDITIVITY-008	Subadditivity under shared support	256	depth_independent
SFT-INFO-MEASURE-COARSENING-MONOTONICITY-009	Monotonicity under observation coarsening	256	depth_independent
SFT-INFO-MEASURE-BALANCE-LEDGER-010	Exact information balance ledger	256	depth_independent
SFT-INFO-MEASURE-RELATIVE-011	Relative information between observations	256	depth_independent
SFT-INFO-MEASURE-DIVERGENCE-012	Information divergence correspondence	256	depth_independent
SFT-INFO-MEASURE-GEOMETRY-013	Information geometry on exact parts	256	depth_independent
SFT-INFO-MEASURE-MULTISCALE-014	Multi-scale information decomposition	256	depth_independent
SFT-INFO-MEASURE-UNIT-CUSTODY-015	Measure conversion and unit custody	256	depth_independent
SFT-INFO-MEASURE-COMPLETENESS-016	Information-measure completeness certificate	256	depth_independent

Signal (signal)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-SIGNAL-ORDERED-RECORD-001	Signal as an ordered observation record	256	depth_independent
SFT-INFO-SIGNAL-AMPLITUDE-SUPPORT-002	Amplitude-label and support separation	256	depth_independent
SFT-INFO-SIGNAL-SAMPLING-SELECTION-003	Sampling as retained position selection	256	depth_independent
SFT-INFO-SIGNAL-FINITE-SUFFICIENCY-004	Exact finite sampling sufficiency	256	depth_independent
SFT-INFO-SIGNAL-ALIASING-005	Aliasing as closed distinction	256	depth_independent
SFT-INFO-SIGNAL-QUANTIZATION-PARTITION-006	Quantization as observation partition	256	depth_independent
SFT-INFO-SIGNAL-QUANTIZATION-ERROR-007	Quantization error custody	256	depth_independent
SFT-INFO-SIGNAL-RECONSTRUCTION-008	Reconstruction from complete retained samples	256	depth_independent
SFT-INFO-SIGNAL-INTERPOLATION-BOUNDARY-009	Interpolation correspondence boundary	256	depth_independent
SFT-INFO-SIGNAL-TANSFORM-REPRESENTATION-010	Transform representation of finite signals	256	depth_independent
SFT-INFO-SIGNAL-TIME-FREQUENCY-011	Time-frequency support correspondence	256	depth_independent
SFT-INFO-SIGNAL-SPATIAL-MULTIDIMENSIONAL-012	Spatial and multidimensional sampling	256	depth_independent
SFT-INFO-SIGNAL-PROVENANCE-013	Signal-to-record provenance	256	depth_independent
SFT-INFO-SIGNAL-COMPLETENESS-014	Sampling and reconstruction completeness certificate	256	depth_independent

Comp (comp)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-COMP-LOSSLESS-RECONSTRUCTION-001	Lossless code and exact reconstruction	256	depth_independent
SFT-INFO-COMP-PREFIX-TREE-002	Prefix-tree compression structure	256	depth_independent
SFT-INFO-COMP-DICTIONARY-003	Dictionary compression structure	256	depth_independent
SFT-INFO-COMP-RUN-RECURRENCE-004	Run and recurrence compression	256	depth_independent
SFT-INFO-COMP-TRANSFORM-005	Transform compression correspondence	256	depth_independent
SFT-INFO-COMP-SOURCE-MODEL-BOUNDARY-006	Source-model compression boundary	256	depth_independent
SFT-INFO-COMP-REDUNDANCY-007	Redundancy as retained excess distinction	256	depth_independent
SFT-INFO-COMP-MINIMUM-DESCRIPTION-008	Minimum description within a fixed grammar	256	depth_independent
SFT-INFO-COMP-LOSSY-COARSENING-009	Lossy compression as declared coarsening	256	depth_independent
SFT-INFO-COMP-DISTORTION-010	Exact distortion relation	256	depth_independent
SFT-INFO-COMP-RATE-DISTORTION-BOUNDARY-011	Rate-distortion correspondence boundary	256	depth_independent
SFT-INFO-COMP-SUCCESSIVE-REFINEMENT-012	Successive and layered refinement	256	depth_independent
SFT-INFO-COMP-SIDE-INFORMATION-013	Compression under side information	256	depth_independent
SFT-INFO-COMP-COMPLETENESS-014	Compression completeness and adverse-control certificate	256	depth_independent

Chan (chan)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-CHAN-INPUT-OUTPUT-RELATION-001	Channel as an exact input-output relation	256	depth_independent
SFT-INFO-CHAN-DETERMINISTIC-TRANSPORT-002	Deterministic channel transport	256	depth_independent
SFT-INFO-CHAN-OBSERVATION-EQUIVALENCE-003	Observation-relative channel equivalence	256	depth_independent
SFT-INFO-CHAN-SINGLE-USER-CAPACITY-004	Single-user channel capacity	256	depth_independent
SFT-INFO-CHAN-RESOURCE-CAPACITY-005	Resource-bounded channel capacity	256	depth_independent
SFT-INFO-CHAN-NOISELESS-COMPOSITION-006	Noiseless channel composition	256	depth_independent
SFT-INFO-CHAN-CASCADE-BOUNDARY-007	Cascaded channel capacity boundary	256	depth_independent
SFT-INFO-CHAN-PARALLEL-COMPOSITION-008	Parallel channel composition	256	depth_independent
SFT-INFO-CHAN-FEEDBACK-009	Feedback channel correspondence	256	depth_independent
SFT-INFO-CHAN-MULTIPLE-ACCESS-010	Multiple-access channel support	256	depth_independent
SFT-INFO-CHAN-BROADCAST-011	Broadcast channel support	256	depth_independent
SFT-INFO-CHAN-RELAY-012	Relay channel support	256	depth_independent
SFT-INFO-CHAN-INTERFERENCE-013	Interference-channel support	256	depth_independent
SFT-INFO-CHAN-BIDIRECTIONAL-014	Bidirectional channel support	256	depth_independent
SFT-INFO-CHAN-NETWORK-CUT-015	Network channel and cut boundary	256	depth_independent
SFT-INFO-CHAN-FINITE-USE-SUCCESSION-016	Finite-use capacity succession	256	depth_independent
SFT-INFO-CHAN-SIMULATION-017	Channel simulation correspondence	256	depth_independent
SFT-INFO-CHAN-COMPLETENESS-018	Channel-family completeness certificate	256	depth_independent

Noise (noise)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-NOISE-DISTINCTION-CLOSURE-001	Noise as unrecorded distinction closure	256	depth_independent
SFT-INFO-NOISE-SOURCE-OUTPUT-MISMATCH-002	Error as source-output mismatch	256	depth_independent
SFT-INFO-NOISE-PATTERN-SUPPORT-003	Deterministic noise-pattern support	256	depth_independent
SFT-INFO-NOISE-COMPOSITION-004	Noise transport through composition	256	depth_independent
SFT-INFO-NOISE-DETECTION-005	Error detection condition	256	depth_independent
SFT-INFO-NOISE-LOCALIZATION-006	Error localization condition	256	depth_independent
SFT-INFO-NOISE-ESTIMATION-007	Error estimation condition	256	depth_independent
SFT-INFO-NOISE-ERASURE-SUBSTITUTION-008	Erasure and substitution distinction	256	depth_independent
SFT-INFO-NOISE-BURST-CORRELATION-009	Burst and correlated error structure	256	depth_independent
SFT-INFO-NOISE-ADVERSARIAL-SUPPORT-010	Adversarial error support	256	depth_independent
SFT-INFO-NOISE-BUDGET-LEDGER-011	Noise budget and exact error ledger	256	depth_independent
SFT-INFO-NOISE-COMPLETENESS-012	Noise-family completeness certificate	256	depth_independent

Code (code)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-CODE-INJECTIVE-REPRESENTATION-001	Code as injective representation relation	256	depth_independent
SFT-INFO-CODE-SEPARATION-CORRECTION-002	Minimum separation and correctability	256	depth_independent
SFT-INFO-CODE-BLOCK-STRUCTURE-003	Block-code structure	256	depth_independent
SFT-INFO-CODE-LINEAR-CORRESPONDENCE-004	Linear-code correspondence	256	depth_independent
SFT-INFO-CODE-PARITY-SYNDROME-005	Parity and syndrome records	256	depth_independent
SFT-INFO-CODE-REPETITION-MAJORITY-006	Repetition and majority structure	256	depth_independent
SFT-INFO-CODE-ERASURE-CORRECTION-007	Erasure-correcting structure	256	depth_independent
SFT-INFO-CODE-SUBSTITUTION-CORRECTION-008	Substitution-error correction	256	depth_independent
SFT-INFO-CODE-BURST-CORRECTION-009	Burst-error correction	256	depth_independent
SFT-INFO-CODE-CONVOLUTIONAL-CORRESPONDENCE-010	Convolutional-code correspondence	256	depth_independent
SFT-INFO-CODE-TREE-SEQUENTIAL-DECODING-011	Tree and sequential decoding	256	depth_independent
SFT-INFO-CODE-PRODUCT-CONCATENATED-012	Product and concatenated codes	256	depth_independent
SFT-INFO-CODE-SPARSE-CHECK-013	Sparse-check code correspondence	256	depth_independent
SFT-INFO-CODE-NETWORK-014	Network coding structure	256	depth_independent
SFT-INFO-CODE-LIST-DECODING-015	List-decoding boundary	256	depth_independent
SFT-INFO-CODE-ADVERSARIAL-BOUNDARY-016	Adversarial coding boundary	256	depth_independent
SFT-INFO-CODE-COMPOSITION-RATE-017	Code composition and rate custody	256	depth_independent
SFT-INFO-CODE-COMPLETENESS-018	Coding-family completeness certificate	256	depth_independent

Rel (rel)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-REL-JOINT-SUPPORT-001	Joint information support	256	depth_independent
SFT-INFO-REL-CONDITIONAL-RESTRICTION-002	Conditional information by exact restriction	256	depth_independent
SFT-INFO-REL-MUTUAL-SHARED-003	Mutual information by shared distinctions	256	depth_independent
SFT-INFO-REL-CHAIN-RULE-004	Chain rule for retained information	256	depth_independent
SFT-INFO-REL-DATA-PROCESSING-005	Data-processing monotonicity	256	depth_independent
SFT-INFO-REL-CONDITIONAL-INDEPENDENCE-006	Conditional independence support	256	depth_independent
SFT-INFO-REL-COMMON-INFORMATION-007	Common information correspondence	256	depth_independent
SFT-INFO-REL-DIRECTED-INFORMATION-008	Directed information on ordered records	256	depth_independent
SFT-INFO-REL-INTERACTION-009	Interaction information correspondence	256	depth_independent
SFT-INFO-REL-MULTI-TOTAL-DEPENDENCE-010	Multi-information and total dependence	256	depth_independent
SFT-INFO-REL-PARTIAL-DECOMPOSITION-011	Shared, unique and synergistic decomposition boundary	256	depth_independent
SFT-INFO-REL-CAUSAL-FLOW-012	Information flow on causal paths	256	depth_independent
SFT-INFO-REL-REPRESENTATION-CHANGE-013	Relative information under representation change	256	depth_independent
SFT-INFO-REL-COMPLETENESS-014	Relational-information completeness certificate	256	depth_independent

Coarse (coarse)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-COARSE-SUFFICIENT-RECORD-001	Sufficient record for a declared observation	256	depth_independent
SFT-INFO-COARSE-MINIMAL-SUFFICIENT-002	Minimal sufficient record	256	depth_independent
SFT-INFO-COARSE-BOTTLENECK-003	Information bottleneck as lawful coarsening	256	depth_independent
SFT-INFO-COARSE-PARTITION-004	Coarse-graining partition structure	256	depth_independent
SFT-INFO-COARSE-REFINEMENT-ORDER-005	Refinement and abstraction order	256	depth_independent
SFT-INFO-COARSE-LOSS-LEDGER-006	Loss ledger for aggregation	256	depth_independent
SFT-INFO-COARSE-FEATURE-SELECTION-007	Feature selection by retained distinction	256	depth_independent
SFT-INFO-COARSE-STATE-LUMPING-008	State aggregation and exact lumping	256	depth_independent
SFT-INFO-COARSE-RECONSTRUCTION-BOUNDARY-009	Summary record reconstruction boundary	256	depth_independent
SFT-INFO-COARSE-COMPOSITION-010	Multi-stage coarse-graining composition	256	depth_independent
SFT-INFO-COARSE-REVERSIBILITY-011	Coarse-graining reversibility condition	256	depth_independent
SFT-INFO-COARSE-COMPLETENESS-012	Sufficiency and bottleneck completeness certificate	256	depth_independent

Retr (retr)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-RETR-INDEX-LOCATION-001	Index as a source-bound location relation	256	depth_independent
SFT-INFO-RETR-EXACT-MEMBERSHIP-002	Exact retrieval and membership	256	depth_independent
SFT-INFO-RETR-FORWARD-INVERTED-003	Inverted and forward index correspondence	256	depth_independent
SFT-INFO-RETR-QUERY-OBSERVATION-004	Query as a declared observation	256	depth_independent
SFT-INFO-RETR-RANKING-ORDER-005	Ranking as an exact finite order	256	depth_independent
SFT-INFO-RETR-TIE-IN-COMPARABILITY-006	Tie and incomparability custody	256	depth_independent
SFT-INFO-RETR-RELEVANCE-RELATION-007	Relevance as a registered relation	256	depth_independent
SFT-INFO-RETR-PRECISION-RECALL-008	Precision and recall exact-part correspondence	256	depth_independent
SFT-INFO-RETR-TAXONOMY-009	Knowledge organization and taxonomy	256	depth_independent
SFT-INFO-RETR-CROSS-REFERENCE-GRAPH-010	Cross-reference and citation graph	256	depth_independent
SFT-INFO-RETR-UPDATE-STALE-BOUNDARY-011	Update and stale-index boundaries	256	depth_independent
SFT-INFO-RETR-COMPLETENESS-012	Retrieval-family completeness certificate	256	depth_independent

Infer (infer)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-INFER-SUPPORT-RESTRICTION-001	Inference as support restriction	256	depth_independent
SFT-INFO-INFER-DETECTION-002	Detection as observation-class decision	256	depth_independent
SFT-INFO-INFER-ESTIMATION-003	Estimation as retained representative selection	256	depth_independent
SFT-INFO-INFER-FILTERING-004	Filtering as sequential record update	256	depth_independent
SFT-INFO-INFER-SMOOTHING-005	Smoothing as whole-record reconstruction	256	depth_independent
SFT-INFO-INFER-PREDICTION-006	Prediction as registered transition support	256	depth_independent
SFT-INFO-INFER-EVIDENCE-ACCUMULATION-007	Evidence accumulation	256	depth_independent
SFT-INFO-INFER-LIKELIHOOD-CORRESPONDENCE-008	Likelihood correspondence on exact support	256	depth_independent
SFT-INFO-INFER-STRUCTURAL-THRESHOLD-009	Decision threshold without fitted scalar	256	depth_independent
SFT-INFO-INFER-ERROR-CUSTODY-010	False-positive and false-negative custody	256	depth_independent
SFT-INFO-INFER-IDENTIFIABILITY-011	Identifiability and observational equivalence	256	depth_independent
SFT-INFO-INFER-SEQUENTIAL-STOPPING-012	Sequential detection and stopping boundary	256	depth_independent
SFT-INFO-INFER-MULTI-SENSOR-FUSION-013	Multi-sensor information fusion	256	depth_independent
SFT-INFO-INFER-COMPLETENESS-014	Inference-family completeness certificate	256	depth_independent

Priv (priv)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-PRIV-LEAKAGE-001	Information leakage as retained adversary distinction	256	depth_independent
SFT-INFO-PRIV-PERFECT-SECURITY-002	Perfect secrecy correspondence	256	depth_independent
SFT-INFO-PRIV-SIDE-INFORMATION-003	Side-information and disclosure boundary	256	depth_independent
SFT-INFO-PRIV-ANONYMITY-004	Anonymity and indistinguishability classes	256	depth_independent
SFT-INFO-PRIV-LEAST-DISCLOSURE-005	Access observation and least disclosure	256	depth_independent
SFT-INFO-PRIV-COMPOSITION-006	Composition of privacy losses	256	depth_independent
SFT-INFO-PRIV-QUERY-RECONSTRUCTION-007	Query disclosure and reconstruction risk	256	depth_independent
SFT-INFO-PRIV-UTILITY-BOUNDARY-008	Privacy-utility exact trade boundary	256	depth_independent
SFT-INFO-PRIV-CRYPTO-HANDOFF-009	Cryptographic ownership handoff	256	depth_independent
SFT-INFO-PRIV-COMPLETENESS-010	Privacy-interface completeness certificate	256	depth_independent

Therm (therm)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-THERM-PHYSICAL-CUSTODY-001	Information record as a physical custody obligation	256	depth_independent
SFT-INFO-THERM-IRREVERSIBLE-MERGE-002	Irreversible merge and missing predecessor label	256	depth_independent
SFT-INFO-THERM-REVERSIBLE-RETENTION-003	Reversible record retention	256	depth_independent
SFT-INFO-THERM-ERASURE-BOUNDARY-004	Erasure correspondence boundary	256	depth_independent
SFT-INFO-THERM-MEASUREMENT-RECORD-005	Measurement record cost	256	depth_independent
SFT-INFO-THERM-MEMORY-RESET-006	Memory reset and provenance custody	256	depth_independent
SFT-INFO-THERM-MAXWELL-DEMON-007	Maxwell-demon information ledger	256	depth_independent
SFT-INFO-THERM-LANDAUER-BENNETT-008	Landauer and Bennett correspondence boundary	256	depth_independent
SFT-INFO-THERM-NON-EQUIVALENCE-009	Information-energy non-equivalence boundary	256	depth_independent
SFT-INFO-THERM-COMPLETENESS-010	Information-thermodynamics completeness certificate	256	depth_independent

Corr (corr)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-CORR-CLASSICAL-SYMBOL-001	Classical symbol-information correspondence	256	depth_independent
SFT-INFO-CORR-DETERMINISTIC-PROBABILITY-002	Deterministic-support probability correspondence	256	depth_independent
SFT-INFO-CORR-PROBABILISTIC-MIXTURE-003	Probabilistic mixture as hidden-support description	256	depth_independent
SFT-INFO-CORR-QUANTUM-SUPPORT-004	Quantum support correspondence	256	depth_independent
SFT-INFO-CORR-QUANTUM-UNIT-005	Quantum information unit correspondence	256	depth_independent
SFT-INFO-CORR-PRODUCT-STATE-006	Composite support and product-state correspondence	256	depth_independent
SFT-INFO-CORR-ENTANGLEMENT-BOUNDARY-007	Joint support and entanglement boundary	256	depth_independent
SFT-INFO-CORR-PHASE-HANDOFF-008	Phase information ownership handoff	256	depth_independent
SFT-INFO-CORR-MEASUREMENT-SUPPORT-009	Measurement support and record correspondence	256	depth_independent
SFT-INFO-CORR-CLASSICAL-QUANTUM-ENCODING-010	Classical-to-quantum encoding boundary	256	depth_independent
SFT-INFO-CORR-QUANTUM-CLASSICAL-OBSERVATION-011	Quantum-to-classical observation boundary	256	depth_independent
SFT-INFO-CORR-NO-CLONING-BOUNDARY-012	No-cloning information boundary correspondence	256	depth_independent
SFT-INFO-CORR-QUANTUM-CHANNEL-013	Quantum channel support correspondence	256	depth_independent
SFT-INFO-CORR-QUANTUM-CODING-HANDOFF-014	Quantum coding ownership handoff	256	depth_independent
SFT-INFO-CORR-NONDUPLICATION-015	Operational-dynamics nonduplication boundary	256	depth_independent
SFT-INFO-CORR-COMPLETENESS-016	Classical-probabilistic-quantum completeness certificate	256	depth_independent

Sem (sem)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-SEM-SYMBOL-CONTENT-BOUNDARY-001	Symbol count and semantic-content boundary	256	depth_independent
SFT-INFO-SEM-REFERENCE-INTERPRETATION-002	Reference and interpretation record	256	depth_independent
SFT-INFO-SEM-CONTEXT-DISTINGUISHABILITY-003	Context-dependent distinguishability	256	depth_independent
SFT-INFO-SEM-BIOLOGICAL-HANDOFF-004	Biological information ownership handoff	256	depth_independent
SFT-INFO-SEM-GENETIC-CORRESPONDENCE-005	Genetic representation correspondence	256	depth_independent
SFT-INFO-SEM-NEURAL-COGNITIVE-HANDOFF-006	Neural and cognitive information handoff	256	depth_independent
SFT-INFO-SEM-CONSCIOUS-REPORT-HANDOFF-007	Conscious access and reportability handoff	256	depth_independent
SFT-INFO-SEM-SOCIAL-SHARED-RECORD-008	Social communication and shared-record handoff	256	depth_independent
SFT-INFO-SEM-TRANSLATION-PRESERVATION-009	Meaning-preserving translation boundary	256	depth_independent
SFT-INFO-SEM-PRAGMATIC-ACTION-BOUNDARY-010	Pragmatic action-information boundary	256	depth_independent
SFT-INFO-SEM-CROSS-DOMAIN-PROVENANCE-011	Cross-domain provenance and nonduplication	256	depth_independent
SFT-INFO-SEM-COMPLETENESS-012	Semantic-handoff completeness certificate	256	depth_independent

Valid (valid)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-VALID-SYMBOL-REPRESENTATION-001	Symbol and representation complete validation vector	256	depth_independent
SFT-INFO-VALID-RECORD-PROVENANCE-002	Record and provenance complete validation vector	256	depth_independent
SFT-INFO-VALID-SOURCE-MEASURE-003	Source and measure complete validation vector	256	depth_independent
SFT-INFO-VALID-SIGNAL-SAMPLING-004	Signal and sampling complete validation vector	256	depth_independent
SFT-INFO-VALID-COMPRESSION-DISTORTION-005	Compression and distortion complete validation vector	256	depth_independent
SFT-INFO-VALID-CHANNEL-CAPACITY-006	Channel and capacity complete validation vector	256	depth_independent
SFT-INFO-VALID-NOISE-CODING-007	Noise and coding complete validation vector	256	depth_independent
SFT-INFO-VALID-RELATIONAL-COARSE-008	Relational and coarse-graining complete validation vector	256	depth_independent
SFT-INFO-VALID-RETRIEVAL-INFERERENCE-009	Retrieval and inference complete validation vector	256	depth_independent
SFT-INFO-VALID-PRIVACY-THERMAL-CORRESPONDENCE-010	Privacy, thermodynamics and correspondence validation vector	256	depth_independent
SFT-INFO-VALID-ADVERSE-BOUNDARY-011	Adverse, absent, unresolved and boundary vector	256	depth_independent
SFT-INFO-VALID-GRAND-LOCK-012	Information Science empirical and formal Grand Lock	256	depth_independent

Hand (hand)

Claim ID	Claim title	Candidates	Closure
SFT-INFO-HAND-ONE-OWNER-DOWNSTREAM-001	Information Science one-owner downstream handoff	256	depth_independent
SFT-INFO-HAND-MEASUREMENT-BOUNDARY-002	Information Science measurement-boundary handoff	256	depth_independent
SFT-INFO-HAND-FORMAL-EMPIRICAL-003	Information Science formal-to-empirical handoff	256	depth_independent
SFT-INFO-HAND-CONVENTIONAL-CORRESPONDENCE-004	Information Science conventional-correspondence handoff	256	depth_independent
SFT-INFO-HAND-OPEN-EXTENSION-005	Information Science open-extension handoff	256	depth_independent
SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006	Information Science cross-branch completeness certificate	256	depth_independent

The Classical and Reversible/Quantum Computation narratives above identify their foundational and landmark claim IDs directly. Their complete 657-row title index remains in the scientific audit volumes, where it can be read beside the corresponding machine records without overwhelming the conceptual handoff to Physics.

Part II. From physical law to chemical identity and organised matter

13. Physics: quantities, constants, fields and measured nature

The preceding branches establish exact mathematics, records, executable computation and reversible and quantum processes. Physics begins when those formal objects must be connected to physical quantity, dimension, interaction, source, response and measurement. This is a genuine change of evidential boundary. A mathematically unique survivor is not yet an observation of nature; a measured value is not permitted to become a hidden premise of the derivation that it later tests. The Physics branch therefore carries two constitutions at once: an exact derivational constitution for the generated physical structure, and a source-bound empirical constitution for comparisons with the world.

The active Physics paper is version 1.3.0. Its current categorical inventory contains **368 of 368 registered Physics claims**, generated from **257,776 candidates and 257,776 decisions**, with **368 unique survivors**, **1,472 of 1,472 adverse controls passed** and an independent reconstruction for every claim. All 368 closures are classified as `depth_independent`. The evidence surface distinguishes **238 sealed empirical-validation claims** from **130 independently replicated formal claims without empirical packages**. Those categories are not pooled into a generic claim of empirical confirmation.

13.1 Scope, ownership and present status

Physics owns physical quantities and dimensions, mechanics, fields, forces, waves, thermodynamic and vacuum relations, the physical quantum and relativistic interfaces, constants and scales, matter and interaction structures, atomic and nuclear physical relations, gravitation and physical cosmology. It consumes mathematical, informational and computational objects without taking ownership of their upstream proofs. Chemistry owns the chemical identity and reaction systems constructed from physical carriers; Materials Science owns collective structure-property relations in organised matter; Astronomy owns celestial-system observations; and Engineering owns artefact, safety and deployment claims. A physical law may be used by all of those branches while retaining one scientific owner.

Physics family	Claims	Scientific role
Measurement and metrology	13	Quantity, unit, comparison, uncertainty and physical evidence custody
Mechanics and dynamics	24	State, change, momentum, energy and dynamical response
Fields, forces, waves and geometry	41	Local relation, propagation, orientation, source and spatial response
Thermodynamics and vacuum	31	Multiplicity, work, temperature, entropy, vacuum floor and global share
Physical quantum and relativistic structure	24	Quantum weights, measurement boundary, causal order and relativistic relation
Constants, scales and precision	9	Exact generated constants, dimensional interfaces and precision comparison
Matter, interactions and flavour	62	Particle families, charges, couplings, masses and interaction structure
Atomic and molecular physics	15	Bound-state, orbital, transition and molecular physical relations
Nuclear and hadronic physics	17	Nuclear closure, hadronic structure, decay and binding relations
Spacetime and gravitation	34	Dimension, inverse-square response, gravitation, metric and hierarchy
Continua and collective matter	18	Collective modes, continua, radiation and macroscopic response
Physical cosmology boundary	16	Primordial, matter-share, expansion and vacuum-scale relations
Post-seal empirical validation	64	Registered measurements, holdouts, corrections and returned physical families
Total	368	Current registered Physics categorical scope

The Physics result also closes **488 of 488 Physics-owned V1/V2 atoms at the same declared strength**. That reconciliation means that the older obligations have a current owner and current proof or evidence route; it does not make the older papers authoritative proof premises. Grand Locks 075 and 076 preserve a 349-claim pre-extension snapshot. Within that historical lock, the formal certificate covers 347 claims and 534 graph nodes, while the empirical lock retains 234 pre-lock empirical claims, 147 distinct source IDs, fourteen unfavourable or scope-boundary claims and six legacy receipt shapes. Nineteen later

claims-including the Unified Constants Object, the Tesla resonance return, the vacuum/inertia family and the new-sector and Smithion family-are separately audited. They are not attributed backwards to the earlier lock.

13.2 The physical admission boundary

The branch preserves the Foundation distinction between structural absence and numerical zero. A missing relation may be admitted as a structural absence; the derivation may not silently import zero as an already available number. Proof values are generated positive whole or rational values. Direction and opposition remain held labels until an admitted relation acts on them. The formal grammar does not take irrational or imaginary values, floating-point approximations, completed infinity, an ungenerated continuum, fitted coefficients or free parameters as premises.

Every Physics claim is processed by the same admission engine. Its package must identify the complete candidate grammar and cardinality, decide every candidate, retain exactly one survivor, explain minimality and named-shape uniqueness, pass four registered controls and reconstruct independently. Where the claim touches observation, the package must additionally isolate the empirical target, preserve source and custody, retain every row and specify the falsifier. External numerical values may enter only after the formal result has sealed. Comparisons use exact rational intervals wherever the source permits them; displayed decimals are readings of those exact records, not replacement proof objects.

13.2.1 Prediction capability and target custody

The first three Physics claims do not state a conventional physical law. They establish the capability boundary required before any later law may be called a sealed prediction. The prediction form is a finite data-only interpreter. It accepts registered exact inputs and admitted operations, holds state by single assignment, emits one declared terminal result and exposes the complete step trace. It has no ambient capability from which a target could leak. The portable target-commitment claim then binds a complete named support, a nonce and a content identity to a distinct custodian. Only a matching experiment and matching execution envelope may receive the one post-seal release. The hostile package claim protects the model-admission receipt and candidate census from a data file that attempts to add an instruction, extra field or alternate route.

These claims explain what target blindness means in the V3 corpus. It is not a statement that the author had never heard of the phenomenon. It is a machine relation: the executable process that chooses and evaluates the formal survivor cannot access the target content, and a distinct custody record establishes when the content was released. Conversely, where a result was developed while the target was visible, the result may remain a useful reconstruction or correspondence but is not relabelled blind. The empirical status is carried by the receipt, not inferred from confident prose.

13.2.2 Observation, quantity and dimension

Physical observation begins as a retained distinction between a system, a reference and a record. A quantity is a reference-equivalence carrier: two physical records are comparable only where the declared equivalence preserves the kind being measured. Dimension supplies a canonical composition path and an independence test. A unit is a reversible scale transport within one dimension; it does not change the physical kind. Reference realisation adds the operational trace from the definition to the apparatus or procedure that instantiates it.

This ordering blocks several common conflations. A number without its quantity kind is not yet a physical value. Two quantities with the same displayed number need not be comparable. Two measurements with the same nominal unit need not possess the same realisation, uncertainty or source identity. A lawful conversion requires a complete dimension path and a reversible factor, while a calibration requires a chain of comparisons whose links and uncertainties are retained. Dimensional consistency is therefore derived as a relation among typed carriers rather than imposed as a visual check on symbols.

The measured-value record is kept separate from the proof value. Its fields identify the source, method, condition, value, uncertainty or interval, version and observation boundary. Measurement uncertainty is represented by the complete retained set of admissible alternatives under the source's method; it is not a licence to change the sealed prediction. Calibration closure composes the registered comparison chain. If a link is missing, incompatible or outside its versioned scope, the result halts rather than assuming that the source's institutional standing fills the gap.

The exact source-boundary growth discriminator is the bridge from this metrology to spatial law. It compares positive distance and response ratios from the same retained source, propagates their complete uncertainty support and searches the positive successor-power family. Its stopping certificate must show that no later positive integer can re-enter the admissible interval. That is why the inverse-square comparison can say that two is the sole positive integer in the interval, rather than merely reporting that a fitted exponent is near two.

13.2.3 Exact interval arithmetic and source roles

Finite source decimals are parsed as exact fractions. An uncertainty statement becomes exact lower and upper endpoints; multiplication and division propagate those endpoints through the registered monotone relation. A published decimal followed by an ellipsis is bounded at its next decimal place. This arithmetic prevents binary floating equality from deciding admission and prevents an unreported rounding choice from manufacturing overlap.

The principal external custodians have different evidential roles. BIPM and its metrological records provide definitions, realisations and traceability; NIST and CODATA provide constants and their stated uncertainties; NIST ASD provides atomic spectra; PDG supplies versioned particle summaries and measurement ranges; IAEA and AME records supply nuclear closure and mass surfaces; IAPWS supplies thermophysical water records; GWOSC supplies gravitational-wave observations; CERN Open Data supplies collider-bound records; and NASA LAMBDA supplies cosmological source material. None is cited as proof by prestige. Each enters only through the rows and discriminators registered for a particular claim.

This division matters most where familiar physical language invites an unlicensed inference. A structural correspondence may identify the form of an inverse-square relation without predicting a dimensional coefficient. A post-seal value inside a registered measurement interval is an exact numerical correspondence at that boundary, not automatically a prospective blind prediction. A successful source comparison validates the registered relation and protocol; it does not confirm all of Physics or every extension of the grammar.

13.3 From the Fold to spatial and dynamical structure

The minimal Fold supplies a binary distinction, reversible return and ordered carrier. Physics asks what physical organisation becomes necessary when that carrier must support location, direction, propagation and restoration. The generated orbit with multiplier three,

$$\frac{1}{7} \rightarrow \frac{2}{7} \rightarrow \frac{4}{7} \rightarrow \frac{1}{7},$$

has three distinct positions before return. In the Physics grammar this closes the spatial direction count at three. A boundary has rank two, and the corresponding spreading relation has inverse-square exponent two. The empirical exponent comparison is retained separately: the tested form is $2+q$, with $q=(2.7\pm3.1) \times 10^{-16}$, and two is the sole positive integer inside the registered interval. The observation tests the exponent; it does not serve as the premise from which the exponent was selected.

The same direction structure gives six cubic neighbours: three independent directions, each with two orientations. Propagating collective response then separates into one longitudinal and two transverse modes. These results later become part of the Materials derivation of lattice neighbourhood and acoustic branches. Their physical ownership remains here; the material branch owns the way a specimen and its organisation realises them.

Mechanics and dynamics develop ordered state change, conservation and response from this carrier. Fields and forces add a local source relation and the boundary through which an influence propagates. Waves require repeated return with a phase or orientation record; continua and collective matter retain the scale and condition under which a discrete derivation may be used as a collective description. Nothing in that sequence licenses an ungenerated continuum as a foundational premise. The continuum is a physical description at its admitted boundary, not an erasure of the generated records beneath it.

Radiation closes a fourth-power structural dependence. A doubling therefore produces the exact structural factor

$$2^4 = 16.$$

The result identifies the exponent and its generated scaling. It does not, by itself, supply a universal dimensional coefficient or replace a source-specific radiometric measurement.

13.3.1 Events, recurrence and mechanical quantity

The mechanics family begins with an event: a retained physical state and a generated change at a declared boundary. Duration is not inserted as an already flowing real line. It is obtained from ordered recurrence, so a clock is a physical system whose repeated relation can be compared with another event trace. Location is a relation to retained support; separation compares two locations; displacement adds a held orientation. Speed compares separation with recurrence, while velocity retains orientation. Acceleration is a change of that oriented velocity over a further ordered recurrence.

Inertial persistence follows when a closed motion trace has no registered transfer that changes it. Momentum is the carrier that composes inertial weight with oriented motion. Force is a constrained rate of momentum transfer, not an unexplained primitive push. Work composes force with displacement; kinetic change records the corresponding change of the motion carrier; energy is

the retained capacity for a permitted transfer. Power orders that transfer by duration. Angular motion adds cyclic orientation, and angular momentum and torque carry the associated conserved and transferred distinctions.

Composite motion requires a centre relation without importing numerical-zero coordinates. The branch constructs the complete weighted part relation among positive carriers and holds orientation separately. Conservation is then a bijection across the closed transfer network: every departing distinction has a named receiving relation, and no unregistered source or sink appears. Constrained motion adds a restoration relation. Oscillation is the resulting reversible recurrence around the held constraint, rather than a sinusoid assumed as a primitive mathematical object.

The order matters. A formula such as work equals force through displacement would be circular if force, displacement and transfer had not first acquired distinct carriers. The claim sequence prevents that circularity by admitting event, duration, separation, velocity, acceleration, inertia, momentum, force, energy, power and cyclic response in turn. Each later law names the earlier receipt it consumes. The empirical measured-value successors for momentum, force and related quantities test those compositions through exact unit and source adapters; they do not define the formal relation retrospectively.

13.3.2 Phase, spectrum and coupled dynamics

Once a cyclic orientation can be retained, a free carrier can accumulate Fold phase. The free-particle relation is an exact finite dispersion over the registered carrier. Potential evolution composes sequential rotations as an additive energy rotation. A stationary spectrum is the set of closed recurrent states admitted by the boundary, not an appeal to a continuous eigenvalue domain.

The later mechanics family extends this construction to odd-lattice all-region occupancy, exact recurrence vectors, half-One pair synchronisation and coupled ensembles. Expansion and information-rate carriers are related at their declared interface. A coupled-map synchronisation threshold is normalised exactly; orbital restoration is tested across the generated dimension family; and finite Fold symmetry is connected to conservation and least action. These are formal dynamics results. Where a conventional apparatus or dataset is used, the relationship is a separately registered correspondence.

The resulting picture is neither a continuum premise nor a denial that continuum models can be useful. The primary object is a finite exact recurrence trace. A continuum description appears later as a coarse-grained correspondence when the preserving coordinates and error boundary are named. This is why a numerical integration, a measured trajectory and an exact Fold orbit retain three different evidence statuses even when their graphs look similar.

13.3.3 Sources, fields and interaction

A field is introduced only after source and response have distinct physical carriers. The source is the retained origin of a response relation; the field records how that response is available across support. Geometric dilution follows from the growth of the source boundary. Gravitation first appears at the weak relational boundary, while electric distinction adds held charge orientation. Electric potential records the work relation, magnetic response arises with oriented change, and induction records changing linked support. Electric and magnetic relations then close compositionally.

Gauge-equivalent descriptions are admitted when distinct descriptions retain the same complete observable relation. The gauge relation does not declare that every symbolic rewriting is physically identical; it names the transformation and the retained observable. Local propagation adds causal accessibility. Radiative transport occurs when a response detaches from the source while preserving its energy-momentum ledger. Source continuity and field conservation prevent an apparent local description from deleting a source or transfer. Reciprocal interaction records the distinction transferred between participants.

The branch then extends the exact carrier to charge, Coulomb potential and Gauss closure; magnetic relativistic correction; Lorentz transfer partition; and planar and three-space Maxwell curvature. Prime-sector structure gives the force and mediator ladder and the complete finite prime-sector inventory. The same exact operations support wave, polarisation, interference and nonlinear composition. A finite-loop closure carries the declared floor and stopping boundary; it is not an appeal to an infinite divergent object subsequently repaired by an imported prescription.

The empirical interaction-class claim asks whether the registered source records distinguish the formal classes. Later comparisons cover relativistic and electromagnetic fields, finite-loop and renormalisation boundaries, particle sectors and scattering. Every comparison keeps its source, version, adapter and adverse control. Agreement with one field relation does not establish an unseen mediator, and a source that constrains a model-dependent extension does not automatically falsify the wider exact carrier.

13.3.4 Waves, propagation and resonance

Period is a completed recurrence and frequency counts recurrence against a reference duration. A generated wave propagates a retained change through connected support. Propagation speed, wavelength and frequency are three typed coordinates of that one relation. Superposition composes compatible carriers; interference records their predecessor merging; diffraction records how a constrained support redistributes the propagated relation. Polarisation is a held transverse orientation. Dispersion occurs when the support makes propagation condition-dependent. Resonance is recurrence matching between a driver and an allowed response. Wave energy and momentum remain transported quantities rather than being inferred from visual periodicity alone.

Bounded-cavity recurrence makes the allowed modes discrete. The exact round-trip relation is $2q$, and positive whole recurrence gives $2qn$. The quarter-wave family retains the odd harmonics. A common recurrence can carry distinct longitudinal and transverse modes, so the existence of a shared frequency does not make their physical speeds or supports identical. Exact resonant transfer closes only with a complete energy ledger: input, stored response, output and loss must all have a named route.

That ledger is the reason the Tesla return can be stated persuasively without erasing physical boundaries. Its five records support the declared resonance classes, but an Earth-cavity response remains different from an ideal quarter-wave resonator; material and radiative losses remain; longitudinal and transverse propagation are not silently assigned one speed; and no missing energy is converted into a claim of source-free power. The family extends the wave laws through measurement while retaining the exact class that each measurement can actually test.

13.3.5 Thermodynamic carriers and the direction of observation

Thermodynamics begins when a physical observer cannot retain every microstate distinction in the macrostate record. Temperature orders the accessible thermal carrier. Equilibrium is an equivalence relation between systems that share the relevant thermal boundary. Heat and work are distinct transfer traces even when both alter the same energy ledger. Closed energy accounting requires every transfer to be retained.

Entropy is generated from unresolved support. It counts the distinctions that remain open to the declared observation class; it is not introduced as an already available logarithmic continuum. Irreversible observation acquires an orientation because the later record has distributed distinctions that the declared observer cannot reverse without restoring the complete trace. Microscopic determinism is not contradicted: the microtrace remains held in the full state even where the macro-observation cannot recover it.

The finite unattainability boundary replaces an appeal to an actually completed limit with a positive finite stopping statement. Statistical weight is an exact ratio over generated support. A thermodynamic state relation then composes the retained quantities; phase coexistence distinguishes multiple stable organisations on common conditions; kinetic transport and collision mixing describe redistribution; fluctuation remains finite; and susceptibility, response and transport are typed correspondences between a perturbation and the resulting record.

The one-distinction erasure law closes with a Maxwell-demon ledger. Erasure is not simply deletion from the universe: a distinction is transferred into the environmental record, and the cost is evaluated at the declared thermal boundary. The post-seal Landauer comparison tests that relation. A favourable comparison supports the registered erasure cost; it does not turn an information-theoretic statement into a universal licence to ignore apparatus, temperature or implementation.

13.3.6 Vacuum, extraction and restoration

The vacuum sequence begins with the half-One floor and its oscillator spectrum, then constructs the perpetually live odd-denominator recurrence and the direction of vacuum polarisation. Vacuum and inertia acquire an exact unity relation at their common exchange boundary. A positive asymmetric beat can make a local transfer available, but any returned cycle must restore the full energy ledger. This is the point at which a theoretical drive relation and a claim of free energy separate decisively.

The terminal family carries a local resonant drive channel, driven vacuum-to-inertia co-variation, a finite-depth positive lower bound and the complete six-label restoration ledger. It admits a local state change where the pump and boundary are present. It does not admit an unpowered cycle. The apparatus question remains an empirical extension: without the registered public driven measurement, implementation is unresolved even though the formal relation closes. Ground-state response, boundary response and a unity anchor have narrower evidential roles and remain named as such.

13.4 The fine-structure derivation

The fine-structure result is the clearest compact example of the branch's method because every component is generated before the measured comparison. The binary Fold supplies $b=2$; the three-position return supplies $c=3$. The least binary covers of $3^3=27$ and $3^4=81$ satisfy

$$2^4 < 27 \leq 2^5, \quad 2^6 < 81 \leq 2^7,$$

so the selected cover depths are five and seven. The seventh binary power is $2^7=128$, the square of the three-position carrier is $3^2=9$, and the complete directed cover is $2 \cdot 5^3=250$. The admitted return ladder is

$$(125, 175, 245, 343).$$

Its exact finite correction is

$$C = 250 + \frac{1}{175 + \frac{1}{245 + \frac{1}{343}}} = \frac{3676744786}{14706643}.$$

The inverse fine-structure value and its reciprocal are then

$$\alpha^{-1} = 128 + 9\frac{C+1}{C} = \frac{503846395469}{3676744786} = 137.035999177180855...,$$

$$\alpha = \frac{3676744786}{503846395469} = 0.007297352564321794....$$

The formal package enumerates 1,024 candidates, retains one survivor, passes four controls and reconstructs independently. The exact derivation is claim SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001, with receipt `sha256:b7b1663a8ed78da756a2624d0bca690ac2ade089a9396060c96f141a6f2896a2`. The empirical successor uses a distinct 256-form comparison grammar. The sealed value lies inside the registered CODATA interval `[137.035999156,137.035999198]`. That evidence record is

SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001, with receipt

`sha256:13cf55d69e81157f6a00ec4968703133953f817c337147c7d3f160a86d52210a`. The source interval validates the sealed numerical correspondence; it is not used to choose the formal survivor.

13.4.1 Internal convergence and falsification

The four exact promotion stages are retained, not only the terminal decimal. Their inverse values are

$$\frac{34259}{250}, \quad \frac{5995462}{43751}, \quad \frac{1468922449}{10719245}, \quad \frac{503846395469}{3676744786}.$$

They record how the finite cover changes as each permitted direction is promoted from depth five to depth seven. The terminal stage is selected by the complete ladder and stopping rule. The measured target does not decide which stage to publish.

The 1,024-form grammar varies the registered structural choices. Omission of a required carrier, an alternate cover, an incomplete ladder or an extra rule is decided explicitly. Minimality asks whether any retained part can be removed; named-shape uniqueness asks whether a differently named candidate preserves the same complete relation. The four adverse controls test dependency, census, tamper and reconstruction boundaries as registered in the package.

The empirical successor has its own 256 candidates because measurement custody is a different question from formal derivation. It requires the sealed formal value, exact source interval, matching target commitment, complete row and tampered control. If the sealed fraction had fallen outside the CODATA interval, the empirical claim would have halted without changing the formal receipt. If a new source later reports a disjoint interval, that result would be adverse evidence and would not authorise selection of a different ladder by prose.

13.4.2 Parameter freedom and dimensional restraint

No numerical coefficient in the exact fraction is fitted to the CODATA target. The numbers two and three are admitted upstream carriers; five and seven are least-cover depths; 128, nine and 250 are exact descendants; and the ladder is the complete finite promotion family. This dependency route is what the phrase "zero free parameters" means for the claim.

The result is dimensionless. It can therefore enter later ratios without a unit adapter. When it is composed with a measured dimensional reference-as in the Higgs or proton-radius result-the reference is named post seal. The exact dimensionless derivation and the dimensional numerical consequence retain separate source and uncertainty status.

13.5 One constants object rather than a catalogue of fitted numbers

The branch does not treat the principal constants as disconnected guesses. The Unified Constants Object binds one rooted Fold geometry to $b=2$, $c=3$, spatial and boundary ranks three and two, cover depths five and seven, the fine-structure value, lepton and quark structures, cosmological shares, the proton-Planck hierarchy and the vacuum relations. Its current identifier is SFT-PHYS-UNIFIED-CONSTANTS-OBJECT-077; its receipt is

sha256:e6112bcfa663b3d24727bd03688491f55fdb5fc32359542b92acee7e551fbac8. "Unified" here names the declared dependency object. It does not mean that a single successful comparison empirically confirms every member, or that contingent initial conditions have been derived.

Several exact descendants illustrate how the common object is tested at different physical boundaries.

Result	Exact SFT form or value	Current empirical classification
Inverse fine-structure constant	503846395469/3676744786	Sealed numerical value inside the registered CODATA interval
Charged-lepton construction	Cubic invariants (1,1/6,1/485,3/1454), terminal α^3 refinement and Koide value 2/3	Adjacent mass ratios lie inside the CODATA-derived intervals
Higgs ratio and mass	$m_H/v=1/2+6\alpha/5$; with post-seal $v=246.22\text{ GeV}$, $m_H=31557437733819647/251923197734500\text{ GeV}=125.266104978...$ GeV	Inside the registered PDG interval [125.09,125.31] GeV
Higgs self-coupling	$\lambda=0.1294167525...$	Derived descendant of the same sealed relation and stated inputs
Electroweak share	1930922298157999/8642477221479757=0.223422318471252882...	Inside [0.22333,0.22351]; compatible W/Z test passes, all-input tension retained
Planck/proton squared hierarchy	$2^{127}(1-2\alpha/3)=1.693134633261878984... \times 10^{38}$	Inside the exact propagated CODATA range [1.693112648161510904...,1.693187333165341130...] $\times 10^{38}$
Electron magnetic anomaly	0.00115965218046558296...	Inside the registered CODATA interval
Muon magnetic anomaly	0.00116592071499941707...	Inside the registered Fermilab interval
Proton-radius coefficient	$4(1-\alpha/10)$, giving approximately 0.840621761 fm at the registered dimensional interface	Current intervals pass; the historical CODATA-2014 mismatch remains visible
Primordial scalar and tensor shares	31/32=0.96875 and 1/32=0.03125	Scalar inside Planck [0.9607,0.9691]; tensor below the registered $r<0.032$ boundary by 0.00075

The Higgs validation is retained under SFT-PHYS-VALIDATION-HIGGS-SYMMETRY-TERMINAL-066; the electroweak result under SFT-PHYS-VALIDATION-ELECTROWEAK-TERMINAL-003; the hierarchy under SFT-PHYS-VALIDATION-PROTON-PLANCK-TERMINAL-003; the muon anomaly under SFT-PHYS-QED-MUON-MAGNETIC-ANOMALY-004; the proton-radius return under SFT-PHYS-PROTON-RADIUS-TERMINAL-029; and the cosmological return under SFT-PHYS-COSMO-HUBBLE-CALIBRATION-001. These IDs identify distinct registered boundaries. Their numerical agreement is not merged into a single unqualified "match".

13.6 Matter, interactions, atoms and nuclei

Once quantity, direction, propagation and coupling are available, the next problem is persistent physical identity. The matter and interaction families derive held structures, charge kinds, coupling relations, mediator counts and mass or scale relations. The flavour work separates family structure from the measured masses that later instantiate it. The charged-lepton construction, for example, uses exact cubic invariants (1,1/6,1/485,3/1454) and a terminal α^3 refinement; the Koide relation closes at exactly 2/3. Only after that structure seals are the adjacent ratios compared with CODATA-derived intervals.

Atomic and molecular physics then owns bound-state, orbital and transition relations before Chemistry assigns chemical identity or reaction meaning. Nuclear and hadronic physics own nuclear carriers, closures and binding relations. The nuclear closure sequence is

$$(2, 8, 20, 28, 50, 82, 126, 184).$$

Its empirical ranking package examines all 2,548 registered AME2020 rows and retains nickel-62 as the unique maximum at the declared binding boundary. This is a complete registered source comparison, not a statement that every nuclear observable has been reduced to one number. The sequence becomes a critical upstream input to Chemistry's g-block and Smithium returns.

The physical quantum constitution remains deterministic at the level of the complete Fold state. Quantum or probability weights are exact ratios over an unresolved support, not fundamental floating-point randomness inserted as an axiom. No-signalling is separately derived and preserved. Consequently, determinism does not licence controllable signalling through unresolved state,

nor does a probability rule by itself establish a measurement outcome. The branch keeps carrier, unresolved support, measurement record and empirical apparatus in their distinct layers.

13.6.1 Physical quantum states and measurement

A physical quantum state is the correspondence between an exact reversible carrier and the physical preparation and observation records that instantiate it. Reversible evolution preserves the complete carrier. An observable is a declared question posed to that carrier; measurement creates a separate held record. The weight assigned to an outcome is an exact ratio over the complete branch support compatible with the retained preparation and question.

Incompatible observations are those whose complete records cannot be composed without changing the preparation or observation boundary. Their uncertainty relation expresses the retained alternatives, not a fundamental absence of a complete Fold state. Spin is a finite cyclic label action. Identical constituents retain no hidden name through exchange; antisymmetric composition therefore excludes the duplicated state. Barrier traversal is evaluated on the complete support, and bound-state spectra are the finite recurrent states that close under the declared boundary.

Entanglement is nonfactorisable joint support. The Bell family identifies the local-hidden-record factorisation assumption that fails when the complete preparation, setting and joint record are retained. Context-dependent joint observation makes the setting part of the state description. Decoherence is the distribution of a quantum record into the environment, and the quantum-to-classical limit is operational: a declared observer can no longer reverse or distinguish the distributed alternatives. The no-signalling claim is separate. Complete enumeration over the remote held fibre preserves the local marginal, so the joint dependence cannot be used as a controllable message channel.

The two-hand and full three-plus-one Dirac squares establish the finite carrier needed for spinorial physical structure. Their descendants include the bare gyromagnetic value, the finite turn projection, electron and muon anomaly relations, spin-statistics, occupation and Bose condensation, and the preparation-depth and joint-correlation laws. Post-seal experiments test these relations at their own boundaries: a superconducting-circuit Bell comparison tests the factorisation statement; Pauli-blocking, Bose-condensation and two-turn-spinor records test their named phenomena. No one experiment is made to carry the status of the entire quantum family.

13.6.2 Electroweak, colour and flavour organisation

The matter family first separates mass-energy correspondence, particle-antiparticle orientation, fermion and boson composition and conserved labels. The observed elementary-particle spectrum is an empirical inventory; the formal particle carrier and its possible classes are upstream. Composite hadrons add colour-singlet organisation. Scattering records a source, preparation, outgoing support and cross-section measure. Transition and decay retain every channel, while mixing and oscillation distinguish a propagation carrier from the state in which it is prepared or observed.

Electroweak Fold-fibre mixing, the exact PMNS squared-support triple and the exact squared W-to-Z relation occupy different levels. The on-shell electroweak share is the terminal numerical relation. Its complete binary support has sixteen cells; the construction retains three generator directions, charged support 15^2 , four neutral pair channels of 14^2 and a terminal $\alpha/17$ self-return. This gives the exact fraction already reported in Table 13.5. The PDG on-shell interval is the matching validation surface. A compatible-input W/Z relation passes, while the all-input tension remains an explicit adverse or boundary record rather than being averaged away.

The colour family derives a self-sourced running direction, maximal self-antipodal phase, exact colour coupling and a finite fixed-point boundary. Strong-carrier confinement and masslessness at One speed are retained together at the declared carrier. Finite accumulated coupling separation, two-fibre scattering, reciprocal mean free path, preserved and broken channels and inverse mediator range build the interaction description without importing a fitted running function. The colour-singlet excitation gap has a formal positive boundary and a separate post-seal lattice-spectrum comparison.

Quark channels possess exact invariants and cover depths. Two mass cubics and a terminal self-coupling dressing generate the quark mass relations; exact CKM fibre alignment and its mass-root physical mixing law then specify the mixing surface. The current source packages compare the dressed relations and CKM rows after seal. The same discipline applies to neutrinos: an exact splitting ladder and positive absolute-mass structure precede PMNS and CP comparisons. The single-hand Majorana discriminator and neutrinoless-double-beta noncancellation are standing empirical boundaries. Present status records do not convert an unavailable or null search into a detected Majorana carrier.

The common quark/lepton construction, depth-independent mass-ratio family, mirror-closed lepton carrier, mixing-row coupling return, common generation depth and lightest-quark confinement lift explain how the separate flavour results inhabit one dependency object. Terminal CKM, baryon-to-photon, proton/electron and scattering/decay/running claims close their

declared relations. The baryogenesis family retains the dependency and residue needed for an asymmetry without pretending that every contingent cosmological initial condition has been reconstructed.

13.6.3 The Higgs, charged leptons and precision descendants

The charged-lepton cubic is formulated through symmetric exact invariants so that irrational roots do not enter proof. Sum, pair sum, leading product and sharpened product are fixed before comparison. The terminal self-coupling refinement holds the unique complete electromagnetic contribution at the registered depth. Rational brackets then carry the predicted adjacent squared mass ratios to the CODATA intervals. The Koide invariant $2/3$ is an independent exact consequence of the Fold normalisation, not a fitted summary of the displayed masses.

The scalar family begins with half-One as the displaced ground and the leading rungs $1/2$, $1/4$ and $1/8$. Six-cell scalar support, its return and cover depth five force $m_H/v=1/2+6\alpha/5$. The dimensional reference $v=246.22\text{ GeV}$ is added only after the ratio seals. The resulting mass lies within the current registered PDG interval. The self-coupling is the native descendant $\lambda=(m_H/v)^2/2$. Individual ATLAS and CMS offsets and the broad direct self-coupling constraint remain in the complete record; the mass correspondence is not represented as a precision direct measurement of λ .

The electron anomaly uses the exact terminal turn $355/113$ with the admitted α to produce one exact rational result. The muon relation adds the exact second-generation correction

$$\alpha^2\left(\frac{2}{17} + \frac{\alpha}{106} + \frac{\alpha^2}{853}\right).$$

Both values are compared with their separate source intervals. The fact that the same α appears in both is a dependency fact, not permission to use the electron result to tune the muon correction.

The proton-radius relation similarly separates its exact dimensionless coefficient from the post-seal dimensional scale. Its present comparisons include electronic hydrogen, muonic hydrogen, CODATA 2022 and a conservative PRad interval. The disjoint CODATA-2014 interval remains historically adverse. That chronology is scientifically important: a current pass is not evidence that the earlier measurement never conflicted with the relation.

13.6.4 Atomic existence, spectra and periodic structure

Atomic physics begins with orbit-cell capacity obtained from three-space and boundary orientation. An exact existence boundary then limits the admitted atomic carrier. Cubic support supplies neighbour balance. The hydrogen ladder is a depth-independent exact recurrence; gross, fine and first-bound-return hierarchies refine the same carrier. Selection and field-splitting laws specify which transitions preserve the required held labels.

The terminal atomic family separately closes the Lamb-shift successor, fine-structure successor, hyperfine and twenty-one-centimetre successor, reduced-mass and Rydberg-scale relation, shell filling and periodicity, transition rates and lifetimes, multipole structure, and Zeeman and Stark splitting. NIST ASD and other registered records test the spectral consequences through source-bound intervals and line identities. A measured line is an observation of an atomic transition, while the formal selection rule is the generated relation that says which transition class is permitted. Neither is silently substituted for the other.

Molecular rotational and vibrational hierarchies appear in Physics where they are physical bound-state modes. Chemistry later owns the identity of the molecule, its chemical state and the cross-property evidence. The Physics terminal family includes rotational, vibrational and isotope spectroscopy at the physical carrier boundary. This one-owner distinction prevents the same spectral row from being counted as two independent confirmations merely because it is useful to two branches.

The shell and closure results provide Chemistry with capacities $2, 6, 10, 14, 18$ and the physical atomic boundary used in the g-block return. Physics does not claim the chemical separation, oxidation state or synthesis route of a future element. Those are genuinely downstream questions.

13.6.5 Nuclei, hadrons, decay and reaction channels

Nuclear binding begins as a relation between retained nucleon content and the bound whole. Nuclear levels and transitions require recurrent internal states; radioactive support records a state-changing route; reaction accounting retains every entrance and exit channel. Fission decomposes one nuclear carrier, while fusion composes carriers across the registered barrier. The formal direction, threshold and release relations are distinct from measured cross-sections or yields.

The exact colour-sector coupling and nuclear closure sequence lead to terminal nucleon composition, binding dominance and mass ordering. The complete AME2020 comparison evaluates the positive-composite binding-per-nucleon surface and finds nickel-62 uniquely maximal among 2,548 rows. That result is stronger than selecting a familiar example, because every

registered row participates; it is narrower than a universal nuclear theory, because the target and ranking statistic remain exactly those registered.

Light-hadron support follows the squared trajectory

$$Q(J) = \frac{6J - 3}{5}.$$

All five registered carriers lie inside their resonance-support intervals without a fitted slope or intercept. The dimensional Regge carrier is a later interface. Residual nuclear interaction retains an inverse-mediator range. Radioactive topology and exact survival distinguish a structural decay route from a source-bound half-life. Fusion and fission families retain direction, barrier, release, yield ordering and distinct threshold carriers. The deuteron claim separates binding and spin from the exclusion of the unsupported dinucleon states.

Decay widths, branching partitions and lifetimes are admitted only where the complete positive channel ledger and required dimensional information exist. A missing positive width or unit is a mandatory halt, not a numerical zero. This same rule becomes central in Chemistry's Smithium lifetime record.

13.6.6 New sectors as falsifiable extensions

The penta and hepta families continue the prime-sector carrier beyond the known three- and eight-sector anchors. Their exact couplings, charge kinds, mediator counts and beta slopes are generated independently. The category-clean particle census then separates sector gauge carriers, named nonsector bosons, known fermions and Smithion kinds before producing the total of 110. The no-extra-sector claim supplies the outside-particle falsification boundary, and the Smithion interaction claim specifies search signatures.

This is a standing-prediction structure, not a retrospective relabelling of a known particle list. The known sectors and fermions serve as registered anchors. The seventy-two new gauge carriers and twelve Smithion kinds await the declared searches. Model-dependent exclusions may constrain part of the surface; an unobserved carrier remains unobserved; an event becomes a discovery only under the registered identity and falsification protocol. These distinctions preserve both the force of the prediction and the possibility of its failure.

13.7 Vacuum, gravitation and the cosmological boundary

The vacuum family distinguishes a local unresolved floor from a global share. The local floor is $1/2^{20}$; the global vacuum share is $11/16$; and the normalised cosmological relation is

$$\Lambda(c/H)^2 = \frac{33}{16}.$$

The registered Planck interval comparison passes. The dimensional form enters only after the structural result has sealed, so the measured value cannot select the rational relation. The principal vacuum-scale validation is
SFT-PHYS-VALIDATION-VACUUM-DENSITY-SCALE-036.

The matter-share construction gives dark and baryonic shares $27/32$ and $5/32$, their exact ratio $27/5$, and a refined ratio $279/52$. The expansion family gives $13/12$ and its refinement $3305/3048$. The registered Planck and SH0ES comparisons pass at their declared interfaces. The scalar-tensor construction gives $31/32$ and $1/32$. The paper does not convert those fractions into an equality with an imported number of e-folds; that stronger statement is not admitted.

Gravitation inherits the three-dimensional carrier, rank-two boundary and inverse-square response, then adds physical source, hierarchy and measurement. The proton-Planck result is a squared hierarchy and retains that exponent in its statement. Physical cosmology is the outer ownership boundary of this paper: it supplies exact physical relations that Astronomy may use, while observations of particular celestial systems and a complete cosmic history remain with Astronomy and Cosmology.

13.7.1 Relational spacetime and causal structure

Spacetime begins from a relational event rather than a container existing before every physical relation. An invariant event interval compares two events while preserving causal kind. The accessibility cone separates events that can participate in a lawful propagation chain from those outside the declared causal support. Inertial frames are equivalent descriptions of a complete relation, and the limiting propagation speed is invariant under the admitted transformation.

Relative clock rate and relative spatial extent are derived as paired consequences of the same frame relation. The exact positive velocity composition law preserves the One-speed boundary without a negative proof quantity. Fizeau evidence is used as a post-seal discriminator for that composition. The experiment tests the physical adapter; it does not provide the algebraic form as a premise.

Inertial-gravitational equivalence introduces the common response boundary. Source-linked relational curvature records how the physical relation changes with source, and free propagation follows that curved relation. Field-source closure preserves the complete source ledger. A weak potential and Gauss closure connect the local relation to boundary flux. Static Fold-covariant clock and redshift claims test the same geometry from timing, while the gravitational-redshift and acceleration equivalence keeps those observational routes distinct.

The lattice-curvature family constructs exact finite curvature. Gravity self-sources through its field-energy ledger, producing a finite contraction towards the registered fixed point rather than an unbounded imported iteration. The four-coordinate symmetric curvature-source relation preserves conservation. The static spherical exterior family closes coefficient, flux and clock at one boundary; the quadrupole third-difference family closes the radiated-power structure at another. Their conventional correspondences and dimensional calibrations occur after the formal seal.

13.7.2 Gravitational radiation, horizons and nonstandard geometries

Gravitational-wave propagation retains one limiting speed and two graviton polarisations at the registered boundary. Quadrupole structure supplies the radiative carrier. The terminal observation law orders inspiral with rising chirp, merger and damped ringdown. Complete GW151226 and GW190521 records test that three-stage ordering while retaining the short-signal and alternative-source limitations. A correct order is not represented as a complete numerical reconstruction of every waveform parameter.

A causal horizon is the boundary beyond which the declared observer lacks an accessible return route. Strong-field structure retains a finite floor. Horizon information closure requires the complete record to remain in the underlying Fold state even where a local observation class cannot reconstruct it. Compact-object and horizon-thermodynamic claims then state their exact registered relations and post-seal comparisons.

Wormhole, warp-support and closed-timelike-recurrence claims are admissibility boundaries, not observations that such systems exist. Each asks which support, source and causal relations would be required and where the finite positive constitution halts. The post-seal nonstandard-spacetime record preserves the absence of a qualifying realisation where appropriate. Compatibility of a formal geometry does not become empirical confirmation of an artefact.

The quantum-gravity claim composes one finite Fold quantum carrier with one finite gravitational carrier. It does not assert that an imported continuum has been quantised, or that a numerical experiment proves nature at an unmeasured scale. Its status is exact within the declared composition grammar and open at further empirical boundaries.

13.7.3 Continua, fluids, plasmas and collective order

Finite coarse graining provides the continuum correspondence used for macroscopic matter. Density is exact content over support. Pressure and stress are transferred-response relations; flow conservation keeps the carrier ledger closed. Inviscid and viscous descriptions differ by the retained momentum-transport channel. Turbulence is a finite cascade with a declared scale and stopping boundary, not a completed infinity of eddies.

Plasma collective response adds charge-carrier organisation, oscillation and screening. Magnetofluid composition combines the flow and electromagnetic relations. Lattice order and excitation, band support and transport, collective phase transition, coherent superconducting transport and topological collective state provide the physical upstream objects for Materials. Exact lattice operators, causal cones and phase modes state the formal carrier. Radiation, acoustic, laser, plasma and Alfvén response laws retain their separate propagation and source conditions.

The criticality, universality and turbulence family distinguishes an exact scale relation from a measured exponent. Its current correction record remains visible where a measured-value route changed. Post-seal comparisons retain blackbody, acoustic, laser, plasma, Alfvén, critical-exponent, universality and turbulence rows. These sources test a complete collective-response surface; they do not collapse distinct media into a single universal measured value.

13.7.4 The physical cosmology handoff

Propagation redshift, relational expansion, background radiation, lensing, causal distance and finite structure growth are physical observation relations. The Physics-to-Astronomy handoff states exactly which carriers move downstream. Astronomy owns the identity and evolution of stars, galaxies and other celestial systems; Physical Cosmology retains the universal physical relation used to measure them.

The current Physics paper includes returned stellar hydrostatic, nuclear-chain, collapse, heavy-element, galactic-response and tidal-lock families because they validate declared physical laws. Their observations are not used to take ownership of the complete astronomical system. The stellar and galactic validation records retain target chronology and source scope. When a

result is a blind terminal validation, it retains that status; where a conventional model or already opened value shaped development, it remains a comparison or reconstruction.

The spatial-flatness family records a complete partition with an absent curvature remainder; "absent" here is structural, not the imported number zero. The depth-five cosmic energy budget and the composite Fold-orbit topology retain their exact boundaries. Inter-universe transport is an admissibility boundary, not a measured passage. Thermal history, nucleosynthesis, recombination, primordial support and structure-growth claims retain their separate source records and measured-value corrections. Thus the cosmological chapter can be ambitious without turning every formal extension into an observed history.

13.8 Returned physical families and standing predictions

Later field-wide work returned three major physical families after the pre-extension Grand Lock. Each remains separately dated and audited.

The Tesla resonance family derives the bounded round-trip length $2q$, mode sequence $2qn$, odd quarter-wave harmonics, one longitudinal plus two transverse components and a complete energy ledger. Five source records test the registered relationships. The claim is SFT-PHYS-VALIDATION-TESLA-RESONANCE-FAMILY-082, with receipt sha256:270786783cb68b4737d35faf5385079767c22036b03dc8f83965bb1eb2467b92. The record preserves the difference between an Earth-cavity response and an ideal quarter-wave class, retains propagation losses and distinct speeds, and makes no source-free or unlimited-power claim.

The vacuum/inertia family derives a local drive from one retained state to another, a ratio of One, a floor $1/2^{k+1}$ and a six-label restoration cycle. Its current family record is SFT-PHYS-VALIDATION-VACUUM-INERTIA-DRIVE-FAMILY-087, with receipt sha256:ae878e3fe8666a5471b9d19a2f67ab235b05f2187f3e9934523cc0e1661f4ca1. The official evidence preserves NIST ground-state and boundary-response records and the CNES unity result at their actual roles. It also preserves the fact that the proposed apparatus remains theoretical, that no public prototype has supplied the required driven test, that pump energy is required and that no source-free gain has been established. The CNES record is not silently reclassified as the absent driven experiment.

The new-sector family closes exact penta and hepta structures. The penta sector has coupling $4/5$, five charge kinds, twenty-four mediators and slope four; the hepta sector has coupling $6/7$, seven charge kinds, forty-eight mediators and slope six. The combined inventory gives 110 fundamental kinds. Seventy-two new gauge carriers and twelve Smithion kinds remain standing predictions. Their current family record is SFT-PHYS-VALIDATION-NEW-SECTOR-COMPLETE-FAMILY-095, with receipt sha256:2f13335a3bfdd7a395689ff113676a6c3287d90c7b4c8e49eac53730d5c5f2bd. Model-dependent exclusions and null searches are relevant evidence records; they are not discoveries and do not, without the registered falsifier, retire the complete prediction family.

13.9 Physics evidence reconciliation

The Physics Grand Lock is SFT-PHYS-VALIDATION-GRAND-LOCK-076, with receipt sha256:93f4497d6f7ef3c477246079f62c21f96a7ae27fd9516fa876e9d413bbab569e. Its purpose is custody, not rhetorical aggregation. The current paper therefore retains all fourteen pre-lock unfavourable or scope-boundary records, the electroweak all-input tension, the historical CODATA-2014 proton-radius mismatch, the unavailable driven vacuum/inertia test, losses and class distinctions in the Tesla family, and the unobserved status of the penta, hepta and Smithion sectors. A current interval pass does not erase a historical adverse comparison. A later, stronger source can change the current status only where the evidence constitution permits reclassification; the chronology and earlier record remain visible.

The present closure is consequently precise. Physics is complete for its 368 registered current categorical claims, their candidate censuses, controls, dependencies and declared empirical packages. It does not claim contingent initial conditions, a physical completed infinity, an imported continuum, all of astronomy, or downstream chemical and material ownership. The 238 empirical-validation claims are not universal confirmation, and the 130 formal or independently replicated claims are not relabelled empirical merely to produce a simpler headline.

13.9.1 Validation order and measured-value successors

The post-seal family is arranged in the same order as the physical dependencies. Molar Planck, momentum, force, electric potential, electric field, frequency, wave energy and molar thermal records first test the basic measurement adapters. The inverse-square, inverse fine-structure, nuclear closure, Koide, electroweak and proton-Planck packages then test exact headline relations. Each successor names the formal claim it evaluates; the empirical receipt cannot overwrite that formal package.

The vacuum series distinguishes a nonempty floor, running direction, a unity anchor and a boundary extraction test. Dynamics and spectra, relativistic and electromagnetic fields, optical operations, finite-loop boundaries, clocks, equivalence, gravitational waves and horizons follow. The nonstandard spacetime row is an admissibility and realisation boundary rather than a positive observation.

Quark dressing, CKM, proton/electron, neutrino, Majorana, magnetic-anomaly, atomic, velocity-composition, vacuum-density and thermal-history packages each retain their own source class. Collective radiation, acoustics, laser, plasma and Alfvén records do not become particle-physics evidence simply because they share upstream wave laws. The empirical graph makes the dependency visible while preserving the ownership and observation type.

Several claims are explicitly labelled blind in the active paper, including the terminal electroweak, proton-Planck, Higgs, stellar and compact-object validations. Others are post-seal, versioned, corrected or development-bound. The chronology field, not the prominence of the result, determines which word is lawful. A later measurement inside a prediction interval can validate the current consequence while the original date and target custody remain fixed.

13.9.2 Measured-value corrections

The final validation surface includes named measured-value corrections for the electroweak relation, common Fold scale, cosmic transport, criticality and turbulence, thermal history and helium, and the hadron Regge family. A correction is not hidden polishing. Its record identifies the earlier value or adapter, the corrected route and whether the formal law, evidence target or status changed.

The proton-radius chronology illustrates a lawful update without historical erasure. The current coefficient and scale translation agree with the registered present intervals, but the earlier CODATA-2014 comparison remains unfavourable. The electroweak family similarly preserves an all-input tension while the on-shell target passes. Criticality and turbulence retain condition-bound measured exponents rather than forcing them into one exact number. Hadron support preserves the distinction between a dimensionless exact trajectory and the dimensional source adapter.

Corrections to source transport are treated the same way. If a previously unavailable record later becomes accessible, current access may be updated, but the failed transport remains in chronology when required. If the content does not test the registered discriminator, successful download does not become successful evidence.

13.9.3 What the Grand Lock establishes

Formal Grand Lock 075 binds the pre-extension ownership surface and its acyclic 534-node dependency dictionary. It verifies root reachability and generator dependence. The generator-change control moves twenty-one of twenty-one generator-dependent headline values under the registered $3 \rightarrow 4$ perturbation, while half-One, spatial rank three and boundary rank two retain their independent status. This tests dependency sensitivity; it is not an experiment on nature.

Empirical Grand Lock 076 reconciles 234 pre-lock empirical claims and 147 distinct source identities, preserving fourteen unfavourable or scope-boundary claims and six legacy receipt shapes. The later nineteen claims have their own completion records. Consequently, the Grand Lock establishes that no row is omitted at its frozen boundary and that every retained row has an auditable status. It does not declare every outcome favourable, retrospectively seal later work or prevent a future lawful falsification.

The active inventory is

output/release/physics-1.3.0/05_Physics-Current-Categorical-Inventory.json, whose registered identity is sha256:203e84420470fd8e1adc81f260629aa1fadbcb7bd33a2e268202c12f6b04ff5b. The release manifest is output/release/physics-1.3.0/03_Physics-Paper-001-v1.3-Manifest.json, with SHA-256 6123007fa373352f299abe86c0dfa2a6b0a08d6ae8006aa55d24ff418e578560. Full claim receipts, candidate decisions and source rows remain in the audit and machine layers rather than being repeated mechanically here.

13.10 Handoff to Chemistry

Physics hands Chemistry a physical carrier with exact direction and boundary, charge and coupling structures, atomic and nuclear closures, orbital and transition relations, energy and scale interfaces, and measurement custody. What remains unresolved is chemical rather than merely physical: which physically possible carriers count as the same chemical species; how composition and bond establish a molecule; how reaction changes one chemical identity into another; how solvent, temperature and environment condition a chemical outcome; and how those claims can be tested without treating a database value as an axiom. Chemistry is required because physical possibility does not by itself supply chemical identity.

14. Chemistry: identity, composition, reaction and molecular evidence

Chemistry begins from the physical elements, interactions, orbit structures and measurement interfaces already admitted. Its new carrier is the chemical individual: an identity stable enough to be composed, bonded, transformed, separated and measured, yet sensitive to charge, stereochemistry, phase, environment and reaction route. The branch must therefore distinguish a formal chemical constitution from the many source- and condition-bound values through which real substances are observed.

The active Chemistry paper is version 1.3.0. Its frozen discipline census has **272 of 272 registered obligations**. Nine later lineage and Smithium returns bring the live paper to **281 claims**. Their packages enumerate **71,936 candidates**, retain **281 unique survivors**, pass **1,124 of 1,124 controls** and reconstruct independently in all 281 cases. Closure is `depth_independent` for 276 claims and `finite_complete` for five. The evidence surface contains **273 post-seal empirical packages** and **eight formal-only claims**; those eight are not presented as measured results.

14.1 Scope, ownership and family census

Chemistry owns chemical identity, elemental and isotopic chemical roles, composition, bond, acid-base and redox relations, reaction, catalysis, stereochemistry, molecular property, chemical phase and transport, kinetics, organic and inorganic organisation, electrochemistry, radiochemistry, analysis, computational chemical representations and polymer chemistry. Physics remains the owner of the upstream particle, atomic and nuclear laws. Materials owns bulk specimen response, microstructure and phase-property relations. Biology owns living function and biochemical organisation; Medicine owns clinical outcome; Earth Sciences owns environmental history; and Astronomy owns cosmic context.

The field census is not a list assembled after the results were known. It is a registered surface divided into foundation, complete-field, validation and handoff families.

Chemistry family	Claims	Principal boundary
Measurement	8	Chemical quantity, specimen, method, uncertainty and comparison
Elements	10	Elemental identity, ordering, isotope and periodic relation
Composition	7	Constituent count, ratio, formula and conservation
Bonding	12	Retained chemical relation, bond order, geometry and change
Acid/base and redox	10	Transfer, oxidation state, electron/proton custody and balance
Reaction	12	Reactant-product identity, stoichiometry, pathway and completion
Catalysis	8	Route change, retained catalyst identity and conditional response
Stereochemistry and polymer foundation	8	Orientation, isomer identity, recurrence and chain relation
Analytical foundation	8	Signal, calibration, separation and identification
g-block and Smithium return	11	Periodic extension, returned element and identification protocol
Molecular electronic structure	16	Electronic carrier, state, orbital and correlation
Molecular structure and properties	14	Geometry, symmetry, conformation and cross-property identity
Thermodynamics, phase and transport	19	State functions, phase change, mixture and transport
Kinetics	13	Rate, order, activation, isotope effect and mechanism
Inorganic chemistry	17	Coordination, solid molecular identity and inorganic reaction
Organic chemistry	16	Functional relation, aromaticity, synthesis and transformation
Electrochemistry	13	Electrode, cell, potential, current and chemical work
Nuclear and radiochemistry	12	Isotope transformation, decay chemistry and source custody
Analytical chemistry	22	Complete measurement, separation, spectroscopy and uncertainty surface
Computational chemistry	14	Representation, model, approximation, verification and resource boundary
Polymer chemistry	13	Repeat unit, distribution, architecture, transition and degradation
Validation	12	Empirical reconciliation and Chemistry Grand Lock
Ownership handoffs	6	Typed transfers to Materials, Biology and later branches

Chemistry family	Claims	Principal boundary
Live family total	281	272 frozen discipline obligations plus nine separately dated lineage and Smithium returns

14.2 From physical element to chemical identity

The physical nucleus and electron structure do not yet define every chemical question. Chemistry first establishes what must be held fixed for two records to name the same element, isotope, ion, molecule or stereochemical state. It then derives composition as a counted relation between retained identities. Stoichiometry is not imported as a convention: it is the exact preservation ledger required when a chemical transformation changes relations without silently deleting the carriers that enter or leave the declared boundary.

Bonding follows because a composition alone does not determine a chemical individual. The bond family adds a retained relation, order, orientation and allowed change. Molecular geometry and stereochemistry become necessary when the same constituent inventory supports non-identical arrangements. Acid/base and redox families then track distinct transfer relations and their custody; they do not collapse proton transfer, electron transfer, oxidation state and measurement into a single word. Reaction families specify the complete reactant and product identities, stoichiometric balance and route. Catalysis changes the available route while retaining the catalyst's declared identity; it is not treated as unexplained acceleration.

This sequence is the Chemistry narrative spine: elemental identity permits composition; composition plus relation permits bond; bond plus orientation permits molecular and stereochemical identity; a typed change of those relations permits reaction; and a comparison between routes permits catalysis and kinetics. Each later family inherits the prior custody rather than redefining its inputs.

14.2.1 Chemical measurement and retained identity

A chemical entity is the least carrier whose relevant constituents and relations remain distinguishable through the declared observation. A species is an observation-equivalence class of such entities; a substance adds a composition identity at the sample boundary. Amount of substance is a counted reference carrier. A chemical formula is the reversible exact encoding of the registered composition, and nomenclature is admissible only where the name can be traced back to that identity without loss.

This sequence separates an object from the language used to report it. Two names may be equivalent if they decode to the same complete chemical identity; the same short name may be inadequate when isotope, charge, stereochemistry or phase changes the tested claim. Chemical measurement uncertainty retains every alternative allowed by the method, and reference traceability records sample, standard, calibration and source. The observation does not alter which entity the formal grammar selected.

A sample is not assumed homogeneous merely because it carries one label. Analyte and matrix are separate carriers. The method must state what portion was sampled, which species or property it can discriminate, its calibration, interferences and uncertainty. A result outside the method's discrimination boundary is unresolved or out of bound, not evidence that the chemical distinction is absent.

14.2.2 Elements, isotopes and the periodic carrier

Element identity is retained nuclear charge at the chemical boundary. Atomic number orders those identities. Isotopes preserve the element while changing the nuclear composition; an atomic-weight record is consequently a source-bound relation to isotopic composition, not an immutable exact value of the element in every specimen. Periodic order preserves nuclear identity and records the recurrence of outer chemical organisation. Group and period are compositional coordinates of that recurrence.

Valence availability supplies the boundary at which an atomic carrier can enter a chemical combination. Ion formation changes the held charge state while preserving elemental identity. The known-periodic-table observation claim stops at the observed elements through 118. The generated g-block, Smithium and endpoint returns are later standing predictions, deliberately kept outside the known-table observation statement.

The distinction between an exact ordering and a measured atomic weight is important. The ordering is formal once the element carrier is admitted. The weight depends on isotope distribution, source and reporting convention. A new isotope measurement can lawfully update the observed record without rewriting the element identity or periodic grammar.

14.2.3 Composition, mixtures and solutions

Exact composition counts retained constituents. Conservation through reaction requires the elemental identities entering the boundary to have a complete route into the products or declared outgoing support. Stoichiometric coefficients are positive counts of those routes. A limiting component is the first required constituent whose available count cannot support another complete product event. Yield is the retained product share relative to the registered limiting and recovery boundaries.

Mixtures preserve multiple component identities on common support. A solution adds a solute-solvent role distinction; it does not erase the constituents into a featureless whole. Composition may be reported through several lawful coordinates, but a conversion is valid only where all required amounts, volumes, masses, temperatures and reference states are available. A measured concentration is condition-bound. The formal conservation ledger cannot infer an unmeasured loss, side product or evaporation merely to make a preferred yield balance.

This composition grammar becomes a recurring control throughout Chemistry. Electrolysis must conserve charge and elemental identity. Polymerisation must retain repeat-unit and end-group routes. Radiochemical separation must retain the nuclide and its chemical state. Smithium entrance partitions must sum to the registered Z , N and A coordinates. In every case, a missing route halts rather than being represented by an imported numerical zero.

14.2.4 Bond, molecule and supramolecular relation

A chemical bond is a retained interaction closure between identified constituents. Covalent bonding shares support; ionic bonding transfers a held label; metallic bonding distributes collective support. Bond order counts the joining multiplicity. Bond length and energy are empirical or reconstructed coordinates attached to a particular state and method; they do not define the existence of the bond by themselves.

A molecule is a complete connected bonded carrier. Geometry adds adjacency and orientation. Isomers share composition while differing in one or more retained relations. Intermolecular interaction adds residual joining between otherwise complete chemical individuals. Supramolecular organisation requires reversible recognition, and a molecular network is a connected whole whose relations extend beyond a single molecular carrier.

The branch therefore avoids two reductions. A molecular formula is not a complete structure where adjacency or orientation remains unresolved, and an observed bulk association is not automatically a covalent molecule. The claim must preserve the carrier and relation appropriate to its boundary. This is particularly important at the Chemistry-to-Materials handoff, where a network or interface may produce a bulk response without creating a new molecular identity.

14.2.5 Acid/base, redox and electrochemical separation

The conjugate acid-base partition retains the paired states before and after a transfer. Proton-transfer and electron-pair donor-acceptor relations are separate because their carriers and evidence differ. Amphoteric behaviour is a dual-role relation under distinct partners or conditions. A buffer retains a conjugate pair whose composition supports a bounded response; it is not a claim of invariant pH under arbitrary addition.

Electronegativity is an ordering of held affinity. Bond polarity records an unequal relation, while oxidation state is an accounting coordinate. Coupled oxidation and reduction preserves the complete electron-transfer ledger. Separating the two sides of that transfer produces the electrochemical cell. Potential, current and work are downstream measured or derived relations; they do not replace the chemical identity of the half-reactions.

These distinctions prevent oxidation state from being mistaken for a directly localised physical charge in every bonding description. They also prevent a cell voltage from proving one preferred microscopic mechanism. A mechanism requires its own complete intermediate and elementary-step trace.

14.2.6 Reaction, mechanism, equilibrium and catalysis

A reaction is a source-bound transformation from complete reactant identity to complete product identity. A mechanism is an ordered set of elementary steps. An intermediate is retained where it is produced and later consumed on that path. The activation barrier is the boundary between the adjacent states; reaction rate counts completed transitions over an ordered recurrence, and kinetic order records the observed dependency multiplicity at the stated condition.

Equilibrium is balanced reversible support. Thermochemical accounting retains the energy transfer, while reaction direction requires the complete energy and distinction ledger. A chemical phase is a state identity; solubility is an equilibrium across solution and retained phase support. Photochemical reaction adds an excitation channel and therefore requires wavelength or state custody rather than treating illumination as an untyped cause.

A catalyst retains its identity through a completed reaction cycle. Its scientific role is to supply an alternative path. Selectivity compares the generated product routes, and turnover counts completed catalyst cycles under the declared conditions. A reaction network composes multiple paths; autocatalysis closes when a product increases the availability of its own generating route. Adsorption, colloidal dispersion and chemical transfer at an interface extend the reaction carrier to a boundary shared with Materials.

Catalytic evidence is therefore conditional. A higher measured rate can support the alternative-path relation when reactants, products, catalyst, temperature, transport and method are retained. It does not show that the catalyst changed equilibrium, that all competing paths vanished or that its identity was conserved unless those coordinates were measured.

14.2.7 Stereochemical, polymer and analytical foundations

Chirality is a held orientation not superposable under the admissible transformations. Enantiomers form the corresponding pair; diastereomers retain a different relation. Functional-group recurrence provides a local chemical role that can be traced across a reaction family. Organic reaction families compose those roles with the complete entering, leaving and rearranged carriers.

A polymer chain repeats a monomer-derived identity while retaining its connections and ends. A population requires the chain-length distribution; a single nominal molecular size cannot stand for every chain. The earliest Chemistry-to-Biology molecular handoff supplies identity and reaction capability but does not yet supply biological function.

Analytical foundations distinguish sample, analyte and matrix; establish calibration; record selectivity and interference; and derive complete identification surfaces for mass, infrared, electronic and rotational-vibrational spectra. A complete analytical result includes the falsification record. One peak without source, calibration and competing identity support is not sufficient merely because it appears at an expected position.

14.3 Periodic extension, the g-block and Smithium

The periodic return begins from the Physics nuclear closures and orbital capacities. The generated orbit capacities are

$$2, 6, 10, 14, 18,$$

and the nuclear closure sequence is

$$2, 8, 20, 28, 50, 82, 126, 184.$$

The next g-block is registered to open at atomic number 121. That standing prediction is SFT-CHEM-PRED-G-BLOCK-001, with receipt sha256:5e21d5cb1ad9758257dfa1fb732792adf4e9b7aa7ae0300fdaf21cb2426772bb. The returned element Smithium occupies the double closure

$$Z = 126, \quad N = 184, \quad A = 310,$$

with predicted configuration $[Og] 8s^2 5g^6$, eight valence electrons and possible oxidation states from +2 through +8. Its standing prediction is SFT-CHEM-PRED-SMITHIUM-001, with receipt

sha256:8028271d2d1e27a0c1351974b39d09131916ed17cd349b36268eaa6b8b2b2613. The periodic endpoint at $Z=137$ is inherited through the Physics fine-structure result and registered as

SFT-CHEM-PRED-PERIODIC-ENDPOINT-001, receipt

sha256:df9ea26853f6ac0678893d0e005f2a25a4d86ab3084ab3147353432050d1bf16. The observed periodic table remains through element 118; the derived extension is not rewritten as an observation.

The Smithium family goes beyond naming an element. Its synthesis grammar retains 22,875 entrance partitions, with gamma, alpha, beta and fission ledgers. A numerical lifetime cannot be admitted because the current package lacks a positive measured width with the required unit; the calculation halts rather than inventing one. Seven ion rows are retained. The 5g shell has capacity eighteen, occupation six and twelve holes, producing registered g-to-f and g-to-h electric-dipole transition classes. Twenty-one pairwise chemical-state distinctions and a five-record identification protocol specify what a future observation would need to establish.

The formal family and its current source package pass. The synthesis routes, cross-sections, lifetime, branching fractions, lines and separation behaviour remain unmeasured or unresolved. Current non-observation is neither a null result nor retirement of the prediction. That distinction is essential: a failed access, an experiment not run and a completed experiment with a null outcome are three different records.

14.3.1 Synthesis grammar versus feasible route

Every one of the 22,875 entrance partitions is a labelled positive two-body composition whose coordinates sum to $Z=126$, $N=184$ and $A=310$. That census proves completeness of the conservation grammar. It does not promote any partition into an experimentally feasible reaction. Feasibility would additionally require projectile and target availability, beam energy, barrier and compound-state relations, a positive cross-section, survival against competing channels and a measurable yield.

The distinction matters because a conserved route can be physically possible within one grammar and experimentally negligible or unavailable. The formal census identifies the complete search surface. Future measurements can then rank, exclude or observe rows without changing which partitions conserve the registered coordinates.

14.3.2 Decay, lifetime and identification

The gamma, alpha, beta and fission ledgers enumerate permitted channel coordinates while preserving daughter identity and conservation. A channel's presence in that grammar is not a measured branching fraction. Numerical lifetime requires a positive width and unit at the declared source boundary. Because that input is absent, the lifetime package halts explicitly. The halt prevents an arbitrary lifetime estimate from entering the standing prediction.

The ion and oxidation ladder, g-to-f and g-to-h transition classes and twenty-one chemical-state distinctions provide independent identity axes. The five-record protocol requires those axes to be composed under common custody. One unexplained event, one mass coordinate or one spectral feature is not by itself the complete Smithium identification.

The prediction is therefore designed to become more discriminating when data arrive. A measured cross-section, decay line or chemical separation can fill one present unresolved coordinate. A contradictory observation can be adverse. Neither outcome is permitted to rewrite the earlier seal or pretend that the observation was known before the prediction.

14.4 Molecular structure and the cross-property vector

Chemical identity becomes experimentally useful only if the same retained molecule can be traced across different property records. The molecular electronic and structure families establish state, orbital, geometry, conformation, symmetry and correlation carriers. The cross-property vector then tests whether records from distinct sources preserve a common molecular identity rather than comparing unrelated rows.

SFT-CHEM-CROSS-PROPERTY-MOLECULAR-VECTOR-014 covers thirteen property families, 9,025 source rows and 1,104 molecular carriers. Of those carriers, 676 occur on overlapping property surfaces, accounting for 6,676 overlap rows. Its receipt is `sha256:c3953d770e1837fc70536ba386e631a93a812b8e1008fe71d1635af1386999b8`. The count is an exact custody result for the registered source surface. It is not a claim that a molecule possesses a context-free scalar summary or that every property has been measured under the same conditions.

The property packages preserve state, method, temperature, pressure, preparation, source version and uncertainty where available. Molecular-state and correlation records retain adverse and unavailable rows. The NIST SRD 101 torsion surface, for example, contains forty-one species, forty-four torsion targets and 782 path states, with one unresolved row preserved. An unresolved row is not removed from the denominator to improve the apparent outcome.

14.4.1 Molecular electronic and joint-state constitution

The molecular electronic carrier preserves nuclear identity, electron count, held spin and the relation joining the parts. Nuclear-electronic, vibronic and rovibronic states are exact compositions of those typed carriers. Fermionic occupation retains exclusion; ground and excited states are finitely ordered; degeneracy is an equivalence under the stated symmetry rather than a deletion of the distinct supporting records.

Complementary electron-pair exchange and molecular exclusion determine which joint states may close. Nonfactorisable molecular support records correlation between parts, while dissociation names the boundary at which the complete molecular carrier becomes separate products. Multicentre and delocalised support remain connected relations; they are not forced into a pairwise bond description when that description loses the registered distinction.

A molecular-state transformation specifies the initial carrier, terminal carrier and permitted transition. Observation classes and selection rules state which distinction a method can resolve. Configuration order, basin, barrier and path distinguish a stable state from the route between states. Isotopologue laws retain nuclear change while preserving chemical identity and the scale separation relevant to the comparison.

The chemical measurement-reduction law connects the exact joint state to the record available to a real instrument. The operational classical-quantum correspondence names the boundary at which a chemical description may use a classical state

ledger without erasing the underlying quantum support. Neither law asserts that an instrument observes the complete formal state directly.

14.4.2 Geometry and property relations

The H₂/D₂ bond-length and dissociation-path claims test isotope-preserving relations on named molecular carriers. Symmetry can force a bond angle, while a dihedral or torsional state requires a held axis and orientation. Dipole organisation preserves charge separation and uses a squared-magnitude relation only after the vector carrier is defined. Polarisability is a response to a specified perturbation, and its isotropic composition is distinct from a direction-resolved measurement.

Ionisation energy and electron affinity are state-order and magnitude relations with terminal identities. Vibrational recurrence and rotational constants preserve their axes and mass distribution. Intermolecular binding is ordered by the retained states; magnetic response and formation energy each have their own source and condition boundary. The cross-property vector does not claim that one coefficient predicts all thirteen families. Its zero-parameter content is the common source-bound molecular carrier and the complete law by which the distinct property records are composed.

This is why the overlap counts matter scientifically. A table with 9,025 rows could still be a collection of unrelated measurements. The 1,104-carrier identity map and 676-carrier overlap surface test whether the same molecule, state and provenance can be followed across properties. A separate fitted coefficient for each family is a registered falsifier of the claimed common carrier, not an accepted escape route.

14.5 Thermodynamics, phase, mixtures and transport

Chemistry requires a state ledger because the same composition may support different phases, and because work, heat, mixture and transport depend on the declared condition. The entropy construction defines the formal object as the complete ledger of unresolved distinctions. A conventional scalar entropy is introduced only at the post-seal measurement interface. In the registered water surface, thirteen rows are retained and the phase jump is

$$\frac{13637763}{125000} \text{ J mol}^{-1} \text{ K}^{-1}.$$

The independently reconstructed ratio is

$$\frac{4066845420}{37275593}.$$

The record is SFT-CHEM-ENTROPY-MULTIPLICITY-CORRESPONDENCE-005, with receipt sha256:be32db30904fd7974578c60b3e656371d428f718d2b9dec8097277d135f32ce1. The exact source arithmetic establishes the registered correspondence; it does not make every thermodynamic entropy a count free of method or condition.

The colligative family retains composition-dependent response across 276 records: 144 boiling-point rows, thirty-seven freezing-point rows and ninety-five osmotic rows. The claim is

SFT-CHEM-COLLIGATIVE-COMPOSITION-RESPONSE-014, with receipt

sha256:f8c933f2777154b409d33968b254da03d8786cab4bb3dfc87e7686d5aef17177. Because boiling, freezing and osmotic measurements use different methods and conditions, their results remain typed within the joint ledger rather than being compressed into one undifferentiated success rate.

Phase and transport extend the state ledger to boundaries, coexistence and movement through a chemical environment. They supply important upstream objects to Materials, but Chemistry retains ownership only where chemical identity, composition or reaction is the tested variable. Once bulk microstructure, specimen history and collective response determine the claim, ownership passes to Materials Science.

14.5.1 State functions without imported scalar premises

The finite microstate-support law names the complete distinctions available to the chemical carrier. Temperature is a correspondence between thermal ordering and the chemical state. Internal energy composes the retained contributions; heat and work remain different transfer routes. Entropy records unresolved support at the declared observation boundary.

Enthalpy-equivalent state, free-energy-equivalent reaction direction and chemical-potential-equivalent component laws are derived as exact relations among the admitted carriers. The conventional dimensional values and source tables enter after seal. This prevents a handbook free energy from becoming the premise that selects the reaction direction it is later said to validate.

Activity records non-ideal composition at the chemical observation boundary. Fugacity supplies the corresponding real-gas mixture relation. Neither is an arbitrary fitted correction in the formal grammar; empirical values remain condition- and

method-bound. The phase-rule relation counts the independent components and phase constraints. One-component and multicomponent phase boundaries retain their complete state and path.

Colligative response follows from composition while preserving the distinct boiling, freezing and osmotic measurement routes. Solvation and dissolution are free-order relations between the retained states. Diffusion counts molecular transport. Viscosity and thermal conductivity remain composition-bound responses. Coupled mass, heat and charge transport keeps all three ledgers rather than using one phenomenological coefficient as an unexamined primitive.

The empirical consequence is disciplined rather than sterile. Exact composition and state relations can be asserted strongly where their grammar closes. A measured boiling shift, diffusion coefficient or conductivity is reported with its method and condition. Agreement establishes the named correspondence; disagreement remains adverse at that boundary and cannot be removed by choosing a different post hoc convention.

14.5.2 Phase and transport ownership

Chemistry owns phase where the claim identifies chemical state, coexistence, solubility or reaction. Materials owns phase where microstructure, processing and bulk property are the tested consequence. This is not an arbitrary split. The falsifiers differ: chemical phase identity may be disproved by composition or molecular-state evidence, whereas a material phase-property claim may fail because texture, grain, defect or specimen history was not retained.

Transport has the same typed boundary. Molecular diffusion and chemical potential belong to Chemistry. Transport through a porous scaffold, heterostructure or processed electrode belongs to Materials. Organism-level transport belongs to Biology. The one-owner graph lets the downstream claim depend on the upstream law without counting one relation repeatedly as a new derivation.

14.6 Kinetics, mechanisms and aromatic recurrence

A balanced reaction states what changes; kinetics asks how the registered change occupies time and route. The kinetic families derive rate, order, activation and mechanism carriers while preserving the distinction between an elementary step and an inferred multistage reconstruction. A sequential mechanism package retains an adverse one-millisecond contamination record and an unresolved late-stage boundary. Those rows remain part of the mechanism's current scientific status.

The isotope-effect vector contains ninety explicit rate ratios and three direct decay effects, drawn through forty-seven pages and twenty-three worksheets. Normal, inverse, near-equal and adverse outcomes are all retained. The family therefore establishes a complete registered evidence surface, not a rule that every isotopic substitution changes a rate in one preferred direction.

The aromatic recurrence identifies the generated support sequence

6, 10, 14.

Its stability comparison retains an excess of $150.39 \text{ kJ mol}^{-1}$, uncertainty 6.60 kJ mol^{-1} , and lower envelope $143.79 \text{ kJ mol}^{-1}$. The evidence archive comprises fifty-nine tables and 353 rows; three sources were captured before their outcomes were opened. The claim is SFT-CHEM-AROMATIC-RECURRENCE-STABILITY-003, with receipt sha256:5339c34ad1f819bbefa0ebed15404358964ec90488544bdc060c5feff9ea9b28. The chronology of those three sources is preserved, while already available sources retain their retrospective or development status.

14.6.1 Rates, paths and branching

The elementary-transition relation counts complete changes of one declared state into another. Concentration dependence retains which reactant carriers participate in the observed rate relation. Temperature dependence retains the condition and recurrence boundary. An activation-barrier value is attached to the generated path, and the transition-state-equivalent claim identifies the finite path boundary without importing an unobserved conventional state as an axiom.

Competing channels form a complete partition. Sequential mechanisms compose terminal and intermediate states in order; parallel mechanisms preserve distinct routes to their products. A reversible kinetic relation and an equilibrium correspondence share carriers but answer different questions: kinetics concerns transition counts, while equilibrium concerns balanced support. Catalytic turnover counts a closed catalyst cycle. Diffusion-limited reaction identifies the boundary at which transport, rather than the local chemical step, limits the observed rate.

Reaction dynamics adds scattering preparation and product-state observation. The source record must retain what was actually resolved. A bulk rate cannot uniquely establish an elementary mechanism where multiple complete paths remain compatible. Mechanism language is therefore reconstruction language unless the registered evidence discriminates the path.

14.6.2 Isotope effects and chronology

An isotope substitution changes nuclear mass while retaining the relevant chemical identity. The kinetic isotope-effect law traces that controlled change through the path and rate relation. Normal, inverse and near-equal effects are all lawful possibilities; the sign and magnitude depend on the complete state and mechanism. The adverse rows test whether the registered law preserved enough of that boundary.

The ninety ratios and three decay effects are not compressed into a mean. Source page, worksheet, isotope identities, conditions and result class remain available. A source observed during development retains that chronology. A later outcome-unopened source may test the sealed relation prospectively. The paper does not promote the earlier rows to blind status simply because the same law later survived a stronger test.

14.6.3 Aromatic evidence as a model of layered status

The aromatic family shows why structural derivation and energetic evidence must be separated. The recurrence *6,10,14* is the exact generated support. The stability excess and uncertainty are source-bound energetic records. The positive lower envelope tests the registered direction while retaining the measurement spread. The three outcome-unopened captures supply a chronology class different from the already opened tables.

The result can therefore be stated strongly and exactly without claiming more than the evidence. The support sequence is derived at its grammar. The stability surface supports the declared positive relation. It does not prove that every aromatic system has the same energy, or that "aromatic" is one universal scalar property detached from molecule and method.

14.7 Complete-field chemistry

The inorganic and organic families extend the same identity and reaction constitution across coordination, functional groups, aromatic systems, synthesis and transformation. Electrochemistry separates electrode, cell, potential, current, transfer and chemical work. Nuclear and radiochemistry retain both the upstream nuclear transformation and the chemical state in which its products are separated or measured. The branches are adjacent but not interchangeable: a decay law does not automatically establish the chemistry of the daughter species.

Analytical chemistry supplies the complete measurement surface needed to know which chemical individual was present and how strongly the instrument record supports that conclusion. Calibration, separation, signal, interference, uncertainty and source transport remain explicit. A missing file or failed extraction is reported as unavailable, not as a negative measurement.

Computational chemistry owns representations and approximations used to infer chemical structure or response. Its claims name the model, resource boundary, verification route and relationship to observation. A successful implementation verifies the computation it performs; it does not turn the model output into an experimental observation. Low-confidence predictions, conflicting records, unavailable inputs and resource halts remain visible.

Polymer chemistry adds recurrence, repeat-unit identity, chain architecture, distribution, transition and degradation. It does not treat a nominal repeat unit as sufficient to fix a bulk material property. Molecular distribution, processing and morphology create the point at which Materials ownership begins.

14.7.1 Inorganic and coordination organisation

An inorganic coordination entity retains a central carrier and every ligand identity. Coordination number counts the complete central-ligand incidence; denticity counts the connected donor sites contributed by one ligand; chelation adds the topology joining those sites. Geometry is selected from the complete held orientation and adjacency relation. Coordination forms that preserve the same complete relation are equivalent, while isomers retain a difference that the admissible transformations cannot remove.

The ligand-incidence state partition and balanced splitting laws carry the electronic support into complex spin. Spin extrema and crossover are ordered states under declared conditions. Electronic transitions and selective colour have source-specific energy and observation boundaries; unpaired support determines a magnetic-state class but does not supply a context-free measured moment.

Organometallic chemistry adds a direct metal-carbon relation and exact electron-support accounting. Oxidative addition and reductive elimination are inverse changes at the registered carrier. Insertion, migratory insertion and elimination preserve an adjacency trace. Metal clusters require a finite connected carrier and distinguish metal-metal bonding from mere proximity.

Solid-state formula and local coordination remain Chemistry-owned where the claim concerns chemical identity. Defect chemistry and non-stoichiometry track composition and charge compensation. Once defect distribution and microstructure

determine a bulk property, ownership moves to Materials. The inorganic acid/base, redox and network claim closes the complete chemical route without absorbing the specimen-scale response.

14.7.2 Organic structure, recurrence and synthesis

Organic conjugation is a connected alternating support. Multiple symbolic representations may name the same carrier only when the complete molecular identity is preserved. Aromatic, antiaromatic and nonaromatic classes then separate by their recurrence and state relation. The exact aromatic support sequence and the registered stability surface supply a quantitative test; the class label itself is not evidence of a measured stabilisation energy.

Conformers are generated by the permitted orientation changes and quotiented by exact equivalence. Their populations and ordering are condition-bound. Substitution retains entering and leaving carriers; acceptor-driven aromatic substitution adds its directional support. Addition changes joining multiplicity; elimination is its complete inverse family. Rearrangement preserves composition while changing adjacency. Cyclic transitions retain the relative orientation through the path.

Polymerising organic networks require initiation, propagation and termination. Selectivity is a complete partition of the generated product support, not the reporting of one favoured product alone. A protecting group is a reversible state whose restoration must be demonstrated. Retrosynthesis enumerates admissible decompositions and requires exact forward reconstruction. A plausible disconnection is not admitted if the forward route cannot restore the target identity under the declared grammar.

The organic validation vector preserves reaction records that are favourable, adverse, condition-limited or unavailable. It does not average unlike mechanisms into one yield. Source conditions, catalyst, solvent, temperature, time, work-up and analytical identity remain part of the observation where the record supplies them.

14.7.3 Electrochemical relations

The electrochemical family begins with a half-reaction identity and a held transfer orientation. Electrode potential is a chemical relation to the named reference. Cell potential composes the two half-cell relations without losing their orientation. Concentration-dependent potential retains composition and condition. Electrochemical work and reaction direction then relate the cell carrier to the complete chemical transformation.

Electrolysis counts product amount through the completed transfer ledger. Ionic conductivity separates mobile charge, support and response; mobility and transference distinguish the contribution of each carrier. Electrode reaction rate, overpotential and polarisation retain the departure from the named reference route. The double layer is an interfacial charge organisation, not a bulk molecular identity. Coupled corrosion is a reaction network whose anodic and cathodic paths must both close.

Electrochemical storage reaches a one-owner boundary. Chemistry owns species, redox states, half-reactions and chemical work. Materials owns electrode microstructure, phase evolution, mechanical damage and bulk transport. Engineering owns cell architecture, control, safety and deployment. A measured capacity or cycle life cannot be moved between those owners merely to simplify the narrative.

The electrochemical validation vector retains potential references, calibration, concentration, temperature, current protocol, material identity and uncertainty. A voltage agreement may validate the registered composition law; it does not establish the microscopic rate path or device lifetime unless those were the sealed target.

14.7.4 Nuclear and radiochemical custody

A radiochemical claim must preserve both nuclide identity and chemical carrier. The radioactive transformation network records parent, daughter, route and time. Activity relates amount to ordered recurrence. A branching chemical yield retains every radioactive channel and the chemical recovery by which a product is counted. Transient and secular equilibrium are distinct finite relations between production and removal.

Isotope exchange preserves elemental identity while moving the isotope record between chemical carriers. Equilibrium and kinetic fractionation have different path and condition dependencies. A radiotracer is an inference tool: its custody must show that the retained signal follows the declared carrier. Radiochemical separation and decontamination distinguish recovery of the target from removal of other activity.

Fission-product distribution combines an upstream nuclear yield with downstream chemical state and separation. Radiation chemistry adds the network of chemical transformations produced by the deposited physical energy. The branch does not infer a chemical yield from a nuclear decay law alone or treat detector counts as a complete molecular identification.

Where the required half-life, branching fraction, activity calibration or chemical recovery is unavailable, the row remains unresolved. Structural absence in a generated network cannot be replaced by a numerical zero in a source table, and a source-access failure is not a measured non-detection.

14.7.5 Analytical reconstruction

Analytical accuracy and trueness relate the result to a reference, while precision and repeatability compare repeated records. Sensitivity is the response to a change in analyte. Detection and quantification boundaries require registered decision rules. Selectivity retains the complete interference matrix; a method is not selective merely because the target peak is visible in an uncomplicated sample.

The spectroscopy surface distinguishes what each instrument observes. NMR chemical shift, scalar coupling, relaxation and exchange have different carriers. Raman, infrared and ultraviolet-visible relations separate line position from intensity. Fluorescence separates yield and lifetime; phosphorescence adds intersystem structure. Mass spectra retain molecular, isotopic and fragment identities. Rotational spectra preserve the molecular axis and line relation. X-ray, electron and neutron diffraction share structural use but interact with different physical carriers and therefore retain distinct evidential roles.

Chromatography records retention and resolution; electrophoresis records mobility and separation; electroanalysis records the electrical response to a chemical carrier. Multimodal reconstruction composes these independent records into one molecular identity only where their provenance and state are compatible. The complete performance package retains accuracy, precision, sensitivity, selectivity, range, uncertainty and failed or unavailable rows.

This architecture resists the temptation to announce identification from one convenient signal. A source may support composition but leave stereochemistry open; diffraction may support geometry while a charge state remains unresolved; a chromatographic peak may separate a carrier without naming it. The audit layer records exactly which distinctions each method closes.

14.7.6 Computational chemical objects

The canonical molecular graph encodes atoms, bonds and retained labels. Isomorphism decides identity under permitted renaming; substructure identifies a preserved part. Constitutional-isomer, stereoisomer and conformer enumerations operate at successively richer relation boundaries. A complete census must retain the equivalences by which apparent duplicates are removed.

Reaction graphs generate transformations, and atom mapping preserves the constituent route through the balanced reaction. Mechanism search retains the proof trace from generated path to decision. Molecular similarity is an exact declared relation, not a universal surrogate for identity. Database identity and provenance bind a record to source, version, encoding and transformation history.

Symbolic property evaluation executes admitted chemical laws on registered inputs. Prediction uncertainty and applicability state where that execution may be compared with observation. Classical and quantum chemical algorithms are correspondences between implementations and the formal carrier; neither is granted empirical authority solely because it ran successfully.

The current evidence surface preserves conflicting databases, low-confidence results, unavailable packages and resource halts. A computation that cannot finish within its registered resource boundary is a halt, not a favourable prediction. A result developed against a known target remains development-observed. The clean distinction between representation, implementation and measurement is particularly important for the later Protein Fold programme.

14.7.7 Polymer populations, architecture and transition

Degree of polymerisation counts repeat units in a chain. Number-average and mass-weighted molecular sizes are different population relations; dispersity compares them. Chain-growth and step-growth networks have distinct initiation, propagation and composition routes. Copolymer sequence preserves monomer identity and ordering rather than reporting composition alone.

Branched, star and network architectures require their complete connectivity. Crosslinking adds retained joins, and gelation occurs at a finite connected boundary. Polymer conformation and size depend on chain identity and condition. Phase and transition claims distinguish chemical architecture from collective state. Degradation and depolymerisation retain every route by which the chain or network changes.

The polymer validation vector therefore keeps chain distribution, molecular weight method, architecture, processing, temperature and source. A nominal repeat-unit structure cannot be used to assert one glass transition, mechanical response or degradation time for every specimen. Those collective responses pass to Materials through the registered Chemistry-to-Materials ownership records, with each exact canonical claim identity preserved in the branch audit.

14.7.8 Validation vectors as evidence surfaces

Nine substantial validation vectors cover molecular geometry, thermochemistry, equilibrium, kinetics, spectroscopy, electrochemistry, inorganic coordination, organic reaction and polymer chemistry. A cross-source reproducibility vector tests whether independent source routes preserve the same claim-specific relation. The adverse and out-of-bound vector then ensures that unfavourable rows cannot disappear into a favourable summary.

"Substantial" names the registered source surface, not a claim of universal coverage. Each vector retains its row count, source identity, chronology, falsifier and transport status in the audit package. The conceptual paper can therefore explain the result without printing 26,486 evidence lines, while the complete archive remains capable of disproving a misleading summary.

14.8 Chemistry evidence reconciliation and correction history

The Chemistry Grand Lock is `SFT-CHEM-VALIDATION-EMPIRICAL-GRAND-LOCK-012`, with receipt `sha256:be4b72344aaefd342ef43b7660e19fe73fcd6cda30541ca198debcc72f994a9`. Its pre-validation surface contains 263 claims: 255 empirical and eight formal-only. The archive retains 26,486 evidence lines and 1,177 source occurrences. It preserves favourable, adverse, unavailable and unresolved records rather than reducing them to a branch-wide binary label.

The lineage reconciliation exposed a genuine correction. The first one-owner-graph run halted with zero credit because the census named owners but did not contain the dependency routes required by the validator. The corrected route reconstructed all 25,013 edges across 1,484 claims, including 18,553 cross-branch uses. The repair did not change a law, empirical target, admission engine or verifier. It corrected the dependency evidence that the original run had lacked, and both the halt and the lawful successor remain in the history.

Other adverse and unresolved records remain local to their claims: failed capture routes, incomplete source tables, unavailable packages, adverse measurements, low-confidence rows, conflicting interpretations and implementation or resource halts. The current paper does not preserve stale failed-access wording where a later capture succeeded, but it retains the historical transport failure when the evidence constitution requires the chronology. It does not convert the absence of a Smithium measurement into a null experiment or use the prestige of a source as evidence for a chemical claim.

14.8.1 Source identity and evidence role

A Chemistry source is admitted for a declared role: identity, composition, geometry, energy, equilibrium, rate, spectrum, electrochemical response, polymer population or another registered discriminator. Author, custodian, title, release, dataset version, locator, snapshot, access state, source ID and hash remain attached where available. A database may contain several roles, but a row receives credit only for the role its fields actually test.

Human-readable references introduce those sources by scholarly identity. The machine IDs bind the captured bytes and adapters. This dual representation lets a specialist understand the evidence without copying a long hash into every sentence, while letting an auditor detect substitution. Failed access and successful access are source-transport states. Neither says whether the chemical outcome is favourable until the required row is evaluated.

The molecular vector, entropy surface, colligative table, isotope-effect worksheets and aromatic archive illustrate different source topologies. The molecular vector reconciles identities across property families. The entropy surface compares exact source arithmetic. The colligative table preserves three experimental classes. The isotope-effect archive retains normal, inverse, near-equal and adverse rows. The aromatic archive adds outcome-opened and outcome-unopened chronology. One generic word such as "validated" would lose those distinctions.

14.8.2 Empirical status by claim, not by branch

The 273 empirical packages include blind, post-seal, retrospective, development-observed, compatible, supportive, adverse, unavailable and unresolved evidence as registered. The eight formal-only claims remain formal-only. The branch Grand Lock establishes that the evidence surface is complete and retained, not that every claim has the same empirical class.

A chemical result may have exact formal closure and mixed measured outcomes. An observed condition may support one mechanism while leaving another compatible. A source may verify molecular identity but not the claimed state. A prediction may remain standing because the experiment has not run. These compound statuses are preserved in the claim-level table and should be read as scientific information rather than editorial untidiness.

Smithium is the most visible example. Its periodic, electronic and chemical structure is formally admitted. Its synthesis, lifetime, line and separation measurements remain unresolved. The mandatory lifetime halt is a correct implementation outcome at an insufficient-input boundary. Calling the whole family either "confirmed" or "failed" would be false.

14.8.3 Reconciliation, correction and extension

The corrected one-owner route covers 1,484 claims and 25,013 dependency edges, of which 18,553 cross branches. Every route is now reconstructable. The earlier zero-credit halt remains evidence that the first census lacked the dependency fields the validator required. The correction changed the route record, not the law or result.

Future Chemistry may add a molecule, reaction, measurement or new element without reopening the exact histories of the present 281 claims. A new source can supersede a transport or refine an empirical status only through an explicit versioned record. A contradiction between current authoritative files cannot be repaired by choosing the more persuasive paragraph; it must be traced to supersession or reported for Maria Smith.

Chemistry is therefore complete for its 272 registered discipline obligations and nine separately dated returns. That completion concerns the frozen current-evidence census; it is not a declaration that every molecule, reaction, condition or future element has already been observed.

The active release manifest is

`output/release/chemistry-1.3.0/03_Chemistry-Paper-001-v1.3-Manifest.json`, with SHA-256 `e1fdf6657842ab59cec28dcefa059ceeec24696b0f736fd1fb54dc0f7432d26`. The current successor inventory is `publications/inventories/successors/chemistry_v1.3.json`, whose current SHA-256 identity is `sha256:06833d8ae5bd915d5c467834e7b95a034720ce69a59ca1ca3d8f2573b1a08049`.

14.9 Handoffs to Materials and Biology

The Materials handoff is `SFT-CHEM-MATERIALS-OWNERSHIP-HANDOFF-001`, receipt

`sha256:da67f6404c234aeb9db221306817032248d20f581db03732828bafed8d718fdb`. It transfers chemical species, composition, bond, reaction, phase and measurement interfaces without transferring ownership of bulk response. The Biology handoff is `SFT-CHEM-BIOLOGY-OWNERSHIP-HANDOFF-002`, receipt

`sha256:37d9b00972e3dd38e15e09ed6de31b355990e32b6d1044d1b4f7e83487c56ade`. It supplies biochemical identity and reaction while leaving living function, heredity, regulation and organism-level evidence to Biology.

14.9.1 Additional one-owner routes

The six Chemistry handoff claims also route chemical carriers to Medicine, Earth and Environmental Science, Astronomy and the complete cross-branch certificate. Medicine receives drug, metabolite, assay and reaction identity but must add health state, dose, intervention, comparator and clinical outcome. Earth and Environmental Science receives chemical species, partition, reaction, transport and source identity but must add environmental compartment, history and system response. Astronomy receives atomic and molecular species, spectral identities and reaction carriers but must add the celestial source and observational geometry.

These handoffs prevent a chemically correct statement from being made to carry a downstream conclusion. A molecule may have a measured binding or reaction relation without being an effective medicine. A chemical may persist in one laboratory medium without establishing planetary fate. A spectral line may identify a species without reconstructing the star or cloud that produced it.

The cross-branch completeness certificate verifies that every Chemistry-owned claim has one owner and every declared incoming or outgoing use has a root route. It does not create a new chemical law. Its scientific value is the absence of duplicate ownership and missing dependency at the frozen graph boundary.

Materials must come next because chemical identity alone cannot explain why the same composition has different strength, conductivity, optical response or degradation after a change of phase, defect population, interface or processing history. Biology cannot yet begin from chemistry alone either: a living system also depends on organised surfaces, soft matter, scaffolds, mineralised structures and multiscale material boundaries. Materials Science is therefore the final physical-chemical bridge into life.

15. Materials Science: structure, property, processing and interface

Materials Science inherits physical law and chemical identity, then asks how organised matter acquires a reproducible response across scale. Its central object is neither a bare particle nor a molecular formula. It is a specimen whose phase, crystal or amorphous order, defect population, microstructure, surface, interface, processing history, direction, scale and environment are part of the claim. The branch exists because two specimens with the same nominal composition may lawfully have different measured properties.

The active Materials paper is version 1.3.0. It closes **289 of 289 registered obligations**, generated from **73,984 candidates and decisions**, with **289 unique survivors**, **1,156 of 1,156 controls passed** and **289 independent reconstructions**. All closures are `depth_independent`. All 289 claims possess receipt-bound empirical packages and the current external status `empirically_tested_and_independently_replicated`. That branch-level count does not make every row favourable: the evidence packages retain adverse, absent, unavailable, unresolved and out-of-bound outcomes at claim level.

15.1 Scope, ownership and family census

Materials owns organised matter at specimen, phase, microstructure, interface and property scale. It consumes Physics constants and response laws and Chemistry identities, bonds and reactions. Biology owns biological function; Medicine owns clinical outcome; Earth and Environmental Sciences own system fate; and Engineering owns application, design, safety and deployment. A biocompatible material claim may establish an interface condition without establishing a therapeutic outcome. A sustainable-materials claim may establish a registered material life-cycle relation without absorbing the whole environmental system.

Materials family	Claims	Principal boundary
Measurement	7	Specimen, direction, scale, method, condition and uncertainty
Crystal and quasicrystal foundation	12	Periodic order, symmetry, system, lattice and non-periodic order
Defects and microstructure foundation	8	Point, line, surface and volume disorder; scale organisation
Electronic and semiconductor foundation	10	Carrier, band, gap, junction and conditional transport
Superconducting, superfluid and topological foundation	12	Ordered response, transition, topology and evidence boundary
Mechanical foundation	8	Load, strain, strength, fracture and direction
Thermal, magnetic and optical foundation	9	Heat, order, response, propagation and measurement
Classes and bulk foundation	10	Metals, ceramics, polymers, composites and class relation
Processing and degradation foundation	8	History, transformation, damage and lifetime
Advanced functional and sustainable foundation	8	Functional response, resource and declared sustainability boundary
Crystallography	8	Complete crystal-system, Bravais and diffraction surface
Defects and multiscale structure	9	Defect interaction, grain, interface and cross-scale custody
Phase science	10	Phase identity, transition, diagram and path dependence
Mechanical properties	14	Elasticity, plasticity, fatigue, creep, fracture and toughness
Thermal properties	7	Capacity, conduction, expansion and transition response
Electronic and ionic transport	12	Electron, hole and ion response in organised matter
Magnetism and superconductivity	12	Magnetic order, hysteresis, superconducting state and transition
Optical materials	10	Absorption, transmission, emission, index and anisotropy
Material classes	12	Class-specific structure-property organisation
Soft matter	10	Viscoelastic, colloidal, gel and self-organised response
Biomaterials	8	Material-biological interface without transfer of biological function
Nanomaterials	10	Size, surface, confinement and scale-dependent response
Surface and interface science	8	Boundary energy, adhesion, wetting and interfacial transport
Degradation	10	Corrosion, wear, chemical damage and retained history
Processing	10	Process-structure-property route and reproducibility

Materials family	Claims	Principal boundary
Computational materials	12	Representation, model, scale, approximation and validation
Extreme conditions	8	High or low temperature, pressure, field and rate boundaries
Sustainable materials	9	Resource, recovery, durability and bounded life-cycle evidence
Validation	12	Full evidence reconciliation and Materials Grand Lock
Ownership handoffs	6	Typed transfers to Biology, Earth, Medicine and Engineering
Total	289	Frozen current-evidence Materials census

The original pre-source foundation contained eighty-four obligations and 21,504 candidates, sealed under sha256:da97a6cb6a001964a069b45a5a3698e7ea90f334a08d69c62bd09c46d8112035; the returned family added eight obligations and 2,048 candidates under sha256:0e8d7f14a7389b7ec44a37205ce2c9074db65f7b6ed5a466b97f2a01418ef331. The frozen current census has identity sha256:cfcafa0a3021dfdea20bbf026e48acfa0281ec14207e632c766bbcb846406e3b.

15.1.1 Material identity, specimen and sampling

A material system is the least complete constituent-and-relation carrier whose composition, bonding organisation and declared scale remain distinguishable. A specimen is a bounded realisation of that material system. Sampling is the registered relation by which the specimen record is taken to represent a larger batch, component or region. The distinction is indispensable: a law about an ideal carrier, a measurement on one specimen and an inference about a production population have different scopes.

Composition records the chemical constituents; phase identity records their collective organisation; microstructure records the mesoscale arrangement; and a material property is a response relation between the fully identified state and a named perturbation. Materials uncertainty and metrological traceability retain sample selection, preparation, instrument, calibration, direction, scale, condition and uncertainty. If any of those coordinates is required by the falsifier, it cannot be omitted from the evidence row.

This constitution explains why the branch does not derive one universal modulus, conductivity or lifetime for a chemical formula. Those magnitudes are relations of specimens. Exact derivation can still force which coordinates a valid record must retain, which structural classes exist and how a response must compose. The measured magnitude then tests the relation at the declared boundary.

15.1.2 Pre-source chronology and the single engine

The original eighty-four obligations and their 21,504 candidates were frozen before NIST or BIPM source identities were selected. The later V1/V2 audit identified eight nonduplicative Materials-owned omissions; their 2,048 candidates and complete prediction set were likewise frozen before the new sources were selected. Those two seals establish chronology for the original foundation. The complete-field expansion remains separately dated in the current inventory.

Every Materials claim uses the unchanged admission engine. Registration checks the single root, zero axioms, zero free parameters, existing dependency receipts, source identity where applicable and exact candidate identity. The engine then checks one decision per candidate, one survivor, minimality, named-shape uniqueness, four adverse controls and an implementation-distinct reconstruction. Empirical packages additionally require a verified derivation seal, capability isolation, target absence before seal, matching custody release, all-row retention, a falsifier and a tampered unfavourable control.

A failed field is not repaired by the manuscript. The engine preserves a rejection or halt outside the admitted census. An editorial description may make the relation easier to understand but cannot change the carrier, candidate count, source chronology or receipt. This is particularly important where the familiar material-class name tempts a writer to add a conventional constitutive model that the exact grammar did not admit.

15.1.3 Source surface and discriminators

Forty-two registered byte-frozen NIST and BIPM/JCGM records form the independent external boundary of the original and successor seals. Each claim requires at least two claim-specific discriminators. The predictor lacks filesystem, network, clock, environment, subprocess, dynamic-import and foreign-function capability. A distinct custodian verifies the snapshot hash and required features before releasing the observation to the matching sealed predictor.

This protocol assigns sources narrow roles. A crystallography reference may test the seven-system and fourteen-Bravais classifications. A metrology record may test the required uncertainty or calibration structure. A specimen datasheet can serve only if its provenance, method and conditions meet the registered discriminator. It cannot prove a universal material law merely because it comes from a respected producer or standard body.

Later complete-field families introduce additional source records. Their transport state remains attached: direct capture, proxy capture, unavailable, failed, unresolved or out of bound. A proxy is not renamed a direct capture. A document that cannot be fetched is not a null measurement. A row that lacks the property needed by the falsifier cannot receive favourable credit.

15.2 Exact crystal structure

The crystal derivation begins from the physical three-direction carrier. Each direction has two orientations, giving exactly six nearest cubic neighbours:

$$3 \times 2 = 6.$$

The admissible periodic rotation orders are

$$1, 2, 3, 4, 6.$$

Five is the least excluded periodic order and therefore marks the first quasicrystalline boundary rather than a failed crystal system. Length equivalence supplies three classes; angle equivalence supplies four. Filtering their twelve nominal combinations through the admitted metric relations leaves seven crystal systems. Combining those systems with the five centring classes and eliminating duplications or incompatible cases leaves fourteen Bravais lattices.

These are exact structural counts, not fits to a crystallographic table. Their principal claim records are:

- SFT-MAT-CRYST-SYSTEMS-001, receipt
sha256:8eea622f2a962a3e3b995693f3ba38acf8c0b79242bd30c1804eff03952d7637;
- SFT-MAT-CRYST-BRAVAIS-001, receipt
sha256:c04968088f5382d095822785a3b505081f9a9027b7a8d206438deafdb6f93a52;
- SFT-MAT-CRYST-QUASICRYSTAL-001, receipt
sha256:c397a36d1bb97f5bbab246e07e290b327c773ea18ec5c2b074d22794d7d1d094; and
- SFT-MAT-CRYST-PHONON-001, receipt
sha256:574d1650178ebf71cddebdd5128c2d2e9837d02ab67443c66d787cfc246e5b6c.

The acoustic result inherits the physical direction decomposition and closes at three branches: one longitudinal and two transverse. A real dispersion curve, velocity or attenuation coefficient remains material-, direction-, mode-, temperature- and method-bound. The exact branch count does not replace those measurements.

The crystallographic evidence layer retains both direct and proxy captures. Failed captures remain explicit MATERIALS_CRYST...HALT records. A proxy may support the declared structural boundary; it is not silently represented as the direct source that failed to transport.

15.2.1 Lattice, cell and translational order

A lattice is a generated adjacency organisation over the admitted spatial carrier. The primitive unit cell is the least complete repeating carrier from which the translational order reconstructs. Its identity is not a chosen drawing convention: two cells are equivalent only where the permitted basis change preserves the complete lattice relation. A crystal adds the material basis to that periodic support.

Simple-cubic coordination follows directly from three independent directions and two held orientations. Reciprocal support is then the exact relation used to describe periodic diffraction. Diffraction itself remains a physical measurement: wavelength, scattering carrier, specimen orientation, instrument and intensity record enter after the structural support has been defined. Quasicrystalline order preserves long-range organisation without the admitted periodic translation. The least excluded order five marks that boundary.

The quasicrystal substitution family adds exact inflation under the registered finite rule. It does not assert that every observed fivefold motif is a quasicrystal. The evidence must distinguish local motif, periodic twin, aperiodic long-range order and the actual diffraction or real-space record.

15.2.2 Why the rotation orders stop at six

The rank-two integral rotation boundary admits only orders whose relevant Euler-totient degree is no greater than two. The factor certificate is depth-independent. If an odd prime p divides the order, $p-1$ divides the degree, leaving only the odd prime three. The remaining powers satisfy the registered bounds $a \leq 2$, $b \leq 1$, with the simultaneous terminal case excluded. The survivors are one, two, three, four and six.

This proof explains why five is excluded from periodic crystallography without making fivefold organisation impossible. Five becomes the first aperiodic or quasicrystalline boundary. The classification is therefore more informative than a memorised list: it connects periodic admissibility and quasicrystalline extension through one exact stopping certificate.

15.2.3 Seven systems and fourteen Bravais classes

The rank-three metric grammar crosses three length-equivalence classes with four angle-equivalence classes, producing twelve nominal forms. Rotation and metric compatibility retain seven quotient classes. Those are the crystal systems. The centring grammar then crosses the seven systems with five named centring forms, giving thirty-five candidates. Basis equivalence and symmetry compatibility reduce that complete product to fourteen distinct Bravais classes.

The number fourteen is not obtained by consulting the observed table and then designing a grammar of the same size. Candidate identities and eliminations are retained in the claim packages. The post-seal crystallography record tests the seven and fourteen classifications against its registered source. A different conventional nomenclature may group labels differently, but it must preserve the exact metric and centring relation before it can count as the same classification.

15.2.4 Diffraction, texture and non-ideal order

The complete-field crystallography family extends the ideal classifications to measurement. Diffraction amplitude composes scattering contributions with their retained phase, while intensity is the corresponding measured ledger. A finite structure factor keeps every constituent and position. Polycrystal texture is an orientation distribution, not a single-crystal direction copied onto a bulk specimen.

Short-range order and diffuse scattering retain correlations that do not close as a perfect lattice. Stacking faults alter the layer sequence and therefore the diffraction relation. Twinning preserves the relation between domains. Incommensurate and modulated structures require an additional organisation whose period does not collapse into the base cell. Total-scattering pair-distribution reconstruction infers real-space pair organisation from the complete registered scattering record.

These claims preserve reconstruction status. A pair-distribution function is an inference from source-bound scattering, not a direct photograph of every atom. Texture, fault and domain models remain tied to their method and resolution. The validation vector retains alternative fits, absent features, failed transports and out-of-bound source rows.

15.2.5 Acoustic, optical and thermal mode boundaries

One held propagation orientation in rank three separates displacement into one parallel and two independent transverse directions. The acoustic branch count is therefore three. Optical lattice modes and thermal mode limits require the material basis, interaction and occupation boundary in addition to that geometric count. A real dispersion surface can contain many branches when the cell contains more than one constituent; the exact three refers to the long-wavelength acoustic classes, not the total spectrum of every crystal.

Velocity, attenuation, frequency and occupation remain measured or condition-bound. The structural mode count constrains those records but does not supply their dimensional values. This precision is what allows the result to hand physical wave structure into Materials without overstating the measured response.

15.3 Defects, microstructure and phase

An ideal lattice cannot account for the response of a manufactured or natural specimen. The next required carrier is departure from ideal order. Point, line, surface and volume defects are distinguished by the part of the organisation they alter. Their density, orientation and interaction then form microstructure: grains, boundaries, inclusions, pores, dislocations and interfaces whose collective organisation changes the measured response.

The multiscale family specifies how a record at one scale may be related to another without pretending that every microscopic detail survives coarse description. The preserving coordinates-specimen, scale, direction, preparation, condition and method-remain attached to the result. A microstructure source-capture halt and an independent extreme-label halt are retained in the current evidence surface; neither is converted into an absent phenomenon.

Phase science follows because organisation may change discontinuously or coexist under controlled conditions. The branch distinguishes composition from phase, equilibrium boundary from path, and phase identity from the measured property used to detect it. Processing can therefore change a specimen without changing its nominal chemical formula, and two specimens can share a phase label while differing in grain size, defect structure or texture. Those distinctions are what allow a later property claim to be falsifiable rather than anecdotal.

15.3.1 Point defects and constituent balance

A point defect is a local departure from the registered site identity. A vacancy is structural absence at that site; it is not numerical zero inserted into the proof. Interstitial and substitutional defects introduce a constituent on a different retained relation. Defect populations are counted through site fractions, with composition and charge balance preserved where required.

Migration is an ordered path between sites. Its empirical rate depends on temperature, material, path and method, while the formal relation requires the complete starting, intermediate and terminal states. Doping in a semiconductor is a controlled substitution that changes carrier availability; it remains a chemical and structural operation before its electronic response is measured.

Defect chemistry is shared with Chemistry through one-owner dependency. Chemistry owns the constituent and charge-compensation identity. Materials owns the spatial population, microstructural interaction and resulting bulk property. One source row may be used by both branches only through that typed dependency, not counted as two independent observations of the same claim.

15.3.2 Dislocations, grains and interfaces

A dislocation is a line defect whose retained relation supports slip. Dislocation reactions, climb and cross-slip change the line organisation under declared carriers and conditions. Grain identity adds a region of common orientation; a grain boundary is the interface between such regions. Curvature can drive grain growth through a complete transfer ledger.

Segregation records constituent preference at an interface or boundary. Precipitation distinguishes the new phase, its composition and its coherent or incoherent relation to the matrix. Coarsening transfers material among inclusions while conserving the retained carrier. Interface migration has its own mobility record and driving relation.

Microstructure-to-bulk correspondence is therefore a multiscale reconstruction. It must state which grain, defect, inclusion or interface statistics survive the scale change and which response is inferred. A micrograph alone is not the bulk property; a bulk property alone does not uniquely reconstruct the microstructure.

15.3.3 Phase fractions and coexistence

A complete multiphase ledger retains every phase identity and fraction. Tie lines and lever partitions are exact composition relations at their registered boundary. Component potential is handed from Chemistry into the phase coexistence relation. The measured phase diagram remains source-, method- and condition-bound.

Metastability is a retained local state whose path to another state is blocked at the declared timescale or barrier. Spinodal-equivalent instability is a different organisation: the state has no retained local barrier under the registered perturbation. Displacive and martensitic transformations preserve a coordinated relation; reconstructive transformations require bond or adjacency reorganisation. Order-disorder transitions change retained arrangement without necessarily changing composition.

Glass transition is a kinetic-arrest boundary, not a universal equilibrium phase point. Its measured value depends on recurrence scale and history. Transformation kinetics retain the complete time-temperature path. Thus the same nominal composition may lawfully display different phase fractions and properties after different cooling, deformation or annealing histories.

15.3.4 From electronic structure to junctions

Electronic bands and structural gaps arise from the collective support of the material carrier. Conductor, semiconductor and insulator classes are response distinctions at the declared conditions. Electron and hole are complementary carrier descriptions. Exclusion bounds occupation. Controlled substitution changes carrier availability, producing p-type and n-type organisations.

Joining those organisations creates a p-n junction and its depletion region. The rectification claim closes the direction-dependent response at the exact registered boundary. Semiconductor transport then adds mobility, concentration, scattering and condition. Interband optical transition connects electronic support to a photon observation.

No conventional band diagram is treated as a premise merely because it is standard. The exact carrier and its admissible relations must be rooted in the upstream Physics and Chemistry receipts. Measured band gaps, mobilities and junction curves enter through their source packages. A match to a rectifying curve does not by itself identify every defect or interface state that produced it.

15.3.5 Coherent and topological states

A superconducting state is a coherent paired-carrier organisation. Its dissipation-closed transport, flux exclusion, flux quantisation and weak-link transfer are separately registered. A superfluid likewise has dissipation-closed transport and quantised circulation. Topological materials retain an invariant under the admitted transformations and a protected bulk-boundary transport relation.

The empirical surface includes the superconducting isotope response, integer and primary fractional Hall classes and exact boundary-mode count. The isotope record is a measured response with material and condition custody, not a proof of every microscopic mechanism. Hall classes test the registered topological relation. A protected mode remains protected only within the stated disorder, gap and boundary conditions.

Later complete-field claims add critical-field organisation, vortex matter and pinning, coherence length, superfluid excitation and critical flow. Each is a property of an identified specimen and state. The exact state classification does not supply one universal transition temperature or critical field.

15.4 Mechanical, thermal, electronic, magnetic and optical response

The property families all use the same general relation: a fully identified specimen and state are subjected to a named perturbation, and a method-bound response is recorded. The shared grammar does not imply that all properties are the same quantity.

Mechanical claims distinguish elastic return, plastic retention, strength, fatigue, creep, fracture and toughness. Load mode, direction, geometry, strain rate, temperature and preparation are part of the evidence. A modulus is not a universal exact number generated from composition alone.

Thermal claims distinguish stored response, transport, expansion and phase change. Electronic and ionic claims distinguish carriers, transport paths, band or gap relations, junctions and environmental dependence. Magnetic and superconducting claims retain order, field history, hysteresis, transition and measurement protocol. Optical claims retain wavelength, direction, polarisation, absorption, transmission and emission. Exact structural relations may constrain these families, but conductivity, strength, transition temperature, index and attenuation remain values of identified specimens under identified methods and conditions.

This exact-versus-measured distinction is one of the branch's central results. It prevents two opposite errors: fitting a structural law to a preferred specimen value, and dismissing a genuine exact relation merely because real specimens exhibit conditional variation. Formal structure and empirical response meet through an explicit interface; neither replaces the other.

15.4.1 Mechanical response and deformation history

Stress records transferred load over the declared support; strain records the resulting change of shape or extent. Their tensor-resolved relation preserves direction and load mode. Elastic recovery closes when removal of the perturbation restores the retained state. Plastic deformation preserves a changed state. Elastic modulus is a response ratio on the registered interval, not a universal scalar of a chemical formula.

Transverse-to-longitudinal strain adds a directional comparison. Viscoelastic response retains memory and recovery over time; viscoplastic flow retains both rate and non-restoring change. A yield surface is a boundary in the complete loading-path support. Work hardening preserves deformation history rather than assigning the same state to a previously deformed and undeformed specimen.

Strength and hardness are distinct boundaries. Fracture energy and toughness retain crack creation and propagation. Stable and unstable crack growth are separate regimes. Fatigue distinguishes initiation from propagation under cyclic loading; creep distinguishes mechanism and rupture-time custody under a sustained condition. Impact and high-rate response require their recurrence scale. A result measured slowly cannot be copied into the shock boundary.

Friction relates two contacting surfaces and their state. Lubrication adds an intervening carrier; a tribofilm is a generated and retained interface. Rheological response distinguishes flow and relaxation classes. Every mechanical measurement therefore names geometry, orientation, rate, temperature, preparation and method where those fields affect the falsifier.

15.4.2 Thermal storage and transport

Heat capacity is excitation storage per temperature relation at the declared state. Thermal conductivity records energy transport through the specimen, while diffusivity composes conductivity with stored response and density. Thermal expansion is an oriented change of extent. None is inferred solely from the existence of lattice modes.

Interfacial thermal-boundary resistance belongs to the interface, not either bulk constituent alone. Phonon scattering retains the complete mean-path ledger. Radiative transport in a material retains absorption, emission and geometry. Thermoelectric response is a coupled electrical and thermal relation, with its performance boundary explicitly named rather than presented as a context-free efficiency.

Phase-change storage distinguishes sensible storage from the retained phase transition. Thermal shock combines gradient, constraint and mechanical response; thermal fatigue adds recurrence and accumulated damage. The validation vector keeps specimen, temperature range, direction, method and uncertainty. A favourable conductivity row does not validate a different microstructure or temperature regime.

15.4.3 Electronic and ionic transport

Electrical conductivity and resistivity are reciprocal response relations only at the compatible state and method boundary. Carrier mobility and concentration are separated so that one measured conductivity does not uniquely infer both. The Hall response provides a carrier-sign and density ledger under its declared model and field. Dielectric permittivity and loss retain frequency, direction and condition.

Ionic conductivity records mobile chemical carriers through organised matter; transference partitions their contributions. Mixed ionic-electronic transport retains both carrier classes. Tunnelling is a finite barrier relation. Band alignment and interface offset belong to a pair of materials and their interface; heterostructure confinement adds the resulting bounded carrier. Defect and trap states preserve their structural source, while screening and depletion retain the response length.

Electrochemical insertion changes the material state as a chemical carrier enters or leaves. Chemistry owns the species and reaction; Materials owns the phase, strain, defect and transport response. This distinction prevents a successful half-cell reaction from being advertised as a complete electrode material lifetime.

15.4.4 Magnetic, superconducting and superfluid response

Ferromagnetic, antiferromagnetic and ferrimagnetic order differ by how local magnetic carriers compose. Paramagnetic and diamagnetic responses are perturbation-bound classes. Spin-glass freezing retains disorder and history. Domains and walls add mesoscale organisation, and the hysteresis loop is a complete path ledger rather than one magnetisation number.

Magnetocrystalline anisotropy attaches the response to crystal orientation. Magnetoresistance retains electrical and field coordinates. Spin transport and relaxation preserve preparation, path and decay of the held spin record. Critical fields, vortex matter, pinning and coherence length describe distinct boundaries of a superconducting specimen. Superfluid excitation and critical flow likewise retain state and apparatus.

The branch does not infer a universal mechanism from a transition alone. A measured critical temperature is condition- and specimen-bound. Flux quantisation and topological class may be exact structural relations while the observed plateau, field or transition remains empirical. The validation vector keeps those evidence classes separate.

15.4.5 Optical and photonic response

Optical absorption and extinction retain wavelength, path and material state. Reflection and transmission close an energy ledger at the interface. Luminescence separates generated excitation, emitted record and quantum yield; the yield is source- and method-bound. Elastic and inelastic scattering retain whether the material state changed.

Birefringence follows from optical anisotropy. Nonlinear mixing requires the complete participating frequencies, orientations and material response. Waveguides retain confinement and loss. A photonic band gap is an organised propagation boundary; a defect mode is a permitted state inside that boundary. Plasmonic response is collective. Exciton generation, transport and recombination retain separate parts of the state trajectory.

An optical classification does not automatically establish device utility. A measured emission line, propagation loss or yield must be tied to specimen, preparation, direction, wavelength, excitation and detector. Engineering later owns whether those relations satisfy an application requirement.

15.4.6 Evidence across property families

The mechanical and tribological, thermal and transport, electronic/ionic and dielectric, magnetic/superconducting/topological, and optical/photonic validation vectors each preserve their own source rows. Cross-source reproducibility asks whether independent records retain the same structure-property relation. It does not require every specimen to have the same magnitude.

An adverse row may show that a proposed universalisation fails while the narrower condition-bound law remains. An absent feature is different from an unavailable source. An out-of-bound measurement may be accurate but unable to test the registered

claim. The complete adverse vector retains all of these states, which is why the branch-level empirical status cannot be read as "every reported measurement agreed".

15.5 Material classes, soft matter and nanoscale organisation

Material classes group recurring structure-property organisations without making the class name the cause. Metals, ceramics, polymers, composites and other classes are defined through their retained chemical, structural and response relations. A polymer's Chemistry owner defines repeat units and chemical architecture; Materials owns morphology, phase, orientation, processing and bulk response. A composite requires constituent identities, interface and spatial organisation, not merely an averaged composition.

Soft matter adds fluctuation, relaxation, viscoelasticity, colloidal relation, gelation and self-organisation at condition-sensitive boundaries. The branch retains the timescale and protocol under which an object appears solid-like or fluid-like. Nanomaterials add size, confinement and surface-to-volume organisation. "Nano" is not used as a universal property claim; the evidence must name the dimension and the response that changes at that scale.

Surface and interface science is required wherever neighbouring materials do not reduce to either bulk constituent. Adhesion, wetting, boundary energy and interfacial transport depend on both sides of the boundary and on its preparation. This interface carrier is a direct prerequisite for biomaterials and for the Biology handoff.

15.5.1 Metals, ceramics, polymers and composites

A metallic material retains collective electronic support together with its crystal, defect and microstructural organisation. A ceramic material retains its ionic or covalent chemical carrier and the corresponding structural relations. A polymeric material retains the chemical chain architecture and adds morphology, phase, orientation and bulk response. A composite contains distinct constituents and the interfaces through which load, heat, charge or damage transfers.

These classes are generated through their retained relations, not through a list of familiar examples. Glass and amorphous organisation lack long-range periodic order while preserving local structure. Porous materials add a void network whose size, fraction, shape and connectivity affect the response. The molecular-to-bulk correspondence states which chemical and physical coordinates survive the scale change. Anisotropy retains direction. Size and surface effects state when a boundary contribution is no longer negligible.

The complete-field class family then separates solid solutions and alloy phases, ordered intermetallic compounds, compositionally complex alloys, refractory and ultra-high-temperature classes, cementitious and concrete composites, fibre- and particle-reinforced composites, metallic glasses, structural and functional ceramics, thermoplastics, thermosets and elastomers, graded materials and architected cellular materials.

Each class supplies a grammar for what must be retained. "Compositionally complex" does not mean composition is irrelevant; "metallic glass" does not mean crystal defects can be copied unaltered into the description; "functionally graded" requires a spatially retained variation; and an architected material requires geometry in addition to constituent identity. Measured properties remain attached to the actual specimen and processing route.

The water bulk-response ledger provides an explicit molecular-to-bulk case. Hydrogen-bonded molecular identity is inherited from Chemistry, while the collective response belongs to Materials. The source rows test the declared bulk relation; they do not turn one water condition into a universal model of all hydrogen-bonded materials.

15.5.2 Soft matter and timescale

Colloidal stability retains particle identity, medium, interaction and aggregation route. Gelation creates a percolated soft network. A foam retains cells, films and drainage; an emulsion retains droplets and continuous phase. Liquid crystals preserve orientational order without requiring full translational order. Membranes and thin films add a bounded geometry whose surface and transport cannot be treated as bulk.

Granular matter retains packing and force-chain support. Jamming is a boundary between rearranging and mechanically retained organisation under the declared load and recurrence scale. Responsive materials change state under a named stimulus. Active materials consume a retained energy source and remain nonequilibrium organisations; "active" is not permission to omit the energy ledger.

Soft response is inseparable from timescale. The same specimen may be solid-like on one observation interval and fluid-like on another. A complete claim therefore retains recurrence, perturbation, recovery and history. Classifying a material by one photograph or instantaneous modulus would lose the relation the family is designed to preserve.

15.5.3 Nanoscale confinement and collective state

A nanomaterial claim begins with a measured size or confinement coordinate. Nanoparticles retain a size and shape distribution, not one nominal diameter. Nanowires have one-dimensional confinement; two-dimensional materials retain layer and stacking; quantum dots retain finite confinement in every spatial direction. Surface-to-volume dominance is a relation between those exact supports.

Nanoscale phase and melting boundaries remain material- and condition-bound. Quantum materials are classified by their collective state, not simply their small size. Moiré and twisted-layer structures add a superlattice relation. Nanocomposites retain interface density. Aggregation and dispersion custody must distinguish the primary particle from the state actually measured in the medium.

The branch therefore avoids using "nano" as a promotional adjective. A source must name which dimension lies in the registered range, which surface or confinement relation changes and which observation discriminates it. A small particle without a demonstrated scale-dependent relation remains a small specimen, not evidence for every nanoscale law.

15.5.4 Surfaces, interfaces and coatings

A surface free-state and energy relation arises because the boundary lacks the same neighbourhood as the bulk. Wetting and contact angle retain the liquid, solid, vapour, preparation and measurement geometry. Adhesion and work of separation record the complete interface ledger. A coating adds a layer and substrate whose individual identities and joining relation remain distinct.

Roughness is scale-bound: the numerical record changes with sampling length, resolution and filtering. Surface reaction and catalysis are handed to and from Chemistry through the identity and reaction carrier, while Materials owns surface organisation and resulting response. Tribofilm formation retains the generated interfacial layer; interface fracture and delamination retain the path by which that relation fails.

An interface is not merely the average of its neighbouring bulk phases. It can carry a distinct composition, structure, energy and transport relation. This is why interface custody recurs in electronics, composites, corrosion, biomaterials and processing. The one-owner graph permits those uses while preventing five different branch sections from claiming to derive the same primitive interface.

15.6 Biomaterials and the boundary of living function

The biomaterials family tests material conditions at a biological interface without taking ownership of life or health.

Biocompatibility is therefore a typed material-biological relation, not a claim that a material is universally safe or therapeutically effective. Its current record is SFT-MAT-BIO-BIOCOMPATIBILITY-INTERFACE-001, with receipt sha256:a10aa7bcb625a83b2d3ebf9dd8e4ff28df0ae2d5d74ab13268cb4ef711b8bd2f.

Scaffold organisation separates porosity from connectivity and both from a later biological outcome. The registered scaffold claim is SFT-MAT-BIO-SCAFFOLD-POROSITY-CONNECTIVITY-003, receipt

sha256:1fe3a1e81c362902e3c4071d0e9dde9d04e9361616fa13f22c91314c8744f559. Mineralised organisation similarly identifies the material architecture without claiming the developmental process that builds it. Its record is SFT-MAT-BIO-MINERALIZED-ORGANIZATION-007, receipt sha256:bd00986cced114e52724965d3b2775686ee13a2353b6a88760f34404a1b4bd07.

Those distinctions make the downstream handoff scientifically clean. Biology may consume an interface, scaffold or mineralised structure, but it must add function, regulation, reproduction, development and organismal evidence. A medical claim must add clinical population, intervention, comparator and outcome. Neither can be inferred from a favourable materials test alone.

15.6.1 Bioresorption, adhesion and controlled transport

Bioresorption is a material degradation ledger in a biological environment. It requires the initial material, degradation route, products, transport and remaining structure. The biological response to those products is a Biology or Medicine question. A materials record may establish that mass or structure changes under the declared medium; it cannot by itself establish safety.

Cell-material adhesion is a handoff relation. Materials owns surface chemistry, roughness, stiffness and the interfacial state. Biology owns cell identity, attachment machinery and function. Mechanical matching likewise compares a material response with a biological tissue boundary without declaring that the match produces healing.

Controlled release is material transport across a designed boundary. The carrier identity and chemical stability come from Chemistry; pore, phase, degradation and diffusion organisation belong to Materials; biological or therapeutic action belongs downstream. A release curve is therefore not a clinical outcome.

Biologically derived and biofabricated material identity retains provenance and processing. The fact that a constituent came from a living source does not make its current material property biologically owned. Conversely, a scaffold manufactured without living input may later participate in a biological system without acquiring function at the Materials proof boundary.

15.6.2 What the Biology handoff preserves

The handoff preserves porosity, connectivity, interface, transport, mechanical response, mineralised organisation and source custody. It also preserves the limitations of every material row: specimen, preparation, condition, method, uncertainty and adverse state. Biology receives those records as inputs and cannot rewrite them as biological axioms or omit an unfavourable interface merely because a later cellular result is positive.

The scientific opportunity lies precisely in this clean boundary. A living system can be studied as a new organisation acting through exact chemical and material carriers, while its function and history remain genuinely biological. Reduction to chemistry would lose that organisation; separation from chemistry would lose the dependency route.

15.7 Processing, degradation, computation and extreme conditions

Processing supplies the path by which chemical constituents and physical conditions become a specimen. The complete relation is process → structure → property, with intermediate states retained where the evidence permits. A heat treatment, deposition, deformation or synthesis route can change phase, grain, defect, texture or interface and thereby change response. Reproducible materials science therefore requires the path, not only the final label.

Degradation is the time-dependent inverse problem. Corrosion, wear, chemical attack, fatigue and environmental damage alter the retained organisation. Lifetime is a conditional reconstruction or measurement, not an intrinsic number detached from load, environment and method. Missing exposure data or a failed source capture remains unresolved or unavailable, not a favourable durability result.

Computational materials claims preserve representation, model, approximation, scale, resource and validation. Simulation may reconstruct a mechanism or predict a registered response; it becomes empirical evidence only through the declared comparison. Test success establishes implementation identity at the tested boundary, not physical confirmation. Resource halts and low-confidence outputs remain in the audit surface.

Extreme-condition claims retain the high- or low-temperature, pressure, field or rate boundary at which a response was measured. An independently generated label that cannot be resolved from the current source remains a halt rather than being harmonised by prose. Sustainable-material claims similarly state their bounded resource, recovery, durability or life-cycle relation. They do not absorb Earth-system consequences or Engineering deployment decisions.

15.7.1 Degradation as a complete change ledger

Oxidation-scale growth composes chemical reaction with transport through the new layer. Corrosion rate retains the coupled electrochemical path, specimen area, exposure and time. Passivation is the formation of a protective state; film breakdown is a distinct boundary. Stress-corrosion cracking composes a mechanical crack path with the corrosive environment rather than attributing failure to either in isolation.

Hydrogen uptake and embrittlement retain ingress, distribution, microstructural interaction and mechanical consequence. Abrasive, adhesive and erosive wear are different transfer relations. Radiation damage accumulates defects and may also permit recovery. Physical ageing records drift in a retained state without requiring a change of chemical identity. Environmental attack and weathering preserve the complete exposure condition.

Service life and failure time are evidence boundaries. A predicted lifetime is a reconstruction from a named load and environment; an observed failure is a source-bound event; a censored specimen that has not failed by the end of a test is neither an infinite lifetime nor a null degradation result. Those statuses must remain visible in the complete table.

The degradation family therefore carries favourable durability results and early failures under one constitution. A later coating or processing change may lawfully create a new specimen and new status, but it does not delete the adverse history of the earlier state.

15.7.2 Processing as the construction of a specimen

Casting retains the mould-filling and solidification history. Thermomechanical forming adds deformation and texture. Machining creates a surface and possible damage state. Additive manufacture retains every layer, melt-pool relation and build history required by the claim. Thin-film deposition records incoming material and growth; epitaxy adds lattice matching and orientation.

Welding, brazing and joining create an interface whose thermal, chemical and mechanical histories differ from either parent material. Polymer processing retains flow and orientation. Powder processing retains particle identity, packing and compaction. Heat treatment, sintering and densification already introduced in the foundation are composed into the complete process-structure route.

A process window is a source-bound region of conditions that produces the registered state with the stated reproducibility. Provenance includes machine, feedstock, geometry, atmosphere, calibration and post-processing where the falsifier requires them. A nominal process name cannot substitute for those coordinates.

Processing evidence is not merely procedural detail. If two specimens have different histories, treating them as one observation can conceal the actual structure-property relation. Conversely, a reproducible path can explain why different laboratories obtain compatible response without asserting that every uncontrolled specimen must agree.

15.7.3 Computational representation and forward simulation

An exact material-structure representation identifies constituents, sites, relations, phase, boundary and scale. A structure-property computation names the law by which that input produces a response. A finite numerical simulation adds discretisation, update rule, termination and error. Multiscale composition specifies how outputs and uncertainties cross scales.

Numerical stability and error propagation remain implementation evidence. Passing a convergence test verifies the behaviour of the registered discretisation; it does not prove that the underlying physical model describes the specimen. The inverse materials problem reverses the relation from target response to candidate structure and must retain non-uniqueness. Machine-learning inference adds training provenance, representation, applicability and uncertainty. A pretrained model is not an admissible proof premise.

Materials databases require identity and provenance. Two rows can be compared only where material, phase, processing, specimen, direction, method and condition are compatible. Phase-field, molecular-dynamics and electronic-structure correspondences each operate on a different carrier and scale. Their outputs cannot be merged as if they were three measurements of one quantity.

Simulation-to-experiment validation is separately registered. The sealed simulation, input state, comparison metric and complete observation rows must be retained. A model may be useful and compatible without being uniquely confirmed. A failed run, missing potential, unavailable input or resource halt remains in the archive.

15.7.4 Extreme conditions and path custody

High pressure, high temperature and cryogenic response each define a material state under a named boundary. High electric and magnetic fields add direction, magnitude and exposure. High strain rate and shock add a short recurrence scale and propagation path. Extreme radiation adds source, dose, spectrum and recovery. Combined extremes require the complete order in which conditions were applied.

Path custody matters because material state can be hysteretic or metastable. A specimen compressed and then heated may not equal one heated and then compressed. A cryogenic value cannot be extended to room temperature by silence. A shock response cannot be inferred from a quasi-static curve without the registered correspondence.

The independent extreme-label halt records a source whose current information does not resolve the required generated class. The correct scientific action is to preserve the halt. Choosing the closest familiar label would manufacture agreement and destroy the falsifier.

15.7.5 Sustainability and bounded system claims

Embodied material and energy ledgers retain the inputs required to produce the declared specimen. Critical-material availability is a source- and time-dependent boundary. Reuse and remanufacture preserve enough identity and function for a further cycle; recycling separates and recovers constituents; yield retains every residual route.

A circular material flow is a network ledger, not the claim that no material leaves the system. Durability and life extension compare complete service histories. Material toxicity is handed to Medicine and Environmental Science for biological and

system outcomes. Substitution preserves a declared function while changing material identity. End-of-life fate retains recovered, dispersed, stored and unresolved residues.

These claims are deliberately bounded. Materials Science can derive and test the material flow, durability or recovery relation. Earth and Environmental Sciences own the wider ecosystem and planetary consequence. Engineering owns whether a design meets safety and function. Economics and Social Systems own institutions, incentives and access. A favourable recycling yield cannot be promoted into a complete sustainability judgement without those downstream relations.

15.7.6 Processing, class and sustainability evidence

The material-class and processing validation vector tests whether the retained identity and path distinguish the registered responses. The cross-source vector then asks whether compatible independent records reconstruct the same relation. Reproducibility is not numerical sameness stripped of uncertainty; it is agreement under the declared specimen and method coordinates.

Adverse, absent and out-of-bound outcomes remain scientifically informative. An adverse process result may falsify a claimed path. An absent phase may be a completed negative observation. An unavailable process log is a provenance failure. An out-of-bound source may report a different material or condition. The Grand Lock retains all four instead of treating them as interchangeable failures or silently excluding them.

15.8 Materials evidence reconciliation

The pre-validation surface contains 271 claims, 1,040 evidence lines and 334 source occurrences. The complete one-owner graph contains 1,681 predecessor claims and 28,437 edges, including 19,852 cross-branch uses. Every current route traces to the root under reconciliation identity

sha256:4d2a8f47c39d945fbe31a8de2c78d2dc66462db7c4773f1a785b8b139d2c148b. Forty-two byte-frozen NIST and BIPM records support the registered base source surface; later complete-field families retain their additional source records and transport states.

The Materials Grand Lock is SFT-MAT-VALIDATION-EMPIRICAL-GRAND-LOCK-012, with receipt sha256:dbb75828d061c43824bfad454e100d0502d76b5ebc3ee74e9dd03cbcd8967268. It retains favourable, adverse, absent, unavailable, unresolved and out-of-bound rows across all 271 pre-validation claims. Failed direct and proxy crystallographic captures, the microstructure capture halt, unavailable source rows and the extreme-label halt remain visible. A later successful transport can update the present access state while preserving the historical failure required by the evidence chronology.

Historical step 291 contained astronomy content. The current ownership audit routes it to Astronomy rather than enlarging Materials by inheritance. That correction preserves the scientific content while refusing duplicate or convenient ownership. The same boundary prevents the materials paper from claiming biological function, clinical outcome, environmental fate or engineering deployment.

15.8.1 Validation-vector architecture

The twelve validation claims are not duplicates of the 271 scientific claims. They assemble typed evidence surfaces. Crystallography and diffraction, defects and microstructure, phase and transformation, mechanical and tribological response, thermal and transport response, electronic/ionic and dielectric response, magnetic/superconducting/topological response, optical/photonic response, and material-class and processing response each have their own vector. Cross-source reproducibility composes compatible independent records. The complete adverse vector preserves everything that cannot receive favourable credit. The Grand Lock proves that every pre-validation claim is represented.

This architecture answers a difficult editorial problem. Printing the same generic source-custody language 271 times would make the scientific argument unreadable. Omitting the rows would make it unauditable. The vectors permit a compact conceptual account while the machine packages retain every source identity, discriminator, value, status and receipt.

The vector status is not a vote. One favourable source does not overrule a discriminating adverse source, and ten compatible rows do not automatically confirm a non-unique relation. Each row retains its evidence class. The claim status follows the registered protocol and falsifier.

15.8.2 Source transport and historical state

Source transport is itself versioned. A direct byte-frozen capture supplies content identity. A proxy can support only the features it actually preserves. A failed access records that the route failed at the recorded time; it does not establish that the source or phenomenon is absent. An unavailable row lacks the required record. An unresolved row contains evidence that the current classifier cannot lawfully decide.

If transport later succeeds, the current access field may change. The historical failure remains where chronology requires it. If a newer dataset supersedes an older source, the manifest must state the supersession and retain the old identity. A numerical value is not silently replaced merely because a later handbook reports a cleaner number.

The crystallographic capture halts, microstructure halt and extreme-label halt are therefore part of the publication, not embarrassing debris. They show where evidence is incomplete and specify what a future source must supply. That transparency makes a later lawful resolution stronger because the test was visible in advance.

15.8.3 One-owner reconstruction

The Materials dependency graph traverses 1,681 predecessor claims and 28,437 edges. Nineteen thousand eight hundred and fifty-two edges cross branch boundaries. A cross-branch use says that the Materials result requires an upstream admitted object; it does not transfer the object's ownership.

The graph is independently reconstructed from materialised registrations. Every one of the 289 current claims reaches SFT-ROOT-THERE-IS-NO-NOTHING. A branch label alone is insufficient: the route must name the actual dependency receipt. This is why the correction to historical step 291 matters. Its planetary and Tully-Fisher content belongs to Astronomy and Cosmology. Routing it there preserves the question and evidence while preventing Materials from claiming closure by absorbing another branch's work.

The handoff claims to Engineering, Biology, Medicine, Earth and Environmental Science and Astronomy make the outgoing boundary equally explicit. They supply material carriers and evidence coordinates. They do not derive an application, organism, clinical result, planetary fate or astronomical system.

15.8.4 Limitations retained inside completion

All 289 claim packages and empirical packages close at the frozen boundary, but new materials and conditions remain possible. A newly synthesised phase, an improved microscopy method or a resolved transport can extend the branch. Such an extension must receive a new registration and version; it cannot be inserted retroactively into a sealed claim.

The branch also retains conventional-model boundaries. Exact classification does not mean every constitutive coefficient is derivable without specimen input. A simulation can be verified without being uniquely validated. A condition-bound measured law may be robust without being universal. A favourable biomaterial interface does not establish health benefit. These are not rhetorical disclaimers appended to weaken the paper; they are coordinates of the scientific claims themselves.

Current full-field completion therefore means that all 289 registered Materials obligations close with their declared candidates, controls, dependencies, evidence packages and reconstructions. It does not mean that every possible material, preparation, direction, scale or condition has been measured. Exact crystal counts and mode counts remain exact; measured moduli, conductivities, strengths, transition temperatures and lifetimes remain specimen- and protocol-bound.

15.9 Handoff to Biology

The formal Biology handoff is SFT-MAT-BIOLOGY-OWNERSHIP-HANDOFF-002, with receipt sha256:f24161d22e77e070e6bd4b052173fe4072996c01e3a29c7755c1dfa8d3252293. Together with Chemistry's biochemical handoff, it supplies Biology with physical quantities, chemical identities and reactions, phase and transport, soft and interfacial organisation, scaffold and mineralised structures, measurement custody and a complete one-owner dependency route.

15.9.1 Engineering, Medicine, Earth and Astronomy routes

The Engineering handoff supplies material identity, property, uncertainty, processing history, degradation and failure boundaries. Engineering must add a specified function, load case, architecture, control, safety margin, reliability and deployment decision. A material with a high measured strength does not by itself establish that a component made from it is safe.

The Medicine handoff supplies a material-biological interface, degradation or release carrier and every adverse material row. Medicine must add patient or population, indication, intervention, comparator, dose, follow-up and outcome. Biocompatibility

at one assay boundary is not clinical safety. Bioresorption is not therapeutic efficacy. The handoff preserves those distinctions so that a later medical claim cannot cite the Materials receipt at a stronger level than it earned.

Earth and Environmental Science receives material flow, degradation, toxicity handoff, durability, recycling and end-of-life carriers. The downstream branch adds environmental compartment, transport, ecological interaction and system history. A bounded recycling yield or weathering test cannot establish global environmental benefit without those relations.

Astronomy receives material-state, phase, optical, thermal, high-pressure and extreme-condition carriers where they are needed to interpret celestial matter. It must add the astronomical source, distance, observation and system model. Historical step 291 is routed there precisely because planetary and galactic evidence cannot be owned by Materials simply because matter is present.

15.9.2 Handoff evidence does not duplicate evidence

A downstream claim may cite an upstream source through its receipt, but that use is not an independent replication of the upstream observation. Independent replication requires a distinct implementation or source route under the registered protocol. The one-owner graph counts dependency edges, not extra confirmations.

Corrections also travel forward. If an upstream source is superseded or a status becomes adverse, dependent claims must be audited for impact. They do not automatically fail if their relation did not use the changed coordinate, and they do not automatically survive if it did. The explicit dependency route makes that impact analysis possible.

What remains unresolved at this boundary is the defining problem of life. Neither a molecule nor a material specimen explains why a bounded organisation maintains itself, uses inherited information, regulates competing processes, develops, reproduces or evolves. Biology must introduce those relations and their evidence without taking physical constants, chemical reactions or material interfaces back as new biological axioms. The next part therefore begins with the living carrier and the distinction between organised matter and organisation that acts to preserve and reproduce its own conditions.

Part II synthesis: the physical-chemical dependency bridge

Physics, Chemistry and Materials Science contribute **938 live claims**, **403,696 registered candidates**, **938 unique survivors** and **3,752 passed controls** to the current V3 corpus. The aggregate is useful only because its members remain typed. Physics contributes 368 current categorical claims; Chemistry contributes 272 frozen obligations plus nine returned claims; and Materials contributes 289 frozen obligations. Their evidence classifications are not made uniform for presentation: Physics separates 238 empirical validations from 130 formal or independently replicated claims, Chemistry separates 273 empirical packages from eight formal-only claims, and Materials retains 289 receipt-bound empirical packages with claim-level adverse and unresolved rows.

Authoritative freeze for this part

This chapter was reconstructed from the active papers and current repository records, not from remembered summaries or superseded drafts. The active manuscript identities checked at the editorial freeze are:

Authority	Repository path	SHA-256
Publication guidance	publication_guidance.md	5ea6b5b558671b0144a553c1b8d20ca32c30b4e01a7d840f6e17248f085405cc
Physics active paper	output/release/physics-1.3.0/01_From-Fold-to-Physics_Physics-Branch-Paper-001-v1.3.md	3e0c2136752e3683ee6f599f234072b12a6ec01e08a4aa287d267717150c4883
Chemistry active paper	output/release/chemistry-1.3.0/01_From-Fold-to-Chemistry_Chemistry-Branch-Paper-001-v1.3.md	d1880beab7d6eec028167d8f353a86936231dc7a2cfd7c2b318cf5512f121dc1
Materials active paper	output/release/materials-1.3.0/02_From-Fold-to-Materials_Materials-Science-Branch-Paper-001-v1.3.md	79fe7890cf31017536cfc5d76bf816783ab805f484de22b675e311e88d12c1da
Current Physics tree	docs/branch_roadmaps/05-physics.md	511e3571b9184972f3516c0dccf585d507a7300742b64c5b2f3c534a8a1731b5
Current Chemistry tree	docs/branch_roadmaps/06-chemistry.md	5409c24655cc41a7ad481658b6caee2e16f6e5734e3abba43efc8e994964d60
Current Materials tree	docs/branch_roadmaps/07-materials.md	ca7fd43522795997707acbc837367a8badfa326fed39ad407ff753203111a63
Current global claim ledger	census/claims.json	b9fc9409b70a513324ce8595a8a28f51ce10718848cd2fe55d8516e8995de61a

The active-paper hashes identify the scientific prose assessed here. Branch manifests and claim receipts bind the underlying release and machine packages. The roadmap hashes identify current dependency and completion guidance. The global ledger supplies the theory-wide owner and root routes. If any of these files changes before release, the chapter must be audited again rather than assuming that the change was editorial.

The aggregate arithmetic is independently transparent:

$$\begin{aligned}
 368 + 281 + 289 &= 938 \quad \text{liveclaims,} \\
 257,776 + 71,936 + 73,984 &= 403,696 \quad \text{candidates,} \\
 1,472 + 1,124 + 1,156 &= 3,752 \quad \text{passedcontrols.}
 \end{aligned}$$

The sums do not erase branch status. Chemistry's 281 is 272 frozen discipline obligations plus nine returned claims. The empirical package counts remain 238/130 by type in Physics, 273/eight by type in Chemistry and 289 receipt-bound packages in Materials.

How this part can change

A lawful correction begins with a discrepancy against one of the current authorities. If the active paper, inventory and machine record disagree, the records are traced to their source and explicit supersession. A clerical identity can be corrected when the authoritative direction is unambiguous. A scientific disagreement-different count, value, chronology, evidence class or owner-must remain visible for Maria Smith unless the repository itself authorises the resolution.

A lawful extension registers a new question, candidate grammar, dependency route, controls, evidence boundary and receipt. It may add a new physical measurement, chemical species, material class or downstream use. It may not change a failed historical row into success, call a new source an old blind target, or insert a measured value into the earlier proof grammar.

Falsification is likewise local and exact. A disjoint CODATA interval would be adverse to the registered fine-structure empirical claim. A qualifying Smithium experiment that fails the sealed identification protocol would be adverse to that

protocol; no experiment leaves the status unresolved. A crystallographic classification outside the complete registered grammar would challenge the relevant Materials claim. The wider corpus would then require a dependency impact audit rather than a rhetorical declaration that everything either survives or collapses.

The dependency relation among the branches is now explicit:

1. Physics turns exact formal carriers into quantities, dimensions, fields, interactions, constants and physical measurement interfaces.
2. Chemistry uses those carriers to derive stable chemical identity, composition, bond, reaction, periodic extension and source-bound molecular evidence.
3. Materials Science organises physical and chemical carriers across phase, defect, microstructure, surface, process and scale to derive structure-property relations.
4. Biology receives the resulting biochemical and material interfaces but must derive living organisation in its own right.

Three worked dependency routes

The fine-structure route shows how an upstream exact carrier becomes a measured physical constant and then a downstream constraint. Foundation supplies binary distinction and reversible return; Mathematics supplies exact finite ratios and order; Information Science supplies immutable record and custody; Computation supplies enumeration and checking; Reversible and Quantum Computation supplies the reversible process boundary; Physics supplies the three-generator, least covers, ladder and exact α . Chemistry then uses the resulting atomic boundary without re-deriving α , and Materials uses the chemical and physical carriers in its electronic and optical response claims. One law is depended upon many times but owned once.

The Smithium route begins with the Physics nuclear closure (*126,184*), orbital capacity and fine-structure endpoint. Chemistry adds elemental identity, electronic configuration, entrance-partition grammar, decay coordinates, ion and oxidation relations, transition classes, chemical states and identification protocol. The empirical status remains a standing prediction because the required synthesis and identification observations are not present. Materials could later study a Smithium phase or specimen only after Chemistry supplies the observed chemical carrier; it cannot manufacture that observation from a formal materials model.

The biomaterial route begins with physical quantity, field, transport and mechanical response; adds chemical species, bond, reaction, solvent and polymer identity; then adds material phase, pore connectivity, surface, interface, degradation and scaffold organisation. Biology receives those carriers and must add cell, function, regulation, development and living-system evidence. Medicine must further add clinical population, intervention, comparator and outcome. A favourable early interface result cannot leap over those dependency boundaries.

One object, different scientific coordinates

Object encountered downstream	Physics coordinate	Chemistry coordinate	Materials coordinate	Boundary handed onward
Atom or ion	Physical bound state, shell, transition and charge	Element, isotope, ion and chemical role	Constituent in phase, lattice, defect or interface	Biological or device carrier only after further organisation
Energy transfer	Work, heat, radiation and conserved physical ledger	Reaction, activation, thermochemical and electrochemical route	Stored, transported, dissipated or damage-producing specimen response	Metabolic, environmental or engineered use remains downstream
Phase	Collective physical state boundary	Chemical state and coexistence identity	Specimen phase, fraction, microstructure and path	Biological compartment or Earth-system phase needs its owner
Spectrum	Physical transition and detector record	Molecular/chemical identity and selection rule	Band, mode, defect, optical or structural response	Instrument or application claim remains Engineering-owned
Interface	Physical boundary and transferred response	Chemical adsorption, reaction and constituent transfer	Surface, adhesion, segregation, coating and interfacial property	Cell response, clinical safety and system fate remain downstream
Prediction	Sealed physical consequence and measurement adapter	Sealed identity, reaction or property consequence	Sealed specimen relation under complete coordinates	Later branches must preserve the original chronology

The table makes reduction precise. A later coordinate does not contradict the earlier carrier; it adds a relation that the earlier branch did not own. The material phase depends on physical and chemical phase but includes specimen history and microstructure. A biological membrane depends on a material interface but adds function and maintenance. The dependency spine is therefore neither a collection of independent essays nor a claim that one vocabulary can replace every scale.

Exact structure and empirical magnitude

Across the three branches, exactness has a consistent meaning. The fine-structure fraction, cover depths, nuclear closures, periodic capacities, entrance-partition count, rotation orders, crystal systems, Bravais classes and acoustic branch count are exact at their registered grammars. Their exactness does not depend on a displayed decimal or sample mean.

Measured values have a different completeness. A CODATA interval is complete for the named release and uncertainty statement. A molecular property is complete only with species, state, method and condition. A material modulus is complete only with specimen, direction, preparation, scale, method and condition. Exact rational interval arithmetic can preserve those records without promoting them into universal proof values.

This separation also supports persuasion. The paper can state that the exact α^{-1} lies inside the registered CODATA interval, that the aromatic support sequence is generated and its energy surface is positive at the registered boundary, and that seven crystal systems and fourteen Bravais classes survive complete enumeration. It need not weaken those results with generic doubt. It must, however, preserve the different chronology and evidence class that makes each statement true.

Adverse and unresolved evidence through the spine

Adverse evidence is inherited as a constraint, not discarded at a handoff. Physics carries the electroweak all-input tension, historical proton-radius mismatch, unavailable driven vacuum/inertia test, resonance losses and standing unobserved sectors. Chemistry carries failed source routes, unavailable and adverse molecular rows, the mechanism contamination record, unresolved Smithium measurements and the corrected dependency graph. Materials carries direct/proxy capture halts, unresolved source rows, a microstructure halt, an extreme-label halt and all claim-level adverse outcomes beneath the Grand Lock.

Later evidence may lawfully change current status. A new driven apparatus could test the vacuum/inertia family. A qualifying synthesis and five-record identity could change Smithium's empirical state. A recovered source could resolve a Materials transport. In every case, the original status, date and falsifier remain visible. A successful later result cannot become an earlier blind prediction unless the chronology already establishes that classification.

Reproducibility and publication authority

The conceptual chapter presents the dependency argument. The complete branch papers preserve every claim section and status. Candidate censuses, decisions, controls, evidence packages, source snapshots, receipts and manifests preserve the machine archive. A specialist may begin with the result and follow a claim ID into its audit; an independent reproducer may ignore the narrative and reconstruct the machine package. Neither access route is subordinate to credentials.

Publication remains a separate act. Maria Smith alone authorises the paper and its version. A passing engine receipt cannot publish itself, and publication does not strengthen a scientific status. The active paper identities and manifests freeze what was assessed; any later edit that changes a count, hash, source or status requires a new verified release state.

The precise boundary of current-evidence full-field completion

Physics, Chemistry and Materials are field-wide complete for their frozen current-evidence censuses. That statement includes exact candidate enumeration, one survivor per claim, controls, root routes, source custody and retained adverse states. It excludes every unregistered material, molecule, condition, apparatus and future observation. It excludes biological function, clinical outcome, Earth-system fate and engineering deployment.

The remaining openness is productive rather than evasive. Standing predictions have declared measurements that can change their status. Exact structures can be extended by new registered questions. Source transports can be repaired without rewriting history. Downstream branches can add genuinely new laws while preserving their upstream carriers. This is the end of the project's current V3 investigation at these three field-wide boundaries, not the end of science.

Why the order cannot be reversed

Chemistry cannot precede Physics because element, isotope, electron state, energy, charge, spatial orientation, transition and measurement already have physical meanings. Treating them as chemical axioms would duplicate upstream ownership and conceal their derivation. Physics cannot complete Chemistry because a physically possible collection of nuclei and electrons does not by itself select species identity, bond, stereochemistry, reaction route, solvation or analytical equivalence.

Materials cannot precede Chemistry because composition, bond, phase identity and reaction supply the constituents whose organisation is measured. Chemistry cannot complete Materials because chemical formula does not fix grain, texture, defect

population, interface, processing history or specimen response. The same chemical carrier can support materially distinct states, and that difference is not an error to be averaged away.

Biology cannot be inserted before the physical-chemical bridge because living systems use quantity, energy, molecule, reaction, membrane, scaffold, soft matter and material transport. Yet Materials cannot complete Biology because maintenance, heredity, regulation, development, reproduction and evolution are new relations. The handoff is therefore a dependency, not a reduction.

The empirical order has the same direction. A target is not allowed to select the formal grammar. A molecular database cannot define the Physics constant used to structure it. A material measurement cannot be used to revise a chemical identity silently. A biological outcome cannot retroactively make an adverse interface test favourable. Later evidence may test, extend or falsify upstream work; it does not become an earlier premise.

The order also clarifies the role of computation. Enumeration, independent reconstruction and tests execute the registered relations, but successful execution does not own the physical, chemical or material result. The application programmes-including Protein Fold-consume these branches through explicit dependencies. Their eventual successes or failures will be important evidence at their registered application boundaries, not selectors of the foundational laws already sealed.

This dependency discipline is what allows the three branches to form one continuous theory without becoming one undifferentiated vocabulary. Every handoff carries enough structure to make the next problem possible and leaves enough unresolved that the next branch has genuine scientific work to do.

This sequence prevents both reduction by assertion and independence by forgetfulness. Chemistry is constrained by Physics but is not merely renamed particle physics. Materials is constrained by Physics and Chemistry but is not merely a list of compositions. Biology will be constrained by all three but cannot inherit function or life from them without a registered biological relation.

For an unfamiliar specialist, the shortest verification route is therefore claim-specific. Begin with the exact statement and owner; inspect its declared dependencies and grammar; reproduce the candidate census, decisions, survivor and controls; then inspect the evidence package only if the statement makes an empirical claim. Check source identity, target chronology, preserving coordinates, all-row treatment and falsifier before reading the interpretation. Finally, follow the outgoing handoff to see whether the next branch uses the result at the same strength. This order prevents a compelling conclusion from concealing a weaker receipt and prevents a machine success from being mistaken for a measurement.

The same route permits efficient criticism. A critic need not reject or accept the entire theory. They may challenge one dependency, candidate omission, decision, source adapter, uncertainty interval, chronology field or ownership edge. A valid defect remains local until the explicit graph shows its downstream consequences. The public archive is designed to make that impact trace possible and to preserve the correction if the challenge succeeds.

The current completion claim is correspondingly bounded. These three branches are complete at their frozen registered current-evidence scopes. Their exact derivations, machine receipts and evidence ledgers are closed at those scopes; their adverse and unresolved records remain part of the result; their standing predictions remain open to measurement; and their ownership handoffs are explicit. This is completion of the project's present V3 investigation at the declared boundaries, not the end of Physics, Chemistry, Materials Science or science itself.

Part III. Life, Mind, Earth, Cosmos, Society, Engineering and the Computational-Proof Programme

Orientation: from physical organisation to living, observing and constructed worlds

The preceding branches establish the exact mathematical, informational, computational, physical, chemical and material carriers on which this part depends. They do not, by themselves, establish life, illness, experience, planetary history, astronomical observation, collective action or a working engineered artefact. Each of those domains adds an ownership boundary that cannot be obtained by changing the vocabulary attached to an upstream law. A chemical reaction is not thereby a living process; a biological deviation is not thereby a diagnosis; an information-bearing computation is not thereby phenomenal presence; a physical flow is not thereby climate attribution; a correlation between people is not thereby power; and a successful prototype is not thereby validation of either its intended use or the theory from which its design began.

This part follows those boundaries in dependency order. Biology adds bounded self-maintenance, heredity, regulation and evolutionary continuation to the chemical and material substrate. Medicine takes the resulting living carrier as an upstream condition but owns patients, disease, prognosis, intervention and clinical evidence. Consciousness and Cognitive Science require living, informational and computational organisation while adding interior observation, perspective, phenomenal differentiation and cognitive evidence. Earth and Environmental Sciences retain physical, chemical and biological flows within one source-dated planetary system. Astronomy and Cosmology then extend the observation chain beyond the immediate observer while separating universal law from object history, source propagation and catalogue custody. Social and Collective Sciences add identified agents, typed relations, institutions, power, history and normative bridges. Engineering Translation turns admitted knowledge into specified, measured, tested and maintained artefacts without allowing application performance to select a scientific law. Cross-Branch Synthesis composes these typed results only after their owners have admitted them.

The computational-proof programme is reported last because it tests whether selected formal structures can be carried through exact implementations and external comparisons. Its programmes do not constitute hidden premises of the field branches. Their status is deliberately non-uniform. Protein Fold V3 is an active exact search and scientific-gate programme with a large verified parent census and a closed high-attraction slice, but its global fold frontier, whole-chain recurrence and protected AlphaFold parity campaign are not closed. Chess Fold and Go Fold retain historical V2 results while their clean V3 recreations have not begun in the current workspace. Unison Fold AI retains a substantial V2 implementation record and a V3 scaffold whose mathematical closure is `OPEN`, whose empirical validation is `NOT_RUN`, and whose remote deployment is `NOT_AUTHORIZED`.

Current numerical boundary of this part

The seven domain branches and the current Cross-Branch Synthesis represented here contain **555 model-admitted claims**, **142,080 exhaustively registered candidates**, **555 unique survivors** and **2,220 passed required controls**. Every one of the 555 current claims is root-traced and recorded as depth-independent at its declared grammar boundary. Those totals must not be misread as a declaration that every possible question in the corresponding disciplines has been answered. They combine published foundation claims with later, separately registered return families:

Branch	Published foundation	Current additions	Current claims	Candidates	Survivors	Controls	Current completion statement
Biology and Life Sciences	75	7	82	20,992	82	328	Foundation and returned families complete; 82/424 registered full-field obligations complete
Medicine and Health Sciences	72	4	76	19,456	76	304	Foundation and returned placebo family complete; full-field census not frozen
Consciousness and Cognitive Science	72	5	77	19,712	77	308	Foundation and returned family complete; full-field census not frozen
Earth and Environmental Sciences	74	1	75	19,200	75	300	Foundation and returned tipping claim complete; full-field census not frozen
Astronomy and Cosmology	72	5	77	19,712	77	308	Foundation and returned family complete; full-field census not frozen
Social and Collective Systems	72	4	76	19,456	76	304	Foundation and returned family complete; full-field census not frozen
Engineering Translation	72	8	80	20,480	80	320	Foundation and returned protocol family complete; full-field census not frozen
Cross-Branch Synthesis	-	12	12	3,072	12	48	Frozen inherited-return family complete; final global synthesis remains open

The evidence classes are also non-uniform. In Biology, 76 current claims are recorded as empirically tested and independently replicated and six as independently replicated at their stated structural boundaries. Medicine has 73 and three respectively; Consciousness has 73 and four; Earth has 74 and one; Astronomy has 73 and four; Social has 73 and three; Engineering has 72 and eight; and the twelve synthesis claims are independently replicated formal compositions. These labels describe the actual claim packages. They do not license a translation from structural replication into empirical confirmation, nor from a source correspondence into a new exact numerical prediction.

The evidence rule governing every chapter

Formal proof language is used only for exact derivational closure at a stated candidate boundary. *Derivation* names a generated structure or law. *Implementation* names an executable realisation. *Prediction* is reserved for a prospectively sealed consequence. *Observation* remains tied to an external source record, and *measurement* to an instrument, method, unit and uncertainty or interval. *Reconstruction* denotes an inferred history or latent state. *Correspondence* denotes a relation established after derivation; *compatibility* means non-adverse but not discriminating evidence; *support* means relevant but non-unique evidence. *Confirmation* and *validation* are used only where the registered evidence protocol warrants them.

This vocabulary matters most where the record is difficult. The first Earth earthquake catalogue comparison is adverse and remains adverse. The later homogeneous holdout is compatible and does not erase the first result. The first SPARC comparison in Astronomy is adverse and remains adverse, even though a separate uncertainty-aware method record places exact rank four at a reported systematic boundary. The Protein V2 whole-chain result is adverse, while several local windows are strong and a theoretical support construction is stronger still; none of those facts may be substituted for another. Failed source transports remain failed transports, not observed absences. Historical corrections remain part of the scientific record even when a later claim supersedes their current wording.

The handoff rule

Every branch owns the distinctions it introduces and passes only typed outputs downstream. Biology may use Chemistry's reaction identities but owns the boundary between chemistry and life. Medicine may use biological state but owns the clinical relation between a person, a condition, an intervention and an outcome. Consciousness may use neural and computational carriers but owns the distinction between public access and interior presence. Earth may use physical laws but owns a particular planet's source-dated history. Astronomy may use those same physical laws but owns distant-source reconstruction and cosmic chronology. Social science may use cognitive and biological capacities but owns collective relations and institutions. Engineering may use any admitted law but owns requirements, design, safety, testing and deployment.

The handoff is therefore neither reduction nor insulation. Upstream laws constrain what is possible; downstream branches add boundary conditions, identities, histories, relations and evidence constitutions. A downstream adverse application result returns a typed question to the responsible owner. It cannot silently repeal an upstream theorem, and successful application cannot retroactively prove one.

16. Biology and Life Sciences: from organised chemistry to retained life

16.1 Why Biology is required

Chemistry establishes entities, compositions, reactions, stoichiometric conservation, molecular distinction and stereochemical relations. Materials Science establishes organised interfaces, phases, defects and collective structure-property relations. Yet neither branch supplies the criterion by which a process network remains the same living organisation while its material constituents turn over. Biology begins at precisely that unresolved point.

The least living carrier admitted here is a bounded process network that retains its identity by regulated material and energy turnover inside a declared viability window. The classification is joint. Compartment, maintenance, heritable continuation and regulated resource exchange must be retained together. Chemistry alone is not relabelled life, and a dissipative or autocatalytic pattern alone is not assigned every biological property. This boundary gives the subsequent families their order: a living organisation must first have a distinguishable compartment; the compartment must support resource and metabolic relations; those relations must permit retained and copied organisation; inherited copies make variation and evolutionary continuation possible; regulated genomes and proteins realise bounded cellular processes; cells support development, organisms, physiology, populations and ecosystems; and all of these require an evidence handoff that preserves the conditions under which a biological statement was observed.

The published Biology foundation contains 75 claims in twelve ordered families. Seven later claims close the current returned origin, homochirality, somatic, neural, differentiation, ecosystem and validation obligations. The current branch therefore contains 82 claims, 20,992 candidates, 82 survivors and 328 controls. Its foundation and current returned families are complete. Its registered full-field programme is not: 82 of 424 obligations are closed, leaving 342 open. That distinction is the present meaning of "Biology complete" in the repository.

16.2 Living identity, compartment and resource closure

Living identity is identity through controlled turnover, not identity of a fixed material inventory. A cell or organism may exchange molecules, energy and information while preserving the organised relation by which those exchanges remain viable. The boundary cannot be implicit. It must identify what is inside, what is outside, which transfers are permitted, how a change in compartment state is detected and what loss would end the retained organisation.

The compartment families therefore preserve membrane or boundary identity, selective exchange, internal and external states, gradients, transport routes and failures. A boundary is not merely a line drawn around reactions. It has a causal role in maintaining distinguishable conditions. Conversely, a sealed container with no regulated exchange does not satisfy the living relation. The system must couple a maintained interior to resources and waste pathways without hiding an external supply.

Metabolism is consequently expressed through complete ledgers. Source, carrier, transformation, useful retention, export and loss stay visible. Energy and matter are not interchangeable words, and neither becomes biological purpose simply because it flows. A metabolic route belongs to the living organisation only when its inputs and outputs participate in the bounded maintenance relation. Autocatalysis can be an important organising mechanism, but it does not independently supply compartment, heredity or regulated exchange.

The returned origin claim sharpens this point. For a driven complete **m**-role autocatalytic carrier, ignition occurs when **m - 1** roles are internally supported and one declared seed is supplied. The exact statement is conditional on that generated carrier and seed boundary. It identifies the least supplied transition required for the complete role relation to become self-supporting; it is not an observation that one such abstract carrier was the unique historical origin of terrestrial life. The origin-of-life extension remains responsible for connecting exact ignition structures to measured prebiotic chemistry, compartment formation, environmental conditions, chronology and competing mechanisms.

16.3 Inheritance, code and evolutionary continuation

Once a living organisation can retain itself, the next unresolved question is how a structure may be continued through copies that are similar enough to preserve identity and different enough to permit change. The inheritance families distinguish template, copy process, copied record, error, repair, lineage and environment. Replication is not perfect identity, and mutation is not automatically functional novelty. The record must show what was retained, what changed and how that change passed into a continuing lineage.

The branch's exact coding construction is among its clearest finite results. Two held distinctions generate four coding labels. Three ordered positions then generate

$$4^3 = 64$$

complete triplet words. Holding the first two positions produces

$$4^2 = 16$$

first-two-position boxes, each containing four possible third-position words. The exact combinatorial construction precedes its biological naming. The DNA and codon interpretation is admitted only through the biological carrier and source boundary; it does not arise by inserting the known genetic code as a premise. Nor does the construction by itself specify every mapping between codons, amino acids, regulatory context, translation machinery and observed organisms. Those mappings remain biological records.

Genes and genomes are treated as organised, regulated carriers rather than as strings detached from expression. Sequence identity, regulatory state, copy-number relation, chromosomal context, repair, transcription, translation and phenotype are kept distinct. A sequence difference is not automatically a functional difference; an expression association is not automatically a causal mechanism; and a population frequency is not an individual destiny. The same custody permits the branch to distinguish a heritable variant from a somatic event and a selected continuation from drift.

Evolution is derived as heritable variation, differential continuation and a controlled environmental relation. Selection and drift are not synonyms. Fitness is condition-bound rather than an intrinsic scalar permanently attached to a variant. A beneficial

variant does not inevitably fix: finite population history, competing variants, environmental change and stochastic sampling records remain relevant at the empirical boundary. This correction supersedes stronger earlier language while preserving the earlier claim in the historical reconciliation. The V3 result is more specific, not less evolutionary: it states exactly which relation must be present before differential continuation can be attributed to selection.

16.4 Protein structure, function and the Fold handoff

The protein family begins with a sequence-bearing chemical carrier and asks how ordered local interactions, geometry, environment and retained alternatives can constrain a folded ensemble. The admitted biological statement is a directed, condition-bounded fold with a retained ensemble. A protein is not reduced to one universal rigid native shape, and a sequence is not treated as if it carried no environmental or interaction context. Conformation, ensemble, function, binding, modification and observation method remain distinct.

This correction matters to the computational programme. The conceptual law does not promise that any finite search implementation will immediately predict every whole-chain experimental structure. It defines the relations an implementation must preserve: sequence identity, interaction custody, geometric compatibility, environmental condition, alternative-state retention, evaluation boundary and falsifiable comparison. A computational candidate can therefore fail while the broader biological relation remains properly posed. Equally, local success cannot be promoted to global prediction without closing the missing whole-chain relation.

The current Protein Fold V3 programme takes this handoff seriously. It uses an exact sequence representation and a six-axis joint relation, retains every registered interaction class and separates the search census from external structure comparison. Its current results are presented in §24.3. The essential point at the Biology boundary is that the branch has supplied a formal protein-folding constitution and a testable implementation route, while the active programme has not yet closed the global fold frontier or the registered AlphaFold parity campaign.

16.5 Cells, development, organisms and physiology

Cells integrate compartment, metabolism, inheritance, expression and protein function into regulated state transitions. The cell-process family therefore keeps state, signal, response, cycle, repair, death and lineage records separate. Recurrence may be necessary for a continuing cell cycle, but recurrence alone does not diagnose cancer. The returned cancer-state claim requires a persistent non-terminal cycling carrier together with loss of a differentiation transition and escape from the combined controls on division, death and differentiation. It remains a structural classification; clinical diagnosis belongs to Medicine and requires patient-, tissue-, assay- and outcome-specific evidence.

The returned somatic-germline result uses denominator factorisation to separate a finite somatic transient from an odd-denominator recurrent carrier. It provides an exact orbit distinction at its generated boundary. It does not make every measured somatic or germline history a function of denominator parity. Biological mapping still requires cell lineage, inheritance route, developmental timing and measurement.

Multicellular development adds positional, lineage and differentiation relations. A multicellular body is not just a collection of identical cells; its organisation depends on controlled difference, signalling, spatial constraint and retained whole-body coordination. Differentiation is therefore not an optional label added after a cell state is measured. It is an owned transition in the organisation. The current cancer correction follows because loss of that transition changes the reachable state relation, while cycling alone does not.

Organism physiology extends the same rule across coordinated systems. A measurement is retained with organism, tissue or organ, condition, method, time and reference boundary. The returned neural threshold claim identifies Half-One as the least normalised activation support in its exact carrier. It does not declare a universal measured membrane voltage, firing rate or behavioural threshold. Dimensional values remain instrument- and system-specific observations.

16.6 Populations, ecosystems and scale

Population and ecological statements require more than an average over individuals. Population identity, sampling frame, births, deaths, movement, inheritance, resource relations, interactions and observation window must be preserved. An ecological network may be represented by nodes and links, but the representation does not establish a universal topology. The branch's source reconciliation preserves the empirical result that strongly scale-free networks are rare rather than treating scale freedom as a default law of life.

The same discipline corrects a claimed universal three-quarter allometric exponent. Current evidence retains three-quarter, two-thirds, unity and non-universal cases. The exact SFT fractions remain formal carriers at their declared boundaries; they are

not silently converted into one universal measured exponent across taxa, traits and methods. Source population, body size range, regression choice, uncertainty and biological mechanism remain part of any empirical allometry claim.

The returned ecosystem claim provides an exact bounded period-four orbit on the $3/5$ Fold carrier. It demonstrates a finite recurrent structure and supplies a candidate carrier for bounded ecosystem dynamics. It does not forecast every ecosystem, prove stability from cycling alone or replace measured species, forcing, resource and disturbance relations. One periodic orbit cannot identify an ecosystem's empirical stability class without those records.

16.7 Homochirality: observed asymmetry and unresolved origin

Biological homochirality is retained as an observation: terrestrial proteins predominantly use L-amino acids and biological sugars predominantly use the opposite handed convention. The V3 branch does not use that observation to claim that the historical origin of the asymmetry has already been measured. Earlier parity-selection language is corrected accordingly.

The returned homochiral-amplification claim begins with a held Fold parity and a successor-amplification relation. It generates increasing finite dominance of one label without deleting the other label from the complete carrier. This is a structural result about amplification from a retained asymmetry. It does not establish which prebiotic process supplied the initial bias, the magnitude of that bias, the environmental route by which it was amplified or the full history connecting the formal carrier to terrestrial biomolecules. Those are open empirical obligations.

16.8 Biological evidence, corrections and current status

The Biology foundation sealed all 75 foundation predictions and 19,200 candidate forms before source selection. A separate custodian opened the registered sources after sealing, required claim-specific fragments, retained adverse and failed rows and compared the source-derived held label with the registered prediction. The current seven-claim return family was registered separately and is not backdated into that foundation chronology.

The foundation used 17 usable sources and preserved four failed transports: one TLS failure, one HTTP 403 response and two challenge-page failures. A failed transport is a statement about access, not a statement about the scientific content that could not be obtained. It remains visible unless a later, separately registered source capture lawfully adds evidence. Thirty Biology-owned V1/V2 atoms were reconciled; 21 required correction at the V3 boundary. The major corrections are retained here because they prevent the current branch from inheriting stronger but unsupported language:

Topic	Current scientific record
Beneficial variants	Differential advantage does not imply inevitable fixation; drift, finite history and environment remain
Protein state	Condition-bounded ensemble retained; no universal rigid single-shape reduction
Homochirality	Biological handedness observed; historical origin unresolved; amplification result is structural
Allometry	No universal three-quarter exponent; multiple exponents and non-universal cases preserved
Biological networks	Strong scale-free form is not a default; empirical rarity retained
Senescence	Denominator parity alone does not identify or predict organismal senescence
Cancer	Recurrence alone does not diagnose cancer; differentiation loss and escape from controls are required in the structural claim
Ecosystem stability	One bounded orbit is not a universal stability forecast

The seven-claim validation record compares the current origin, homochirality, somatic, neural, differentiation and ecosystem structures with seven registered sources. The source-derived categorical labels match the registered claims at that comparison boundary. Observed counterparts are retained, while exact Fold thresholds and orbits remain structural predictions where no direct dimensional measurement establishes them. The homochirality origin and the quantitative mapping from abstract carriers to organisms remain unresolved.

16.9 Biology's downstream handoff and open frontier

Biology hands Medicine an identified living carrier with compartment, metabolic, hereditary, developmental, physiological and population relations. It hands Consciousness a neural and organismic carrier without claiming that neural activity is already phenomenal presence. It hands Earth Science living stocks, flows and ecological interactions without claiming that biological laws determine a planet's contingent history. It hands the Protein Fold programme a formally constrained structure-function problem without claiming that implementation closure has already occurred.

The current branch is complete at its published foundation and returned-family boundary, and open across 342 registered full-field obligations. The open work includes deeper origin-of-life mapping, quantitative homochiral pathways, expanded

molecular and systems biology, complete protein computational tests, additional organismal and ecological evidence and the remaining registered family census. Completion here means completion of a defined current investigation, never the end of Biology or the end of science.

16.10 Dependency-resolved family ledger

16.10.1 From a living boundary to a continuing lineage

The first six Biology families solve a sequence of different problems. A living boundary identifies the organisation; a compartment gives that organisation an oriented inside and outside; metabolism connects the compartment to finite resources; inheritance makes continuation possible; variation makes descendants distinguishable; and evolution specifies how those differences continue under an environment.

Family	Unresolved problem entering the family	Admitted output
Life Boundary	Chemistry and dissipation do not classify a process as living	Bounded self-maintaining organisation, life/non-life boundary, autocatalytic boundary, turnover identity and loss of closure
Compartment Identity	A living process needs an oriented, continuing boundary	Compartment, inside/outside orientation, membrane organisation, selective permeability, nesting and lineage provenance
Metabolism Resource	A compartment cannot persist without controlled exchange	Metabolic network, uptake, conserved transformation, coupling, export and finite resource closure
Inheritance Replication	Turnover alone does not continue organisation across copies	Information-bearing inheritance, template copying, replication, fidelity/error, heritable variation and lineage
Variation Evolution	Heritable copies require a rule for differential continuation	Variation, heritability, differential reproduction, selection, fixation/loss and drift distinction
Gene Genome Regulation	Inheritance must be carried and regulated in an organism	Four-label nucleic-acid alphabet, 64 triplet words, 16 boxes, gene interval, genome organisation and expression regulation

The ordering prevents a circular definition of life by reproduction alone. A crystal can continue a pattern without satisfying the complete living relation; an autocatalytic chemistry can amplify without heredity; a genetic sequence can be copied outside a self-maintaining organism. The branch admits these structures where their own conditions hold and calls the joint system living only after every required relation is present.

Compartment lineage provenance is particularly important. If a boundary divides, merges or is rebuilt, the record must show how the descendant compartment derives from the predecessor. Mere spatial continuity may be insufficient, while complete material identity is unnecessary because turnover is expected. The lineage is carried by the organised transition relation.

Finite resource closure also distinguishes biological persistence from a perpetual-motion description. A culture, cell or organism may be sustained by an environment, but the supplied resources must be inside the declared system ledger or explicitly cross its boundary. Exhaustion, toxic by-product and failed export are possible biological failures and cannot be omitted from a favourable maintenance trace.

The variation family keeps three events distinct: a new difference arises, it is heritable, and its descendants continue differentially. Only the complete relation warrants selection language. Fixation and loss are population histories, not logical consequences of the word "beneficial". The drift claim preserves finite sampling and lineage contingency as an alternative route to a frequency change.

16.10.2 From regulated code to organism and ecosystem

The second half of the foundation carries regulated inheritance into physical form, cells and larger living organisations:

Family	New carrier or relation	Boundary retained
Protein Structure Function	Sequence, directed fold, ensemble, function, interaction, assembly and homochirality	Sequence is not one rigid shape; function and conformation remain condition-bound
Cell Process	Cell identity, transport, signalling, excitation, division and dysregulated continuation	Cycling does not by itself diagnose cancer
Multicellular Development	Composition, adhesion, tissue, differentiation, lineage, morphogenesis and regeneration	Whole-body organisation is not an aggregate of interchangeable cells
Organism Physiology	Organism identity, process, homeostasis, reproduction, life cycle, senescence and longevity	One orbit or parity cannot determine lifespan
Population Ecology	Population, species, community, trophic network, ecosystem, recurrence, topology and allometry	Individual and population levels, topology classes and exponents remain distinct

Family	New carrier or relation	Boundary retained
Observation Handoffs	Specimen, sampling, condition, control, assay, uncertainty, causality and branch ownership	A favourable assay row is not a universal biological law

Protein identity begins with the ordered residue carrier but persists through condition-bound conformation and modification only under declared identity rules. Folding is directed because the transition relation constrains which states can follow; it is ensemble-bearing because accessible alternatives and dynamics may remain. Function adds a biological relation to another carrier or process. Interaction and quaternary assembly require partner identity, orientation, stoichiometry and condition. The V3 family thereby avoids using "fold", "shape", "state" and "function" as interchangeable words.

Cell signalling requires a sender or source state, a transmitted or local carrier, a receiver relation and a response boundary. A molecular association without direction, timing or response cannot establish signalling. Excitable transition adds a held pre-state, threshold relation and post-state. The returned Half-One result provides the least normalised support at its exact boundary, while Biology retains ownership of real membranes, ion channels, networks and measured thresholds.

Multicellular composition introduces relations that no isolated cell owns. Adhesion supplies a continuing interface; a tissue adds ordered collective organisation; differentiation changes reachable roles; developmental lineage preserves the route by which those roles arose; morphogenesis links local changes to whole form; regeneration requires restoration under a declared damage boundary. These distinctions make the differentiation-loss cancer claim possible without reducing cancer to one molecular event.

Homeostasis is regulated bounded return, not perfect constancy. A physiological variable may vary while remaining inside a viable relation, and similar observed values may arise from different control histories. Senescence and longevity require organism, cohort, environment, cause of loss and observation window. A recurrence in a component is not a lifespan theorem.

The population and ecology family marks a second aggregation boundary. Species identity requires the stated reproductive, phylogenetic or other empirical criterion; the word "species" does not fix one universal classification method. Trophic links require measured resource relations. Ecosystem organisation retains abiotic conditions and external forcing. Network topology and allometry are empirical families with explicit counterexamples, not decorations added to the exact orbit work.

16.10.3 The seven returned Biology claims as a connected programme

The returned claims are not seven isolated analogies. They revisit points at which the historical corpus had proposed a stronger connection between exact Fold structure and living evidence:

1. Autocatalytic ignition gives a least seed relation for a complete driven role carrier.
2. Homochiral amplification shows how a held parity can grow into finite dominance without deleting the other label.
3. Somatic-germline factorisation distinguishes a finite transient from an odd-denominator recurrent carrier.
4. Half-One excitation gives a least normalised event support.
5. Differentiation loss joins continuing division to escape from the full division/death/differentiation control.
6. The $\frac{3}{5}$ period-four carrier supplies an exact bounded ecological orbit.
7. The validation claim freezes the seven-source comparison, including the limit between categorical correspondence and unmeasured exact mapping.

The dependency route runs from ignition to retained asymmetry, from lineage type to excitable transition, from cellular control to ecosystem recurrence, and finally to external comparison. None of the exact values is allowed to enter a source row as if it were already measured. This is why the returned family can be formally complete while some organismal mappings remain unresolved.

16.10.4 Biological falsification surface

Biology's categorical claims fail if another generated candidate preserves all registered dimensions, if a dependency route is missing, if the independent implementation does not reconstruct the survivor or if a registered source contradicts a required empirical relation. Quantitative mappings have their own, narrower conditions. The codon count would fail if its two held distinctions and three ordered positions did not reconstruct four labels and 64 words. The earthquake- or SPARC-style notion of a measured interval does not apply to that finite combinatorial identity. By contrast, a proposed universal allometric exponent must survive measured populations and uncertainties; it cannot be rescued by the exact existence of the fraction in another grammar.

Protein computational claims fail at their implementation and blind-comparison boundaries. A local TM-score does not satisfy a whole-chain criterion. A theoretical support path does not satisfy an achieved sequence-only output. An unopened target is unresolved, not a null result. These separations are the operational bridge from Biology to the computational programme.

17. Medicine and Health Sciences: from living variation to clinical knowledge

17.1 Why Medicine is not reducible to Biology

Biology supplies organisms, variation, dysfunction, repair and population relations. Medicine begins when those relations must be judged for an identified person or population under a health, disease, prevention, intervention and care boundary. A biological difference does not become a disease merely because it can be measured. A group association does not become a diagnosis in one person. A mechanism does not become a demonstrated clinical benefit. A favourable mean does not erase adverse events, non-response, censoring or loss to follow-up.

The Medicine foundation therefore begins with patient and population identity and proceeds through health, variation, dysfunction, disease, injury, symptom, sign, measurement, diagnosis, cause, association, mechanism, confounding, risk, prognosis, outcome, intervention, comparator, exposure, adverse events, efficacy, effectiveness, safety, trials, observation, synthesis, screening, prevention, treatment, rehabilitation, individual inference, consent, ethics, privacy and uncertainty. Its order is clinical rather than merely biological: one must first know whose state is being described, then what observation was made, then which causal or prognostic inference is warranted, and only then what intervention and decision can be evaluated.

The published foundation contains 72 claims. Four current returned claims add the expectation/body fibre, the available-state boundary, the separation of objective physiology from report and their source-bound validation. The current Medicine branch therefore contains 76 claims, 19,456 candidates, 76 survivors and 304 controls. The foundation and this return family are complete; a full-field Medicine census is not yet frozen.

17.2 Patient, population, health and disease

The least clinical carrier retains the patient rather than replacing that patient with an aggregate. Identity includes the relevant person, time, condition, observation and decision context. Population evidence may constrain what can reasonably be believed about a person, but it does not uniquely determine that person's outcome or preferred course. Conversely, a single clinical history does not establish a population effect.

Health is not defined as the numerical absence of disease. Health and disease are organised, condition-bound states whose components may coexist. A person may have a diagnosed condition while retaining substantial function; a person may have no current diagnosis while reporting serious impairment or while an unmeasured disease process continues. Variation, dysfunction, disease and injury are kept distinct because each supports different causal, prognostic and intervention questions.

This prevents molecular language from doing clinical work it has not earned. A stereochemical distinction can be chemically exact and biologically important, but stereochemistry alone cannot establish efficacy, toxicity, appropriate dose or patient benefit. Those require exposure, comparator, outcome and adverse-event records. The V3 reconciliation corrects earlier mixed claims that crossed this ownership boundary.

17.3 Symptom, sign, measurement and diagnosis

A symptom is a participant-reported state. A sign is an observed clinical feature. A measurement is an instrument- and method-bound value. A diagnosis is a classified clinical judgement supported by declared criteria and alternatives. These carriers may inform one another, but they are not interchangeable.

The distinction is especially important when a report is the primary access to pain, mood, expectation or another first-person state. The report remains a real observation of what was reported; it is not automatically an objective measure of every proposed physiological mechanism. Equally, a physiological change does not automatically establish the participant's experienced state. The returned placebo family preserves both channels.

A complete diagnostic record must identify the patient and presentation, retain the available signs and measurements, state the decision rule and its version, preserve relevant alternatives and record missing or unavailable evidence. Classification success within one dataset does not by itself prove causation, future outcome or generality. A non-detection is not an observed absence unless the method, sensitivity and sampling boundary warrant that statement.

17.4 Association, cause, risk and prognosis

Clinical records often begin with association. The Medicine branch refuses to rename association as cause. Causal attribution must retain the proposed mechanism, alternative causes, confounding, selection, timing and an intervention or comparison structure capable of distinguishing the claimed route. A plausible mechanism may support a hypothesis while remaining insufficient to establish the magnitude or existence of a clinical effect.

Risk is tied to a target population, an event definition and a time horizon. Prognosis additionally retains the patient's current state, competing events, treatment history, censoring and uncertainty. A denominator pattern or Fold orbit cannot on its own set a person's prognosis. This is another explicit V3 correction: exact mathematical structure may organise a registered prognostic model, but patient-specific outcome requires clinical evidence.

The same discipline applies to screening. Test performance is not one number. With condition present or absent and test positive or negative, the complete finite carrier contains four cells. Sensitivity, specificity, predictive values and error counts describe different conditionings of that carrier. Their interpretation also depends on the population and prevalence boundary. A screening programme must add access, uptake, follow-up, treatment, harms and missed cases; the test result alone does not establish programme benefit.

17.5 Intervention, comparator and the alternative course

An intervention claim must preserve what happened and what it was compared with. Treatment name, dose, route, exposure, timing, adherence and co-interventions remain part of the administered course. Comparator identity must be equally complete. "Usual care", "placebo" and "no treatment" are not interchangeable labels unless their realised conditions are shown to be the same for the claim at issue.

The finite arm-by-outcome relation again exposes all four cells for a binary comparison: intervention with and without the registered outcome, and comparator with and without it. Favourable responders cannot be counted while adverse outcomes, non-response or missing follow-up are discarded. Efficacy, effectiveness, safety, benefit and harm remain distinct. Efficacy concerns the effect within the registered trial conditions; effectiveness concerns intended use in a wider practice setting; safety requires a scoped account of harms and exposure; benefit and harm additionally depend on affected people, timing and the decision context.

The branch permits deterministic allocation to supply auditable randomisation through concealed held labels. The relevant scientific properties are prospective allocation, concealment before assignment, preserved sequence, absence of target-based selection and complete custody-not an imported claim of metaphysical chance. Blinding is separately recorded. Random allocation does not imply blinded participants, clinicians, outcome assessors or analysts unless those controls are actually present.

Trials, observational studies and evidence syntheses therefore retain different inference structures. A randomised trial can estimate its registered comparison under its conduct and analysis conditions. An observational study requires explicit confounding and selection control. A synthesis must retain the eligibility rules, source studies, heterogeneity and missing evidence; it cannot convert incompatible source questions into one homogeneous result by averaging them.

17.6 Expectation, placebo and the available-state boundary

The returned placebo work derives a complete two-preimage expectation/body fibre. The exact structural values are $\frac{1}{4}$ for the body-labelled route and $\frac{3}{4}$ for the expectation-labelled route; both map to the common $\frac{1}{2}$ image, and together the retained preimages compose to the One:

$$\frac{1}{4} + \frac{3}{4} = 1.$$

This is an exact relation in the generated Fold carrier. It is not a claim that every placebo response is 75 per cent, every bodily response 25 per cent, or every clinical outcome a mixture with those measured weights. The fractions are structural coordinates, not imported clinical effect sizes.

The important clinical consequence is the **available-state boundary**. Expectation may order or select among routes that the organism can reach under the declared conditions. It does not manufacture a physiologically absent state merely because a participant or clinician expects it. The claim thus supports a strong, testable role for expectation without turning expectation into an unlimited causal substance.

The empirical constitution separates report from objective physiology. A participant's account, behavioural response, neural measure, autonomic signal and clinical endpoint remain separate rows with separate comparators and controls. The current

validation record uses nine PubMed-indexed source records and retains objective neural, autonomic and other neurophysiological modulation alongside null, small, context-dependent and unresolved findings. The observed evidence supports the categorical distinction and the existence of condition-dependent physiological routes. It does not measure the exact SFT fractions as universal effect magnitudes.

17.7 Ethics, consent and privacy as operational boundaries

Ethics is not an ornament added after the experiment. Consent, capacity, privacy, data minimisation, equipoise, stopping and access conditions change what may lawfully and scientifically be observed. A result acquired by violating its participant boundary is not repaired by statistical accuracy.

Consent requires information, voluntariness, capacity, an identifiable choice and a record. Capacity is decision- and time-specific rather than a permanent global label. Privacy requires declared collection, access, use, retention and disclosure boundaries. Data minimisation forbids collecting an unlimited record merely because it may later prove convenient. Equipoise and stopping rules preserve the relationship between uncertainty, accumulating evidence and participant welfare.

These requirements also govern computational medicine and any future Consciousness or Unison Fold study involving people. A model output is not a medical decision by itself. Its training source, target population, failure modes, calibration, uncertainty, human responsibility and intended-use validation must remain available to the clinical evidence chain.

17.8 Evidence custody, adverse records and current boundary

The Medicine foundation sealed its 72 claims and 18,432 candidates before its external source comparison. Eleven of thirteen registered authority sources were captured. Two HHS transports failed and remain recorded as transport failures. Purpose-matched ICH content was captured separately; it does not retroactively turn the HHS failures into successes.

All 356 V1 rows and 407 V2 steps were reviewed. The inherited Medicine content was decomposed into owned atoms, and the current record corrects four recurrent overreaches:

Earlier conflation	Current boundary
Placebo named without complete comparator or outcome	Placebo claims retain realised comparator, outcome, expectation, blinding and chronology
Molecular stereochemistry treated as proof of efficacy or toxicity	Chemical identity is upstream; clinical exposure, outcome and harm evidence are required
Denominator parity treated as prognosis	Exact orbit structure does not establish a patient's future clinical event
Cellular recurrence treated as cancer diagnosis	Biology supplies a structural carrier; diagnosis requires clinical, tissue and measurement criteria

Every favourable, adverse, null, missing and unresolved row remains part of the relevant evidence record. The branch is current-evidence closed for its foundation and four returned claims, not for all medicine. Open work includes the unfrozen full-field census, disease- and intervention-specific validations, individual decision boundaries, longer-term harms, diverse populations, transportability and additional prospective tests of the expectation/body carrier.

Medicine hands Consciousness a rigorous distinction between first-person report and objective measure. It hands Social Science population and intervention structures that do not erase individuals. It hands Engineering the intended-use, safety, ethics and validation requirements for health technologies. It preserves clinical ownership whenever a biological, computational or engineering result is proposed for patient care.

17.9 Dependency-resolved family ledger

17.9.1 From patient identity to an interpretable diagnosis

The first Medicine families progressively constrain what a clinical statement can mean:

Family	Required distinction	Principal output
Patient Population Health	Person, episode, population, health state, observation and course	An identified longitudinal clinical carrier that cannot be replaced by an aggregate
Variation Dysfunction Disease Injury	Normal variation, dysfunction, disease, injury, comorbidity, severity and stage	A typed condition relation rather than a numerical deviation labelled disease
Symptom Sign Measurement Diagnosis	Report, observed sign, instrument value, classification, alternatives and accuracy	A reconstructible diagnostic chain
Cause Association Mechanism Confounding	Association, causal route, mechanism, confounder, mediator and modifier	A causal claim with its alternatives and context retained

Family	Required distinction	Principal output
Risk Prognosis Outcome	Risk, incidence/prevalence, prognosis, outcome, censoring and calibration	A time- and population-bound future statement

Patient identity must survive repeated episodes. The same person can occupy different health states; different people can produce similar measurements. A clinical episode therefore links time, presentation, investigation, intervention and outcome without treating one visit as a complete person. Longitudinal course preserves change and the gaps between observations.

Normal variation is reference-bound. A value may be uncommon without being pathological, or common while still harmful. Dysfunction requires impaired organisation under a declared task or viability boundary. Disease adds a clinical classification and evidence route. Injury adds an event or damaging process. Comorbidity preserves multiple interacting conditions rather than forcing all observations into one diagnosis. Severity and stage retain the criteria and version by which the classification was made.

Diagnostic accuracy requires the complete condition-by-test carrier, target condition, reference procedure, population and uncertainty. Differential diagnosis retains live alternatives and the evidence that favours or rejects each. The model may identify a best-supported classification without claiming that all alternatives are ontologically impossible.

Causal mediation and effect modification solve different questions. A mediator lies on a proposed route; a modifier changes the relation across conditions or groups. Confounding supplies an alternative common cause. Failure to distinguish them can turn adjustment into either removal of part of the effect or preservation of a spurious association.

Prognostic calibration compares predicted and observed event frequencies at a declared horizon. Discrimination and calibration are not interchangeable. A model may rank patients while mis-estimating absolute risks, or be calibrated in one population and fail after transport. Competing events and censoring remain in the denominator under their proper class.

17.9.2 From an intervention to a responsible decision

Family	Question entering the family	New obligation
Intervention Comparator Counterfactual	What alternative course makes an effect interpretable?	Intervention, comparator, counterfactual, allocation, adherence and co-intervention
Dose Exposure Response Adverse Event	What was received and what followed?	Dose, actual exposure, response, dose-response, adverse event and toxicity
Efficacy Effectiveness Safety Benefit Harm	Which evaluative claim is warranted?	Separate efficacy, effectiveness, safety, net outcome and heterogeneous effect records
Trial Observational Synthesis	Which design supports the inference?	Trial, deterministic concealed allocation, blinding, observational design, synthesis and missing reporting
Screening Prevention Treatment Rehabilitation	Where in the care pathway does action occur?	Screening, prevention, treatment, rehabilitation, monitoring and withdrawal
Individual Population Inference	How can group evidence inform one person?	Transportability, subgroup and multiplicity controls, decision and uncertainty
Consent Ethics Privacy Uncertainty	What may be done and recorded?	Consent, capacity, privacy, minimisation, equipoise, stopping and handoff

Dose is the intended or administered quantity under a route and schedule; exposure is what reached the relevant carrier. Response may be beneficial, adverse or unrelated. A dose-response claim needs ordered exposures, comparators and uncertainty. An adverse event is temporally associated with an exposure; clinical toxicity is a supported harmful relation. Recording one does not automatically establish the other, but discarding adverse events would destroy the safety record.

Absolute and relative effects must remain related to their denominators. A large relative change can correspond to a small absolute difference, and neither identifies who will benefit. Treatment-effect heterogeneity can be a real interaction or a multiplicity artefact; subgroups and hypotheses must be registered with sufficient control.

The deterministic randomisation claim is consequential for the SFT constitution. Concealed held labels can allocate participants in a sequence that is auditable and target-independent. The scientific protection comes from concealment and prospective custody. If an allocator can see the next label or choose enrolment after seeing it, the design no longer preserves the intended comparison.

Monitoring and stopping retain the evidence available at each time. A trial stopped early for benefit, harm or futility has a different chronology from a completed fixed sample. A patient who withdraws consent cannot be treated as a favourable outcome or simply deleted from the historical record. The permitted retention and analysis follow the consent and protocol boundary.

17.9.3 The expectation/body family as a clinical bridge

The exact fibre is useful because it forces expectation and body-labelled routes to be held simultaneously under one observed image. In a trial, similar reported outcomes can arise from different combinations of expectation, natural history, co-intervention, measurement, physiological response and target treatment. Blinding and objective measurement help discriminate some routes; they do not make expectation unreal.

The available-state claim supplies a direct falsification boundary. If an expectation appears to create a state that the registered biological carrier cannot reach, either the carrier grammar is incomplete, an external input is missing, the measurement is misclassified or the claim fails. The theory does not authorise an appeal to expectation as an explanation immune from testing.

The source comparison's null, small and context-dependent results are thus scientifically productive. They limit generality. A physiological modulation in one outcome or context cannot be promoted to a universal clinical effect. Future prospective tests should freeze expectation manipulation, realised comparator, blinding, report, objective channel, adverse event and follow-up before enrolment.

17.9.4 Medical ownership and falsification

Medicine owns whether a result warrants prevention, diagnosis, prognosis or treatment for people. Biology owns the upstream mechanism; Engineering owns the device; Social Science owns distribution and institutional context. A medical claim fails if the patient or population boundary is lost, the comparator is missing, an alternative course is manufactured after the result, adverse or missing rows disappear, or the decision exceeds the evidence's transport boundary.

This ownership is protective, not restrictive. It permits an exact biological or computational discovery to enter clinical research while preventing the discovery's precision from being mistaken for clinical validation. It also permits an adverse clinical result to identify a missing mechanism or use boundary without being diluted into a generic "mixed evidence" label.

18. Consciousness and Cognitive Science: from public information to interior observation

18.1 The unresolved boundary

Information Science establishes distinctions, records, codes, channels and inference. Computation establishes state transitions and executable processes. Biology supplies a living, regulated neural carrier. None of these facts alone licenses a relabelling of access, report, neural activity or computation as phenomenal presence. The Consciousness branch begins by preserving that gap as a scientific boundary rather than filling it with a synonym.

Its published foundation contains 72 claims across observation, access, binding, subject, self-observation, memory, self-model, attention, cognition, substrate independence, qualia differentiation and the red-of-red. Five returned claims add directional synaesthesia, a nonordinary three-quality orbit, sleep and dream recurrence, cessation and a 27-source validation. The current branch contains 77 claims, 19,712 candidates, 77 survivors and 308 controls. Its foundation and returned family are complete at current evidence; the full-field census remains unfrozen.

18.2 External observation and the interior return

An external observation is represented as a many-to-one state image. Different predecessor states can generate the same public image, report or measurement. The image therefore does not contain a predecessor label that was not retained through the observation relation. This is not scepticism about evidence. It is the exact information boundary that makes evidence custody necessary.

Interior observation is forced when the image re-enters the same organised process that generated or received it. The process does not merely carry a public token; it retains a source-located predecessor relation while the common image is taken up again within the continuing organisation. This produces the branch's structural answer to the hard-problem boundary: interiority is not an extra word pasted onto computation, and it is not inferred solely from an external correlation. It is the self-involving return relation of an organised carrier.

The complete two-preimage fibre is central. The exact preimages $\frac{1}{4}$ and $\frac{3}{4}$ share the image $\frac{1}{2}$ and compose to the One. They can carry observer/observed roles without being collapsed into one predecessor:

$$\frac{1}{4} \mapsto \frac{1}{2}, \quad \frac{3}{4} \mapsto \frac{1}{2}, \quad \frac{1}{4} + \frac{3}{4} = 1.$$

An external observer may receive the common image while the participant's organised process retains a predecessor label. Phenomenal privacy is therefore mathematically intelligible without making first-person evidence unscientific. The public record and the retained interior relation are different evidence carriers.

18.3 Access, report and phenomenal presence

Access means that content is available to a specified process. Report means that a content-bearing output was produced under stated conditions. Behaviour is an observed action. Neural correlation is an association between a neural record and another state. Computation is a realised transition system. Physical measurement is an instrument-bound public record. None of these carriers becomes phenomenal presence by changing its label.

This separation protects both sides of the inquiry. First-person report is not dismissed because it is private; it is retained as a report by an identified participant under an elicitation and memory boundary. Objective measurement is not dismissed because it lacks direct access to the retained predecessor; it provides a source-bound record of neural, behavioural or physiological state. Scientific progress comes from relating the records without erasing their different access conditions.

The same rule limits inference from silence or fluent output. Failure to report does not prove absence of experience when access, memory, motor output or communication may have failed. Fluent language does not prove interiority when the required self-involving relation and causal realisation have not been shown. These are evidence boundaries, not declarations about which biological or artificial systems can ever be conscious.

18.4 Binding, unity, subject and perspective

Binding is derived as joint closure with component distinctions retained. A bound state is not a bag of independent features and is not a featureless fusion. Each component participates in a complete relation that can be traced to the same organised process. Unity is complete mutual participation within that interior process.

The subject is the continuing source-located relation across changing content. Perspective follows from that location and continuation. It is not an unchanging substance added to the carrier, and it is not identical to any one momentary representation. Because the organised relation continues while contents change, subject identity can be retained without requiring every material component or every cognitive state to remain fixed.

Introspection is a further return within this structure. It may reveal a retained content or relation while altering attention, report and memory. It is therefore neither infallible nor reducible to an unrelated external measurement. Error, uncertainty and inaccessible content remain possible inside the same exact observation constitution.

18.5 Finite self-model, memory and temporal identity

The self-model construction begins at $\frac{1}{4}$, reaches $\frac{1}{2}$ and closes at the Fold-fixed One in two non-identity acts:

$$\frac{1}{4} \mapsto \frac{1}{2} \mapsto 1.$$

The sequence supplies a finite self-referential closure. It avoids an infinite regress of a model that would require another complete modeller behind it at every step. The final One is fixed under the Fold, so the closure does not need an additional observer to certify the whole. The construction is exact at its formal boundary; mapping each coordinate onto measured neural events remains an empirical task.

Memory requires retention across change and a route by which a state can be reidentified. The least period-two label-bearing witness is the orbit

$$\frac{1}{3} \leftrightarrow \frac{2}{3}.$$

The orbit preserves difference and recurrence. It is a structural carrier for continued memory and temporal identity, not a universal measured oscillation frequency or a claim that every memory alternates between two neural states. Content, storage medium, retrieval condition, reconstruction and report remain separate.

The returned sleep-dream claim uses the same $\frac{1}{3} \leftrightarrow \frac{2}{3}$ period-two carrier to distinguish a continuing interior return from ordinary waking input. Sleep, dream and physiological-stage labels are attached only after the structural closure is fixed. The claim therefore predicts a recurrence form without importing a sleep-stage taxonomy as a mathematical premise. Measured sleep architecture, reports, neural signals and individual differences remain source-bound evidence.

18.6 Qualia differentiation and the red-of-red

Qualia are treated as differentiated participant-retained qualities rather than as public stimulus labels. Physical stimulus, public word, participant quality and report about that quality are four different records. The branch can therefore investigate their relation without treating a wavelength as an experience or a word as the quality it names.

The red-of-red construction uses the exact period-three support

$$\frac{1}{7} \mapsto \frac{2}{7} \mapsto \frac{4}{7} \mapsto \frac{1}{7}, \quad \frac{1}{7} + \frac{2}{7} + \frac{4}{7} = 1.$$

The three labels provide a finite self-reidentifying carrier whose parts close to the One. The construction represents qualitative differentiation, recurrence and whole participation. It does not equate $1/7$, $2/7$ or $4/7$ with a wavelength, intensity, cortical voltage or universal measured colour coordinate. Those dimensional mappings require registered evidence.

The returned synaesthesia claim adds a directional cross-link. A stable Fold lock from one quality carrier to another need not reverse. This preserves the observed possibility that one class of inducer reliably elicits another quality while the reverse route is absent or different. The claim is structural; the individual mappings, development, prevalence and neural mechanisms remain empirical.

The returned nonordinary-state claim uses a three-quality orbit and preserves the boundary that an internally generated report contains no external information beyond the support supplied to the organised process. A vivid or coherent experience can be scientifically real as an experience without being evidence that an independently existing external event occurred. This protects phenomenal report while preventing experience alone from manufacturing an external observation.

18.7 Cessation and the continuation boundary

The cessation claim states that loss of the integrated half-One occupation releases the unified process while component carriers and records may remain separately held. This gives a structural distinction between loss of integrated interiority and immediate destruction of every substrate component.

The result does not imply survival of the same person after the integrated relation ends. Nor does it establish a medical criterion for death, which belongs to clinical and legal evidence boundaries. It supplies a formal carrier by which integration, remaining components and historical records can be distinguished. Any empirical application must identify the organism, measurement, reversibility, time and clinical context.

18.8 Substrate independence and artificial systems

Substrate independence is not established by copying the superficial input and output behaviour of one system into another. The current claim requires a complete bidirectional realisation map preserving state roles, transitions, relations and causal participation. A one-way behavioural imitation or fluent language model does not satisfy that criterion by itself.

The boundary remains open to artificial realisation. It does not decree that only biological tissue can carry consciousness. It specifies what must be shown before a different substrate can be said to realise the same organised relation rather than merely emulate selected outputs. The distinction is directly relevant to Unison Fold AI: architecture, checks and language output may establish implementation facts, but they do not certify phenomenal presence. The Consciousness foundation explicitly records that no current application, including Unison, was used to select or confirm its laws.

18.9 External evidence and current closure

The foundation registered 15 source identities, 61 required source features, 58 present features, three absent full-text features and 18 failed transport or content rows. Those absent and failed rows remain visible. The returned validation family compares the new claims with 27 registered sources. At the categorical comparison boundary, the source-derived distinctions match the registered structures. Heterogeneity, null rows, non-purpose-matched material and unresolved rows remain part of the record.

Observed evidence includes directional and asymmetric synaesthetic relations, nonordinary reports with retained source boundaries, recurrent sleep and dream organisation, and evidence relevant to integration and cessation. It does not measure the exact Fold fractions as universal physiological magnitudes or times. The validation is therefore evidence for the categorical carriers and their distinctions, not a licence to report structural numbers as instrument readings.

The branch reconciled 46 Consciousness-owned inherited questions, including the stronger separately registered qualia-resonance and red-of-red obligations. Corrections remain visible wherever earlier wording treated access, report, correlation or computation as sufficient for experience. The current foundation and returned family are complete; the full-field census, quantitative neural mappings, broader prospective tests and artificial- realisation programme remain open.

Consciousness hands Social Science identified agents with report, belief, perspective, memory and action kept distinct. It hands Medicine a strict boundary between first-person and objective evidence. It hands Engineering a realisation criterion and a human-participant protocol boundary. It hands the computational programme a question that cannot be answered by test counts or fluent output alone.

18.10 Dependency-resolved family ledger

18.10.1 Observation, access and binding

Family	Problem	Admitted distinctions
Observation and Interior Observation	How can an observation belong to the observing process?	Act, external image, interior return, observer/observed roles, equivalence class, atomic moment and measurement boundary
Access, Report and Phenomenal Presence	What does public availability establish?	Access, report, presence, reportability, unavailable processing and altered-state evidence
Binding and Unity	How can several contents form one experience?	Plural contents, binding, unity, integration, synchrony distinction and fragmentation
Subject, Perspective and Interiority	What continues across changing content?	Perspective, interiority, subject carrier, self invariant, evidence perspectives and privacy

The atomic observational moment is the least complete source-relation-image unit at the branch boundary. It does not imply that measured experience is instantaneous or that time has no further structure. Observation equivalence classes collect states that share a public image while preserving the fact that their predecessor labels may differ.

Active unavailable processing is evidence that information can influence a system without becoming reportable content. It prevents "not reported" from being read as "not processed" and prevents processing from being read as phenomenal presence. Altered-state evidence adds condition, induction, physiology, behaviour, report and recovery; the unusualness of a report does not by itself decide its external reference.

Synchrony is neither necessary nor sufficient by name alone for binding. A measured temporal correlation requires participating contents, causal route and joint closure. Fragmentation and divided access demonstrate that one organism may support partially separated access and report routes without the observer declaring multiple or unified subjects from one behavioural sign.

First- and third-person evidence remain complementary. A participant has the retained interior relation; an external observer has the public source path. Phenomenal privacy does not make reports beyond criticism, and public measurement does not exhaust what the participant retains.

18.10.2 Self, memory, attention and cognition

Family	New question	Resulting carrier
Self-Observation and Introspection	Can a process completely inspect itself?	Self-observation closure, blind spot, reconstructive history, loss, confabulation and deterministic self-opacity
Memory, Temporal Continuity and Identity	How is a changing process reidentified?	Memory carrier, persistence, recall, continuity, identity and cessation boundary
Finite Self-Model and Self-Simulation	Must self-reference regress indefinitely?	Representation, application, finite depth, simulation boundary and self-verification limit
Attention, Selection and Availability	How is one content made available among alternatives?	Focus, availability, capacity and switching
Cognition, Inference and Representation	How does the system carry and act on represented relations?	Representation, intentional relation, inference, anticipation, expectation and determined agency

An introspective blind spot follows because the observation act changes or selects the very process it observes. Accumulated introspection loss means that a sequence of self-observations may compress predecessor detail. Confabulation is not equated with deliberate falsehood; it is a reconstructed self-history whose evidential route does not preserve the claimed predecessor. Deterministic self-opacity is therefore possible: a determined process need not contain a complete usable prediction of its own future state.

Recall is reconstructive rather than a perfect reopening of the past. A memory record, retrieval cue, current state and report combine to produce the recalled content. Temporal continuity can persist through change, while identity continuity requires the source-located organisational route. Cessation names loss of that continuing relation, not absence of all material or archival records.

Attention is selection under capacity and switching constraints. Focus does not establish phenomenal presence by itself, and unattended processing can remain active. Cognitive representation has an intentional relation to what it represents, but an

internal token can be mistaken, incomplete or detached from an external referent. Prediction and anticipation are internal forward relations; expectation can shape selection among reachable states. Determined agency preserves the system's own reason- and state-mediated action without requiring ontological randomness.

18.10.3 Realisation, quality and red-of-red

Family	Boundary	Current result
Substrate Independence and Realisation	When do two implementations realise the same conscious organisation?	Carrier, structural criterion, bidirectional map, feed-forward limit, equivalence and AI evidence boundary
Qualia Differentiation, Resonance and Composition	How are qualities distinct, similar, recurrent and jointly bound?	Qualitative identity, similarity, resonance, recurrence, composition and cross-modal binding
The Red-of-Red	How can one quality reidentify itself without becoming its stimulus or word?	Stimulus/experience boundary, qualitative identity, period-three reidentification, recurrence, controls and empirical boundary

Feed-forward organisation can transform inputs into outputs, but the current structural criterion requires the relevant return and self-involving causal relations. Realisation equivalence is bidirectional: each state role and transition in one carrier must have a preserving correspondent in the other. If the map preserves outputs but discards internal causal organisation, it is a simulation or emulation at that boundary, not an established equivalent realisation.

Qualitative similarity does not erase identity. Two experiences may share a relation while remaining distinguishable. Resonance and recurrence require a return path and retained label; composition requires a whole in which parts remain. Cross-modal binding preserves modality-specific qualities while admitting one integrated episode.

The red-of-red contrast controls are essential. A physical red stimulus without the participant quality, the public word without stimulus, a report copied without experience and another quality under similar task conditions must not all be counted as the same event. The empirical boundary specifies which records can support the period-three structural correspondence and which remain unmeasured.

18.10.4 The returned family and open artificial-realisation test

Directional synaesthesia, nonordinary experience, sleep-dream recurrence and cessation exercise four different limits of the foundation: cross-modal direction, internally generated content, state-dependent recurrence and loss of integration. Their 27-source comparison is deliberately heterogeneous because the claims do not share one instrument or effect size.

A future artificial-realisation test must do more than ask a system to say it is conscious. It must freeze the candidate architecture and bidirectional realisation map, identify causal interventions capable of distinguishing it from an output-equivalent feed-forward system, preserve negative and unavailable evidence, and keep report, behaviour, internal state and external measurement separate. Participant-like rights and welfare questions must be handled through a declared ethical bridge before potentially harmful testing.

This is why Unison's current NOT_RUN empirical state is scientifically appropriate. There is no lawful Consciousness result to infer from a scaffold, from inherited V2 certificates or from a language output alone.

19. Earth and Environmental Sciences: one world, many coupled ledgers

19.1 Why a planetary branch is required

Physics supplies universal quantities, fields, matter and propagation laws. Chemistry supplies entities, reactions and composition. Materials Science and Biology add heterogeneous structures and living processes. Earth Science is still required because a planet is a source-dated, bounded history in which all of those carriers interact. Universal law does not determine the actual inventory, layering, forcing history, observation network or future trajectory of one world without initial conditions and measured boundary records.

The Earth foundation therefore proceeds as a series of connected ledgers: Earth-system identity and observation; energy, matter and information budgets; geological material and history; interior and geodynamics; stress, rupture, seismicity and volcanism; hydrosphere and cryosphere; atmosphere, weather and Earth-ionosphere recurrence; oceans and coasts; climate, forcing, feedback, attribution and tipping; biogeochemical and ecological coupling; environmental transport and quality; and field, remote-sensing, forecast and hazard handoffs.

The published foundation contains 74 claims. One returned tipping claim brings the current total to 75 claims, 19,200 candidates, 75 survivors and 300 controls. The foundation and returned claim are complete at current evidence. A field-wide Earth and Environmental Sciences census has not yet been frozen.

19.2 Earth identity, reservoirs and complete budgets

An Earth-system statement first requires a declared system boundary. Reservoir identity, location, phase, composition, observation time and interface must be retained before transfers can be calculated. A stock is not a flow. A measured concentration is not a total inventory. A local sample is not a global mean. A modelled field is not an instrument reading.

Energy, matter and information budgets must therefore identify sources, stocks, flows, transformations, sinks, times and residuals. A budget that omits an external supply or hides an unmeasured sink is not closed because its displayed columns happen to balance. Likewise, a numerical residual is not automatically a new physical effect: uncertainty, sampling, calibration, processing and missing components remain alternative explanations until they are tested.

The geological ledger adds order and provenance. Layer, material, process, age, deformation, exposure and sampling history remain linked. A present outcrop is evidence about a retained geological record, not the past event itself. Reconstruction therefore remains distinct from observation. The planetary-interior and geodynamic families make the same distinction between surface measurement, inferred structure, dynamic model and historical claim.

19.3 Water, ice, atmosphere, oceans and living coupling

Hydrosphere and cryosphere claims retain phase, reservoir, transfer route, residence time, forcing and observation. A glacier, sea-ice field, aquifer, river or vapour column occupies a different carrier and cannot be substituted for the others in a planetary ledger. Melting, freezing, accumulation, flow and loss require source- and time-bound records.

Atmosphere and weather add coupled motion, composition, radiation and measurement networks. Weather describes state and evolution on its declared spatiotemporal boundary; climate describes distributions and persistent relations over another boundary. Neither is made more exact by collapsing one into the other. The Earth-ionosphere result derives bounded distinct recurrence modes, but it does not derive a universal dimensional frequency in hertz. A measured resonance requires Earth geometry, propagation conditions, instrumentation and uncertainty.

Ocean and coast claims similarly preserve basin, depth, circulation, heat, salt, chemistry, biology, boundary exchange and observation. A surface record does not uniquely reconstruct an unobserved deep-ocean history without a model, and a model projection is not a measured future. Coastal hazard depends on exposure and vulnerability as well as the physical event.

Biogeochemical and ecological coupling connect living stocks and processes to planetary reservoirs. Carbon, nutrient and contaminant relations retain chemical identity and phase. Transport is not transformation; a substance may move without changing identity, change identity without leaving the system or become unavailable to one method without being absent from the environment.

19.4 Climate, attribution and tipping

A climate forcing is not a feedback, an observed trend is not by itself a causal attribution, and a projection is not a forecast unless its conditions and issue time are fixed. The branch retains forcing source, response carrier, feedback sign and path, uncertainty, counterfactual or comparison and the observational record used to discriminate alternatives.

The foundation derives a two-basin tipping structure. The returned claim, `SFT-EARTH-TIPPING-FOLD-LOCK-002`, identifies $1/4$ and $3/4$ as exact normalised basin preimages of the common $1/2$ boundary. That is a structural Fold lock. It does not establish that every physical Earth subsystem tips at a measured value of one-half. Dimensional thresholds depend on the actual system, forcing path, state variables, rate, hysteresis, spatial scale and measurement method.

This distinction makes the claim more useful. It supplies a finite grammar for two retained basins and their common boundary while leaving empirical science responsible for identifying the correct variables and values. It also prevents a post hoc choice of units from manufacturing an apparent universal fraction.

19.5 Earthquake magnitude-frequency prediction: adverse and compatible records

The Earth branch contains a prospectively registered quantitative test. Exact candidate enumeration selected magnitude-frequency exponent **1** before the catalogue was opened. For one unit increase in magnitude, that exponent gives the registered exceedance share **1/10**.

The first test used a global 2010-2020 mixed-magnitude catalogue. It recorded

$$N_{\geq 5} = 18,469, \quad N_{\geq 6} = 1,494,$$

with the registered interval

$$0.076998990840681139 \text{ to } 0.084918164588559525.$$

The interval excludes **1/10**. This result is **adverse** and remains adverse. It is not reclassified because a later test used a different, more homogeneous record.

The later preregistered holdout used a 2020-2026 homogeneous m_{ww} catalogue. It recorded

$$N_{\geq 6} = 253, \quad N_{\geq 7} = 27, \quad \frac{N_{\geq 7}}{N_{\geq 6}} = \frac{27}{253},$$

with interval

$$0.085657503235220589 \text{ to } 0.13089188889528297.$$

That interval contains **1/10**, so the holdout is **compatible** with the registered exponent at its declared boundary. Compatibility is not confirmation, and it does not erase the earlier adverse result. The two rows together show why catalogue composition, magnitude scale, completeness and chronology must remain part of the evidence.

19.6 Observation, retrieval, proxy, reconstruction and forecast

The Earth evidence constitution distinguishes direct field observation, instrument retrieval, proxy, reconstruction, model, forecast, non-detection, censoring, missingness and observed absence. A satellite retrieval is a processed estimate tied to an algorithm and version. A proxy is an observed carrier related to another state through a calibrated interpretation. A reconstruction estimates a past state. A forecast is issued before its target period under stated initial conditions. These distinctions cannot be collapsed into the single word "data".

Hazard is the possible damaging event or process. Risk additionally includes exposure, vulnerability, probability or scenario and consequence. A high physical hazard does not imply equal risk for all populations; a low recent loss record does not prove absence of hazard when exposure or observation is limited. Engineering owns the designed mitigations, while Social Science owns collective decisions and unequal exposure.

The foundation registered 21 source identities and 91 required features: 67 were present and 24 absent. The original Smithsonian transport returned HTTP 403 and remains recorded as a failed transport. A separately registered WFS addendum supplies its own evidence and provenance; it does not rewrite the historical failure as success.

19.7 Earth handoffs and open frontier

Earth hands Astronomy a local observation, instrument and planetary-history constitution without making Earth history universal. It hands Social Science hazard, resource and environmental records without deciding policy or ethics. It hands Engineering measured loads and environmental conditions without certifying a design. It receives biological and chemical carriers while owning their planetary coupling.

Current closure covers the 74-law foundation and returned tipping claim. The field-wide census, additional prospective quantitative predictions, regional and system-specific threshold mappings, long-horizon source updates and expanded hazard and environmental validations remain open. The permanent adverse earthquake row and the distinct compatible holdout are part of that frontier, not editorial problems to be removed.

19.8 Dependency-resolved family ledger

19.8.1 The planet as a sequence of constrained ledgers

The Earth branch is organised as twelve ledgers because each later claim depends on distinctions preserved by the earlier one. The ledger structure also prevents a climate, hazard or environmental conclusion from floating free of the source and observation system by which it was obtained.

Ledger family	Carrier added	Evidence boundary preserved
1. Earth-System Identity, Reservoirs, Boundaries and Observation	One identified Earth system, reservoirs, interfaces, state and observation	Absence, non-detection and missingness remain different
2. Planetary Energy, Matter and Information Budgets	Sources, sinks, stocks, flows, storage and residual	No hidden input or unclassified residual is called closure

Ledger family	Carrier added	Evidence boundary preserved
3. Geological Material, Layer, Provenance and History	Material identity, stratigraphic order, cycles and dated provenance	Present observation remains distinct from past reconstruction
4. Planetary Interior and Geodynamics	Interior compartments, heat sources, transport, crust-mantle and plate relations	Inferred interior remains tied to geophysical observations and models
5. Stress, Rupture, Seismicity and Volcanism	Stress/strain, rupture, propagation, catalogue, magnitude-frequency and eruption	Event, signal, catalogue row and population law remain distinct
6. Hydrosphere and Cryosphere	Water reservoirs, phase handoff, cycle, groundwater, surface networks and ice	Phase, location, transfer route and observation time retained
7. Atmosphere, Weather and Earth-Ionosphere Recurrence	Composition, radiation, moisture, circulation, forecast and bounded recurrence	No dimensional resonance is derived from the structural mode alone
8. Ocean and Coast	Ocean identity, salinity, stratification, circulation, exchange and coastal response	Surface, depth, basin and boundary processes cannot be substituted
9. Climate, Forcing, Feedback, Attribution and Tipping	Climate state, timescale, forcing, feedback, variability, attribution and basin transition	Structural half-boundary is not a universal measured threshold
10. Biogeochemical and Ecological Coupling	Carbon, nitrogen, phosphorus, biosphere exchange and biodiversity boundary	Chemical identity and biological process remain owned upstream
11. Environmental Transport, Transformation and Quality	Pollutant identity, source-path-receptor, persistence, accumulation and quality	Transport, transformation, non-detection and harm remain different
12. Field Evidence, Remote Sensing, Forecasts and Hazard Handoffs	Sampling, retrieval, proxy, assimilation, validation, hazard and risk	Observation, reconstruction, model and forecast retain chronology

The first ledger makes state conditional on system boundary. An atmospheric inventory and a whole-Earth inventory answer different questions. A reservoir is identified by contents, location and interface; the same substance can move between reservoirs without violating matter conservation. An Earth-system state is a source-dated composition of retained reservoir and relation states, not a timeless label.

The budget ledger makes closure falsifiable. Source, sink and storage are roles within a time window. A component can change role under another boundary. The residual must remain as a measured or calculated difference with its uncertainty. Calling the residual "zero" because a model imposes balance would erase precisely the evidence that could reveal missing flux or measurement error.

Geological order adds directional history. Stratigraphic succession, cross-cutting relation, material transformation and provenance can constrain a past sequence while leaving gaps. A cycle in geological material does not mean that the same specimen returns unchanged. Provenance carries a path through sources, transformations and deposits.

Interior science depends on surface and orbital observations, experiments and models. Internal heat sources and transport are not direct views into the planet. Plate motion, deformation and boundary type require coordinate, velocity, uncertainty and time. The branch can therefore derive a complete geodynamic claim without claiming that one measurement uniquely reconstructs the entire mantle history.

19.8.2 Event populations and the seismic falsification route

Stress and strain describe state; rupture is a transition; seismic propagation connects the event to a sensor; an earthquake catalogue classifies records; magnitude-frequency is a population relation. These stages make the exponent test interpretable. A mismatch can arise because the proposed population law is wrong, the catalogue is heterogeneous, magnitude types are mixed, the completeness threshold is misidentified or the interval rule is unsuitable. The first registered global test remains adverse because its own boundary was fixed. The homogeneous holdout asks a related but not identical question and is compatible.

Volcanic process and eruption record follow the same event-observation split. Unrest, eruption, deposit, remote signal and catalogue entry are different carriers. A period without a detected eruption is not proof of absent activity when monitoring is incomplete.

19.8.3 Coupled water, air, ocean and climate histories

The hydrological cycle is not one circulating token. It composes many reservoir-specific transitions with residence times and phase changes. Groundwater may be hidden from direct observation; rivers and lakes form directed networks; the cryosphere stores and releases water while changing albedo, sea level and habitat. Each coupling is typed rather than inferred from the common word "water".

Atmospheric composition affects radiative transfer, while radiation, moisture and circulation influence weather state. A forecast must be sealed before its verification period. Reanalysis or hindcast can reconstruct past conditions but cannot become a

prospective forecast by later wording. The Earth-ionosphere cavity adds a bounded recurrent carrier whose observed frequencies remain condition- and dimension-bound.

Ocean stratification separates layers with different density and composition. Circulation moves properties through a basin; air-sea exchange crosses a boundary; tides and coasts add forcing and geometry. These relations feed the climate ledger but remain independently observable. An ocean heat uptake estimate cannot be inferred solely from a surface air-temperature series.

Climate detection asks whether an observed change exceeds a declared variability and uncertainty boundary. Attribution asks which forcing or causal combination best accounts for it. Internal variability remains an alternative component, not a universal explanation. Tipping adds basin, state variable, path and hysteresis. The exact Fold lock supplies a structural relation but must be mapped to measured systems separately.

19.8.4 Environmental consequence and decision handoff

A pollutant remains chemically identified across a source-path-receptor relation. Transformation may create a new entity with different persistence or toxicity. Accumulation requires a time and compartment ledger. Environmental quality is a measured state relative to a declared use, organism, regulation or ecological condition; it is not intrinsic to one concentration without context.

Field sampling records location, time, specimen, method and custody. Remote sensing adds propagation, instrument and retrieval. A proxy adds calibration to another state; assimilation combines observation with model state; forecast validation compares a pre-issued result with later observation. Hazard then passes to Engineering and Social Science with the physical event kept distinct from exposure, vulnerability, mitigation and choice.

The Earth branch can be falsified at every stage. A budget may fail closure, a source attribution may fail against alternatives, a forecast may miss its registered interval, a tipping mapping may lack a stable basin or a source transport may be unavailable. Keeping those failures typed is what permits theory revision without rewriting planetary history.

20. Astronomy and Cosmology: from one sky to a source-custodied cosmos

20.1 The astronomical observation chain

Astronomy begins where a source beyond the immediate observer must be known through propagation and instrumentation. The least complete relation is

source → path → observer → instrument → record.

A brightness, coordinate, spectrum or catalogue label detached from that chain is not an admitted astronomical fact. Propagation can transform the signal; instruments have calibration and selection boundaries; processing can change a record; catalogues have releases, row identities and censoring. Each stage must therefore be retained for reconstruction and falsification.

The published foundation contains 72 claims across observation geometry, object identity, radiation, motion and time, stellar systems, planetary and small bodies, interstellar enrichment and formation, galaxies and populations, compact and high-energy multimessenger astronomy, large-scale expansion, early-universe background and abundance, dark and horizon boundaries and prediction handoffs. Five returned claims add solar, atomic, planetary and lithium structures and a source-bound validation. The current branch contains 77 claims, 19,712 candidates, 77 survivors and 308 controls. The foundation and returned family are complete; the field-wide census remains unfrozen.

20.2 Object identity, populations and cosmic history

An astronomical object is identified through source position, epoch, uncertainty, association rule and retained alternatives. A match across catalogues or epochs is an inference, not an identity conferred by a similar name. Blends, transients, proper motion, resolution and changing source state can all alter the relation. A population additionally requires a selection function, detection boundary, censored and missing rows, volume or exposure and release version.

The branch separates cosmic law from cosmic history. Physics owns universal propagation, gravity, matter and constant relations admitted upstream. Astronomy owns source-dated objects, initial conditions, populations and chronology. An instrument cannot select a universal law, while a universal law cannot by itself reconstruct which stars, galaxies or structures actually formed in this cosmos.

Stars and planetary systems are consequently described through source, composition, structure, energy and evolutionary records. Interstellar matter, enrichment and formation retain transport and chemical provenance. Compact objects and multimessenger events retain channel, path, timing, localisation and association. Large-scale expansion and early-universe evidence retain the observation-to-model bridge rather than presenting a fitted historical model as a direct observation.

20.3 The rank-four luminosity-rotation prediction and its adverse test

The branch fixed structural rank **4** before opening the galaxy data. Its first preregistered SPARC comparison used an all-positive-row unweighted fit and obtained the slope

$$3.486344605219551....$$

That registered comparison is **adverse** to exact rank four and remains adverse. It is not overwritten by a different analysis.

A separate primary-source method record reports an uncertainty-aware orthogonal slope of

$$3.85 \pm 0.09,$$

with systematic movement from **3.5** to **4.0**. Exact rank four lies at the reported systematic boundary. This is relevant context and support for the importance of method choice, but it does not reclassify the first test. The two records retain different selection and fitting boundaries.

The result demonstrates the value of a chronology-preserving theory. A prediction is meaningful because it can fail. A later compatible method or boundary may motivate a sharper preregistration, but cannot make the earlier measurement disappear. Future work must specify the sample, variables, uncertainty model and fitting method before the target is inspected.

20.4 Returned solar, atomic, planetary and lithium structures

The returned solar claim identifies a unit magnitude-frequency relation as an inverse binary share at its exact carrier. The atomic claim records a one-Fold burst completion. The planetary claim generates a depth-seven binary ladder from **1/128** to the One. The lithium-7 claim maps **3/16** through one Fold to **3/32**, with doubling restoring the prior share. Each is a formal structure registered before its source-bound categorical comparison.

The 25-source validation preserves several distinct empirical outcomes. Tipping and multistability are observed across relevant systems, but measured thresholds are system-specific. Solar, radio and other transient classes show heterogeneous, frequently non-unit magnitude-frequency exponents. Binary-like spacing occurs in some planetary records but is not universal. The lithium direction is retained, while a factor-of-few mismatch and mechanism uncertainty remain. These are not minor caveats: they are the current evidence classification.

The categorical validation passes at the registered comparison boundary, but some exact quantitative mappings remain unresolved or adverse. The returned validation does not alter the protected earthquake, Parker, Tully-Fisher or ringdown records and does not backdate later evidence into an earlier prediction.

20.5 Dark sectors, horizons and generated alternatives

The branch keeps direct detection, dynamical inference, lensing, background evidence, model-dependent reconstruction and unobserved alternatives separate. Dark matter and dark energy statements are therefore not reduced to either "directly seen objects" or "mere absence". Their evidential roles depend on the specific observation, model and alternative being tested.

A horizon limits an observation relation; it does not make an unobserved state equal to nothing. A mathematically generated alternative is not an observed world. The multiverse boundary is therefore explicit: formal multiplicity may define alternatives within a model, but observation and empirical status require a source relation that the generated alternative does not obtain by definition.

Observation also cannot recover an unheld predecessor. When distinct Fold preimages share one image, the image alone cannot tell which predecessor occurred. A retained historical or experimental record may make one route reconstructible; inference cannot manufacture the missing distinction after it was discarded.

20.6 Sources, corrections and handoff

The foundation registered 15 primary source identities. Thirteen transports succeeded and two failed transports remain visible. Fifty-three Astronomy- owned V1/V2 questions were mapped to current claims without importing their old answers. The returned validation used 25 sources and retained heterogeneity, mismatches and unresolved mechanisms.

Astronomy hands Cross-Branch Synthesis source-custodied cosmic correspondences only after Physics and Astronomy have each preserved their ownership. It hands Engineering instrument and observation requirements, not a certified device. It hands Social Science no authority to turn cosmic analogy into social law.

Current closure covers the foundation and five returned claims. Open work includes the field-wide census, sharper preregistered quantitative tests, additional source captures, object- and population-specific histories, dark- sector discrimination, system-specific planetary ladders and the unresolved or adverse mappings preserved above. Completion of this branch boundary is not completion of observational astronomy or cosmology.

20.7 Dependency-resolved family ledger

20.7.1 From source geometry to stellar and planetary histories

Family	New unresolved problem	Admitted surface
Observation Geometry	How does a remote source become a record?	Source, observer, signal, frame, distance, survey selection and non-detection
Object Identity and Classification	Which records belong to one object or population?	Identity, class, population, morphology, composition inference and catalogue
Radiation, Motion and Time	What transforms across the observation path?	Flux/luminosity, spectrum, redshift, astrometric/radial motion, variability and transient distinction
Stellar Systems	How does a star form, persist and end?	Formation, structure, energy generation, evolution, population and terminal state
Planetary and Small Bodies	How are bound systems and their members classified?	System identity, orbit, resonance/tide, planet/satellite, exoplanet observation and small-body population
Interstellar Enrichment and Formation	How do matter and composition pass between generations?	Medium, dust, cloud, enrichment, formation environment and stellar feedback

Astronomical distance is never an unqualified number. Its method, calibration, reference and uncertainty are part of the record. Angular position requires a frame and epoch. Survey selection and non-detection require coverage, sensitivity and processing. These constraints precede object classification because the visible sample has already been shaped by the observation system.

Morphology is a source- and resolution-bound description. Composition is inferred through spectra, models and calibration rather than directly handled at the telescope. A catalogue preserves source rows, release version, selection, cross-match and changes. Removing a row from a later release does not delete its historical presence.

Flux is an observed received relation; luminosity is a reconstructed source property requiring distance and propagation. Redshift can contain several physical and observational contributions. Variability is change across measurements; periodicity requires a retained recurrence and alternatives; a transient requires onset, duration, non-detection and event classification.

Stellar energy generation uses admitted Physics and Chemistry but Astronomy owns the source history and population. Stellar formation, evolution and terminal state require composition, mass or structural parameters, environment and time. An inferred sequence across a population is not automatically a direct longitudinal observation of one star.

An orbit is a relation among identified bodies under a coordinate and time boundary. Resonance and tidal state require measured periods or dynamics and uncertainty. Exoplanet observation keeps transit, radial velocity, imaging, microlensing or another method separate from the reconstructed planet. Small- body populations preserve detection and orbit-selection effects.

Interstellar enrichment closes the material history between generations. Dust, gas and molecular clouds occupy different observational and chemical carriers. Stellar feedback changes the environment from which later objects form, but the existence of a feedback channel does not uniquely reconstruct the formation history of a particular source.

20.7.2 Galaxies, compact objects and large-scale structure

Family	Principal relation	Critical evidence distinction
Galaxies and Populations	Identity, structure, dynamics, luminosity-rotation, evolution and environment	Registered scaling test remains distinct from later methods
Compact, High-Energy and Multimessenger	Remnant and black-hole candidate classes, transient, channel association and population	Horizon inference is not direct inspection of an interior
Large-Scale Expansion	Statistical organisation, cosmic web, expansion, distance ladder, lensing and history	Universal law remains separate from contingent history and fitted reconstruction
Early Background and Abundance	Thermal history, microwave background, acoustic structure, abundances, recombination and perturbations	Present signals are evidence for reconstructed earlier states
Dark, Horizon and Unobserved	Dark matter, dark energy, horizon, topology, multiverse and fate	Generated or inferred carriers are not relabelled direct observations
Inference, Prediction and Handoffs	Law/history distinction, model boundary, standing prediction, anomaly, astrobiology and ownership	Prediction chronology and non-detection remain explicit

Galaxy identity must survive substructure, mergers, changing morphology and catalogue release. Dynamics relate source motion and mass inference under a model. The luminosity-rotation family becomes a quantitative test only after variables, sample and fit are frozen. This is why the unweighted slope and the orthogonal uncertainty-aware slope cannot be averaged into a retrospective success.

Compact-object classification distinguishes observed signals from inferred source. A black-hole candidate and horizon evidence retain the alternatives and measurement route. Multimessenger association requires independent channels, timing, localisation, common-source model and chance-coincidence control. A coincident alert is not automatically one event.

Large-scale statistical organisation retains survey mask, selection, distance inference and covariance. The cosmic web is reconstructed from tracers rather than directly observed in an unmediated volume. A distance ladder is a composition of calibrations; its uncertainty and shared systematics must propagate. Gravitational lensing is an observation relation that supports mass and geometry inference under retained alternatives.

The cosmic microwave background is a measured present radiation field used to reconstruct an earlier state. Acoustic structure, abundance and recombination claims retain model and source identities. Early perturbations are inferred through later records. The branch can therefore make exact history claims only at the boundary established by those records; it does not turn model parameters into premise-free observations.

Dark-matter and dark-energy evidence are typed families, not single verdicts. Dynamical, lensing, background and expansion records may jointly support an inferred sector while direct detection remains unresolved. Cosmic topology and ultimate fate are model- and observation-bound. Multiverse admissibility is a formal boundary: mathematical generation establishes a possible model member, not an observed universe.

20.7.3 Returned claims and next observations

The solar/radio, atomic, planetary and lithium returns probe whether exact Fold structures recur across four observational classes. Their value lies partly in the non-uniform evidence. Heterogeneous exponents prevent a universal unit law; some binary-like spacing prevents a universal absence claim; lithium retains the predicted direction while exposing a magnitude and mechanism gap.

A next quantitative campaign must therefore register one source population, release, event definition, completeness surface, exact statistic, interval and decision rule for each claim. Reusing a structural label across solar bursts, planetary spacing and atomic recurrence cannot create one common measurement protocol. Typed common form is the job of Synthesis; dimensional evidence remains with Astronomy and Physics.

21. Social and Collective Systems: from one relation to society

21.1 The least social carrier

Social inquiry begins neither with an anonymous average nor with an institutional name. Its least admissible carrier retains identified people or agents, typed relations, context, time, capabilities, selection and available alternatives. A group is not one indivisible person. An institution is not the word printed on its building. A population is not the favourable respondents who remained visible after non-response and attrition.

This branch depends on Biology, Medicine and Consciousness because collective relations involve living, vulnerable, perceiving and acting participants. It also depends on Information Science and Computation because messages, records,

networks and decisions must be represented. None of those upstream branches owns the new relation among agents, the institution that persists across its members, the distribution of resources and power, the source-custodied history, or the bridge between measured facts and declared values.

The published foundation contains 72 claims in twelve families. Four returned claims add preferential-flow inequality, consensus/polarisation, dissipative social recurrence and a 21-source validation. Together the two current Social ledgers contain 76 claims, 19,456 candidates, 76 survivors and 304 controls. The foundation and returned family are complete; a full-field census is not yet frozen.

21.2 Aggregation and the individual-collective boundary

Aggregation is a many-to-one operation. A collective total or mean may be lawful and useful, but it cannot recover individual predecessors that were not retained. This closes both directions of a common inference error. A population average cannot silently become the state of each member, and one person's record cannot silently become a population distribution.

The observation surface must therefore retain eligibility, recruitment, participation, non-response, measurement, context and time. Selection is part of the result. A platform's visible users are not automatically its affected population; recorded actions are not automatically beliefs; institutional records are not automatically complete histories of those excluded from the institution.

The returned preferential-flow claim derives exact complementary shares that conserve the One. It supplies a finite carrier for cumulative advantage and unequal flow without losing the remainder. The structural relation does not declare one universal empirical wealth distribution or inequality exponent. Observed inequality depends on the resource, access rule, inheritance, network, policy, time and measurement boundary.

21.3 Communication, belief and coordinated action

Social observation is intervention-sensitive. The observer, method, setting, consent and the participants' knowledge of observation can alter the relation. This is not a reason to abandon empirical study; it is a reason to record the intervention made by observation.

Communication requires a complete route. Sending does not prove receipt; receipt does not prove understanding; understanding does not prove belief; belief does not prove action; action does not by itself establish intent. Each transition has its own carrier, loss and alternative explanations. Digital counts such as impressions, clicks or reposts are therefore evidence only for their registered event definitions.

Coordination retains participants, shared or competing aims, available actions, information, timing, rules and outcomes. Conflict retains opposed relations, resources, constraints and costs. A coordinated outcome need not indicate consensus, and conflict need not indicate total social breakdown. Both can occur inside one continuing collective organisation.

21.4 Norms, institutions, power and legitimacy

Norm, rule, institution, enforcement, compliance and legitimacy are different structures. Majority behaviour alone does not establish a norm; a written rule does not prove enforcement; compliance does not prove agreement; and effective coercion does not establish legitimate authority. Institutional identity requires roles, rules, records, succession, resources and boundaries that can persist while particular office-holders change.

Power is directional and dependency-bound. It requires an actor, a target, a resource or gate, alternatives, costs, resistance and the capacity to change the target's available actions. Status, wealth, title or a favourable outcome can be evidence relevant to power, but none establishes the complete relation alone. Exclusion similarly requires an eligibility or access boundary and the path by which an agent's options were removed.

Scientific gatekeeping is included in this empirical surface. Criteria, credentials, costs, decisions, reviews, appeals and lost contributions may be observed and criticised. Institutional prestige is never a proof operator. At the same time, criticism of a gate does not automatically admit the critic's scientific claim. SFT's anti-institutional position is therefore operational rather than merely rhetorical: no credential, consensus, journal, laboratory or public authority can replace the derivation and evidence record; and no declaration of independence can replace it either.

21.5 Consensus, polarisation and recurrent collective dynamics

Consensus is a social event, not a truth operator. A collective decision must retain eligibility, options, information, aggregation rule, participation, dissent, implementation and affected parties. Majority outcome is not unanimity, factual truth or ethical legitimacy.

The returned consensus/polarisation claim makes the information boundary exact. Distinct retained preimages can share the common $1/2$ image. A public aggregate at the common image therefore cannot distinguish whether the collective arrived there through agreement, balanced opposition or another retained composition. Polarised preimages may survive under an apparently centred result. Individual and subgroup records are required to tell the difference.

The returned dissipative-social claim uses the period-two orbit

$$\frac{1}{3} \leftrightarrow \frac{2}{3}.$$

The carrier never rests at the balance point. It provides an exact form for a continuing boom-bust, expansion-contraction or alternating collective process. It is not a universal business-cycle period, electoral forecast or measured social constant. Synchronisation, hysteresis, external intervention, institutional constraint and heterogeneous timing remain empirical.

Network diffusion and cascade claims follow the same rule. A connection does not prove influence, temporal succession does not prove contagion and a cluster does not prove common belief. The record must distinguish directed exposure, common causes, adoption, non-adoption and observation windows. No universal social tipping constant is admitted.

21.6 Knowledge, history, causality and ethics

Knowledge claims preserve source, record, transmission, access, correction and uncertainty. Reputation can guide attention but cannot substitute for evidence. Culture and language are transmitted through people, practices and records; neither becomes an unchanging substance independent of those carriers.

Historical claims require dated, source-custodied paths. A later state is not its own cause and cannot uniquely reconstruct the route by which it arose. Archives preserve perspectives and gaps as well as content. Causal social claims retain assignment or comparison, confounding, spillover, attrition and target population. Interventions retain delivery, uptake, adaptation, harms and missing outcomes.

Ethical judgement is neither erased nor disguised as measurement. A normative claim must identify affected agents, rights, harms, benefits, power and disagreement. Frequency does not establish permission; popularity does not establish justice; and a moral commitment does not become a measured causal effect. Empirical and normative claims can interact only through a declared bridge.

This is integral to the Ernos Labs open-science mission. The platform separates credentials from evidence and criticism from admission. Public access, reproducibility and contestability are institutional commitments, while the truth status of an individual scientific claim still depends on its derivation, evidence class and falsification record.

21.7 Evidence boundary and open continuation

The foundation registered 14 primary custodians. Ten source transports succeeded and four failed transports remain failed. The inherited five Social- owned questions were reconciled without importing their prior answers. The returned family used 21 sources and preserves evidence for cumulative and preferential advantage, controversy and dissent, boom-bust behaviour, hysteresis, variable synchronisation, intervention, reversal and institutional dependence. Unresolved causal relations remain unresolved.

Social Science hands Engineering identified users, institutions, access and governance relations without certifying a technical system. It hands Cross-Branch Synthesis typed collective carriers without allowing physical or biological metaphors to become social identities. It receives medical and cognitive evidence while preserving privacy, agency and population boundaries.

Current completion covers the foundation and four returned claims. Open work includes the field-wide census, broader populations, prospective interventions, institutional and historical source capture, cross-cultural transport, quantitative mappings and the explicit normative programme. Society remains extension-open by constitution.

21.8 Dependency-resolved family ledger

21.8.1 Agents, records and coordinated institutions

Family	Owned relations	Conflation prevented
Agents, Groups and Identity	Agent, relation, group, collective state, organisation and population	Group is not one person; population is not visible sample
Observation, Action and Communication	Observation, action, message, shared record, common knowledge and noise	Send, receive, believe and act remain distinct
Coordination and Conflict	Coordination, cooperation, competition, conflict, bargaining and collective action	Common outcome does not imply common aim
Norm, Institution and Enforcement	Norm, rule, institution, enforcement, compliance/resistance and change	Frequency, rule, enforcement and legitimacy remain separate
Trust, Reputation and Knowledge	Trust relation, reputation, testimony, collective knowledge, expertise and false information	Prestige and repetition are not proof
Exchange, Allocation and Distribution	Exchange, valuation, allocation, production, distribution/inequality and mobility	Price, value, access and welfare are not interchangeable

A social agent is identified by the capability and relation relevant to the claim. The branch can include people, organisations or artificial agents only when their level and available actions are explicit. Group identity adds a membership and continuation rule. Organisation identity adds roles, records and succession. Population identity adds selection and coverage.

Common knowledge is stronger than several agents separately receiving a message. It requires the recursively shared relation established by the communication conditions. Noise can enter transmission, interpretation or record. A shared document is not proof that every eligible agent read or accepted it.

Bargaining retains alternatives and costs; cooperation retains a jointly continued relation; competition retains incompatible claims on a resource or outcome; conflict retains active opposition. Collective action adds the path from individual capabilities to a joint effect. These relations can overlap without becoming synonyms.

Norms may be inferred from expectations and sanctions, while rules have an explicit source and content. Enforcement is an action, compliance an observed relation, resistance a contrary or evasive action and institutional change a transition in rules, roles or practices. Legitimacy requires a normative and social bridge beyond effectiveness.

Trust is directional: one agent accepts vulnerability with respect to another under a task and evidence boundary. Reputation is a transmitted social record. Testimony has source, content, chain and credibility. Expertise is relevant evidence about capability but never an operator that makes a proposition true. False and unsupported information are classified by evidence, not by the status of the speaker.

Exchange retains what each party gives and receives, the rule and the available alternatives. Social valuation may differ among agents and from exchange price. Allocation identifies the gate by which resources are assigned; production identifies contributors and inputs; distribution identifies who holds the result. Inequality and mobility require time and population, not one snapshot or one anecdote.

21.8.2 Power, networks, culture and multi-level causation

Family	Added surface	Evidence limit
Power, Authority and Exclusion	Power, authority, dependency, exclusion, stratification and gatekeeping	Title, wealth or outcome alone does not establish the directional relation
Network, Diffusion and Transition	Network, diffusion, synchronisation, cascade, transition and cycle	Association and common cause remain alternatives to influence
Culture, Language and Transmission	Culture, symbol, language community, education, media and memory	Recorded text is not the whole living practice
Multi-Level Causation	Individual-collective composition, downward constraint, emergence, mixed-agent collective, decision and aggregation boundary	Neither level may silently replace the other
History, Intervention and Ethics	Event, contingency, causal inference, intervention, replication and value boundary	Later state is not its own cause; frequency is not moral authority
Ownership Handoffs	Mathematics, computation, biology, medicine, Earth and engineering routes	Imported analogy cannot acquire social ownership

Authority is institutionally recognised power under a role and rule. Dependency identifies what one agent requires from another. Stratification is a continuing ordered access relation. Scientific gatekeeping becomes measurable through criteria, decision, appeal, cost and exclusion while scientific admission remains governed by evidence.

Network representation retains node and edge definitions. Diffusion requires ordered exposure and adoption; synchronisation requires measured phase or state relation; cascade requires dependency through events; collective transition requires a before, path and after. Socio-economic cycling remains condition-bound, which is why the returned period-two carrier is a structural model rather than a universal calendar.

Culture continues through practice and transmission while changing. A symbol has a community-bound meaning relation. Education and media create structured access and selection, not direct implantation of belief. Collective memory is a maintained public and social record that may differ from each participant's memory and from source-custodied history.

Emergence names a collective relation that is not attributable to one member alone, but it does not remove the member-level carrier. Downward constraint changes available individual actions through a collective relation. Mixed human-artificial systems must retain which agent generated, selected or acted on a proposal. Collective decisions preserve dissent and implementation.

Cross-context replication asks whether the relation survives a new population, institution, time and measurement practice. A numerical match without institutional transport may be superficial. Ethical evaluation then declares the value bridge rather than pretending it was read from the data.

21.8.3 Returned exact relations and their empirical limits

Preferential-flow inequality, the polarisation lock and dissipative recurrence form a coherent sequence. Unequal flows alter retained shares; a common image can hide opposed predecessors; an alternating collective can persist without resting at nominal balance. The source comparison finds real cumulative advantage, controversy, cycles, hysteresis and reversals while retaining institutional dependence and unresolved causation.

Future validation must resist the temptation to choose only famous social examples. A prospective programme should define resource, population, edge, event, time and intervention before observing the target. It should retain failed replications, policy changes, platform changes and populations for which the proposed Fold carrier is not discriminating.

22. Engineering Translation: from an admitted law to a working world

22.1 Why Engineering comes after the sciences

Engineering may use every admitted upstream branch, but it cannot choose which scientific law is true by building the artefact it desires. It begins with a need or purpose, translates that need into verifiable requirements, generates design alternatives, accounts for resources and failure, implements a system, and then distinguishes verification against specification from validation against intended use.

The published Engineering foundation contains 72 claims governing requirements, functions, components, interfaces, architecture, resources, efficiency, reliability, measurement, calibration, tolerance, alternatives, optimisation, control, safety, verification, validation, accessibility, lifecycle, scientific evidence and anomaly handoff. Eight returned claims close the current Tesla, vacuum/inertial, vacuum-beat, sector-five/seven, Smithium, complete protocol-family and consciousness/placebo protocol obligations. The current branch contains 80 claims, 20,480 candidates, 80 survivors and 320 controls.

The eight returned claims are **formal protocol closures**. No experiment is reported as run by those packages. Their empirical status is not inferred from their complete candidate censuses or independent reconstructions.

22.2 Requirements, composition and complete resource accounts

A requirement is not a wish. It retains source, purpose, version, priority or conflict and a verifiable acceptance condition. A function retains declared input, transformation, output, loss and operating boundary. A product name or demonstration cannot substitute for either.

An engineered system is a composition record. Components keep identity and version. Interfaces retain endpoints, units, timing, permitted states and errors. Architecture retains responsibility and failure propagation. Integration retains build order, rollback and failed intermediate states. This makes success reproducible and failure diagnosable.

Efficiency cannot exceed a complete resource account. Input, useful output, external supply, internal stock, flow, time and loss must close within an explicit boundary. A hidden cloud service, pretrained model, operator action or stored resource is still an input. Capacity, reliability and availability are load-, population-, duration- and observation-window bound rather than universal properties of a named system.

22.3 Measurement, tolerance, test and acceptance

A display is not the measurand. Measurement requires an instrument, calibration state, method, unit, resolution, uncertainty and traceable record. Adjustment is not calibration, because changing an instrument does not establish its relation to a reference. Tolerance must be fixed before the outcome; widening it afterwards would change the test.

A demonstration shows that an artefact performed an observed action under some conditions. A test compares a result with a frozen criterion. Acceptance requires every required criterion, permitted exception and authority to be specified. Institutional authority cannot convert a failed frozen criterion into a passing result without an openly versioned requirement change.

Finite design optimisation is admitted only over a generated design space. Alternatives, constraints, violations, ties and failures remain visible. Trade-offs separate measured consequences from declared values. An opaque oracle or hidden scalar preference cannot manufacture a unique optimum and then present it as if it were forced by science.

22.4 Control, safety, resilience and lifecycle

A control claim retains the whole loop: plant, sensor, controller, actuator, delay, bound, disturbance, noise and fault route. One settled trace does not prove global or infinite-time stability. A merged output cannot reconstruct an unheld internal state. Control performance therefore remains tied to the tested operating envelope.

Safety is a scoped evidence case, not the absence of previously observed harm. Hazard source, path, exposure, control, detection and consequence remain. Fail-safe behaviour must reach a tested bounded safe state or visibly halt. Defence layers are independent only when common causes, shared supplies, software dependencies and bypass routes have been assessed.

Verification asks whether the built artefact satisfies its frozen specification. Validation asks whether the verified artefact serves its intended use and users under the actual use conditions. Bidirectional traceability connects need, requirement, design, build, test and result. Maintenance, versioning, decommissioning, environmental impact and recovery remain part of the lifecycle rather than being omitted once a prototype works.

Public reproducibility includes macOS, Windows and Linux where the application claims cross-platform operation. Platform, dependency, command, output and difference must be recorded. A one-command route must traverse all declared stages, show progress and duration, preserve failure and avoid mutating the protected authority.

22.5 The current six-obligation experimental protocol family

The returned engineering work closes a formal protocol surface for proposed tests. Its purpose is to prevent a favourable trace from selecting its own apparatus, controls or interpretation after the event.

The Tesla resonant-transfer protocol requires a sealed, calibrated and connected path; declared phase and load; a complete power ledger; and adverse controls. Coupling, transmission and delivered load power remain distinct. An apparent receiver output cannot become evidence of anomalous transfer while an unaccounted conductive, radiative, stored or measurement path remains.

The vacuum/inertial-response protocol requires a registered drive, paired measurement channels, confound controls, restoration and complete source accounting. The vacuum-beat protocol separately distinguishes an outward 1/6 structural share from returned-cycle restoration. Any net-gain claim must halt if its ledger remains open. A structural fraction is not reported as a measured excess.

The sector-five/seven protocol seals the relevant **p**-fibre before target access and preserves background, systematics, nulls and outcomes outside the registered list. It is a blind-detection design, not a report that a new sector has already been discovered.

The Smithium protocol retains the exact **126/184/310** registered structure and requires apparatus, yield, linked decay, mass, ion, oxidation and spectroscopic records. Official discovery criteria remain distinct from the SFT formal carrier. The protocol does not report a synthesis or discovery.

The human consciousness/placebo/cross-binding protocol retains participant report, physiology, modality, expectation, blinding, the available-state boundary, consent, privacy and adverse events. It cannot infer phenomenal presence from physiology alone or physiological effect from report alone. The no-omission addendum establishes the complete six-obligation experimental family.

These eight current engineering additions have 2,048 candidates, eight survivors and 32 controls. Their exact closure is meaningful: the proposed tests have a complete registered structure. It is not empirical validation. At the current repository state, **no experiment in this returned protocol family is claimed as performed.**

22.6 Scientific anomaly handoff and current status

Application performance cannot select or retroactively prove a fundamental law. A failed build may falsify the artefact's own operating claim, expose a translation error or generate a question for an upstream branch. It does not automatically falsify every upstream theorem. Likewise, a working artefact may validate its intended use but cannot substitute for the derivation and evidence of the laws it uses.

Simulation remains simulation, demonstration remains demonstration and test remains test. A validated anomaly is returned to the categorical owner-Physics, Chemistry, Biology, Consciousness or another branch-with its complete apparatus and evidence record. Engineering does not acquire the anomaly by naming it.

The foundation registered 13 source custodians. Twelve transports succeeded; the FDA-DESIGN-CONTROLS-001 transport failed and remains failed. Seven Engineering-owned inherited questions were reconciled. Current completion covers the foundation and the eight protocol claims. The full-field census, physical execution of the protocols, application-specific validation, deployment and long-term lifecycle evidence remain open.

22.7 Dependency-resolved family ledger

22.7.1 From purpose to a controlled design

Family	Engineering output	Failure made visible
Requirements, Function and Boundary	Requirement, function, constraint, operating boundary, use/misuse and acceptance	Desire, scope and pass criterion cannot be selected after the build
Components, Interfaces and Architecture	Components, interfaces, architecture, system, modules and integration	Hidden dependency and failed intermediate state remain traceable
Resources, Efficiency and Reliability	Resource, capacity/load, efficiency, reliability, availability and failure mode	External supply, time window and loss cannot be omitted
Measurement, Calibration and Acceptance	Measurement, calibration, tolerance, test, acceptance and uncertainty	Display, adjustment, demonstration and passed test remain distinct
Alternatives, Trade-offs and Optimisation	Alternatives, trade-off, bounded optimum, multi-criteria record, robustness and design freeze	Hidden weights and target-selected designs are rejected
Control, Feedback and Stability	Action, loop, operational stability, observability, controllability and fault response	One favourable trace cannot prove global control

Use and misuse cases are included before design because foreseeable operation outside the intended happy path changes requirements and safety. Constraints may conflict; the conflict and authorised resolution must be versioned. An acceptance boundary is frozen in observable terms so that a design does not define success by whatever it happens to achieve.

Interface errors are first-class records. A component can satisfy its own specification while the integrated system fails because units, timing, orientation, version or error semantics do not match. Modular decomposition is scientifically useful only when the composition relation reconstructs the whole behaviour at the stated boundary.

Reliability is the probability or frequency of performing a required function under declared conditions and duration; availability concerns readiness when needed. Capacity and load must be paired. Failure modes include detected, latent, common-cause and cascading routes. A reported uptime without demand, observation and maintenance boundaries is incomplete.

Robust design asks whether acceptable performance persists across registered variation and disturbance. Design freeze prevents the target result from quietly selecting a new parameter or architecture. Change control may reopen the design, but it creates a new version with an impact record rather than rewriting the frozen test.

Observability and controllability are system-bound. A sensor record may be insufficient to distinguish internal predecessors, and an actuator may not reach every state. Fault response must identify detection, isolation, degraded mode, safe halt and recovery. The control family therefore supplies the apparatus logic required by the returned physical protocols.

22.7.2 From safety case to public lifecycle

Family	New responsibility	Boundary
Safety, Hazard and Resilience	Hazard, risk, safety case, fail-safe, resilience and defence in depth	No prior harm is not proof of safety
Verification, Validation and Traceability	Verification, validation, traceability, reproducibility, independent implementation and end-to-end test	Specification and intended use remain different
Cross-Platform Accessibility	Platform behaviour, portable records, interface, deployment, one-command verification and learnability	Local expert operation is not public reproducibility
Lifecycle, Maintenance and Sustainability	Configuration, maintenance, impact, decommissioning, lifecycle and bounded sustainability	Prototype success does not close future operation or disposal
Science, Design and Evidence	Law translation, performance, failure boundary, prototype, simulation and demonstration	Application cannot select or prove the law
Ownership and Anomaly Handoffs	Anomaly return and computation, physical, biological/social and self-hosted routes	An unexplained output returns to its scientific owner with apparatus intact

A safety case is an argued and evidenced composition, not a single test. It links hazards to controls, controls to verification and residual risks to decisions. Resilience concerns continued or restored function after disturbance and cannot be inferred from nominal operation. Defence in depth must survive a common-cause analysis.

Implementation-distinct checking requires a genuinely separate reconstruction route, not the same function called twice. End-to-end verification traverses the public workflow and catches composition failures that unit checks cannot. Reproducibility retains environment and dependency versions as part of the result.

Accessibility is technical and social. A portable data format without a usable interface may exclude the intended public; a polished interface backed by an opaque service is not self-contained. Lightweight deployment and one-command verification reduce barriers only when the command exposes failure and does not mutate source authority.

Configuration identity connects a deployed system to its tested version. Maintenance changes can invalidate previous evidence; change impact determines which tests must be repeated. Decommissioning preserves data, hazards and responsibility. Sustainability claims remain bounded by included resources, lifecycle and affected environment.

Simulation can test a model implementation; demonstration can exhibit one behaviour; prototype evidence can support feasibility. None establishes general operation or empirical truth beyond its boundary. A scientific anomaly requires restoration tests, alternative explanations, calibration and source accounting before it returns upstream.

22.7.3 Returned protocols as executable preregistrations

Each returned protocol can be read as an executable preregistration. The Tesla protocol freezes the power path. The inertial protocol freezes paired channels and restoration. The vacuum-beat protocol freezes a no-net-gain halt condition. The sector protocol freezes the candidate fibre and outside-list outcomes. The Smithium protocol freezes joint chemical and physical identification. The human protocol freezes report, physiology, expectation and ethics. The family and no-omission claims ensure that none can be presented without its complete controls.

Before any experiment, Engineering must create apparatus identities, calibration certificates, operator and environmental records, raw immutable outputs, analysis code and prospective decision rules. A pilot may develop the apparatus, but its observed targets cannot then be called blind. A later blind run requires separate custody. A null result means the registered instrument and analysis found no qualifying effect under adequate conditions; a failed transport or broken apparatus is unresolved, not null.

This is the direct engineering lesson of the computational programme as well. Thousands of tests can verify code while a protected empirical target remains unopened. Formal and implementation readiness are prerequisites for a blind campaign, not substitutes for it.

23. Cross-Branch Synthesis: composition without conflation

23.1 What synthesis may and may not do

Cross-Branch Synthesis owns no primitive law. It composes receipts already admitted by their categorical owners. It cannot rescue a missing derivation, turn analogy into identity, erase an adverse record or move ownership to a more convenient branch. A shared Fold relation may be exact while the physical, biological, conscious and social carriers remain different.

The current frozen synthesis family contains 12 claims, 3,072 candidates, 12 survivors and 48 controls. Its pre-Synthesis dependency graph contained 1,460 claims and 24,076 edges, with one root and no root-trace failure. All twelve claims replayed independently and five focused test groups passed. This family was frozen on 29 July 2026 before the current 2,751-claim graph was complete. It is therefore a completed inherited-return synthesis, not the final terminal synthesis of the present expanded corpus.

23.2 The twelve current compositions

The prime/vacuum claim preserves a multiplicative-order identity across its typed owners. The common-lock claims preserve the same Fold form across physical, conscious and social carriers while refusing to identify their meanings. Folding, fixation and optimisation share a descent structure, but a protein fold, a dynamical fixed point and an engineering optimum remain categorically distinct.

Longitudinal and transverse recurrence share a formal relation while their media, dimensional speeds and observations remain separate. Fold doubling supplies a second-harmonic structure. The prime-denominator vacuum period- divisor result remains a formal standing prediction, not an empirical measurement. Positive observable and structural absence are distinguished so that a non-record is not mistaken for the numerical or ontological absence of a carrier.

The Tesla synthesis composes Physics, Earth and Engineering records while keeping Earth recurrence modes distinct from the quarter-wave sequence. The Unified Constants Object remains Physics-owned and does not introduce new adjustable dials. The premeasurement ledger binds prediction, falsification, chronology and source custody. The one-owner/no-omission claim prevents a result from acquiring multiple inconsistent authorities. The root-traced terminal assembly proves that the frozen twelve-claim family reaches the common root at its declared boundary.

23.3 What remains before final synthesis

Since the frozen synthesis was built, the corpus expanded from 1,460 to 2,751 claims. The current programme therefore still requires the global one-owner lineage merge, canonical publication reconciliation and a final heavy verification of the expanded graph. The last completed heavy `verify-all` run, on 28 July 2026, covered 1,319 claims with 1,319/1,319 receipt replays and 969/969 tests. The 2,751-claim corpus has passed its current lightweight global validator, but the final heavy run has not been repeated at that scale.

This distinction sets the publication boundary of the preliminary Theory of Everything. The current synthesis demonstrates that typed cross-branch composition is possible and root-traced. The final publication update must repeat that work against the complete contemporary graph rather than treating an earlier terminal receipt as timeless authority.

23.4 Dependency-resolved family ledger

The twelve synthesis claims form four connected groups. The first establishes typed shared structures: prime/vacuum order, common Fold locks, folding-fixation-optimisation descent and wave recurrence. The second registers new formal consequences: second-harmonic doubling, the vacuum period-divisor prediction and the positive-observable/structural-absence boundary. The third controls physical translation: Tesla composition and the Physics-owned Unified Constants Object. The fourth controls scientific governance: the premeasurement ledger, one-owner/no-omission and terminal root trace.

The governing test for every composition is whether the shared relation can be stated without deleting the owners' distinguishing coordinates. A protein fold and an engineering optimisation may both descend through a generated candidate order, but protein sequence and physical interaction do not become design criteria. Conscious and social half-locks may share a two-preimage fibre, but a participant's interior relation does not become a survey distribution. Wave recurrence can be common while medium, polarisation, dimensional speed and instrument remain branch-specific.

The synthesis family would fail if it introduced an unregistered primitive, if any input receipt were absent, if a claim could not reach the common root, if one result acquired two categorical owners, or if analogy were used where the typed relation did not commute. Its completed 29 July graph satisfies those conditions for the then-frozen 1,460-claim input. The expanded 2,751-claim corpus requires a new terminal assembly rather than an editorial assertion that the earlier graph already contained later work.

24. The computational-proof programme

24.1 Purpose and evidence hierarchy

The computational-proof programme asks whether selected SFT structures can be implemented as public, exact and falsifiable systems in domains whose state spaces are large enough to defeat informal inspection. Protein folding, Chess, Go and Unison Fold AI test different boundaries. Protein tests sequence, interaction, geometry and protected structural comparison. Chess tests exact legality, strategic search and play against rated opposition. Go tests rules, state census, search and play across board scales. Unison tests whether the formal architecture can be carried into a language-capable artificial system without importing a pretrained answer oracle as the native model.

These are computational proofs only in a typed sense. An exhaustive census can prove that one registered survivor remains within its complete finite grammar. A passing test suite can verify implementation behaviour against its specification. A blind external comparison can provide empirical evidence at its registered target boundary. A match in one task does not prove the entire Theory of Everything, and a failure in one implementation does not by itself repeal its upstream branch. Each result must keep its own formal, implementation and empirical status.

The programme also preserves chronology. Historical V2 results remain part of the evidence even where a V3 recreation is planned. They are not silently called V3 results. A clean V3 scaffold is not called a completed recreation. Protected targets and comparator outputs stay unopened until the registered gate authorises access.

24.2 Protein Fold V2: the strong local result and the adverse whole-chain result

Protein Fold V2 produced a mixed but scientifically valuable record. Its whole-chain test on a 76-residue target was adverse:

V2 whole-chain measure	Recorded result
TM-score	0.0269927379497572
C-alpha dRMSD	52.89314678065376 Å

That result is not hidden inside a favourable average and is not superseded by the existence of strong local windows. It states that the achieved sealed sequence-only implementation did not recover the registered whole-chain structure at that boundary.

The same programme recorded strong local-window matches:

Local window	TM-score
HLV	0.9914591922414299
RLI	0.9656795312222751
RGG	0.9059580746307327

These results show that the implementation captured substantial local geometric structure on the tested windows. They do not turn the adverse whole-chain result into success. A separate theoretical support construction contains a complete trace with TM-score **0.9891211351**. That construction shows that the formal support can carry a high-similarity route; it is not the achieved sealed sequence-only whole-chain output and must never be reported as if it were.

The combined record motivates V3. Local success indicates that the interaction and geometry programme is not empty. The whole-chain failure identifies the missing global constraint, recurrence or search closure that a superior version must solve. The support construction demonstrates capacity inside a declared theoretical route while preserving the need for an independently achieved blind output.

24.3 Protein Fold V3: current exact result

Protein Fold V3 is a clean, active programme rather than a cosmetic renaming of V2. Its current scientific gate records:

Current Protein V3 item	Verified state
Scientific gate v20 suite	893 passed; 0 failed; 0 errors
Publication full suite v21	898 passed; 0 failed; 0 errors
Exact parent candidates	42,173,082
Parent leaves	375,014
Certified prunes	41,798,068
Frontier states	205

Current Protein V3 item	Verified state
Shards	11,012
Symbolic fallbacks	0
High-attraction slice	583 states
Dominated states in slice	573
Retained states in slice	10
New nondominated states	8
Protected target accesses	0
AlphaFold output accesses	0
Registered blind targets opened	0

The exact parent census is complete and verified at its registered boundary. The separately registered 583-state high-attraction slice is closed and independently verified. Those are substantive current results: V3 has enumerated and certified a 42,173,082-candidate parent search without symbolic fallback, and it has reduced the tested high-attraction slice to ten retained states, including eight new nondominated states.

The remaining attraction subband census and seeded-parent high-band reconciliation are active and incomplete. No result may be imported from an unfinished process merely because files or temporary states exist. Lower bands are not closed. The complete fixed-geometry sector is not closed. The global frontier is not closed. Arbitrary-sequence recurrence and the full predictor are not closed.

24.4 The AlphaFold-generalised blind parity campaign

The intended empirical gate is an AlphaFold-generalised blind parity campaign. "Generalised" means that the V3 system must confront the same broad kind of sequence-to-structure task without choosing a target because it favours the method. "Blind" means protected target structures and AlphaFold outputs remain inaccessible until the relevant derivation, implementation, scoring and failure-handling records are sealed. "Parity" requires a preregistered comparison surface and cannot be claimed from an illustrative match.

At the current gate, the campaign is **not permitted**. AlphaFold parity is **unavailable**, not adverse and not successful. There have been zero protected target accesses, zero AlphaFold-output accesses and zero opened targets. This is the correct status because the remaining exact search and global predictor obligations have not closed.

The planned campaign must retain sequence custody, target custody, release and model version, comparator settings, prediction timestamp, structure source, alignment and scoring rules, uncertainty, failure cases and all targets. It must report local and whole-chain outcomes separately. Any target that cannot be processed, any comparator failure and any missing experimental structure must remain in the denominator under its registered classification.

The current preliminary Protein paper is version **0.9.4**, DOI **10.5281/zenodo.21717581**. It reports the gate that exists now and is intended to receive a lawful version update when the remaining exact work and blind campaign are complete. Preliminary publication does not authorise a forecast to be rewritten as an achieved result.

24.5 Chess Fold: historical V2 evidence and V3 boundary

Chess Fold V2.2 retains exact legality anchors, **22/22** main tests and **8/8** endgame tests. Its recorded match results are:

Opposition	Record	Score
Elo 1700	3 wins, 7 draws, 2 losses	54.2%
Elo 1900	6 wins, 3 draws, 3 losses	62.5%
Elo 2100	2 wins, 5 draws, 1 capped draw, 4 losses	41.7%

These are implementation and measured play results at their historical test boundaries. They show legal, functioning play and non-trivial performance, including a positive score in the registered 1900 series. They do not by themselves establish universal playing strength or a solved game.

A later auxiliary probe examined 36 positions. All 36 completed; nine choices changed, four changes matched the Stockfish-best move, and none of the four named death-slide positions changed. This probe was agent-authored and lacks the programme's publication governance, so it remains auxiliary evidence rather than a promoted main result.

The current repository contains no clean V3 Chess recreation workspace. Accordingly, V3 Chess is **not started**. Its programme remains to rebuild the legality, search, evaluation, resource and opponent-custody stack without silently importing the V2 runtime; repeat exact anchors and regression tests; pre-register rated and tactical comparisons; and preserve every draw, cap, timeout, illegal move and loss.

24.6 Go Fold: exact finite censuses, measured play and V3 boundary

Go Fold V2.2 includes exact legal-state censuses across small boards:

Board	Exact census
1 × 1	1
1 × 2	5
2 × 2	57
2 × 3	489
3 × 3	12,675
4 × 4	24,318,165

The 2 × 2 search retains **17,038,501 nodes**. These counts are exact implementation records and strong foundations for a V3 rules and state-space reconstruction.

Two independently replayed 9 × 9 batches each recorded **2-0** against GNU Go. The recovered 19 × 19 KataGo harness record reports **2-0** at a 128-ply point cut-off. Round 1 also contains two rejected play responses after which the harness continued; Round 2 contains no rejected response. The cut-off and Round 1 defect remain part of the result. They cannot be edited into completed ordinary games or a clean two-game run.

As with Chess, the present repository contains no clean V3 Go recreation workspace. V3 Go is **not started**. The intended programme is to rebuild exact rules and legality, reproduce the finite censuses, seal search and scoring, then run governed comparisons whose resignation, pass, illegal move, rejected response, cut-off and final scoring rules are fixed in advance.

24.7 Unison Fold AI: implementation strength and generation weakness

Unison V6.3 retains a large historical implementation record: **326 suites, 2,002 checks**, zero main-corpus failures and **326/326** certificate identities. Its current standalone verifier passes **21/21** checks. The historical broader verifier passes **43/47**, so four historical checks remain outside a complete pass and must not be hidden by the current standalone total.

Generation evidence is weak. The recorded MMLU runs moved from **9/128** to **0/128**. Free generation produced **1/12** successful cases. The Codex generation probes for baseline, F1 and F3 each produced **0/12**. The correct interpretation is that Unison V2 established substantial architecture, implementation and certificate machinery but did not establish competitive general answer generation.

The V3 workspace is a clean version **0.1.0-scaffold**. It does not import the V2 runtime. Pretrained answer weights are not allowed to serve as the native model. The proposed primary interface is a Discord text bot, but remote operation is not thereby authorised. The present statuses are exact:

V3 Unison dimension	Current status
Workspace	Clean scaffold
Mathematical closure	OPEN
Empirical validation	NOT_RUN
Remote deployment	NOT_AUTHORIZED
V2 runtime imported	false
Pretrained answer weights as native model	prohibited

These statuses are not adverse empirical results because the empirical programme has not run. They are also not implementation success. V3 must first close its mathematical architecture, implement it, verify its certificate and resource surfaces, establish safety and governance, and only then conduct registered evaluations. Any human-participant or public deployment must also satisfy the Medicine, Consciousness, Social and Engineering boundaries already derived.

24.8 Programme comparison and next gates

Programme	Strongest current record	Standing limitation	Next lawful gate
Protein Fold V3	Verified 42,173,082-candidate parent; closed 583-state high-attraction slice; scientific gate v20 893/893 and full suite v21 898/898	Remaining bands and global predictor open; no target or AlphaFold access	Complete exact search and global recurrence, then authorise sealed blind parity
Chess Fold	V2.2 legality and endgame tests; measured play through Elo 2100	No V3 workspace; auxiliary probe not publication-governed	Clean V3 rebuild and preregistered tactical/rated evaluation
Go Fold	V2.2 exact censuses; two 9×9 2-0 batches; qualified 19×19 record	No V3 workspace; 19×19 cut-off and rejected-response defect retained	Clean V3 rebuild, census replay and governed full-game comparison
Unison Fold AI	V2 architecture with 2,002 main checks and complete certificate identities	Weak generation; V3 mathematical closure open and validation not run	Close V3 mathematics and implementation before authorised evaluation or deployment

The programme's strongest general result is methodological. It has made failure localisable. A false move, bad fold, rejected Go response or failed generation probe no longer has to be interpreted as a verdict on an entire world-view. The registered layers show whether the failure belongs to a candidate grammar, an implementation, a resource limit, a comparison method, an external transport or the scientific claim itself.

24.9 Technical ledger

24.9.1 Common constitution of a computational proof

Every programme begins with a problem carrier and a boundary. Protein begins with an ordered residue sequence and a registered structural target. Chess and Go begin with game rules, legal histories and an opponent or exact census. Unison begins with a native architecture and a registered task surface. The programme then separates six layers:

1. **Registration.** The claim, candidate grammar, resource boundary, evaluation rule, target custody and stopping rule are fixed.
2. **Enumeration or construction.** Candidates or executable states are generated without consulting protected answers.
3. **Elimination.** Every rejected candidate receives a reconstructible reason; pruning must conserve the census.
4. **Implementation verification.** Unit, property, regression, independent and end-to-end tests verify what the software actually does.
5. **External comparison.** Only an authorised custodian opens targets or opponents and preserves every outcome.
6. **Interpretation.** Formal, implementation and empirical status are reported independently, with ownership of failure retained.

The conservation equation for a complete finite search is simple but non-negotiable:

$$N_{\text{generated}} = N_{\text{rejected}} + N_{\text{retained}} + N_{\text{unresolved}}.$$

If pruning is hierarchical, every parent count must reconstruct from its children and certified prunes. A hash binds file identity, not scientific correctness; correctness additionally requires the generator, decision rule, controls and independent reconstruction.

Target blindness is a capability condition rather than a statement of good intent. The derivation and search process must be unable to read protected structures, comparator outputs, hidden solutions or later evaluation records. Opening a target during development changes chronology. The resulting work may remain useful as development-observed evidence, but it cannot later be called a blind prediction.

Negative outcomes have different meanings. A programme may return a valid but poor-scoring result, fail to return an output, exceed its registered resources, encounter an invalid external response, or be unable to access a target. Poor score is adverse to the performance criterion; no output is an implementation failure; resource exhaustion is a bounded unresolved result unless the claim included that resource; invalid external response is a transport or harness failure; unopened target is not tested. These classes must not be compressed into "failed".

24.9.2 Protein V3 representation and six-axis relation

Protein V3 starts from the complete ordered sequence rather than from a target structure. Residue identity and position are preserved. Local and non-local interactions must be distinguishable, and any environment, chain boundary, modification or partner included in the claim must be registered. A representation that discards sequence order or imports target coordinates would fail the Biology handoff.

The six-axis joint relation scales the formal Fold programme by evaluating candidate relations together rather than allowing one attractive local quantity to select a geometrically impossible global state. The exact axis names and machine values remain in the Protein archive; scientifically, their roles cover the retained sequence carrier, interaction relation, geometric compatibility, attraction or ordering, recurrence/continuation and evidence boundary. The joint relation is important because whole-chain folding is a constraint-composition problem. A conformation may satisfy many local preferences while producing global collision, topological inconsistency or an incorrect long-range arrangement.

The current parent search generated **42,173,082** candidates. Its account closes as **375,014** leaves plus **41,798,068** certified prunes. The search records **205** frontier states across **11,012** shards and uses **zero** symbolic fallbacks. A symbolic fallback could have hidden an unevaluated route behind an expression; its absence means the registered parent result is an explicit exact computation at this boundary.

Pruning is a scientific result only if it preserves every candidate that could satisfy the frozen relation. Each certificate must identify the parent, subspace, rejecting condition and reconstructible count. The equality between parent, leaves and prunes is necessary but not sufficient: a systematically wrong pruning predicate could conserve counts. Independent verification and adverse controls therefore test both arithmetic and semantics.

The high-attraction slice asks a narrower question of **583** states. It classifies **573** as dominated and retains **10**, of which **8** are new nondominated states. Dominance means that another registered state is at least as good on every relevant frozen comparison and better on one; it does not mean that a dominated state has been empirically observed to be biologically impossible. The slice is complete and independently verified at this bounded relation.

The remaining attraction subbands cannot be inferred from the high slice. A lower-attraction state could participate in a globally superior chain after other constraints are composed, unless the proof shows it cannot. The seeded-parent reconciliation must also demonstrate that no seed-specific route escaped the parent accounting. This is why the programme has not opened its protected targets despite a large completed census.

24.9.3 From local geometry to whole-chain recurrence

The V2 record demonstrates a central distinction. Local windows can align very strongly while the whole chain is wrong. A chain is not the arithmetic mean of independent local folds. Long-range contact, chain closure, steric consistency, secondary-structure packing and competing interactions compose the global problem.

V3 must therefore prove or implement a recurrence that carries admissible local relations into a complete fixed-geometry sector. Each extension step should preserve the prefix or partial carrier, record newly introduced interactions, reject geometric impossibilities and retain alternatives that remain non-dominated. If two partial states share a compressed image but differ in an unheld interaction relevant later, merging them would destroy a possible fold. The information-loss law thus has a direct algorithmic consequence: state compression is sound only when discarded predecessor distinctions are proven irrelevant to every future extension.

The global frontier closes only when every registered full-chain candidate is either retained, certified dominated, rejected by an exact constraint or classified unresolved under the frozen resource boundary. Arbitrary-sequence recurrence adds a generality claim beyond one target length or composition. The full predictor then connects the exact frontier to one sealed output or ensemble under a prospective decision rule.

Potential success is supported, but not guaranteed, by three current facts. First, V2 achieved very high local-window similarity, so the representation can capture real local geometry. Second, the theoretical support trace achieved a high whole-chain TM-score inside a declared construction, so the formal support is capable of carrying a globally close route. Third, V3 has already scaled exact enumeration and dominance reasoning to more than 42 million parent candidates without fallback. Together these make continued development scientifically motivated and make success a credible prospective expectation. They do not establish the missing sealed sequence-only whole-chain result.

An adverse final comparison would still be informative. If the exact implementation closes but blind performance remains poor, the candidate grammar or mapping from formal relation to protein evidence is incomplete. If the search cannot close within the

registered resources, the implementation generality claim must narrow. If performance is strong only in local windows, the global recurrence remains the identified problem. Each outcome directs a different next revision.

24.9.4 Blind parity protocol and scoring constitution

Before the AlphaFold-generalised campaign opens, the following records must be sealed:

Record	Required content
Target manifest	Target IDs, sequence source, chain boundaries, release/version and exclusion rules
Custody manifest	Who can access structures and AlphaFold outputs; capability separation and access log
SFT build identity	Code, dependencies, parameters fixed by derivation, resource envelope and execution command
Comparator identity	AlphaFold implementation or service, model/version, inputs, settings and output custody
Scoring plan	Whole-chain and local metrics, alignment, missing residues, multimers, ensembles and tie rules
Decision plan	Success, parity, adverse, unresolved and transport classifications fixed before opening
Completeness plan	All targets, timeouts, invalid outputs, failed accesses and comparator failures retained
Chronology	Seal time, prediction time, comparator time, target opening and analysis time

TM-score, RMSD and dRMSD measure different aspects of agreement. TM-score is length-normalised and more tolerant of local deviations; RMSD can be dominated by a few large discrepancies; distance RMSD tests internal distances without requiring the same rigid alignment. No one metric should be selected after the outputs are seen. Local-window metrics remain secondary unless local accuracy is itself the preregistered claim.

Parity must specify whether it concerns mean, median, target-wise win/loss, threshold achievement, non-inferiority interval or another complete statistic. Averages alone can hide catastrophic targets. Per-target results and confidence or uncertainty procedures must remain. If AlphaFold cannot process a target, that row is a comparator failure under the frozen rule, not an automatic SFT win. If SFT cannot process one, it is not deleted.

No target sequence should be selected because an experimental structure is known to favour the SFT grammar. Exclusions may be scientifically necessary—for example, unresolved residues, complexes outside the programme's carrier or non-standard chemistry—but must be fixed from source metadata available under the registered custody rule, not from observed performance. Any development set must be permanently distinguished from the blind holdout.

24.9.5 Chess V3: exact rules before strength

Chess Fold V3 must begin by reproducing the game, not by tuning an evaluation against an engine. The position carrier retains board occupancy, side to move, castling rights, en-passant state, repetition-relevant history and move count where the rule requires it. Legal move generation must preserve check, promotion, castling constraints and every terminal condition. A visually valid board without history may not determine the legal successor set.

The V2 22/22 main and 8/8 endgame anchors are the first regression surface, but V3 should extend them with property tests: every generated move must produce a legal successor; reversible moves should reconstruct their predecessor where history permits; independently generated perft counts should match fixed positions; and illegal moves should be rejected without mutating state. Draw by repetition, move-count rules, stalemate, insufficient material and capped external games must have separate records.

Search must retain node identity, depth or resource, transposition conditions, ordering and score provenance. A Fold compression may merge positions only when every rule-relevant and evaluation-relevant distinction is preserved. Evaluation should be derived or frozen before opponent play. Stockfish may serve as an external comparator or analysis tool after sealing; its move cannot be imported as a hidden label into the native chooser.

The V2 match record suggests non-trivial play but also a decline at the 2100 boundary. V3 should not promise a monotonic Elo curve from three small series. A governed campaign needs a fixed opponent version, hardware or node budget, time control, colour balance, opening policy, adjudication, disconnection rule and complete PGNs. Score intervals and game-level records should accompany the headline percentage.

Tactical and strategic probes answer different questions. A named death-slide suite can test whether a specific failure mode has changed; a broad position suite can test move agreement. The prior auxiliary run's four Stockfish-best changes and zero death-slide improvements show why both must be reported. V3 governance should register probe authorship and freeze expected properties without turning engine agreement into the sole definition of a good move.

24.9.6 Go V3: history, territory and harness custody

Go positions require board occupancy, side to move and sufficient history for ko or superko under the selected rule set. Capture, suicide, pass, termination and scoring conventions must be fixed. A state census depends on those choices; changing them changes the grammar. The V2 small-board totals therefore serve as versioned anchors, not rule-independent constants.

The exact sequence from 1×1 through 4×4 demonstrates that the V2 implementation could enumerate rapidly expanding legal spaces. The 2×2 node count of **17,038,501** is especially useful for a complete independent replay because the final state count alone cannot reveal duplicated histories, wrong ko handling or unsound transposition merging.

The V3 engine should expose legal-move generation, capture groups and liberties, history identity, evaluation, search and time/resource accounting separately. Property tests can assert conservation of stone changes through placement and capture, legal predecessor/successor relations, hash consistency and score reconstruction. Adverse cases must include suicide-like moves, ko recapture, double pass, rejected external protocol responses and timeouts.

The V2 9×9 results are complete at their recorded two-game batch boundaries. The 19×19 result is weaker because it stops at 128 plies and Round 1 continued after two rejected play responses. A V3 harness must treat an invalid or rejected response according to a frozen rule-retry, forfeit, unresolved transport or abort-and may not continue silently. Ordinary final scoring, resignation and cut-off adjudication must remain separate.

A strong V3 result would first reproduce the exact censuses, then pass independent legality tests, then conduct complete games against fixed public opponents. A win is an empirical implementation result under the stated rules and budget, not a proof that Go is solved. A loss preserves a position record from which search, evaluation or protocol failures can be analysed.

24.9.7 Unison V3: native architecture before borrowed fluency

The clean V3 constraint forbids the V2 runtime from entering as an unexamined foundation. It also forbids pretrained answer weights from serving as the native model. This does not deny the utility of external models as development tools, evaluators or interfaces when their role is declared. It prevents borrowed statistical content from being presented as a consequence of the native Fold architecture.

Mathematical closure should specify the state carrier, Fold operations, memory/record relation, learning or update rule, inference, generation and resource bounds. Every operation needs a typed input and output, invariants and failure state. Implementation closure then requires certificate identity, independent reconstruction, deterministic or recorded variability and end-to-end execution on the public platforms claimed.

The V2 **326/326** certificate identities and **2,002** main checks are strong evidence that a large formal implementation can remain internally auditable. The **43/47** historical verification and poor generation results prevent that strength from being converted into a claim of general intelligence. V3 should preserve the certificate discipline while treating generation as an empirical problem.

Evaluation must separate factual question answering, structured reasoning, free generation, instruction following, safety, memory and conversational continuity. MMLU is one source-bound benchmark and can be contaminated or misused; a zero score still records a concrete failure at the chosen interface. Free-generation success needs a frozen rubric or independent judge custody. External language-model judging must record model, prompt, version and every disagreement.

The proposed Discord bot adds a deployment system: authentication, message transport, privacy, moderation, rate and resource limits, error handling, logging, data retention, abuse response and rollback. Engineering validation must establish intended public operation. Social and Medicine boundaries govern vulnerable users and advice. Consciousness claims remain unavailable from fluent behaviour alone. Until those stages are complete, remote deployment properly remains `NOT_AUTHORIZED`.

24.9.8 Cross-program evidence and publication gates

The four programmes should share a common release grammar without pretending their metrics are comparable:

Gate	Protein	Chess	Go	Unison
Formal carrier closed	Partial: parent and high slice closed; global open	V2 only; V3 not started	V2 only; V3 not started	V3 open
Implementation suite	Scientific gate v20 893/893; publication full suite v21 898/898	Historical V2 anchors	Historical V2 censuses and play harness	V2 main 2,002 checks; V3 not run
Protected external target	Unopened	Future governed opponents/positions	Future governed opponents/games	Future registered benchmarks/users

Gate	Protein	Chess	Go	Unison
Current empirical class	Active, incomplete; parity unavailable	Historical measured V2 record	Historical measured V2 record with qualified 19 × 19 row	V2 weak generation; V3 NOT_RUN
Deployment authority	Local scientific programme	None asserted for V3	None asserted for V3	NOT_AUTHORIZED

A version update is warranted when a programme has a materially new, independently verified record, even if the result is adverse. Versioning must state exactly which earlier result is preserved, corrected or superseded. A new version may update current status while retaining the historical chronology of the old release. It must not create a new record merely to hide an adverse version.

For Protein, the next publishable milestone is closure of the remaining exact search or a scientifically informative narrowing of that search, followed by the authorised blind campaign when its gate passes. For Chess and Go, the next milestone is a clean V3 reproduction of rules and exact anchors before rated play. For Unison, it is mathematical and implementation closure before any claim of empirical capability.

The completed computational-proof programme will contribute a set of independent, domain-specific application results to the final Theory of Everything. It will not be used to rewrite the upstream derivations. Its role is stronger: it makes those derivations confront exact execution, public failure and protected external comparison.

24.10 Integrated status, corrections and final extension frontier

24.10.1 What has become derivable

Across this part, the SFT dependency spine now reaches living organisation, genetic coding structure, inheritance, evolution, condition-bounded protein ensembles, development, physiology and ecology; patient-centred clinical knowledge, causal and prognostic boundaries, intervention evidence and an expectation/body fibre; interior observation, binding, self-model, memory, qualitative recurrence and substrate-realisation criteria; one Earth's coupled budgets, histories, climate and hazard records; a source-to-record cosmos; identified collective agents, institutions, power, ethics and historical causality; and a complete engineering constitution for requirements, resources, safety, validation and anomaly handoff.

The major conflation resolved is the repeated substitution of one evidence carrier for another. Chemistry is not life. Biological difference is not clinical diagnosis. Public access is not phenomenal presence. A model is not an observation. A social aggregate is not a person. Consensus is not proof. A prototype is not validation. A test suite is not empirical confirmation. Cross-branch similarity is not cross-branch identity.

24.10.2 What current evidence establishes

The current 555 claims in this span each have a complete candidate census, one survivor, four required controls and a root trace at their registered formal boundary. Their source comparisons and independent reconstructions establish the empirical or implementation status recorded in each package. The evidence also establishes that the corpus preserves findings which do not favour the strongest proposed numerical reading.

The standing adverse, corrected and unresolved records include:

Record	Current status
Biology universal three-quarter allometry	Corrected; multiple exponents and non-universal cases retained
Biology strong scale-free default	Corrected; empirical rarity retained
Homochirality origin	Observed asymmetry; historical origin unresolved
Medicine prognosis from denominator parity	Corrected; clinical horizon and evidence required
Consciousness from access, report or fluent output alone	Not admitted; complete interior or realisation relation required
Earth first earthquake catalogue	Adverse and unclassified
Earth homogeneous earthquake holdout	Compatible; does not erase first result
Astronomy first SPARC rank-four comparison	Adverse and unclassified
Astronomy returned exact mappings	Mixed: categorical pass with heterogeneous, mismatched, unresolved or adverse quantitative rows
Engineering returned protocols	Formally closed; experiments not run
Protein V2 whole-chain fold	Adverse; strong local windows and theoretical support kept separate
Protein V3 blind AlphaFold parity	Unavailable; gate not permitted and targets unopened

Record	Current status
Chess and Go V3	Not started in current workspaces
Unison V3 validation and deployment	NOT_RUN and NOT_AUTHORIZED

Failed source access also remains a positive historical fact about custody: four Biology transports, two HHS Medicine transports, the Consciousness absent/failed source rows, one protected Earth transport with a separate addendum, two Astronomy transports, four Social transports and one Engineering transport remain visible under their exact records.

24.10.3 What remains open

The final extension frontier has several layers.

First, Biology has 342 open obligations in its registered 424-obligation full-field census. The later domain branches have complete foundations and returned families but do not yet have frozen full-field censuses. Their next work must expand coverage without rewriting foundation chronology.

Second, Protein Fold V3 must complete the remaining attraction bands, seeded-parent reconciliation, fixed-geometry sector, global frontier, arbitrary-sequence recurrence and full predictor before the 100-target AlphaFold-generalised blind parity campaign can be authorised. Chess Fold and Go Fold require clean V3 recreations. Unison Fold AI requires mathematical and implementation closure before empirical evaluation or deployment.

Third, the 2,751-claim graph requires a final global one-owner lineage merge, canonical publication reconciliation and heavy verification. The completed twelve-claim synthesis must be regenerated against that contemporary graph.

Fourth, every numerical structural correspondence that has not yet been measured remains a prediction or open mapping at its registered chronology. No editorial completion pass can lawfully transform it into an earlier blind prediction or empirical confirmation.

24.10.4 Contribution to the complete dependency spine

These branches turn the abstract Fold into a disciplined account of the worlds in which science is performed. Life retains itself while matter turns over. Medicine retains the person while populations inform care. Consciousness retains interior difference while public images merge. Earth Science retains one planet's history while universal laws constrain it. Astronomy retains the source path by which a cosmos becomes observable. Social Science retains people and power while aggregates compress them. Engineering retains requirements, loss and failure while knowledge becomes an artefact. The computational programme then exposes the exact point at which a formal relation does-or does not yet-survive implementation and blind comparison.

This is foundational completion at declared boundaries and active extension beyond them. "Complete" always names the end of the registered current project investigation, never the end of science. The preliminary Theory of Everything will be version-updated as the computational programmes, downstream censuses, global synthesis and final verification close. Its present value is not that open work has been hidden. It is that derivation, implementation, evidence, failure and future test are now connected in one public dependency record.

24.11 Reproducibility and authority note for this part

The conceptual narrative above is controlled by the repository's active-paper inventory, exhaustive content matrix, current programme-status audit, branch claim packages, source manifests, evidence receipts, verification reports and programme gates. Complete claim IDs, candidate decisions, controls, source identities, full SHA-256 receipts and executable traces remain in the scientific audit layer and machine archive. Short displays in this chapter do not replace those records.

The active published foundation papers for the domain branches carry these DOIs: Biology [10.5281/zenodo.21630203](https://doi.org/10.5281/zenodo.21630203); Medicine [10.5281/zenodo.21630785](https://doi.org/10.5281/zenodo.21630785); Consciousness and Cognitive Science [10.5281/zenodo.21636397](https://doi.org/10.5281/zenodo.21636397); Earth and Environmental Sciences [10.5281/zenodo.21640810](https://doi.org/10.5281/zenodo.21640810); Astronomy and Cosmology [10.5281/zenodo.21640812](https://doi.org/10.5281/zenodo.21640812); Social and Collective Sciences [10.5281/zenodo.21640814](https://doi.org/10.5281/zenodo.21640814); and Engineering Translation [10.5281/zenodo.21640816](https://doi.org/10.5281/zenodo.21640816). Protein Fold V3 preliminary results carry current version DOI [10.5281/zenodo.21717581](https://doi.org/10.5281/zenodo.21717581) and retained concept DOI [10.5281/zenodo.21713536](https://doi.org/10.5281/zenodo.21713536).

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24.12 Evidence custody, chronology and historical reconciliation

24.12.1 Foundation seal and returned-family chronology

The domain foundations and the later returns are two publication layers. The foundation papers were sealed, generated and externally compared at their own candidate boundaries. The returned families were registered later to answer specific unresolved or corrected V1/V2 obligations. A present total may add the two layers, but it may not make the later claim part of the earlier blind seal.

Branch	Foundation result	Later result	Chronology that must remain visible
Biology	75 claims, 19,200 candidates, 300 controls	7 claims, 1,792 candidates, 28 controls	Foundation source comparison precedes origin/homochiral/somatic/neural/differentiation/ecosystem return
Medicine	72, 18,432, 288	4, 1,024, 16	Clinical constitution precedes expectation/body and physiological comparison
Consciousness	72, 18,432, 288	5, 1,280, 20	Hard-problem foundation precedes synaesthesia/nonordinary/sleep/cessation comparison
Earth	74, 18,944, 296	1, 256, 4	Two-basin foundation precedes normalised Fold-lock return
Astronomy	72, 18,432, 288	5, 1,280, 20	Foundational observation and SPARC test precede solar/atomic/planetary/lithium family
Social	72, 18,432, 288	4, 1,024, 16	Collective constitution precedes inequality/polarisation/cycle return
Engineering	72, 18,432, 288	8, 2,048, 32	Translation foundation precedes proposed experimental protocols

Each claim nevertheless follows the same minimum custody pattern. A registration identifies the statement, owner, dependencies, candidate grammar, count, controls, falsification condition and evidence boundary. The engine generates the complete grammar without target access. One survivor is admitted only when every registered preservation coordinate is satisfied and every alternative receives a decision. An independent implementation reconstructs the result. Source registration and capture occur under the claim's chronology. The evidence record compares the source-derived held relation with the sealed statement. A measurement receipt or validation report binds the conclusion to the exact artefacts.

The repeated 256-candidate grammar in these branches is not a claim that every scientific question has 256 natural possibilities. It is the complete Cartesian product of eight registered binary preservation dimensions for each stated obligation. The depth-independent status establishes uniqueness within that explicit grammar. Generalisation beyond the grammar requires a new claim and cannot be obtained merely by repeating the receipt.

24.12.2 Source transport is part of evidence, not a clerical footnote

A source identity records the custodian, title or dataset, release, locator, snapshot or access date and hash where available. Transport status records whether the required material was actually acquired. Content status records whether the captured source contains the preregistered feature. These are different questions.

Biology preserved four failed transports while using 17 usable source records. The TLS failure, HTTP 403 and two challenge pages describe failed access paths. They do not show that the inaccessible sources disagree, agree or lack a feature. Medicine preserved two failed HHS routes and separately captured purpose-matched ICH material. The ICH record has its own identity; it cannot impersonate the inaccessible HHS documents.

Consciousness registered 61 source features across 15 identities. Fifty-eight were present, three full-text features absent and 18 transport or content rows failed. A missing feature in an accessed document is different from an unopened document. Earth registered 91 features across 21 identities, with 67 present and 24 absent. Its original Smithsonian 403 remains historical even though a WFS addendum supplies separately registered content.

Astronomy retained two failed transports among 15 primary sources. Social retained four among 14. Engineering retained the failed FDA-DESIGN-CONTROLS-001 transport among its 13 custodians. The cumulative record therefore does not pretend that open science guarantees every external server will remain available. It guarantees that failure is visible and that a replacement route receives a new provenance rather than changing history.

Source prestige has no formal role in candidate selection. Primary or official sources are preferred because they reduce ambiguity in the observed record, not because institutional status proves a claim. A source may be authoritative for its own measurement or rule while still supplying adverse evidence for the SFT statement.

24.12.3 Corrections are current results

The V3 historical audit reviewed the V1 corpus of 356 rows and V2 corpus of 407 steps. Attribution to a branch was done at the atomic claim level because older rows often combined several domains. A corrected claim retains its historical text and the current replacement route. It is not erased and does not continue as a second live authority.

Biology's 30 owned atoms include 21 corrections. They remove inevitability from beneficial fixation, rigidity from protein state, a solved-origin claim from homochirality, universality from one allometric exponent and strong scale-free topology, and diagnostic or stability power from denominator parity or one orbit. Medicine's five owned inherited atoms are corrected where placebo lacks a comparator, stereochemistry substitutes for clinical evidence, parity substitutes for prognosis or recurrence substitutes for diagnosis.

Consciousness reconciles 46 owned questions at a boundary that refuses to identify access or report with phenomenal presence. Earth reconciles three owned questions while preserving the adverse earthquake test. Astronomy reconciles 53, including the quantitative galaxy test and unobserved-sector classifications. Social reconciles five without turning prestige or consensus into evidence. Engineering reconciles seven without allowing a desired device to select the physical law it is meant to test.

A correction may narrow a claim and still increase scientific value. "Every biological network is scale-free" is easy to repeat but empirically false as a universal. A typed topology claim with a sampling and fit boundary can be tested. "Cycling is cancer" is memorable but clinically unsafe. A continuing cycle plus differentiation loss and escape from three controls is more specific and falsifiable. The current corpus treats this increased specificity as progress.

24.12.4 Adverse, compatible, null and unresolved chronology

The status vocabulary is operational:

- **Adverse** means the registered evidence conflicts with the registered decision condition.
- **Compatible** means the evidence lies within the permitted region but is not discriminating enough to establish unique support.
- **Null** means an adequate registered test observed no qualifying effect or difference under its declared rule.
- **Unresolved** means required evidence or closure is incomplete, unavailable or censored.
- **Failed transport** means the evidence route could not deliver its required source or response.

The first earthquake test is adverse because its interval excludes 1/10. The second is compatible because its separately registered interval contains 1/10. They cannot be pooled after the fact without a new synthesis claim. The first SPARC fit is adverse to exact rank four; the separate 3.85 ± 0.09 method record and 3.5-4.0 systematic range provide relevant context without changing the first classification.

Protein V2's whole-chain output is adverse because it produced a valid output far outside a successful structural comparison. Protein V3's unopened parity campaign is unresolved or unavailable at its gate, not adverse, because no protected target has been tested. Unison V3 NOT_RUN is a programme state, not a null benchmark result. Chess V3 and Go V3 are absent as clean current workspaces; their V2 measurements remain historical evidence.

When later evidence changes current status, the historical row remains. A resolved adverse result would require a new claim or a protocol-authorised reclassification rule that was present before the later evidence. Merely obtaining a favourable later result is not enough.

24.13 One-owner dependency and handoff matrix

The following matrix identifies the objects exchanged across this part and the claims each recipient is forbidden to infer without further work:

Upstream owner	Typed handoff	Downstream owner	Not supplied by the handoff
Chemistry and Materials	Reaction, molecule, interface and structure-property carriers	Biology	Living organisation, heredity or evolutionary continuation
Biology	Organism, physiology, variation, population and mechanism	Medicine	Diagnosis, prognosis, efficacy, safety or patient decision
Biology, Information and Computation	Neural, informational and transition carriers	Consciousness	Interior presence, qualitative identity or artificial realisation

Upstream owner	Typed handoff	Downstream owner	Not supplied by the handoff
Physics, Chemistry and Biology	Planetary stocks, flows and coupled processes	Earth	Earth's actual initial conditions, history, attribution or hazard exposure
Physics and Earth observation	Propagation, reference and instrument relations	Astronomy	Remote object identity, catalogue population or cosmic chronology
Biology, Consciousness, Medicine and Information	Agents, capabilities, reports, communication and health constraints	Social	Institution, power, legitimacy, collective causation or ethical verdict
All scientific branches	Admitted laws, measurements and use constraints	Engineering	Requirement, safe design, intended-use validation or deployment authority
Categorical owners	Immutable admitted receipts	Cross-Branch Synthesis	Permission to rename analogy as identity or create a primitive law
Biology and Engineering	Protein carrier and verified implementation boundary	Protein Fold	Successful whole-chain prediction or AlphaFold parity
Computation and Engineering	Rule and software constitutions	Chess/Go	Playing strength, solved-game status or clean V3 completion
Computation, Consciousness, Social and Engineering	Architecture, realisation, interaction and deployment boundaries	Unison	Consciousness, empirical capability or public deployment permission

One-owner does not mean one contributor or one file. It means one categorical authority for the scientific kind of claim. A medical device may require Physics, Biology, Medicine, Social Science and Engineering records, but the Engineering acceptance test cannot decide the medical benefit and the medical trial cannot redefine the physical measurement. Cross-branch work succeeds when all owners' records commute without erasure.

An anomaly follows the matrix in reverse. The application first verifies that its own requirement, code, apparatus and calibration are sound. It then returns the complete record to the owner of the challenged scientific relation. If the anomaly crosses several branches, Synthesis may assemble the typed question; it cannot pre-assign the answer.

24.14 Publication completion and version-update rule

Publication readiness has four independent dimensions. A **formal** result has a complete registered grammar, survivor and proof receipt. An **implementation** result has an executable realisation and verification record. An **empirical** result has a source- or target-bound comparison with chronology. A **publication** result has passed editorial, identity, rendering, manifest and authority checks. Completion in one dimension cannot be imported into another.

The active foundation papers remain the scholarly entry points for their branches. Their current file identities are:

Branch paper	Active SHA-256
Biology	af18711f0520f03a6d2e7fe5004166856d2203ad370a4932178c59eb00abddb5
Medicine	1f3acb2f7b0100da9e2530779f0a8e556570ae2bab46302554af8ef3fb15fd22
Consciousness and Cognitive Science	62f473b326aa9cf0ecfab2dcaaf763aa752df468210aef52b0d12c40f3aebaf9
Earth and Environmental Sciences	044d381d988635cb69a368ff9dbfe2fad8d87b31cec7f4edd76faee959c60354
Astronomy and Cosmology	135b2ada93caf788571b9e7c0d384d60ec0851d9c47a85fe432d6ad2618ef643
Social and Collective Sciences	83105339ba4dd4f419f06935ff7926264510e62588f78bc58bdd79d1788c9ef8
Engineering Translation	e232febfaadd820c6ac4557b42bf9b9fc0b894c316c272e87e19b34a6bc44d0

Where a current V3 work already has a standalone concept record, an update should preserve that record and create a new version under it rather than a second unrelated paper. This preliminary Theory of Everything is different: it is the first theory-wide publication of the V3 rebuild and therefore requires a new standalone record, not a version of the deprecated V2 ToE. After that first V3 record exists, later V3 ToE updates should remain on its new concept lineage. Every release package must identify the exact manuscript and PDF, metadata, licence, authorship, repository revision and machine manifest. An updated abstract and headline status must be reconstructed from active evidence rather than copied from an earlier version.

For the preliminary Theory of Everything, publication does not wait for every future scientific question. It reports the current completed foundations, field-wide branches, return families and computational gates exactly. The planned later version will incorporate complete Protein, Chess, Go and Unison results when they lawfully exist, repeat the global synthesis and heavy verification, and preserve this preliminary state as historical chronology.

For Protein, the existing concept was established by version 0.9.3 and the current version 0.9.4 lawfully continues that concept. Additional changes belong in later versions of that paper, not a new standalone concept. The release must not imply that the

active search process has yielded a result absent from its scientific-gate receipt. It may report complete work, active work and reasoned expectations, provided the evidence class is explicit.

The final authority gate belongs to Maria Smith. Automated checks can verify that files render, counts match and hashes reproduce. They cannot decide to alter a scientific claim whose meaning is ambiguous, and they cannot grant remote publication authority to themselves.

24.15 Current returned-claim and synthesis inventory

The following compact inventory makes the later-than-foundation layer visible without repeating full machine receipts. "Independent" in the evidence column means independently reconstructed formal status at the present claim boundary; it does not mean an external experiment has confirmed the claim. "Empirical comparison" means the branch's registered source comparison has been performed with its stated adverse, null and unresolved rows retained.

Biology additions

Claim ID	Current content	Evidence status and limit
SFT-BIO-ORIGIN-AUTOCATALYTIC-IGNITION-002	Exact m - 1 internally supported roles plus one supplied seed for ignition of the complete driven m -role carrier	Independent formal reconstruction; historical terrestrial origin not established
SFT-BIO-HOMOCHIRAL-AMPLIFICATION-002	Held Fold parity with successor amplification produces increasing finite dominance while retaining both labels	Independent formal reconstruction; origin of biological parity and quantitative pathway unresolved
SFT-BIO-SOMATIC-GERMLINE-ORBIT-SPLIT-002	Denominator factorisation separates a finite somatic transient from an odd-denominator recurrent carrier	Independent formal reconstruction; no universal organismal lineage mapping claimed
SFT-BIO-NEURAL-HALF-ONE-THRESHOLD-002	Half-One is the least normalised support for the registered excitable event	Independent formal reconstruction; not a dimensional voltage or universal neural threshold
SFT-BIO-DIFFERENTIATION-LOSS-CANCER-002	Persistent non-terminal cycling plus lost differentiation and escape from division/death/differentiation control defines the structural cancer-state carrier	Independent formal reconstruction; clinical diagnosis remains Medicine-owned
SFT-BIO-BOUNDED-ORBIT-ECOSYSTEM-002	The 3/5 Fold carrier has an exact bounded period-four orbit	Independent formal reconstruction; not a universal ecosystem forecast
SFT-BIO-VALIDATION-PRIOR-MECHANISMS-COMPLETE-FAMILY-002	Seven-source comparison of the complete returned Biology family	Empirical comparison passes categorically; exact organismal mappings and homochirality origin remain open

Medicine additions

Claim ID	Current content	Evidence status and limit
SFT-MED-PLACEBO-EXPECTATION-FIBRE-002	Body-labelled 1/4 and expectation-labelled 3/4 are the two preimages of 1/2 and compose to One	Independent formal reconstruction; fractions are not clinical effect sizes
SFT-MED-PLACEBO-AVAILABLE-STATE-BOUNDARY-002	Expectation may order reachable routes but cannot add a state absent from the biological carrier	Independent formal reconstruction; reachability requires patient- and condition-specific evidence
SFT-MED-PLACEBO-OBJECTIVE-REPORT-SEPARATION-002	Objective physiology and participant report are separate evidence channels with comparator and control	Independent formal reconstruction; neither channel replaces the other
SFT-MED-VALIDATION-PLACEBO-NOCEBO-COMPLETE-FAMILY-002	Nine-source comparison retaining objective modulation and null, small, context-dependent and unresolved rows	Empirical comparison at categorical boundary; no universal magnitude confirmed

Consciousness and Cognitive Science additions

Claim ID	Current content	Evidence status and limit
SFT-CONSC-SYNAESTHESIA-DIRECTIONAL-LOCK-002	A stable directed cross-quality Fold lock need not reverse	Independent formal reconstruction; individual mapping and mechanism remain empirical
SFT-CONSC-NONORDINARY-THREE-QUALITY-ORBIT-002	A held three-quality orbit supports internally generated nonordinary content without adding external information	Independent formal reconstruction; report is not evidence of an unobserved external event
SFT-CONSC-SLEEP-DREAM-PERIOD-TWO-002	Exact 1/3 ↔ 2/3 recurrence supplies the sleep-dream period-two carrier	Independent formal reconstruction; physiological stage labels are post-closure mappings
SFT-CONSC-CESSATION-LOCK-ANCHOR-002	Cessation releases integrated half-One occupation while components and records may remain separately held	Independent formal reconstruction; no continued person or clinical death criterion inferred
SFT-CONSC-VALIDATION-NONORDINARY-COMPLETE-FAMILY-002	Twenty-seven-source comparison of the four returned structures	Empirical comparison passes categorically; heterogeneity, null and unresolved rows retained

Earth and Environmental Sciences addition

Claim ID	Current content	Evidence status and limit
SFT-EARTH-TIPPING-FOLD-LOCK-002	1/4 and 3/4 are exact normalised basin preimages of the common 1/2 boundary	Independent formal reconstruction; dimensional thresholds remain system-specific measurements

Astronomy and Cosmology additions

Claim ID	Current content	Evidence status and limit
SFT-ASTRO-SOLAR-RADIO-UNIT-RELEASE-002	Unit magnitude-frequency release gives an inverse binary share at its exact carrier	Independent formal reconstruction; observed exponents are heterogeneous
SFT-ASTRO-ATOMIC-BURST-COMPLETION-002	One Fold closes the registered atomic burst carrier	Independent formal reconstruction; dimensional atomic process mapping remains source-bound
SFT-ASTRO-PLANETARY-BINARY-LADDER-002	A depth-seven binary ladder runs from 1/128 to One	Independent formal reconstruction; binary-like spacing is not universal
SFT-ASTRO-LITHIUM-SEVEN-ONE-FOLD-DEPLETION-002	3/16 maps by one Fold to 3/32 , with doubling restoring the share	Independent formal reconstruction; direction retained but factor-of-few and mechanism uncertainty remain
SFT-ASTRO-VALIDATION-PRIOR-COMplete-FAMILY-002	Twenty-five-source Earth/Astronomy comparison of the returned structures	Empirical comparison passes categorically with mixed quantitative status; protected adverse records unchanged

Social and Collective Systems additions

Claim ID	Current content	Evidence status and limit
SFT-SOCIAL-PREFERENTIAL-FLOW-INEQUALITY-002	Preferential flow produces exact complementary unequal shares conserving One	Independent formal reconstruction; no universal empirical distribution claimed
SFT-SOCIAL-CONSENSUS-POLARIZATION-LOCK-002	A common half-image cannot distinguish retained polarised preimages	Independent formal reconstruction; aggregate balance is not consensus
SFT-SOCIAL-DISSIPATIVE-PERIOD-TWO-CYCLE-002	1/3 ↔ 2/3 supplies an exact collective period-two carrier that never rests at balance	Independent formal reconstruction; not a universal cycle duration or forecast
SFT-SOCIAL-VALIDATION-EXACT-COMplete-FAMILY-002	Twenty-one-source comparison of inequality, polarisation and recurrence	Empirical comparison passes categorically; institutional dependence and unresolved causation retained

Engineering Translation additions

Claim ID	Current content	Evidence status and limit
SFT-ENG-TESLA-RESONANT-TRANSFER-PROTOCOL-002	Sealed calibrated connected resonant-transfer path with phase, load, complete power ledger and adverse controls	Independent formal protocol; experiment not run
SFT-ENG-VACUUM-INERTIA-RESPONSE-PROTOCOL-002	Driven response with paired channels, confound controls, restoration and source accounting	Independent formal protocol; experiment not run
SFT-ENG-VACUUM-BEAT-RESTORATION-PROTOCOL-002	Outward 1/6 structure separated from returned-cycle restoration, with halt on open net-gain ledger	Independent formal protocol; no measured excess reported
SFT-ENG-SECTOR-FIVE-SEVEN-DETECTION-PROTOCOL-002	Blind p -fibre detection preserving backgrounds, systematics, nulls and outside-list results	Independent formal protocol; prediction is not discovery
SFT-ENG-SMITHIUM-SYNTHESIS-IDENTIFICATION-PROTOCOL-002	Joint 126/184/310 structure with apparatus, yield, decay, mass, ion, oxidation and spectroscopy custody	Independent formal protocol; synthesis and official discovery not claimed
SFT-ENG-NOVEL-TRANSLATION-S-COMplete-FAMILY-002	Completeness record for the five physical and chemical protocol families	Independent formal reconstruction; no empirical promotion
SFT-ENG-CONSCIOUSNESS-PLACEBO-CROSS-BINDING-PROTOCOL-002	Human protocol separating report, physiology, modality, expectation, blinding, reachability and ethics	Independent formal protocol; experiment not run
SFT-ENG-NOVEL-TRANSLATION-S-NO-OMISSION-ADDENDUM-002	No-omission record for all six experimental obligations	Independent formal reconstruction; verifies protocol inventory only

Cross-Branch Synthesis claims

Claim ID	Typed synthesis	Current limit
SFT-SYNTH-PRIME-VACUUM-ORBIT-IDENTITY-001	Prime-order and live-vacuum period identity	Formal typed identity; no empirical vacuum observation added
SFT-SYNTH-COMMON-LOCK-IDENTITY-001	Common Fold lock across distinct owners	Carriers and meanings remain distinct
SFT-SYNTH-COMMON-DESCENT-IDENTITY-001	Folding, fixation and optimisation descent identity	Biological, dynamical and engineering criteria remain owner-specific
SFT-SYNTH-WAVE-MODE-RECURRENT-IDENTITY-001	Longitudinal/transverse common recurrence	Media, speed, dimension and observation remain distinct
SFT-SYNTH-FOLD-SECOND-HARMONIC-IDENTITY-001	Fold doubling as second-harmonic operation	Formal operation; no dimensional frequency imported
SFT-SYNTH-VACUUM-PERIOD-DIVISOR-PREDICTION-001	Prime-denominator period-divisor prediction	Standing formal prediction, not empirical confirmation
SFT-SYNTH-POSITIVE-OBSERVABLE-ABSENCE-BOUNDARY-001	Positive observable distinguished from structural absence	Missing or failed record cannot become an observed absence
SFT-SYNTH-TESLA-CORRESPONDENCE-ASSEMBLY-001	Typed Physics-Earth-Engineering Tesla composition	Earth modes remain distinct from quarter-wave sequence
SFT-SYNTH-UNIFIED-CONSTANTS-ASSEMBLY-001	Cross-branch assembly of the Physics-owned Unified Constants Object	No new free parameter or transferred ownership
SFT-SYNTH-PREDICTION-FALSIFICATION-LEDGER-001	Complete premeasurement chronology and decision ledger	Later correspondence cannot be backdated into prediction
SFT-SYNTH-ONE-OWNER-NO-OMISSION-LEDGER-001	One categorical owner and no omitted inherited claim	Governance closure at frozen input graph
SFT-SYNTH-ROOT-TRACED-TERMINAL-ASSEMBLY-001	One-root terminal assembly of the frozen synthesis graph	Applies to 29 July frozen input, not automatically to expanded 2,751 claims

24.16 Current completion gates in plain scientific language

Biology

The paper may say that the 75-law foundation and seven returned claims are complete and that all 82 current claims are root-traced. It may also say that the returned categorical source comparison passed. It must say that the full-field census stands at **82/424**, with **342** obligations open. It must not call the whole of Biology field-wide complete. Protein's conceptual family is admitted, while Protein Fold's global computational and blind empirical work remains open.

Medicine

The paper may say that the 72-law clinical foundation and four placebo/nocebo returns are complete. It may report the source-observed physiological and report distinctions. It must preserve the null, small and context-dependent rows and state that exact fractions are structural. It must not claim a frozen full-field census, a universal placebo magnitude or a patient treatment effect from the formal fibre alone.

Consciousness and Cognitive Science

The paper may say that the 72-law foundation and five returns are complete at their current evidence boundary. It may present interior observation, phenomenal privacy, self-model, memory, red-of-red and returned orbit structures as exact derivations. It must not report access, neural correlation, fluent AI output or a Fold coordinate as direct measurement of phenomenal presence. The full-field and artificial-realisation programmes remain open.

Earth and Environmental Sciences

The paper may say that the 74-law foundation and one returned tipping lock are complete. It may report exact exponent 1 as the preregistered earthquake prediction. It must state the first mixed-catalogue comparison as adverse and the later homogeneous holdout as compatible, preserving both intervals and chronology. A universal measured tipping value is not admitted, and the full-field census remains open.

Astronomy and Cosmology

The paper may say that the 72-law foundation and five returns are complete. It may report exact rank four as the pre-data structural prediction and the later source method context. It must report **3.486344605219551...** as the adverse first fit and avoid reclassifying it. It must preserve heterogeneous exponents, non-universal spacing, lithium mismatch and mechanism uncertainty. The field-wide census remains open.

Social and Collective Systems

The paper may say that the 72 foundation claims and four returned claims are complete. It may report exact preferential shares, polarisation lock and period-two carrier. It must not convert those structures into universal distributions, social thresholds or forecasts. Consensus remains non-evidential for truth, prestige non-evidential for proof and ethics explicitly bridged. The full-field census remains open.

Engineering Translation

The paper may say that the 72-law foundation and eight protocol claims are formally complete and independently reconstructed. It must state that the returned experiments have not been run. Formal protocol closure is not implementation, observation, discovery or validation. The full-field census, apparatus execution and intended-use validations remain open.

Cross-Branch Synthesis and the final Theory of Everything

The paper may say that the twelve-claim inherited-return synthesis is complete against its frozen input graph. It must state that the graph pre-dates the current 2,751-claim corpus. A final one-owner merge, publication reconciliation, terminal synthesis and heavy verification remain. The last heavy run's **1,319/1,319** replays and **969/969** tests cannot be silently relabelled a heavy run of all 2,751 current claims.

Computational programmes

The paper may report Protein's completed parent census, completed high slice, scientific gate v20 **893/893** tests and publication full suite v21 **898/898** tests. It must report every remaining search gate and zero protected accesses. It may report V2 Chess and Go results only as V2 historical results. It must state that their V3 clean recreations are not started. It may report Unison V2 certificate strength and weak generation evidence together. It must preserve V3 `OPEN`, `NOT_RUN` and `NOT_AUTHORIZED` exactly.

These completion statements are the controls on the persuasive voice of the monograph. They allow the work to state its real scale and novelty directly while ensuring that a reader can distinguish what is derived, what is built, what is observed and what the next version is intended to test.

24.17 Worked dependency sequences

24.17.1 From chemical sequence to a blind protein prediction

The protein programme illustrates why the dependency spine is more than a contents list. Chemistry first owns amino-acid and bond identity. Materials Science owns collective structure-property and interface relations. Biology then owns the ordered protein sequence, condition-bounded fold, ensemble, interaction, assembly and function. Computation owns exact generation, state identity, pruning and resource accounting. Engineering owns the executable pipeline, test suite, target custody and public comparison. Only after every handoff is complete can an empirical structural prediction be issued.

Suppose an implementation generates a compact candidate with favourable local interactions. Chemistry can verify that its residue and interaction labels are well formed. Biology can verify that it belongs to the registered folding carrier. Computation can verify that it survived the exact grammar. Engineering can verify that the code returned it under the sealed command. None of those facts establishes agreement with an external structure. That requires target opening, alignment and a registered whole-chain metric.

Conversely, suppose the blind comparison is adverse. The target record first shows that the empirical output failed its criterion. The state and execution trace then permit the failure to be localised. An omitted interaction is a Biology or translation-grammar problem. An unsound prune is a computational problem. A unit or chain-index error is an engineering implementation problem. A mismatched experimental chain or comparator release is an evidence-custody problem. The adverse row remains whatever the diagnosis; the next version can then correct the identified owner without rewriting the earlier result.

This sequence explains why AlphaFold output access remains forbidden now. If comparator outputs were visible while the remaining attraction bands or global recurrence were being designed, they could select the representation or pruning rule. The programme would still be capable of development and retrospective correspondence, but its eventual output could not honestly carry the intended blind status.

24.17.2 From bodily state to expectation, report and clinical decision

The placebo/nocebo programme begins with a living body's reachable states. Biology owns the physiological carrier and its constraints. Consciousness owns expectation as an interior and cognitive relation and owns the difference between experience and public report. Medicine owns the patient, comparator, outcome, adverse event and decision. Engineering owns the instrument and trial system. Social Science owns institutional expectation, communication and access where they affect the setting.

The exact 1/4, 3/4 and 1/2 fibre organises the possibility that distinct retained routes share a common observed image. It does not tell a clinician that three-quarters of an observed change was psychological. To make that claim, a study would need an identifiable physiological measure, expectation manipulation, realised comparator, blinding, timing, report and appropriate causal analysis. The available-state boundary additionally asks whether the response was physiologically reachable without the target treatment.

If a participant reports improvement while the registered objective measure does not change, the report is not deleted and the physiological channel is not fabricated. If physiology changes without a reported experience, the objective row stands without being renamed a phenomenal effect. If both change, their association and chronology can be tested. If neither changes under an adequate protocol, that is a null result at the registered outcome. If an instrument fails, the result is unresolved rather than null.

This sequence protects patient evidence from two opposite reductions. It prevents first-person experience from being dismissed as unreal because it is not the instrument value, and it prevents subjective report from being used as proof of any proposed objective mechanism. A future version can strengthen the empirical connection only by preserving both.

24.17.3 From exact exponent to earthquake catalogue evidence

The magnitude-frequency sequence begins in Mathematics with exact ratio and order, continues through Physics with measured quantity and uncertainty, and enters Earth Science through rupture, propagation, magnitude scale and catalogue population. The exact candidate enumeration selects exponent 1 before the source is opened. The Earth evidence protocol then defines which catalogue rows count above each threshold and how the exceedance interval is calculated.

In the first test, the observed share derived from 1,494 of 18,469 events and the entire registered interval lay below 1/10. The proper result is adverse. The finding raises specific scientific questions about mixed magnitude types, catalogue completeness and population boundary, but those questions do not alter the measurement already made.

The later holdout narrows the source to homogeneous *mww* records and shifts the thresholds. Its 27 of 253 share and interval include 1/10. Because the source and rule were registered separately, this is a lawful compatible result. It suggests that catalogue homogeneity may matter and warrants further prospective testing. It does not establish that the first test was mistaken; the first question and source were different.

A future synthesis could register a hierarchical model of catalogue types or a new multi-source decision rule. It would be a new claim with its own chronology. The current monograph instead preserves the two tests in the order they occurred, which is the only basis on which readers can evaluate the theory's predictive conduct.

24.17.4 From source light to galaxy scaling

The SPARC sequence begins with astronomical source identity, distance, flux, motion and catalogue selection. Luminosity is reconstructed from the received record and distance; rotation is reconstructed from source motion under a measurement and model boundary. The rank-four result is fixed in the formal branch before the external rows are fitted.

An unweighted all-positive-row fit is one operationalisation and produces the adverse **3.486344605219551...** slope. An orthogonal uncertainty-aware method is another and produces **3.85 ± 0.09**, with reported systematic movement to 4.0. These analyses do not differ merely by typographical precision. They weight uncertainty and geometry differently, so the exact method is part of the claim.

The standing evidence therefore supports three simultaneous statements: rank four was a pre-data SFT prediction; the first registered operational test was adverse; and a separate established method places four at its stated systematic boundary. Removing any one of those statements would distort the current record. Future work can improve the test by freezing a physically justified sample and uncertainty model prospectively.

24.17.5 From social evidence to an engineered public platform

The open-science platform is itself an Engineering and Social Science object. Engineering owns repository structure, portable formats, tests, release identity, reproducibility commands and deployment. Social Science owns access, gatekeeping, community interaction, institutional dependence and unequal costs. The scientific branches own the claims and evidence hosted by the platform.

Open access is therefore not achieved merely by placing code on a public URL. A reader must be able to identify the active paper, distinguish it from a superseded draft, locate its claim package, reproduce its counts and understand its licence. A public command must fail visibly. Large assets require stable archival identities. Accessibility includes documentation and platform support, while privacy and consent constrain any participant data.

The anti-institutional mission follows from this operational design. A reader does not need a credential or journal subscription to inspect a candidate census or adverse record. Criticism can be registered against a specific claim. Yet openness does not mean automatic scientific admission: an asserted replacement must satisfy the same dependency, evidence and falsification requirements. The platform challenges monopoly over access and authority without abandoning standards of proof.

24.17.6 From field branches to the preliminary and final ToE

The preliminary monograph assembles current authoritative results in dependency order. It is not preliminary because the existing foundations are informal. They are receipt-backed and published. It is preliminary because the active full-field, computational and terminal-synthesis programmes have not all reached their final registered gates.

The final ToE update requires four types of closure. The domain programme must finish Biology's registered census and freeze the later full-field boundaries selected for completion. The computational programme must produce governed V3 Protein, Chess, Go and Unison records, whether favourable or adverse. The lineage programme must reconcile one owner for every historical and current claim. The machine programme must run terminal synthesis and heavy verification against the final graph.

Those requirements do not make the preliminary paper scientifically empty. They make its version status precise. It already reports a root-traced corpus of 2,751 admitted claims and the current branch results, together with the limits of the verification and application programmes. The future version will add evidence and closure; it will not retroactively create the present chronology.

24.18 Final programme outlook

The immediate priority remains Protein V3 because it is the only clean V3 computational programme with an active large exact search and a protected blind target design. Its next report should state one of three outcomes without embellishment: the remaining exact gates closed; the search was lawfully narrowed with an explicit unresolved remainder; or a registered obstruction showed that the current grammar cannot close. Any of these can be publishable if the machine and scientific records are complete.

Chess and Go follow with clean reconstruction rather than patching historical runtime state into a V3 label. Their V2 records provide strong regression and performance baselines. Reproducing them is necessary but not sufficient: V3 must show its own derivation and implementation identities. Governed external play can begin only after rule and harness closure.

Unison requires the strictest separation between architecture and performance. The V2 system demonstrates an unusually large auditable certificate surface, while its generation scores show that auditability alone did not yield useful general language behaviour. V3's opportunity is to preserve exact internal custody while discovering whether a native Fold architecture can learn, reason and generate under public tests. The answer remains unknown until the mathematics closes and evaluation runs.

The programme should publish adverse results as readily as favourable ones. An exact theory becomes more persuasive when the reader can see where an implementation failed, which assumption changed and which result survived. The current Protein whole-chain, earthquake and SPARC records already show that this is possible. Their continued visibility is part of the evidence that the platform does not select only its successes.

The ultimate computational contribution will therefore be more than four headline demonstrations. It will be a reusable public method for moving from premise and dependency, through complete candidate accounting, to executable artefact, protected comparison and versioned correction. Protein, Chess, Go and Unison occupy different domains precisely so that a success in one cannot hide a weakness in another. Together they test the generality of the Fold as a scientific programme.

24.19 Registered unresolved work and discriminating next tests

Open work becomes scientifically useful when its next observation can distinguish alternatives. The following register states the main unresolved questions in this part and the kind of evidence that would materially change their status.

Owner	Unresolved question	Discriminating next evidence
Biology	Does the autocatalytic ignition carrier map quantitatively onto a plausible prebiotic network?	Prospective chemistry with registered roles, seed, drive, compartment, product custody and competing non-autocatalytic routes
Biology	Which physical route fixed and amplified terrestrial homochirality?	Source-dated stereochemical experiment or natural record separating initial bias, amplification and racemisation
Biology/Protein	Can the six-axis Fold relation recover complete structures from sequence alone?	Closed global recurrence followed by protected whole-chain comparison on the complete target manifest
Medicine	When does expectation measurably alter objective physiology and patient benefit?	Registered expectation manipulation with realised comparator, blinding, report, objective channel, adverse events and follow-up
Consciousness	Which physical organisations satisfy the bidirectional realisation criterion?	Causal intervention study preserving state roles and return relations, not output correlation alone
Earth	Is exponent 1 stable across prospectively homogeneous seismic populations?	New catalogue holdouts with frozen magnitude type, completeness threshold, region/time boundary and interval rule
Earth	Where do particular coupled systems tip?	System-specific state variables, path, hysteresis and prospective threshold observations in dimensional units
Astronomy	Does rank four survive a prospectively frozen galaxy sample and uncertainty model?	Pre-sealed SPARC or independent-source analysis with selection, distances, variables, covariance and fitting rule fixed
Astronomy	Which returned binary and lithium structures have exact quantitative counterparts?	Source-specific forecasts with dimensional mapping and decision intervals registered before release
Social	When do preferential flow and polarised common images causally govern collective outcomes?	Cross-context interventions or natural experiments retaining institutions, networks, dissent and reversals
Engineering	Do the six returned protocols yield their registered physical or human effects?	Built, calibrated apparatus and blinded execution preserving every null, adverse, transport and restoration row
Synthesis	Does the one-owner terminal graph close after all 2,751 current claims are included?	New global lineage merge, synthesis build and heavy verification at the release freeze
Chess	Does a clean V3 Fold engine retain or improve V2 play?	Rule-complete recreation and governed matches with fixed opponents, time control, openings and adjudication
Go	Does a clean V3 Fold engine reproduce exact censuses and complete external games?	Independent census replay and unbroken harness runs under frozen rules and scoring
Unison	Can a native V3 Fold architecture generate useful answers without borrowed answer weights?	Mathematically closed implementation followed by preregistered factual, reasoning and generation suites

These tests have different possible conclusions. A quantitative miss can be adverse even when the categorical carrier survives. A source may be compatible without uniquely supporting SFT. A failed apparatus run may be unresolved rather than null. A new successful implementation may validate an application without confirming the universal law it uses. Each next protocol must state in advance which conclusion follows from which outcome.

The tests also have different protection requirements. Protein and benchmark programmes require target custody. Human studies require consent, privacy, blinding and adverse-event care. Environmental and astronomical studies require release and selection custody. Engineering anomaly studies require calibration, restoration and complete source ledgers. Social interventions require affected-agent and ethical boundaries. One generic "validation" label cannot replace these constitutions.

24.20 Specialist reading and reproducibility routes

A domain specialist need not begin with machine hashes. The conceptual route is the relevant branch chapter followed by its dependency-resolved family ledger and returned-claim inventory. The specialist can then use the active foundation DOI to inspect the complete paper, the scientific audit volume to locate claim IDs and evidence status, and the machine package to replay the candidate census and controls.

A reviewer concerned with an adverse result should begin with its chronology and measurement receipt, then inspect the source manifest, decision rule and later evidence as separate records. A reviewer concerned with formal uniqueness should begin with the candidate grammar, verify the 256-form census and inspect independent reconstruction. A reviewer concerned with a computational result should inspect the build identity, test classes, resource ledger, target-access log and per-target outputs rather than relying on the headline score.

The active-paper SHA-256 values identify the precise Markdown authorities used for this part. Full source captures and receipts may be larger or more numerous; their identities remain in the release manifest. A changed paper hash does not by itself indicate a scientific change, because editorial formatting also changes bytes. The version record must therefore state whether scientific content, evidence status or only presentation changed.

Reproducibility is complete only when a fresh environment can reach the same registered result or produce a classified failure. The public route should report dependency versions, commands, elapsed time and output paths. It should never silently download a target that the scientific gate keeps protected, never replace a missing source with an unregistered copy and never alter an authoritative receipt in place.

This layered reading route permits accessibility without dilution. A newcomer can follow the dependency argument; a specialist can challenge the scientific boundary; an auditor can reconstruct the evidence; and a developer can replay the machine process. All four encounter the same current status rather than different stories written for different audiences.

24.21 Authority identities used for this part

This part was reconstructed from the current repository rather than from an earlier narrative summary. The publication rules were read from `publication_guidance.md`, SHA-256

5ea6b5b558671b0144a553c1b8d20ca32c30b4e01a7d840f6e17248f085405cc. The active-paper, claim-count and programme map came from `AUTHORITATIVE_CORPUS_INVENTORY.json`, SHA-256 382ad55bd6e26930d9b2c49fe44b35bf97c95859fdac49cb331b5254adb7dac3, and `EXHAUSTIVE_TOE_CONTENT_MATRIX.json`, SHA-256 8a7c13c51a1e13d3527eab827d1f1be76b3ecd0103a157ad4b120c2995a2993f.

The current global completion and heavy-verification boundary came from

`CURRENT_PROGRAMME_STATUS_2026-07-29.md`, SHA-256

a5abf1a17e1ca8bc3dea524b85f1dad90829e6a0628027f7e4c3723e43ca77fa. Protein's active scientific gate came from `current_scientific_gate_v20.json`, SHA-256

b9483b91c07d0ae2e70125d48dac11fa5a3a489202e894886db3c0f0e4ac6221. Unison V3's status came from its `workspace_manifest.json`, SHA-256

9a6811ab7bbbd2c43dff5149aba9bb1ca00faedcc77c29857918870f8e26dfb2. These identities authorise the exact OPEN, NOT_RUN, NOT_AUTHORIZED, target-access and search-completion language used above.

The Cross-Branch account was checked against the branch roadmap, SHA-256

b83ded3d7aaf2ac24a69d675a6d6ffe89f653b6ca951b15e569e74696f5e34c1, the prior-identities completion record, SHA-256 6f0023cb0a5aa3c25f27b2107927049cddb07ab020afb7f4ab4edd3e2935dded, the whole-subcategory boundary, SHA-256

9dfefb55c9e60a8aac36502f9b27243c7c2269a781f211e78bf44e3ac46eb38b, and the frozen prior input, SHA-256 dad7434bcb576fb37ea4f9c66848e4969b48121da291698faf0e3976b73607ec. Together they establish both the twelve-claim completion and the reason it cannot serve as the terminal assembly of the later 2,751-claim graph.

The seven active branch-paper hashes are recorded in §24.14. Their complete claim packages, evidence manifests and audit volumes were used to recover the family order, exact current counts, source-transport histories and correction boundaries. This authority note is a reproducibility index, not a substitute for those scientific records. If any listed identity changes before release, the affected counts and status language must be reconstructed and this part must receive a new manuscript hash.

The active human-readable authorities were selected by the inventory rather than by filename recency alone. Biology is output/release/biology-1.0.0/02_From-Fold-to-Life_Biology-and-Life-Sciences-Foundation-Paper-001-v1.0.md; Medicine is output/release/medicine-1.0.0/02_From-Fold-to-Medicine_Medicine-and-Health-Sciences-Foundation-Paper-001-v1.0.md; Consciousness is output/release/consciousness-cognitive-science-1.0.0/02_From-Fold-to-Consciousness_Consciousness-and-Cognitive-Science-Foundation-Paper-001-v1.0.md; Earth is output/release/earth-environment-1.0.0/02_From-One-World-to-Earth_Earth-and-Environmental-Sciences-Foundation-Paper-001-v1.0.md; Astronomy is output/release/astronomy-cosmology-1.0.0/02_From-One-Sky-to-Cosmos_Astronomy-and-Cosmology-Foundation-Paper-001-v1.0.md; Social is output/release/social-collective-sciences-1.0.0/02_From-One-Relation-to-Society_Social-Collective-Sciences-Foundation-Paper-001-v1.0.md; and Engineering is output/release/engineering-translation-1.0.0/02_From-One-Law-to-a-Working-World_Engineering-Translation-Foundation-Paper-001-v1.0.md.

Those files preserve the published foundation chronology. The later claim packages and exhaustive audit volumes authorise the larger current totals. Neither layer supersedes the other in every respect: the foundation paper is the publication authority for its original version, while the current package is the authority for a returned claim and its present evidence status. This monograph reports both and labels the transition.

Historical Chess, Go, Protein and Unison results were checked through the programme records referenced by the authoritative inventory. Their record- manifest SHA-256 identities are, respectively,

4d161dcf64981a4360f09eeebdef11abba724b31d473dc02de7f8f6f8721defe,
382d0bd935ef013987e45f584506516638149202a92d9e067fb04b3571b8b764,
26a7efdb5ebe1930c12e0a2084288d52276efd71845bb393cf9809e6d64024c4 and
e7e99d024be9321c00e24912bec199f9f88709e85fc27ae3afa7e9104551d09b. The complete current Protein
and Unison identities are printed above because they directly authorise present V3 status.

At final release freeze, the build must recompute every full hash rather than copy these displays, identify any file whose bytes changed, and decide from the diff whether the change is editorial, scientific or evidential. Scientific or evidential changes require a new status audit. Editorial changes still require a new release-manifest identity but do not by themselves change claim status.

Part IV. Evidence reconciliation, limitations, reproducibility and conclusion

25. Corpus-wide evidence reconciliation

The theory-wide result is not a sentence assembled from the strongest result in each branch. It is the complete retained state of every branch at the release freeze. The global audit therefore carries formal, computational, empirical, historical and publication status as separate coordinates.

At the current claim boundary the reconciliation is exact:

- 2,751 registered claims are model-admitted;
- 892,246 generated candidates are retained in the aggregate census;
- 2,751 unique survivors are retained;
- 11,004 of 11,004 registered controls pass;
- 2,751 candidate-census identities reconstruct;
- 2,751 claim receipt identities match;
- 2,751 admitted claims trace to the single root; and
- no declared dependency is missing from the current claim ledger.

These results establish current claim-package closure. They do not erase the separate final-ToE obligations recorded by the programme ledger. In particular, the all-branch heavy verification has not been rerun since the corpus expanded beyond the 1,319-claim all-foundations state, the canonical one-owner lineage merge remains older than the current graph, and several branches remain complete only at their foundational rather than field-wide boundaries.

25.1 Formal reconciliation

Formal reconciliation asks whether each admitted statement has exactly one owner, a complete dependency route, a registered candidate grammar, the declared number of decisions, the required survivor state, controls and a receipt matching the accepted package. The complete claim inventory binds these fields compactly. The branch audit volumes preserve the complete record without forcing thousands of repeated receipt blocks into the conceptual narrative.

An exact formal result remains limited by its grammar. The global total does not convert every depth-independent branch statement into a theorem about an unregistered extension. Where the claim is a bounded census, the bound is retained. Where it is a correspondence, the correspondence class is retained. Where an external observation is required, the formal survivor does not replace that observation.

25.2 Computational reconciliation

Computational reconciliation asks whether the registered source reconstructs the stated census, decisions and receipt; whether an implementation-distinct checker agrees; whether controls fail closed; and whether the public archive identifies the producing source and artefacts. Repository tests are evidence for those implementation questions. They do not manufacture empirical confirmation.

The current lightweight repository validator passes at the 2,751-claim boundary. The canonical engine seal verifies sixteen runtime files, and the verification-authority seal verifies thirty-three protected files. The last complete heavy run at the earlier all-foundations boundary replayed 1,319 of 1,319 derivations and passed 969 of 969 tests. This paper does not silently promote that earlier heavy run into a 2,751-claim heavy verification. The final global run remains an explicit open operation.

25.3 Empirical reconciliation

Empirical reconciliation preserves the method and chronology attached to each external result. A current result may be:

- independently replicated;
- empirically tested and independently replicated;
- a sealed prediction awaiting measurement;

- a compatible but non-discriminating observation;
- a post-target reconstruction;
- a development-observed result;
- adverse;
- null;
- unavailable because transport or access failed; or
- unresolved because the required record is incomplete or has not been run.

The same words are not used interchangeably. A claim may possess exact formal closure and an unresolved empirical extension. A branch may have field-wide closure at its current-evidence boundary while a proposed later application remains open. A historical application may show measured success while its clean V3 recreation has not started.

25.4 Publication reconciliation

Publication is a separate state. The current V3 paper set contains sixteen authoritative published papers from Methods through Engineering Translation. The seven field-wide papers from Mathematics through Materials were published as new versions on their existing concept DOI lineages on 29 July 2026. Their public files and lineages were checked against the approved local release identities. The Protein Fold preliminary record was published separately on 31 July 2026; its current v0.9.4 successor remains an application-level preliminary publication.

This ToE version is authorised for first publication as a new standalone V3 record after a frozen package and final local verification. Zenodo draft 21717584 reserves version DOI 10.5281/zenodo.21717584 and concept record

21717583. The deprecated V2 ToE concept remains historical and does not receive this V3 rebuild as another version.

26. Corrections, adverse results and historical custody

26.1 Correction is part of the result

V3 treats a correction as a relationship between records, not as permission to rewrite history. The corrected record must identify the earlier artefact, the source of the mismatch, the new authoritative direction and the scope of the change. If the correction affects only a file identity or clerical reference, the record must say that no scientific count or evidence class changed. If it changes a grammar, output or empirical conclusion, both the earlier result and the lawful successor status remain visible.

The clean-room reconstruction itself is a corpus-scale reconciliation. Earlier SFT versions remain historical provenance but are not proof premises for V3. Their observations and applications can be compared after the corresponding V3 result seals. Agreement is not assumed, and disagreement is not removed to make the history appear linear.

26.2 Protein-folding chronology

Protein folding provides the clearest current example. The V2 lineage retained strong local three-residue geometry correspondences while its sealed 76-residue whole-chain ubiquitin construction was adverse at the global structure boundary, with TM-score 0.0269927379497572 and C-alpha dRMSD 52.89314678065376 Å. Those exact measurements remain historical implementation evidence. They are neither erased by the local successes nor treated as the result of the new V3 six-axis system.

V3 then rebuilt representation, coordinate and interaction custody. A historical 4APD whole-chain control exposed an inter-residue carbonyl-frame defect. The corrected constructor preserved the complete defective record, reconstructed every unordered atom pair and shared-atom correlation, and removed the sub-angstrom collision. The correction did not improve the C-alpha structure score: the retained control remains adverse. Later sequence-only and partial-recurrence development tests contain both improvement and regression. Their exact target chronology remains visible, and none is silently relabelled as a primary blind test of the incomplete full V3 system.

The stronger work supersedes specific implementation obstacles only at the boundaries it closes. It does not turn the historical adverse rows into confirmation. The registered AlphaFold-generalised campaign remains future evidence until the capability gate permits target access and all matched rows have been evaluated.

26.3 Historical application results

The V2 Chess Fold, Go Fold and Unison Fold AI records are retained with their producing-source identities, measurement protocols and known defects. Chess retains its measured match tallies and exact legality controls. Go retains its exact small-board censuses, GNU Go results and the recovered KataGo harness record, including the disclosed round-one synchronisation defect and point-at-cutoff status. Unison Fold AI retains successful verification surfaces and adverse development benchmarks in the same archive.

These records motivate and constrain the clean V3 recreations. They do not authorise a statement that a recreation exists when its workspace is absent, or that an empirical campaign has been run when its registered status is `NOT_RUN`.

26.4 Field evidence

The branch papers preserve their own adverse, corrected, failed-transport and unresolved rows. Examples include adverse astronomical ranking comparisons, an adverse first global mixed-magnitude Earth catalogue, unavailable or superseded source transports, bounded mathematical correspondences that do not establish an unbounded theorem, and corrected chemical and materials evidence routes. Their branch chapters state the exact outcome and lawful successor status. The global paper does not compress them into a claim that all external tests were favourable.

27. Limitations and open frontier

27.1 Meaning of completion

Every use of *complete* in this monograph names its object and boundary.

- **Formal closure** means that the registered derivation and candidate grammar close.
- **Implementation closure** means that the executable construction and its declared checks close.
- **Foundational branch closure** means that the registered foundation of a branch closes.
- **Field-wide branch closure** means that the frozen current-evidence discipline census closes.
- **Empirical validation** means that the named evidence protocol has been run and met.
- **Publication completion** means that Maria Smith has approved and released the bound artefact.

None means the end of science. A frozen current inventory can be complete while new measurements, discovered phenomena, improved source captures and lawful extensions remain possible.

27.2 Open branch work

The present open branch programme is explicit:

1. Biology continues from 82 closed obligations within its frozen 424-obligation full-field census.
2. Medicine, Consciousness and Cognitive Science, Earth and Environmental Sciences, Astronomy and Cosmology, Social and Collective Systems, and Engineering Translation retain complete foundations and current return families, but their complete full-field censuses are not yet uniformly frozen and closed.
3. The current Cross-Branch Synthesis closes its twelve-claim inherited-return family, while the final current-graph one-owner merge and terminal global reconciliation remain open.
4. The final global heavy verification must be rerun on the release-frozen complete graph rather than inferred from the earlier 1,319-claim run.

27.3 Open computational-proof work

Protein Fold V3 must complete and independently verify its remaining exact fixed-geometry bands, reconcile the global frontier, compose the complete torsion-protonation-solvent recurrence, pass registered development and adverse controls, freeze all target and comparator infrastructure, preregister the primary 100-target panel and then run the matched AlphaFold campaign. Parity is decided only from the full retained result table.

Chess Fold V3 and Go Fold V3 require clean workspaces, V2 provenance reconciliation, target-free law reconstruction, exact legality or state custody, controlled development, preregistration and new measured campaigns. Their historical V2 victories remain evidence but not substitutes for V3 completion.

Unison Fold AI requires mathematical closure of its V3 architecture, executable implementation, development controls, preregistered empirical validation, adverse-case retention and Maria Smith's separate deployment authority. The present scaffold is not a validated model.

27.4 Source and transport limitations

Some scientific evidence assets exceed GitHub's ordinary 100 MB object limit. Their omission from normal Git storage is declared, and their byte identities and archival routes remain in the relevant manifests. A later large-file distribution may improve access without changing the claims. Failed source access and unavailable transport are retained as transport states, not converted into absence or null evidence.

The release must also be self-contained with respect to its publication authority. Historical ToE text stored outside this repository may be cited by published DOI or imported as a checksum-bound archival input; it cannot remain an unbound local prerequisite of the public package.

27.5 Editorial and interpretive limitations

The monograph spans formal science, natural science, computation, biological and social systems, and engineering. A single narrative cannot replace the specialist detail retained in each complete branch paper and audit package. The three-layer design is therefore part of the scientific method: conceptual argument here, complete audit in the branch volumes and complete execution in the machine archive.

Readers should not infer that an unfamiliar vocabulary indicates an imported conventional object, or that a familiar word retains every conventional assumption. Each SFT object is defined by its carrier, dependency route and relation. Conversely, a structural correspondence to a conventional object is not automatically numerical identity or empirical confirmation.

28. Reproducibility

28.1 Release artefacts

The publication candidate contains three explicit levels.

1. **Conceptual monograph:** the complete dependency-ordered paper, rendered PDF and source Markdown.
2. **Scientific audit:** the complete 2,751-row claim inventory, seventeen branch audit volumes, source and evidence status, chronology, correction and publication tables.
3. **Machine archive:** the frozen corpus inventory, content matrix, claim packages, receipts, candidate censuses, controls, certificates, source identities, tests and release manifest.

The release manifest records the size and SHA-256 identity of every uploaded file. The immutable Git revision binds the repository-resident machine archive. Files excluded from ordinary Git because of size retain their separate archival identity and are not implied to be part of the Git commit.

28.2 Reproduction sequence

A verifier begins from the release manifest rather than from prose:

1. verify every uploaded file identity;
2. verify the authoritative active-paper identities;
3. reconstruct the global claim ledger and branch totals;
4. verify all candidate-census and receipt identities;
5. verify one root trace and no missing dependency for every claim;
6. run the current lightweight corpus verifier;
7. run branch-specific verification commands recorded in the branch packages;

8. compare the generated paper, claim inventory and audit volumes against the frozen inputs; and
9. render the paper and inspect every page for layout defects.

The final approval dossier records commands, environment, exit status, counts, hashes, page totals and any skipped or unavailable heavy operation. A missing dependency, stale hash, malformed table, unresolved placeholder or scientifically material discrepancy halts publication rather than being silently repaired in prose.

28.3 Independent criticism

Reproduction is not restricted to the included verifier. Anyone may construct an independent implementation, challenge candidate completeness, identify an unregistered alternative, dispute a source classification, rerun an empirical comparison or expose a defect. A valid challenge enters through a public record and receives the same source, control and chronology custody as a favourable result.

29. Data and code availability

The current repository is:

<https://github.com/MettaMazza/ernos-labs-sft-platform>

Papers and documentation are licensed under CC BY 4.0. Repository code is licensed under Apache-2.0. Claim packages, receipts, tests, source manifests, publication records and reproducibility instructions accompany the repository. The release manifest identifies any evidence asset distributed separately due to ordinary GitHub object limits.

The release package records the current repository HEAD separately from the SHA-256 identity of every release file; uncommitted files are therefore not misrepresented as Git-resident objects. Maria Smith authorised the standalone V3 publication and matching `main`-branch Git operation. Zenodo draft 21717584 reserves the first version DOI, and publication creates the new V3 concept lineage. Final transaction identities may be added mechanically without changing the scientific text or its frozen evidence state.

30. Conclusion

Smithian Fold Theory V3 now exists as one public dependency system rather than as a collection of disconnected claims. It begins from the operational impossibility of presenting nothing as a counterexample, derives the structural One and the minimal Fold, and develops exact mathematical, informational and computational structures before approaching physical measurement. Physics reconstructs quantities, fields, matter, constants and their evidence interfaces; Chemistry reconstructs identity, composition, bonding and reaction; Materials reconstructs collective structure and property. Biology and the later branches then receive these objects without being permitted to alter their ownership.

The present corpus contains 2,751 admitted claims generated from 892,246 candidates, with one retained survivor per claim, 11,004 passed controls, matched receipts and complete root traces. Seven branches from Mathematics through Materials close their frozen current-evidence full-field scopes. Biology and the later natural, cognitive, collective and engineering branches retain completed foundations and returned families while their wider full-field programmes remain explicit. Cross-Branch Synthesis closes the current twelve-claim inherited-return surface without pretending to own the primitive laws it composes.

The major conflation resolved by the V3 structure is the conflation of a law with its measurement, implementation, reputation or later application. A formal survivor is not an observation. A successful test is not automatically a derivation. A conventional equation is not an SFT premise. A measured decimal is not permitted to select the exact law it later compares with. A historical program result is not silently promoted into the clean V3 system. These separations permit strong claims where the exact record warrants them and equally visible adverse or unresolved status where it does not.

The computational-proof programme is the next empirical edge. Protein Fold V3 has already advanced from local historical results to exact whole-system representation, interaction custody, certified search and progressively larger closed physical boundaries. Its remaining frontier and AlphaFold campaign are declared before completion. Chess Fold, Go Fold and Unison Fold AI retain their earlier measured histories while the V3 recreations are required to earn new status independently.

The open work is therefore named rather than hidden: Biology continuation, the downstream full-field censuses, the final one-owner lineage merge, the release-frozen heavy verification and the four computational-proof programmes. Their

dependency positions and present evidence states are preserved so that a later result can extend this edition without rewriting its chronology.

This preliminary edition is therefore neither a short announcement nor the claim that inquiry has ended. It is the full current scientific object, presented at the scale required to expose its dependency structure and audit surface. Later versions will add the remaining field-wide and computational results, corrections and new evidence while preserving the history reported here. Completion means completion of the registered project investigation and release boundary-not the end of science.

References and authoritative records

The complete human-readable reference lists remain in the branch chapters and their source registries. The following corpus-level records govern this edition:

1. Smith, Maria. *There Is No Nothing: A Premise-Free Operational Foundation and an Open Verification Platform for Smithian Fold Theory*. Version 0.3.0. DOI: 10.5281/zenodo.21627646.
2. Smith, Maria. *From Nothing to Fold: A Premise-Free, Parameter-Free and Machine-Closed Foundation for Smithian Fold Theory*. Version 1.3.0. DOI: 10.5281/zenodo.21627656.
3. Smith, Maria. *From Fold to Mathematics*. Version 1.5.0. DOI: 10.5281/zenodo.21688766.
4. Smith, Maria. *From Distinction to Information*. Version 1.4.0. DOI: 10.5281/zenodo.21688817.
5. Smith, Maria. *After Turing: The Fold Machine*. Version 1.4.0. DOI: 10.5281/zenodo.21688837.
6. Smith, Maria. *The Quantum Fold Machine*. Version 1.4.0. DOI: 10.5281/zenodo.21688860.
7. Smith, Maria. *From Fold to Physics*. Version 1.3.0. DOI: 10.5281/zenodo.21688879.
8. Smith, Maria. *From Fold to Chemistry*. Version 1.3.0. DOI: 10.5281/zenodo.21688899.
9. Smith, Maria. *From Fold to Materials*. Version 1.3.0. DOI: 10.5281/zenodo.21688923.
10. Smith, Maria. *From Fold to Life*. Version 1.0.0. DOI: 10.5281/zenodo.21630203.
11. Smith, Maria. *From Fold to Medicine*. Version 1.0.0. DOI: 10.5281/zenodo.21630785.
12. Smith, Maria. *From Fold to Consciousness*. Version 1.0.0. DOI: 10.5281/zenodo.21636397.
13. Smith, Maria. *From One World to Earth*. Version 1.0.0. DOI: 10.5281/zenodo.21640810.
14. Smith, Maria. *From One Sky to Cosmos*. Version 1.0.0. DOI: 10.5281/zenodo.21640812.
15. Smith, Maria. *From One Relation to Society*. Version 1.0.0. DOI: 10.5281/zenodo.21640814.
16. Smith, Maria. *From One Law to a Working World*. Version 1.0.0. DOI: 10.5281/zenodo.21640816.
17. Smith, Maria. *From Sequence to an Auditable Fold: Smithian Fold Theory V3 Protein Fold Computational Proof-Preliminary Results*. Version 0.9.4. DOI: 10.5281/zenodo.21717581. This is the current standalone Protein paper at this ToE freeze; a later ToE version must cite any lawfully published Protein successor without backdating it into this record.
18. Smithian Fold Theory V3 repository, `census/claims.json`, current branch papers, claim packages, receipts and source manifests, frozen by the local release manifest and their recorded file identities.

Appendix A. Publication authority and change control

Maria Smith reviewed and authorised this first standalone V3 preliminary version. Editorial agents may correct grammar, formatting and unambiguous clerical mismatches. They may not change a scientific statement, claim status, evidence class, chronology, branch owner, exact value, DOI, licence or publication authority without an authoritative record and Maria Smith's approval.

The release approval dossier must distinguish:

- changes already authorised by current machine and evidence records;
- clerical reconciliations with no scientific effect;

- unresolved discrepancies preserved for Maria Smith;
- files excluded from the release and why;
- tests and renders actually completed; and
- authorised remote actions and their resulting receipts.

The local dossier does not itself perform publication; Maria Smith's separate action-specific authorisation controls the remote transaction.

Appendix B. Machine identity register

The generated machine identity register accompanies this source. It includes the SHA-256 identities and byte sizes of the active papers, manuscript parts, complete claim inventory, scientific audit volumes, corpus inventory, content matrix, render outputs, verification reports and final release package. The local register and approval dossier are authoritative for this version. The reserved version DOI and authorised public repository revision are publication identities; inserting their final transaction receipts is not a scientific revision.

Appendix C. Complete claim inventory

The generated complete claim inventory follows this conceptual monograph as a separate audit appendix and release file. It contains all 2,751 claims in dependency order and exposes each claim's ID, name, branch, formal and external status, closure, dependencies, candidate count, survivor count, control count and receipt identity. The seventeen branch audit volumes retain the remaining claim-level fields and full evidence records required by `publication_guidance.md`.